Multivariate Data Analysis

Lecture Notes (Ch. 1–11)

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Chapter 1. Multivariate Data and the Geometry of the Sample

Data matrices, descriptive statistics, projections, sphering, Mahalanobis distance, eigendecomposition

1 Multivariate Data and the Goals of Analysis

Multivariate (multidimensional) data are measurements of two or more variables on each individual or experimental unit, for example a multiple regression data set with one response and several explanatory variables measured on each individual. The goals of analysis fall into two groups:

(i) **Inference about an unknown population**: understand and infer features of the population from which the data were drawn.

(ii) **Prediction of future observations**: predict the value or category of new observations.

Whatever the goal, the first step is to understand the data themselves and the information or patterns hidden in them. Exploratory data analysis (EDA) assesses the quality of the data, finds potential errors and outliers, and reveals the basic structure; visual exploration then gives the insight needed for further analysis and modelling. Numerical summaries are descriptive statistics (mean vector, covariance matrix, correlation matrix); graphical summaries are scatterplot matrices, principal component plots, heatmaps, correlation plots and trellis graphs (scatterplots or line plots drawn separately for subgroups of the data). After EDA we study the probability structure of multivariate data and statistical inference about it.

Example 1.1: Palmer penguins (slides Example 1.1)

Body measurements of penguins from the Palmer archipelago. Observed variables: year (one of 2007, 2008, 2009); species (Adelie, Chinstrap, Gentoo); island (Biscoe, Dream, Torgersen); sex (female, male); and the four numerical variables bill length, bill depth, flipper length and body mass. For n penguins with p numerical and q categorical variables the data form an $n \times (p+q)$ array; here $n = 333$, $p = 4$, and year may be treated as numerical or categorical. Only the numerical variables go into the data matrix $\mathbf{X}$; a categorical variable of interest, such as species, is kept as a separate vector. This is the typical case $n > p$.
A scatterplot of flipper length against body mass, coloured by sex, shows that heavier penguins tend to have longer flippers. Factors that affect overall size (year, species, island) are not shown in this plot.

Example 1.2: Golub gene expression data (slides Example 1.2)

A representative early example (Golub et al., 1999) of using genetic information to predict disease. For 72 leukemia patients the expression levels of 7129 genes (numerical variables) were measured, together with one categorical variable giving the leukemia type (ALL or AML). This is high-dimensional, low-sample-size data ($p = 7129 \gg n = 72$). Scientific question: can the leukemia type be predicted from gene expression alone?
A heatmap shows a numerical data matrix as colours: in the slides' heatmap the 22 columns are selected genes and the 72 rows are patients, and a colour bar on the left marks ALL or AML. A heatmap of all 7129 variables would be far too wide to read.

2 The Data Matrix and Descriptive Statistics

Motivation. Multivariate data are an $n \times p$ matrix, and almost every method in the course is a statement about this matrix. Writing the sample mean and covariance as matrix products lets us use linear algebra instead of indices.

2.1 Data matrix

Each of the $p \geq 2$ numerical measurements on an item is called a **variable**, since its value varies from item to item. Measurements of p variables on n items form an $n \times p$ matrix

$$\mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} = [\mathbf{x}_{(1)}, \ldots, \mathbf{x}_{(p)}],$$

where x_{ij} is the value of variable j on item i. The ith row $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top \in \mathbb{R}^p$ is the **observation vector** of item i (vectors are column vectors, so the row is $\mathbf{x}_i^\top$); the jth column $\mathbf{x}_{(j)} = (x_{1j}, \ldots, x_{nj})^\top \in \mathbb{R}^n$ is the **variable vector** of variable j. Observations are written as vectors so that multivariate statistics can be computed with matrix algebra, and all variables in $\mathbf{X}$ are taken to be numerical; categorical variables are handled separately.

2.2 Sample mean, covariance and correlation

For univariate data the distribution is summarized by the mean, median, variance, standard deviation, range, skewness and kurtosis. For multivariate data we likewise compute the mean and variance of each variable; since the variables are related, the covariance and correlation of each pair are also key summaries. The covariance measures the direction and size of a linear relation, and the correlation is the standardized covariance, taking values in $[-1, 1]$.

Definition 1.3: descriptive statistics

$$\bar{\mathbf{x}} = \begin{pmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_p \end{pmatrix}, \quad \bar{x}_j = \frac{1}{n} \sum_{i=1}^{n} x_{ij},$$

$$\mathbf{S} = (s_{jk}), \quad s_{jk} = \frac{1}{n-1} \sum_{i=1}^{n} (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k),$$

$$\mathbf{R} = (r_{jk}), \quad r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}}\sqrt{s_{kk}}}, \qquad \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2},$$

with $\mathbf{D} = \operatorname{diag}(s_{11}, \ldots, s_{pp}) = \operatorname{diag}(s_1^2, \ldots, s_p^2)$ and $\mathbf{D}^{-1/2} = (\mathbf{D}^{1/2})^{-1} = \operatorname{diag}(s_1^{-1}, \ldots, s_p^{-1})$. $\mathbf{R}$ is defined only when all sample variances s_j^2 are positive. The textbook also uses $\mathbf{S}_n = \frac{n-1}{n}\mathbf{S}$ (divisor n).

Theorem 1.4: properties of the sample covariance

1. $\mathbf{S}$ is symmetric: $s_{jk} = s_{kj}$ for all $j, k \in \{1, \ldots, p\}$.
2. $\mathbf{S} = \dfrac{1}{n-1} \displaystyle\sum_{i=1}^{n} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$.

3. With the centered data matrix $\mathbf{X}_C = (x_{ij} - \bar{x}_j)_{ij}$, $\mathbf{S} = \frac{1}{n-1}\mathbf{X}_C^\top \mathbf{X}_C$.

Proof. (1) The defining sum is symmetric in j and k.

(2) The (j,k) entry of $(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$ is $(x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$.

(3) The ith row of $\mathbf{X}_C$ is $(\mathbf{x}_i - \bar{\mathbf{x}})^\top$, and $\mathbf{X}_C^\top \mathbf{X}_C = \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$. ■

Remark 1.5: a Gaussian with covariance S always exists

$\mathbf{S}$ is symmetric and positive semidefinite, so a symmetric square root $\mathbf{S}^{1/2}$ exists (§5). With $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\mathbf{X} = \boldsymbol{\mu} + \mathbf{S}^{1/2}\mathbf{Z}$, $\mathrm{E}\mathbf{X} = \boldsymbol{\mu}$ and $\mathrm{Cov}\mathbf{X} = \mathbf{S}^{1/2}\mathbf{I}_p\mathbf{S}^{1/2} = \mathbf{S}$. So there is always a Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\mathbf{S}$.

2.3 The centering matrix

Let $\mathbf{1}_n = (1, \ldots, 1)^\top$ be the vector of n ones, $\mathbf{0}_n$ the zero vector, $\mathbf{I}_n$ the $n \times n$ identity, $\mathbf{J}_n = \mathbf{1}_n \mathbf{1}_n^\top$ the $n \times n$ matrix of ones, and $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{J}_n$.

Theorem 1.6: matrix forms of $\bar{\mathbf{x}}$ and S (slides Exercise 1.1)

$$\bar{\mathbf{x}} = \tfrac{1}{n}\mathbf{X}^\top \mathbf{1}_n, \qquad \mathbf{X}_C = \mathbf{X} - \mathbf{1}_n \bar{\mathbf{x}}^\top = \mathbf{X} - \tfrac{1}{n}\mathbf{1}_n \mathbf{1}_n^\top \mathbf{X} = \mathbf{H}_n \mathbf{X},$$
$$\mathbf{S} = \tfrac{1}{n-1}\mathbf{X}^\top \mathbf{H}_n \mathbf{X} = \tfrac{1}{n-1}\mathbf{X}_C^\top \mathbf{X}_C,$$

and $\mathbf{H}_n^\top = \mathbf{H}_n$, $\mathbf{H}_n^2 = \mathbf{H}_n$.

Proof. The jth entry of $\frac{1}{n}\mathbf{X}^\top \mathbf{1}_n$ is $\frac{1}{n}\sum_i x_{ij} = \bar{x}_j$, giving the first identity. Hence $\mathbf{X}_C = \mathbf{X} - \mathbf{1}_n \bar{\mathbf{x}}^\top = \mathbf{X} - \frac{1}{n}\mathbf{1}_n \mathbf{1}_n^\top \mathbf{X} = \mathbf{H}_n \mathbf{X}$. Symmetry is clear, and idempotency follows from $\mathbf{1}_n^\top \mathbf{1}_n = n$:

$$\left(\mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n\right)^2 = \mathbf{I}_n - \tfrac{2}{n}\mathbf{J}_n + \tfrac{1}{n^2}\mathbf{1}_n(\mathbf{1}_n^\top \mathbf{1}_n)\mathbf{1}_n^\top = \mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n.$$

Therefore $\mathbf{X}_C^\top \mathbf{X}_C = \mathbf{X}^\top \mathbf{H}_n^\top \mathbf{H}_n \mathbf{X} = \mathbf{X}^\top \mathbf{H}_n \mathbf{X}$, and $\mathbf{X}_C^\top \mathbf{X}_C = \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top = (n-1)\mathbf{S}$. ■

[Supplement] A symmetric idempotent matrix has eigenvalues 0 or 1, so $\mathrm{rank}(\mathbf{H}_n) = \mathrm{tr}(\mathbf{H}_n) = n - \frac{1}{n}\mathrm{tr}(\mathbf{J}_n) = n - 1$: $\mathbf{H}_n$ is the orthogonal projection onto the $(n-1)$-dimensional subspace $\{\mathbf{v} \in \mathbb{R}^n : \mathbf{1}_n^\top \mathbf{v} = 0\}$.

3 Sample Geometry in p-Space

3.1 The data ellipse

Each $\mathbf{x}_i$ is a point in Euclidean space $\mathbb{R}^p$, and the points can be shown in a scatterplot. For bivariate data ($p = 2$) the scatterplot shows the relation between the two variables and their distribution; in the slides' penguin plot of bill length against bill depth, the three species form visibly separate clouds, with marginal histograms on the sides.

The mean vector is the center of gravity of the points (in the penguin plot the group mean vectors are marked by triangles) and the covariance matrix describes their spread. The **data ellipse**

$$\{\mathbf{x} \in \mathbb{R}^p : (\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x} - \bar{\mathbf{x}}) = c\}$$

for a fixed constant $c > 0$ summarizes the location, shape and size of the distribution; for $p \geq 3$ it is an ellipsoid. Its size reflects the spread and its shape the correlations. The penguin plot uses $c = 5.991 = \chi^2_2(0.05)$, one ellipse per species. If the cloud is not elliptical, the mean vector and covariance matrix alone describe it poorly.

Example 1.7: data ellipses

With mean vector $\mathbf{0}$ and covariance matrices $\mathbf{S}_1 = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}$, $\mathbf{S}_2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, draw the data ellipses and check that they reflect the sizes of the variances and covariances.
[Supplement] For $\mathbf{S}_1$ the axes of the ellipse are the coordinate axes with half-length ratio $3:1$. $\mathbf{S}_2$ has eigenvalues $9, 1$ with eigenvectors $(1,1)^\top/\sqrt{2}$, $(1,-1)^\top/\sqrt{2}$, so its ellipse is the same ellipse rotated by $45°$: the positive covariance tilts it along the line $x_2 = x_1$. In both cases the half-axis lengths are $\sqrt{c\lambda_j}$ (§5).

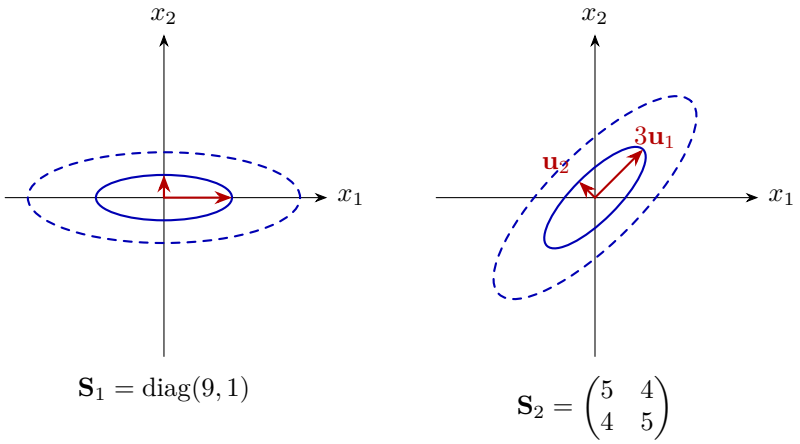


Figure. *Data ellipses* $\mathbf{x}^\top\mathbf{S}^{-1}\mathbf{x} = c$ *for* $c = 1$ *(solid) and* $c = 4$ *(dashed). Both matrices have eigenvalues* 9 *and* 1*, so the ellipses have the same shape; the covariance* 4 *in* $\mathbf{S}_2$ *rotates the axes to* $\mathbf{u}_1 = (1,1)^\top/\sqrt{2}$ *and* $\mathbf{u}_2 = (-1,1)^\top/\sqrt{2}$.

3.2 Subspaces, orthogonal projection and projection scores

The observations $\mathbf{x}_i$ are vectors of the Euclidean space $\mathbb{R}^p$, so transformations and groupings of the observations, and the computation of statistics, can be done with vector operations.

Definition 1.8: subspace, line, basis, affine subspace

- **Subspace**: a line or plane through the origin. For a vector $\mathbf{v}$, the line $\{c\mathbf{v} : c \in \mathbb{R}\}$ is a 1-dimensional subspace of $\mathbb{R}^p$; for linearly independent $\mathbf{v}_1, \ldots, \mathbf{v}_q$, $\mathcal{L}_q = \mathrm{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_q\} = \{\mathbf{V}\mathbf{c} : \mathbf{c} \in \mathbb{R}^q\}$, $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_q]$, is a q-dimensional subspace, said to be spanned by $\mathbf{v}_1, \ldots, \mathbf{v}_q$.
- **Line**: an infinitely extended straight line, as opposed to a line segment, which is a part of it. A flat, infinitely extended surface is a **plane** ($q = 2$) and, for $q = p - 1$, a **hyperplane**.
- **Basis**: the basis $\mathbf{v}_1, \ldots, \mathbf{v}_q$ spanning $\mathcal{L}_q$ is not unique. A basis of mutually orthogonal unit vectors is an **orthonormal basis**.
- **Affine subspace**: lines and planes in Euclidean space, not necessarily through the origin, e.g. the line $\{\mathbf{x} + t\mathbf{v} : t \in \mathbb{R}\}$ through $\mathbf{x}$ in the direction $\mathbf{v}$, or the plane $\{\mathbf{x} +$

$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 : c_1, c_2 \in \mathbb{R}\}$. The affine subspace $\mathcal{L}_q(\mathbf{x}) = \mathbf{x} + \mathcal{L}_q$ is the subspace $\mathcal{L}_q$ translated by $\mathbf{x}$. It is not a vector space in general (it need not contain the origin) but has the same shape as $\mathcal{L}_q$; for every affine subspace there is a parallel subspace of the same dimension. Its representation by $(\mathbf{x}; \mathbf{v}_1, \ldots, \mathbf{v}_q)$ is not unique: affine subspaces generated by different $(\mathbf{x}; \mathbf{v}_1, \ldots, \mathbf{v}_q)$ and $(\mathbf{x}'; \mathbf{v}'_1, \ldots, \mathbf{v}'_q)$ may coincide.

Definition 1.9: orthogonal projection and projection score

The orthogonal projection of $\mathbf{x}$ onto the line $\text{span}\{\mathbf{v}\}$, and the projection score, are

$$P_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}\mathbf{v}^\top}{\mathbf{v}^\top\mathbf{v}}\mathbf{x} = s_{\mathbf{v}}(\mathbf{x})\frac{\mathbf{v}}{\sqrt{\mathbf{v}^\top\mathbf{v}}}, \qquad s_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}^\top\mathbf{x}}{\sqrt{\mathbf{v}^\top\mathbf{v}}},$$

where $\mathbf{v}/\sqrt{\mathbf{v}^\top\mathbf{v}}$ is the unit vector in the direction of $\mathbf{v}$. The $p \times p$ matrix $\mathbf{P}_{\mathbf{v}} = \mathbf{v}\mathbf{v}^\top/(\mathbf{v}^\top\mathbf{v})$ is the **projection matrix**: multiplying any $\mathbf{x}$ by it gives the projection onto $\text{span}\{\mathbf{v}\}$. The score $s_{\mathbf{v}}(\mathbf{x})$ is the distance from the origin to $P_{\mathbf{v}}(\mathbf{x}) \in \text{span}\{\mathbf{v}\}$, with sign: positive in the direction of $\mathbf{v}$, negative in the opposite direction (a signed distance).
The projection onto the subspace $\mathcal{L}_q = \text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_q\}$ is $P_{\mathcal{L}_q}(\mathbf{x}) = \mathbf{P}_{\mathbf{V}}\mathbf{x}$ with $\mathbf{P}_{\mathbf{V}} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$, $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_q]$. If $\mathbf{v}_1, \ldots, \mathbf{v}_q$ are orthonormal ($\mathbf{V}^\top\mathbf{V} = \mathbf{I}_q$),

$$P_{\mathcal{L}_q}(\mathbf{x}) = \mathbf{V}\mathbf{V}^\top\mathbf{x} = s_{\mathbf{v}_1}(\mathbf{x})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x})\mathbf{v}_q,$$

so $\mathbf{V}^\top\mathbf{x} = (s_{\mathbf{v}_1}(\mathbf{x}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}))^\top$ are the coordinates of the projection when the basis vectors are used as coordinate axes of $\mathcal{L}_q$.

- The scatterplot of variables i and j is the projection of the data $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ onto the 2-dimensional subspace spanned by the standard basis vectors $\mathbf{e}_i, \mathbf{e}_j$.
- If different bases span the same subspace, $\text{span}(\mathbf{v}_1, \mathbf{v}_2) = \text{span}(\mathbf{v}'_1, \mathbf{v}'_2)$ with $(\mathbf{v}_1, \mathbf{v}_2) \neq (\mathbf{v}'_1, \mathbf{v}'_2)$, the projection of $\mathbf{x}$ is the same but the scores differ: $(s_{\mathbf{v}_1}(\mathbf{x}), s_{\mathbf{v}_2}(\mathbf{x})) \neq (s_{\mathbf{v}'_1}(\mathbf{x}), s_{\mathbf{v}'_2}(\mathbf{x}))$ in general.

Derivation: why $\mathbf{P}_{\mathbf{V}}$ is the orthogonal projection [Supplement]

$\mathbf{P}_{\mathbf{V}}^\top = \mathbf{P}_{\mathbf{V}}$ and $\mathbf{P}_{\mathbf{V}}^2 = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top\mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top = \mathbf{P}_{\mathbf{V}}$. Also $\mathbf{P}_{\mathbf{V}}\mathbf{x} \in \mathcal{L}_q$, and the residual satisfies $\mathbf{V}^\top(\mathbf{x} - \mathbf{P}_{\mathbf{V}}\mathbf{x}) = \mathbf{0}$, so it is orthogonal to $\mathcal{L}_q$. By Pythagoras, for any $\mathbf{V}\mathbf{c}$, $\|\mathbf{x} - \mathbf{V}\mathbf{c}\|^2 = \|\mathbf{x} - \mathbf{P}_{\mathbf{V}}\mathbf{x}\|^2 + \|\mathbf{P}_{\mathbf{V}}\mathbf{x} - \mathbf{V}\mathbf{c}\|^2 \geq \|\mathbf{x} - \mathbf{P}_{\mathbf{V}}\mathbf{x}\|^2$: $\mathbf{P}_{\mathbf{V}}\mathbf{x}$ is the closest point of $\mathcal{L}_q$ to $\mathbf{x}$.

3.3 Projection onto an affine subspace and decomposition of observations

The affine subspace $\mathcal{L}_q(\bar{\mathbf{x}})$ is $\mathcal{L}_q$ translated by $\bar{\mathbf{x}}$, so the projection of $\mathbf{x}$ onto it is done in two steps (for simplicity let $\mathbf{v}_1, \ldots, \mathbf{v}_q$ be orthonormal):

(1) Translate, then project: translate $\mathbf{x}$ and $\mathcal{L}_q(\bar{\mathbf{x}})$ together by $-\bar{\mathbf{x}}$, $\mathbf{x} \mapsto \mathbf{x} - \bar{\mathbf{x}}$, $\mathcal{L}_q(\bar{\mathbf{x}}) \mapsto \mathcal{L}_q$, and project $\mathbf{x} - \bar{\mathbf{x}}$ onto $\mathcal{L}_q$: $P_{\mathcal{L}_q}(\mathbf{x} - \bar{\mathbf{x}}) = \mathbf{P}_{\mathbf{V}}(\mathbf{x} - \bar{\mathbf{x}}) = s_{\mathbf{v}_1}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_q$.

(2) Translate back by $\bar{\mathbf{x}}$:

$$P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{P}_{\mathbf{V}}(\mathbf{x} - \bar{\mathbf{x}}) = \bar{\mathbf{x}} + s_{\mathbf{v}_1}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_q.$$

The Euclidean distance between $\mathbf{x}$ and $\mathbf{y}$ is the length of $\mathbf{x} - \mathbf{y}$:

$$d(\mathbf{x}, \mathbf{y}) = \Big\{\sum_{i=1}^{p}(x_i - y_i)^2\Big\}^{1/2} = \sqrt{(\mathbf{x} - \mathbf{y})^\top(\mathbf{x} - \mathbf{y})} = \|\mathbf{x} - \mathbf{y}\|.$$

An observation splits into its projection onto an affine subspace and the remainder, and by Pythagoras

$$\mathbf{x} = \underbrace{\bar{\mathbf{x}} + \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x}-\bar{\mathbf{x}})\mathbf{v}_j}_{P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x})} + \underbrace{r_{\mathcal{L}_q}(\mathbf{x})}_{\text{remainder}}, \qquad d(\mathbf{x},\bar{\mathbf{x}})^2 = \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x}-\bar{\mathbf{x}})^2 + \|r_{\mathcal{L}_q}(\mathbf{x})\|^2.$$

This decomposition is used repeatedly when we discuss dimension reduction (PCA).

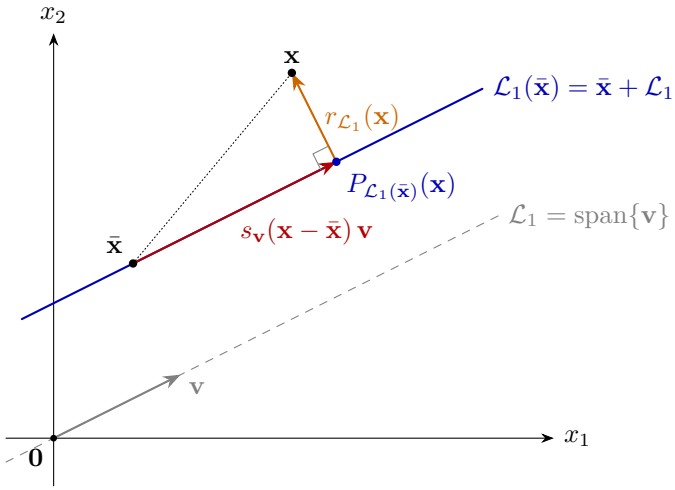


Figure. *Projection onto the affine line $\mathcal{L}_1(\bar{\mathbf{x}})$ with unit direction $\mathbf{v}$: translate by $-\bar{\mathbf{x}}$, project onto the parallel subspace $\mathcal{L}_1$, translate back. The observation splits as $\mathbf{x} = \bar{\mathbf{x}} + s_{\mathbf{v}}(\mathbf{x}-\bar{\mathbf{x}})\mathbf{v} + r_{\mathcal{L}_1}(\mathbf{x})$ with a right angle at the projection, so $d(\mathbf{x},\bar{\mathbf{x}})^2 = s_{\mathbf{v}}(\mathbf{x}-\bar{\mathbf{x}})^2 + \|r_{\mathcal{L}_1}(\mathbf{x})\|^2$ (dotted side).*

Remark 1.10: geometric meaning of variance

Projection scores are related to conditional variances. If $X_1 \sim N(0,\sigma_1^2)$ and $X_2 = X_1 + \epsilon_2$ with independent $\epsilon_2 \sim N(0,\sigma_2^2)$, then

$$\mathrm{Var}(X_2 \mid X_1) = \mathrm{Var}(\epsilon_2) = \sigma_2^2, \qquad \mathrm{Var}(X_1 \mid X_2) = \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2+\sigma_2^2}.$$

[Supplement] The second formula is the conditional-variance formula of Ch. 3, $\mathrm{Var}(X_1 \mid X_2) = \mathrm{Var}(X_1) - \mathrm{Cov}(X_1,X_2)^2/\mathrm{Var}(X_2) = \sigma_1^2 - \sigma_1^4/(\sigma_1^2+\sigma_2^2)$: what remains after removing the part of X_1 explained by (projected on) X_2 is the conditional variance.

3.4 Linear and affine transformations

A $q \times p$ matrix $\mathbf{L}$ defines the **linear transformation** $\mathbf{x} \mapsto \mathbf{L}\mathbf{x} \in \mathbb{R}^q$ of $\mathbf{x} \in \mathbb{R}^p$. In general $q > p$ only increases the dimension without adding information, so it is rarely used in multivariate statistics. Examples:

- the orthogonal projection $\mathbf{P}_{\mathbf{V}}\mathbf{x}$ is the linear transformation $\mathbf{x} \mapsto \mathbf{P}_{\mathbf{V}}\mathbf{x} \in \mathbb{R}^p$ by the $p \times p$ matrix $\mathbf{P}_{\mathbf{V}}$;
- the projection scores $(s_{\mathbf{v}_1}(\mathbf{x}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}))$ are the linear transformation $\mathbf{x} \mapsto \mathbf{S}_{\mathbf{V}}\mathbf{x} \in \mathbb{R}^q$ by the $q \times p$ matrix $\mathbf{S}_{\mathbf{V}} = (\mathbf{v}_1/\|\mathbf{v}_1\|, \ldots, \mathbf{v}_q/\|\mathbf{v}_q\|)^\top$. In the simplest case, $\mathbf{x} = \frac{\mathbf{v}_1\mathbf{v}_1^\top}{\|\mathbf{v}_1\|^2}\mathbf{x} + r(\mathbf{x})$ and $s_{\mathbf{v}_1}(\mathbf{x}) = \mathbf{v}_1^\top\mathbf{x}/\|\mathbf{v}_1\|$.

The projection (or the projection scores) onto an affine subspace is not a linear transformation, because of the translation. It belongs to the similar class of **affine transformations**: for a $q \times p$ matrix $\mathbf{L}$ and $\boldsymbol{m} \in \mathbb{R}^q$, $\mathbf{x} \mapsto \boldsymbol{m} + \mathbf{L}\mathbf{x}$.

Theorem 1.11: affine maps and standardization (slides Exercise 1.2)

1. Orthogonal projection onto $\mathcal{L}_q(\bar{\mathbf{x}})$ is an affine transformation, with $\boldsymbol{m} = \bar{\mathbf{x}} - \mathbf{P}_{\mathbf{V}}\bar{\mathbf{x}}$ and $\mathbf{L} = \mathbf{P}_{\mathbf{V}} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$; so is the score map $\mathbf{x} \mapsto (s_{\mathbf{v}_1}(\mathbf{x}-\bar{\mathbf{x}}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}-\bar{\mathbf{x}}))^\top$.
2. (Standardization) Let the data have mean vector $\bar{\mathbf{x}}$, covariance matrix $\mathbf{S}$ and $\mathbf{D} = \operatorname{diag}(s_{11}, \ldots, s_{pp})$. The standardized observations $\mathbf{x}_i' = \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}})$ have sample mean $\mathbf{0}$ and all sample variances equal to 1, and $\mathbf{x} \mapsto \mathbf{D}^{-1/2}(\mathbf{x} - \bar{\mathbf{x}})$ is an affine transformation.

Proof. (1) $P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top(\mathbf{x} - \bar{\mathbf{x}})$, so $\boldsymbol{m} = \bar{\mathbf{x}} - \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top\bar{\mathbf{x}}$ and $\mathbf{L} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$. The score vector is $\mathbf{S}_{\mathbf{V}}\mathbf{x} - \mathbf{S}_{\mathbf{V}}\bar{\mathbf{x}}$, also affine.

(2) The mean of the transformed data is $\bar{\mathbf{x}}' = \frac{1}{n}\sum_i \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}}) = \mathbf{D}^{-1/2}(\frac{1}{n}\sum_i(\mathbf{x}_i - \bar{\mathbf{x}})) = \mathbf{D}^{-1/2} \cdot \mathbf{0} = \mathbf{0}$. The sample variance of variable j is $S'_{jj} = \operatorname{Var}(s_{jj}^{-1/2}(x_j - \bar{x}_j)) = s_{jj}^{-1/2} s_{jj} s_{jj}^{-1/2} = 1$. Finally $\mathbf{x}' = \mathbf{D}^{-1/2}\mathbf{x} - \mathbf{D}^{-1/2}\bar{\mathbf{x}} = \mathbf{L}\mathbf{x} + \boldsymbol{m}$. ■

3.5 Rotation of coordinates

A linear transformation often used in multivariate statistics is the **linear transformation induced by a change of coordinates**. The usual coordinates of $\mathbb{R}^p$ are given by the standard basis $(\mathbf{e}_1, \ldots, \mathbf{e}_p)$ of mutually orthogonal unit vectors. For $p = 2$, rotating the standard coordinate axes gives new coordinates: for an angle θ (radians) let

$$\mathbf{R}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \mathbf{v}_1 = \mathbf{R}_\theta\mathbf{e}_1, \quad \mathbf{v}_2 = \mathbf{R}_\theta\mathbf{e}_2,$$

the basis rotated counterclockwise by θ. Then $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2]$ satisfies $\mathbf{V}^\top\mathbf{V} = \mathbf{V}\mathbf{V}^\top = \mathbf{R}_\theta\mathbf{R}_\theta^\top = \mathbf{I}_2$. An observation $\mathbf{x} = (x_1, x_2)^\top$ is

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 = [\mathbf{e}_1, \mathbf{e}_2]\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \qquad \mathbf{x} = \mathbf{V}\mathbf{V}^\top\mathbf{x} = [\mathbf{v}_1, \mathbf{v}_2]\begin{pmatrix} \mathbf{v}_1^\top\mathbf{x} \\ \mathbf{v}_2^\top\mathbf{x} \end{pmatrix}.$$

The coordinates of $\mathbf{x}$ are (x_1, x_2) in the standard system and $(\mathbf{v}_1^\top\mathbf{x}, \mathbf{v}_2^\top\mathbf{x})$ in the new system $(\mathbf{v}_1, \mathbf{v}_2)$; the latter are the projection scores onto $\operatorname{span}(\mathbf{v}_1)$ and $\operatorname{span}(\mathbf{v}_2)$. In $\mathbb{R}^p$ one may use any orthonormal basis $(\mathbf{v}_1, \ldots, \mathbf{v}_p)$ instead of $(\mathbf{e}_1, \ldots, \mathbf{e}_p)$: a linear map by an orthonormal basis is a rotation (change) of coordinates.

4 Standardization, Sphering and Mahalanobis Distance

4.1 Univariate standardization and its limitation

For univariate data (see the data-ellipse discussion above): how far is a point $x \in \mathbb{R}$ from the center μ of the normal distribution $N(\mu, \sigma^2)$? The **standardized distance** $|x - \mu|/\sigma$ is more meaningful than the difference $x - \mu$. For univariate data $x_1, \ldots, x_n$ with mean $\bar{x}$ and standard deviation s, the **standardization**

$$x_i \mapsto \frac{x_i - \bar{x}}{s}$$

gives data with mean 0 and standard deviation 1.

Standardization of multivariate data makes the mean and variance of every variable 0 and 1: with $\mathbf{D} = \operatorname{diag}(s_1^2, \ldots, s_p^2)$, the naive transformation is $\mathbf{x} \mapsto \mathbf{D}^{-1/2}(\mathbf{x} - \bar{\mathbf{x}})$. It is used often, to keep results from depending on the units of measurement. But unlike the univariate case, standardization alone **does not give a distance that reflects the distribution**, because it ignores the correlations.

Example 1.12: distances of points in bivariate data (slides Example 1.3)

For bivariate data with $\bar{\mathbf{x}} = \mathbf{0}$, which of the points $\mathbf{y} = (-500, 10)^\top$ and $\mathbf{z} = (1500, 15)^\top$ is farther from the center? The Euclidean distances are $d(\bar{\mathbf{x}}, \mathbf{y}) = \sqrt{500^2 + 10^2} = 500.1$ and $d(\bar{\mathbf{x}}, \mathbf{z}) = \sqrt{1500^2 + 15^2} = 1500.075$, so $\mathbf{z}$ is farther. Yet in the scatterplot there are many observations around $\mathbf{z}$ and almost none around $\mathbf{y}$: relative to the distribution, $\mathbf{y}$ looks farther from the center.

The sample covariance is $\mathbf{S} = \begin{pmatrix} 640000 & 9600 \\ 9600 & 225 \end{pmatrix} = \begin{pmatrix} 800^2 & 800 \cdot 15 \cdot 0.8 \\ 800 \cdot 15 \cdot 0.8 & 15^2 \end{pmatrix}$ (correlation 0.8). After standardization $\mathbf{y}' = \mathbf{D}^{-1/2}(\mathbf{y} - \mathbf{0}) = (-0.625, 0.667)^\top$ and $\mathbf{z}' = \mathbf{D}^{-1/2}(\mathbf{z} - \mathbf{0}) = (1.875, 1.000)^\top$, and still $d(\mathbf{0}, \mathbf{y}') = 0.91 < 2.125 = d(\mathbf{0}, \mathbf{z}')$: the standardized $\mathbf{y}'$ is closer to the center. Each variable now has standard deviation 1, but the standard deviation in the direction of $\mathbf{y}'$ is $sd(s_{\mathbf{y}'}(\mathbf{X}')) = 0.36$ and in the direction of $\mathbf{z}'$ it is $sd(s_{\mathbf{z}'}(\mathbf{X}')) = 1.31$, not 1. Standardization does not standardize the spread in an arbitrary direction.

4.2 Sphering

The spread of the data is described by the covariance matrix $\mathbf{S}$ and the corresponding ellipse (ellipsoid). If the covariance is the identity $\mathbf{I}_p$, the ellipse is the set of $\mathbf{x} \in \mathbb{R}^p$ with $c = (\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{I}_p^{-1}(\mathbf{x} - \bar{\mathbf{x}}) = \|\mathbf{x} - \bar{\mathbf{x}}\|^2$: the distribution is shown as a sphere, and the spread is 1 in **every** direction $\mathbf{v}$. A **sphering transformation** is a transformation after which the covariance matrix is the identity.

Theorem 1.13: isotropy of sphered data (slides homework)

If the data $\mathbf{X}$ have mean vector $\mathbf{0}$ and covariance matrix $\mathbf{I}_p$, then for any unit vector $\mathbf{v}$ the scores $S_{\mathbf{v}}(\mathbf{X}) = [S_{\mathbf{v}}(\mathbf{x}_1), \ldots, S_{\mathbf{v}}(\mathbf{x}_n)]$, $S_{\mathbf{v}}(\mathbf{x}_i) = \mathbf{v}^\top \mathbf{x}_i$, have sample variance 1.

Proof. The scores have mean $\mathbf{v}^\top \bar{\mathbf{x}} = 0$ and sample variance $\frac{1}{n-1}\sum_i (\mathbf{v}^\top \mathbf{x}_i)^2 = \mathbf{v}^\top \mathbf{S} \mathbf{v} = \mathbf{v}^\top \mathbf{I}_p \mathbf{v} = \|\mathbf{v}\|^2 = 1$. ■

Example 1.14: sphering a diagonal covariance (slides Example 1.5)

Let $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n]^\top \in \mathbb{R}^{n \times 2}$ have $\bar{\mathbf{x}} = (10, 10)^\top$ and $\mathbf{S} = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$. The standardization matrix that makes each variance 1 is $\mathbf{D}^{-1/2} = \mathrm{diag}(1/2, 1/3)$, and $\mathbf{x} \mapsto \mathbf{D}^{-1/2}\mathbf{x}$ is the sphering transformation of these data: the transformed covariance is

$$\mathbf{S}' = \frac{1}{n-1}\sum_{i=1}^{n} (\mathbf{D}^{-1/2}\mathbf{x}_i - \mathbf{D}^{-1/2}\bar{\mathbf{x}})(\mathbf{D}^{-1/2}\mathbf{x}_i - \mathbf{D}^{-1/2}\bar{\mathbf{x}})^\top$$

$$\mathbf{S}' = \frac{1}{n-1}\sum_{i=1}^{n} \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{D}^{-1/2} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} = \mathbf{D}^{-1/2}\mathbf{D}\mathbf{D}^{-1/2} = \mathbf{I}_2.$$

Sphering turns the elliptical covariance structure into a sphere.

Definition 1.15: sphering for a general covariance matrix

For a general $\mathbf{S}$, first rotate the coordinates so that all pairwise sample correlations become 0, then standardize. Given $\mathbf{X}$ with covariance $\mathbf{S}$, the sphering transformation is built in three steps.

1. **Rotation**: choose an orthonormal basis matrix $\mathbf{U}_S = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ such that the rotated observations $\mathbf{x}'_i = \mathbf{U}_S^\top \mathbf{x}_i$ $(i = 1, \dots, n)$ have a diagonal sample covariance $\mathbf{D}_S = (d_{jk})$ $(d_{jk} = 0$ for $j \neq k)$, i.e. $\mathbf{D}_S = \mathbf{U}_S^\top \mathbf{S} \mathbf{U}_S = \operatorname{diag}(d_{11}, \dots, d_{pp})$.
2. **Standardization**: with $\mathbf{D}_S^{-1/2} = \operatorname{diag}(d_{11}^{-1/2}, \dots, d_{pp}^{-1/2})$, apply $\mathbf{x}' \mapsto \mathbf{D}_S^{-1/2} \mathbf{x}'$ to the rotated observations.
3. **Back-rotation**: rotate back by the inverse of the first rotation, $\mathbf{x} \mapsto \mathbf{U}_S \mathbf{x}$.

Applying the three steps in turn gives the sphering transformation for $\mathbf{S}$:

$$\mathbf{x} \mapsto \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \mathbf{x}.$$

[Supplement] The sphered covariance is $\mathbf{I}_p$. By the first step $\mathbf{S} = \mathbf{U}_S \mathbf{D}_S \mathbf{U}_S^\top$, so

$$\begin{aligned}
\mathbf{S}' &= \tfrac{1}{n-1} \textstyle\sum_i \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \\
&= \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \, \mathbf{S} \, \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \\
&= \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top = \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{D}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top = \mathbf{U}_S \mathbf{U}_S^\top = \mathbf{I}_p.
\end{aligned}$$

■

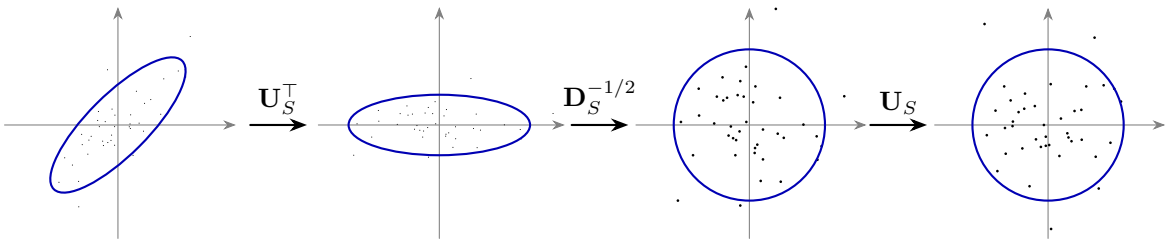
Example 1.16: applying the sphering transformation (slides Example 1.6)

Find the sphering transformation for data $\mathbf{X} \in \mathbb{R}^{n \times 2}$ with $\bar{\mathbf{x}} = \mathbf{0}$ and $\mathbf{S} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$.

First rotate with the orthonormal basis matrix $\mathbf{U}_S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$, for which the rotated observations have diagonal sample covariance $\mathbf{D}_S = \operatorname{diag}(9, 1)$:

$$\mathbf{U}_S^\top \mathbf{S} \mathbf{U}_S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{D}_S;$$

then standardize and rotate back. Note that $\mathbf{U}_S = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ with $\theta = \pi/4$. Hence the sphering transformation is $\mathbf{x} \mapsto \mathbf{U}_S \operatorname{diag}(1/3, 1) \mathbf{U}_S^\top \mathbf{x}$.



(a) data, cov $\mathbf{S}$ (b) rotated, cov $\operatorname{diag}(9, 1)$ (c) standardized, cov $\mathbf{I}_2$ (d) rotated back, cov $\mathbf{I}_2$

Figure. *The three steps of sphering for* $\mathbf{S}$ *of Example 1.16, applied to 40 points with exactly this sample covariance (computed in Python): (a) original data with ellipse* $\mathbf{x}^\top \mathbf{S}^{-1} \mathbf{x} = 4$*; (b) rotation by* $45°$ *clockwise,* $\mathbf{x} \mapsto \mathbf{U}_S^\top \mathbf{x}$*; (c) standardization; (d) rotation back by* $45°$ *counterclockwise. The overall map is* $\mathbf{S}^{-1/2} = \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top$ *and the ellipse becomes a circle.*

Applied with the sample covariance matrix of the data of Example 1.12, the sphering transformation shows the "outlier-looking" point $\mathbf{y}$ as farther from the center than $\mathbf{z}$: the Euclidean distances of

the sphered points $\mathbf{y}'$, $\mathbf{z}'$ to the sphered center $\bar{\mathbf{x}}'$ are

$$\|\mathbf{y}' - \bar{\mathbf{x}}'\| = 2.65 > 2.18 = \|\mathbf{z}' - \bar{\mathbf{x}}'\|.$$

[Supplement] These two numbers come from the covariance matrix of the plotted sample. With the rounded matrix $\mathbf{S}$ quoted in Example 1.12, the same computation gives 2.04 and 2.05 (a Python check): $\mathbf{y}$, a third as far as $\mathbf{z}$ in Euclidean distance, is then as far as $\mathbf{z}$ in Mahalanobis distance. Either way, the distribution-based distance reverses the misleading Euclidean comparison.

4.3 Mahalanobis distance

A distance that takes the full information in the covariance matrix into account is the **Mahalanobis distance**, an example of a statistical distance that reflects the distribution.

Definition 1.17: Mahalanobis (statistical) distance

For a distribution with mean vector $\bar{\mathbf{x}}$ and covariance matrix $\mathbf{S}$, the Mahalanobis distance of a point $\mathbf{y}$ from the mean $\bar{\mathbf{x}}$ is the Euclidean distance between the sphered points $\mathbf{y}' = \mathbf{U}_S\mathbf{D}_S^{-1/2}\mathbf{U}_S^\top\mathbf{y}$ and $\bar{\mathbf{x}}' = \mathbf{U}_S\mathbf{D}_S^{-1/2}\mathbf{U}_S^\top\bar{\mathbf{x}}$:

$$d_S(\mathbf{y}, \bar{\mathbf{x}}) = \|\mathbf{y}' - \bar{\mathbf{x}}'\| = \|\mathbf{U}_S\mathbf{D}_S^{-1/2}\mathbf{U}_S^\top(\mathbf{y} - \bar{\mathbf{x}})\|.$$

Theorem 1.18: quadratic form of the Mahalanobis distance

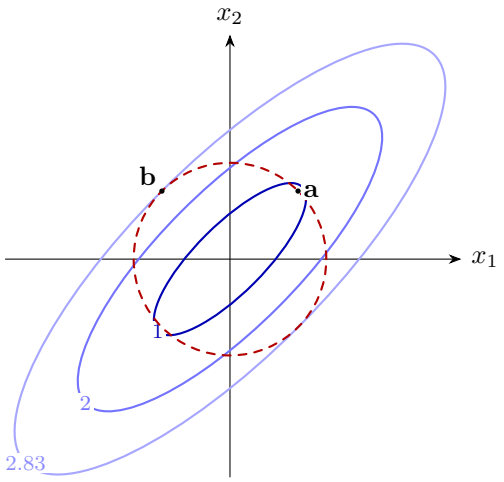
$d_S(\mathbf{y}, \bar{\mathbf{x}})^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top\mathbf{S}^{-1}(\mathbf{y} - \bar{\mathbf{x}})$, and the sphering transformation is $\mathbf{x} \mapsto \mathbf{S}^{-1/2}\mathbf{x}$.

Proof. Using the orthogonality of $\mathbf{U}_S$,

$$d_S(\mathbf{y}, \bar{\mathbf{x}})^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top\mathbf{U}_S\mathbf{D}_S^{-1/2}\underbrace{\mathbf{U}_S^\top\mathbf{U}_S}_{\mathbf{I}_p}\mathbf{D}_S^{-1/2}\mathbf{U}_S^\top(\mathbf{y} - \bar{\mathbf{x}}) = (\mathbf{y} - \bar{\mathbf{x}})^\top\mathbf{U}_S\mathbf{D}_S^{-1}\mathbf{U}_S^\top(\mathbf{y} - \bar{\mathbf{x}}).$$

More simply, $\mathbf{U}_S\mathbf{D}_S^{-1}\mathbf{U}_S^\top = \mathbf{S}^{-1}$, which gives $d_S(\mathbf{y}, \bar{\mathbf{x}})^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top\mathbf{S}^{-1}(\mathbf{y} - \bar{\mathbf{x}})$. Also $\mathbf{U}_S\mathbf{D}_S^{-1/2}\mathbf{U}_S^\top = \mathbf{S}^{-1/2}$, so the sphering transformation can be written $\mathbf{x} \mapsto \mathbf{S}^{-1/2}\mathbf{x}$. ■

The multivariate counterparts of univariate standardization (studentisation) and of the standardized distance are the sphering transformation and the Mahalanobis distance.



$\mathbf{a} = (2, 2)$: $d_S(\mathbf{a}, \mathbf{0}) = 0.94$
$\mathbf{b} = (-2, 2)$: $d_S(\mathbf{b}, \mathbf{0}) = 2.83$
dashed: $\|\mathbf{x}\| = 2.83$
solid: $d_S(\mathbf{x}, \mathbf{0}) = 1, 2, 2.83$

Figure. *Euclidean versus Mahalanobis distance for* $\bar{\mathbf{x}} = \mathbf{0}$ *and* $\mathbf{S} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. *The points* $\mathbf{a}$ *and* $\mathbf{b}$ *lie on the same Euclidean circle (dashed), but* $\mathbf{a}$ *lies along the long axis of the data ellipses* $d_S = 1, 2, 2.83$ *and* $\mathbf{b}$ *against the correlation, so* $d_S(\mathbf{a}, \mathbf{0}) = 0.94$ *while* $d_S(\mathbf{b}, \mathbf{0}) = 2.83$.

5 Eigendecomposition and Positive Definiteness

Motivation. A covariance matrix is symmetric and positive semidefinite, so it has an orthonormal eigenbasis. The eigenvectors are the natural axes of the data cloud and the eigenvalues the variances along them; PCA, Mahalanobis distance and the normal density all rest on this decomposition.

5.1 Eigendecomposition

For a symmetric $\mathbf{A} \in \mathbb{R}^{p\times p}$, a nonzero $\mathbf{u} \in \mathbb{R}^p$ satisfying $\mathbf{Au} = \lambda\mathbf{u}$ is an **eigenvector** and the scalar $\lambda \in \mathbb{R}$ an **eigenvalue**. The left side is the linear transformation of $\mathbf{u}$ by $\mathbf{A}$, and the right side only stretches or shrinks $\mathbf{u}$ by λ: the eigenvectors of $\mathbf{A}$ keep their direction under $\mathbf{A}$, and the eigenvalues give the factor of stretching.

Example 1.19: eigenstructure of the sphering matrix (slides Example 1.7)

Check the eigendecomposition on the sphering matrix $\mathbf{A} = \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{pmatrix}$ of Example 1.16: applying it to the vectors $\mathbf{u}_1 = (1,1)^\top/\sqrt{2}$ and $\mathbf{u}_2 = (-1,1)^\top/\sqrt{2}$ used in the first (rotation) step only rescales their lengths. Indeed

$$\mathbf{Au}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} \frac{2}{3}\cdot 1 + (-\frac{1}{3})\cdot 1 \\ (-\frac{1}{3})\cdot 1 + \frac{2}{3}\cdot 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} = \frac{1}{3}\mathbf{u}_1.$$

[Supplement] Similarly $\mathbf{Au}_2 = \frac{1}{\sqrt{2}}(-\frac{2}{3}-\frac{1}{3},\ \frac{1}{3}+\frac{2}{3})^\top = \mathbf{u}_2$. Sphering rescales the eigen-direction of eigenvalue λ_j of $\mathbf{S}$ by $1/\sqrt{\lambda_j}$ (1/3 and 1 here).

Theorem 1.20: properties of eigenvalues and eigenvectors (slides homework)

Let the symmetric $\mathbf{A} \in \mathbb{R}^{p\times p}$ have p eigenvalues; for convenience assume they are distinct and ordered, $\lambda_1 > \lambda_2 > \cdots > \lambda_p$.

1. If $\mathbf{u}_i$ is an eigenvector for λ_i, so is $c\mathbf{u}_i$ for every $c \neq 0$: eigenvectors are not unique.
2. Among the vectors of length 1 ($\|\mathbf{u}\| = 1$) there are exactly two eigenvectors for λ_i, $\mathbf{u}_i$ and $-\mathbf{u}_i$. Ignoring the sign, the unit eigenvector $\mathbf{u}_i$ for λ_i is unique.
3. Eigenvectors $\mathbf{u}_i, \mathbf{u}_j$ for distinct eigenvalues $\lambda_i \neq \lambda_j$ are orthogonal: $\mathbf{u}_i^\top\mathbf{u}_j = 0$.

Proof of (3). Given $\mathbf{Au}_i = \lambda_i\mathbf{u}_i$ and $\mathbf{Au}_j = \lambda_j\mathbf{u}_j$, compute $\mathbf{u}_i^\top\mathbf{Au}_j$ in two ways. First, $\mathbf{u}_i^\top\mathbf{Au}_j = \mathbf{u}_i^\top(\lambda_j\mathbf{u}_j) = \lambda_j(\mathbf{u}_i^\top\mathbf{u}_j)$. Second, $\mathbf{A}^\top = \mathbf{A}$, so transposing $\mathbf{Au}_i = \lambda_i\mathbf{u}_i$ gives $\mathbf{u}_i^\top\mathbf{A} = \lambda_i\mathbf{u}_i^\top$; multiplying on the right by $\mathbf{u}_j$, $\mathbf{u}_i^\top\mathbf{Au}_j = \lambda_i(\mathbf{u}_i^\top\mathbf{u}_j)$. Hence $\lambda_i(\mathbf{u}_i^\top\mathbf{u}_j) = \lambda_j(\mathbf{u}_i^\top\mathbf{u}_j)$, and since $\lambda_i \neq \lambda_j$, $\mathbf{u}_i^\top\mathbf{u}_j = 0$. ■

Theorem 1.21: eigendecomposition of a symmetric matrix (slides homework)

Let the eigenvalue–eigenvector pairs of the symmetric $\mathbf{A} \in \mathbb{R}^{p\times p}$ be $(\lambda_i, \mathbf{u}_i)$, $i = 1, \ldots, p$, with $\|\mathbf{u}_i\| = 1$. The eigendecomposition of $\mathbf{A}$ is

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top = \sum_{i=1}^{p} \lambda_i\mathbf{u}_i\mathbf{u}_i^\top,$$

where the eigenvector matrix $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ is orthogonal and the eigenvalue matrix $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$ is diagonal.

The most important matrices in multivariate statistics are symmetric covariance matrices $\mathbf{S}$, so we restrict the eigendecomposition to symmetric matrices. It is not unique in general; whether it is depends on the eigenvalues.

- If the p eigenvalues are distinct, $\lambda_1 > \lambda_2 > \cdots > \lambda_p$, the decomposition $\mathbf{U\Lambda U}^\top$ is unique (up to the signs of the eigenvectors: $\mathbf{u}_i$ and $-\mathbf{u}_i$ are the same eigenvector).
- If the eigenvalues are not distinct, an eigendecomposition still exists, but the eigenvector matrix is not unique. For example, $\sigma^2\mathbf{I}_p$ has eigenvalues $\lambda_i = \sigma^2$ for all i, and its eigenvector matrix can be any orthogonal $\mathbf{U}$ ($\mathbf{UU}^\top = \mathbf{I}_p$).
- For a given matrix the eigendecomposition generally has no closed-form solution; it is computed numerically, e.g. by orthogonal iteration.

5.2 Inverse, definiteness, square root

The inverse of a symmetric matrix can be obtained from its eigendecomposition:

$$\mathbf{A}^{-1} = \mathbf{U\Lambda}^{-1}\mathbf{U}^\top, \qquad \mathbf{\Lambda}^{-1} = \operatorname{diag}\Big(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_p}\Big),$$

defined only when no eigenvalue is 0 ($\lambda_i \neq 0$). A symmetric $\mathbf{A}$ whose eigenvalues are all positive ($\lambda_i > 0$) is called **positive definite** (PD); one whose eigenvalues are all non-negative ($\lambda_i \geq 0$) is called **non-negative definite** (NND; positive semi-definite).

Definition 1.22: NND and PD

The definiteness of a matrix $\mathbf{A} \in \mathbb{R}^{p\times p}$ is defined as follows.

1. (NND) If $\mathbf{x}^\top\mathbf{Ax} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{A}$ is non-negative definite.
2. (PD) If $\mathbf{x}^\top\mathbf{Ax} > 0$ for all nonzero $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{A}$ is positive definite.

Theorem 1.23: eigenvalues and definiteness (slides homework)

1. If all eigenvalues of a symmetric $\mathbf{A}$ are positive, $\mathbf{A}$ is PD.
2. If all eigenvalues of a symmetric $\mathbf{A}$ are non-negative, $\mathbf{A}$ is NND.

Proof. We must show $\mathbf{x}^\top\mathbf{Ax} > 0$ for every nonzero $\mathbf{x} \in \mathbb{R}^p$.

Step 1 (eigendecomposition). $\mathbf{A}$ is symmetric, so $\mathbf{A} = \mathbf{U\Lambda U}^\top$ with $\mathbf{U}$ orthogonal and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$.

Step 2 (quadratic form). $\mathbf{x}^\top\mathbf{Ax} = \mathbf{x}^\top\mathbf{U\Lambda U}^\top\mathbf{x}$.

Step 3 (substitution). Let $\mathbf{y} = \mathbf{U}^\top\mathbf{x}$. Since $\mathbf{U}^\top\mathbf{U} = \mathbf{I}$, $\mathbf{y}^\top\mathbf{y} = \mathbf{x}^\top\mathbf{x}$, and $\mathbf{x}^\top\mathbf{Ax} = \mathbf{y}^\top\mathbf{\Lambda y} = \sum_{i=1}^p \lambda_i y_i^2$.

Step 4 (positivity). Since $\lambda_i > 0$ and $y_i^2 \geq 0$, and $\mathbf{x} \neq \mathbf{0}$ implies $\mathbf{y} \neq \mathbf{0}$, at least one $y_i \neq 0$; hence $\sum_i \lambda_i y_i^2 > 0$ and $\mathbf{A}$ is PD.

For (2) the same computation gives $\sum_i \lambda_i y_i^2 \geq 0$. ■

[Supplement] The converses also hold: if $\mathbf{A}$ is PD (NND), then $\lambda_i = \mathbf{u}_i^\top \mathbf{A}\mathbf{u}_i > 0$ (≥ 0), using $\mathbf{A}\mathbf{u}_i = \lambda_i \mathbf{u}_i$ and $\|\mathbf{u}_i\| = 1$. So a symmetric matrix is PD (NND) if and only if all its eigenvalues are positive (non-negative).

- If a symmetric $\mathbf{A}$ is PD, its **inverse exists**. When the sample size n is much larger than the dimension p, the sample covariance matrix $\mathbf{S}$ is PD in "most" cases, so $\mathbf{S}^{-1}$ exists.
- In general the sample covariance matrix is not always PD, but it is always NND.
- An NND matrix that is not PD has a zero eigenvalue, so **its inverse does not exist**.

Theorem 1.24: the sample covariance is always NND, not necessarily PD (slides homework)

Proof. Step 1 (NND). Substituting $\mathbf{S}$, $\mathbf{z}^\top \mathbf{S}\mathbf{z} = \mathbf{z}^\top \Big(\frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \Big) \mathbf{z} = \frac{1}{n-1} \sum_{i=1}^n \{(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{z}\}^2$. Each $(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{z}$ is a scalar, so its square is non-negative and so is the sum: $\mathbf{z}^\top \mathbf{S}\mathbf{z} \geq 0$ for every $\mathbf{z} \in \mathbb{R}^p$.

Step 2 (not necessarily PD). For $\mathbf{S}$ to be PD we would need $\mathbf{z}^\top \mathbf{S}\mathbf{z} > 0$ for all nonzero $\mathbf{z}$. If the data points $\mathbf{x}_1, \ldots, \mathbf{x}_n$ lie in a lower-dimensional subspace of $\mathbb{R}^p$ (e.g. all points on a line or in a plane within $\mathbb{R}^p$), $\mathbf{S}$ has at least one zero eigenvalue, and there is a nonzero $\mathbf{z}$ with $\mathbf{z}^\top \mathbf{S}\mathbf{z} = 0$. So $\mathbf{S}$ is always NND, but not necessarily PD: it can have zero eigenvalues when the points do not spread out in all p dimensions. ■

Definition 1.25: symmetric square root

For NND $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, define $\mathbf{S}^{1/2} = \mathbf{U}\mathbf{\Lambda}^{1/2}\mathbf{U}^\top$, $\mathbf{\Lambda}^{1/2} = \mathrm{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_p})$. The square root $\mathbf{S}^{1/2}$ so defined is symmetric and satisfies $\mathbf{S}^{1/2}\mathbf{S}^{1/2} = \mathbf{S}$. Likewise for symmetric PD $\mathbf{S}$ the negative square root is

$$\mathbf{S}^{-1/2} = (\mathbf{S}^{-1})^{1/2} = \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top, \qquad \mathbf{\Lambda}^{-1/2} = \mathrm{diag}\Big(\frac{1}{\sqrt{\lambda_1}}, \ldots, \frac{1}{\sqrt{\lambda_p}}\Big).$$

5.3 Covariance matrices and ellipses

The shape of a covariance matrix $\mathbf{S}$ can be described by its ellipse $E_S = \{\mathbf{x} \in \mathbb{R}^p : \mathbf{x}^\top \mathbf{S}^{-1} \mathbf{x} = 1\}$. The shape of this ellipse is closely tied to the eigendecomposition: with $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$,

$$\frac{(\mathbf{u}_1^\top \mathbf{x})^2}{\lambda_1} + \cdots + \frac{(\mathbf{u}_p^\top \mathbf{x})^2}{\lambda_p} = 1,$$

so the principal axes of E_S are parallel to the eigenvectors $\mathbf{u}_i$, and the half-length of each axis is the square root $\sqrt{\lambda_i}$ of the eigenvalue. The eigendecomposition of covariance matrices is one of the most important concepts in multivariate statistics; through $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ we have

1. the change to coordinates in which all correlations are 0 ($\mathbf{x} \mapsto \mathbf{U}^\top \mathbf{x}$), and
2. the sphering transformation to covariance $\mathbf{I}_p$ ($\mathbf{x} \mapsto \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top \mathbf{x}$).

Theorem 1.26: covariance of rotated data (slides Discussion)

If the covariance matrix of $\mathbf{X}$ is $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, the rotated data $\mathbf{x}_i' = \mathbf{U}^\top \mathbf{x}_i$, $i = 1, \ldots, n$, have sample covariance matrix equal to the eigenvalue matrix $\mathbf{\Lambda}$. The eigenvalues of a covariance matrix are therefore the variances of mutually uncorrelated rotated coordinates.

[Supplement]. The rotated mean is $\mathbf{U}^\top \bar{\mathbf{x}}$, so the sample covariance is $\frac{1}{n-1}\sum_i \mathbf{U}^\top(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{U} = \mathbf{U}^\top \mathbf{S}\mathbf{U} = \mathbf{U}^\top \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top \mathbf{U} = \mathbf{\Lambda}$. ■

6 Geometry of Principal Component Analysis

To explore data with many variables, displaying and inspecting the scatterplots of all pairs of variables is inefficient. We therefore extract q **principal components** from the data and study the main variation through a few new variables. **Principal component analysis** (PCA) finds, in order, the linear combinations of the original variables that explain the most variation of the data. Here we obtain the principal components from the eigendecomposition of the covariance matrix and study the geometric meaning of projection and dimension reduction and its visualization. PCA is a representative dimension-reduction method and is treated again in Ch. 7.

6.1 Which direction to project on?

Center the observation $\mathbf{x}_i$ at the mean $\bar{\mathbf{x}}$ and project onto a unit vector $\mathbf{a}$; the score is $t_i(\mathbf{a}) = \mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}})$, $\|\mathbf{a}\| = 1$. The sample variance of these projection scores is

$$\frac{1}{n-1}\sum_{i=1}^{n} t_i(\mathbf{a})^2 = \mathbf{a}^\top \mathbf{S}\mathbf{a}.$$

Definition 1.27: principal component directions

The first principal component direction is the direction that maximizes the variance of the projection scores: $\mathbf{u}_1 \in \arg\max_{\|\mathbf{a}\|=1} \mathbf{a}^\top \mathbf{S}\mathbf{a}$. The optimal direction is the eigenvector for the largest eigenvalue of $\mathbf{S}$. Later principal component directions maximize the variance among directions orthogonal to those already found.

[Supplement] $\mathbf{u}_1$ is the top eigenvector. With $\mathbf{z} = \mathbf{U}^\top \mathbf{a}$ ($\|\mathbf{z}\| = 1$), $\mathbf{a}^\top \mathbf{S}\mathbf{a} = \sum_j \lambda_j z_j^2 \le \lambda_1 \sum_j z_j^2 = \lambda_1$, with equality at $\mathbf{a} = \mathbf{u}_1$. Under the constraint $\mathbf{a} \perp \mathbf{u}_1$, i.e. $z_1 = 0$, the maximum is λ_2 at $\mathbf{u}_2$, and so on. ■

6.2 Principal component scores

The ith row of the centered data matrix $\mathbf{X}_C$ is $(\mathbf{x}_i - \bar{\mathbf{x}})^\top$, and the sample covariance is $\mathbf{S} = (n-1)^{-1}\mathbf{X}_C^\top \mathbf{X}_C$. Let $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ with $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ orthogonal and $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$, $\lambda_1 \ge \cdots \ge \lambda_p \ge 0$. Using the eigenvectors as new coordinate axes gives the principal component score matrix

$$\mathbf{T} = \mathbf{X}_C \mathbf{U} = (t_{ij}), \qquad t_{ij} = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{u}_j.$$

Each column of $\mathbf{T}$ has mean 0, and its sample covariance is $\frac{1}{n-1}\mathbf{T}^\top \mathbf{T} = \mathbf{U}^\top \mathbf{S}\mathbf{U} = \mathbf{\Lambda}$. Hence the jth principal component has sample variance λ_j, and different principal component scores are sample-uncorrelated.

Example 1.28: principal components of two test scores

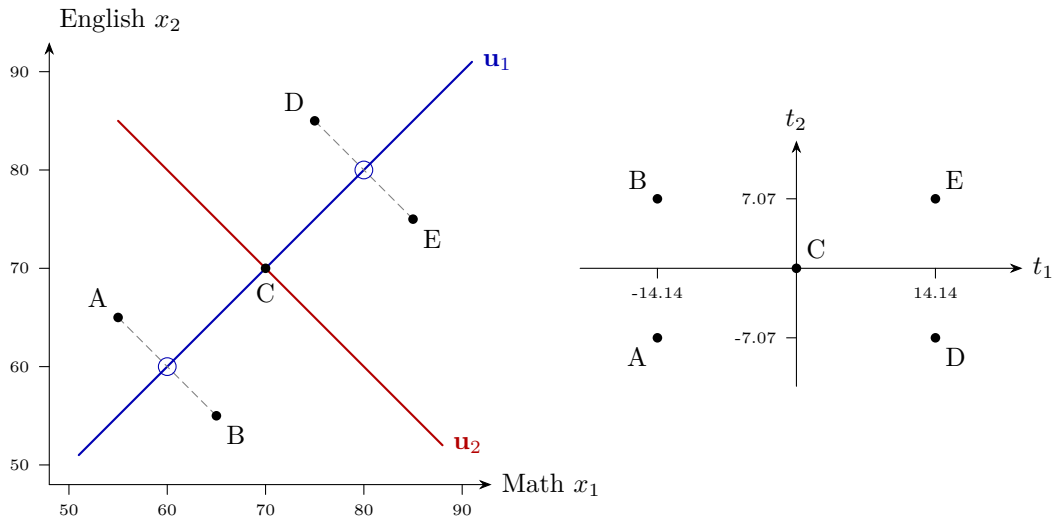
For illustration, take the mathematics and English scores of five students.

Student	Math x_1	English x_2	PC1 score	PC2 score
A	55	65	-14.14	-7.07
B	65	55	-14.14	7.07
C	70	70	0	0
D	75	85	14.14	-7.07
E	85	75	14.14	7.07

The sample mean and covariance are $\bar{\mathbf{x}} = (70, 70)^\top$, $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$, with eigenvalues and unit eigenvectors $\lambda_1 = 200$, $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$, $\lambda_2 = 50$, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$. The scores of each student are

$$t_{i1} = \frac{(x_{i1} - 70) + (x_{i2} - 70)}{\sqrt{2}}, \qquad t_{i2} = \frac{(x_{i1} - 70) - (x_{i2} - 70)}{\sqrt{2}}.$$

The first principal component expresses the **overall level** of the two scores, the second the **relative difference** between mathematics and English. Students A and B have the same first score but different second scores: with the first principal component alone, the difference in their relative strengths cannot be distinguished.



(a) original scores (b) principal component scores

Figure. *(a) The five students with the principal axes through $\bar{\mathbf{x}} = (70, 70)$; dashed segments are the orthogonal projections onto the first axis, and A, B (and D, E) project to the same point. (b) The score plot (t_{i1}, t_{i2}): PC1 (80% of the variance) is the overall level, PC2 (20%) the math–English difference that separates A from B.*

6.3 Dimension reduction and reconstruction error

For student E, $\mathbf{x}_E = (85, 75)^\top$ and the first principal component score is $t_{E1} = 10\sqrt{2}$. Reconstructing with the first principal component only,

$$\hat{\mathbf{x}}_E = \begin{pmatrix} 70 \\ 70 \end{pmatrix} + 10\sqrt{2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 80 \\ 80 \end{pmatrix} :$$

the average level of the two scores is kept, but the difference between the subjects disappears.

Theorem 1.29: reconstruction error

For the reconstruction $\hat{\mathbf{x}}_i = \bar{\mathbf{x}} + \sum_{j \le q} t_{ij}\mathbf{u}_j$ from the first q principal components, the sum of squared reconstruction errors is

$$\sum_{i=1}^{n} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = (n-1) \sum_{j=q+1}^{p} \lambda_j.$$

Among all q-dimensional planes through the mean, PCA chooses the one that minimizes the reconstruction error; maximizing the total retained variance and minimizing the reconstruction error select the same subspace.

[Supplement]. $\mathbf{x}_i - \hat{\mathbf{x}}_i = \sum_{j>q} t_{ij}\mathbf{u}_j$, so $\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = \sum_{j>q} t_{ij}^2$, and $\sum_i t_{ij}^2 = (n-1)\lambda_j$. For any $\mathbf{V} \in \mathbb{R}^{p\times q}$ with orthonormal columns, Pythagoras gives $\sum_i \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{V}\mathbf{V}^\top(\mathbf{x}_i - \bar{\mathbf{x}})\|^2 = (n-1)\{\mathrm{tr}\mathbf{S} - \mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})\}$, so minimizing the error is maximizing $\mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})$, whose maximum $\lambda_1 + \cdots + \lambda_q$ follows from von Neumann's inequality (Ch. 2) or Ch. 7. ■

6.4 Proportion of variance, scale and caveats

- The total sample variance is $\mathrm{tr}(\mathbf{S}) = \sum_{k=1}^{p} \lambda_k$, so the **proportion of variance explained** by the jth principal component is $\lambda_j / \sum_{k=1}^{p} \lambda_k$ and the **cumulative proportion** of the first q components is $\sum_{j=1}^{q} \lambda_j / \sum_{k=1}^{p} \lambda_k$. In the test-score example the first component explains $200/(200+50) = 80\%$. Choose q where the cumulative proportion reaches a chosen level, or from the decline in a **scree plot** of the ordered eigenvalues. For visualization one usually takes $q = 2$ or 3 and checks how much of the total variation these components explain.

- PCA based on the covariance matrix depends on the **units and scales** of the variables: if the English test is scored out of 1000 instead of 100, its variance is multiplied by 100 and the principal directions change. One can instead run PCA on the standardized data $z_{ik} = (x_{ik} - \bar{x}_k)/s_k$, $k = 1, \ldots, p$, $s_k > 0$, whose covariance matrix is the sample correlation matrix of the original data (**correlation PCA**). Standardizing is useful when units are arbitrary, but when differences in variance are themselves important information it may be inappropriate.

- **Does a high explained variance mean good group separation?** Let $Y \in \{-1, 1\}$ with $P(Y = 1) = P(Y = -1) = 1/2$ label two groups, and let $X_1 = 10Z$, $X_2 = Y$, $Z \sim N(0,1)$, $Z \perp\!\!\!\perp Y$. The population covariance and the proportion explained by the first component are $\mathbf{\Sigma} = \begin{pmatrix} 100 & 0 \\ 0 & 1 \end{pmatrix}$, $\frac{\lambda_1}{\lambda_1+\lambda_2} = \frac{100}{101} \approx 99.0\%$. The first component X_1 explains almost all of the variation but is independent of Y, while the second component X_2 alone separates the groups perfectly. PCA looks for directions that preserve much of the variation; a high proportion of variance does not guarantee that the information needed for group separation or prediction is kept.

Example 1.30: PCA of the gene expression data (slides Example 1.8)

Apply PCA to the Golub gene expression data (slides Example 1.2; $p = 7129$, $n = 72$) and draw the scatterplot of the first two principal component scores. Compared with the scatterplot of the first two original variables, the score plot separates ALL and AML much more clearly. PCA does not use the category labels, so this example shows that the principal components can be used to *explore* group differences.

7 Random Vectors and the Geometry of Variable Vectors

Motivation. Viewing each variable as a vector in $\mathbb{R}^n$ gives covariance and correlation a geometric meaning: lengths are standard deviations and angles are correlations. This picture makes projection arguments (regression, partial correlation) transparent.

7.1 Random vectors

In multivariate statistics the data $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n]^\top$ are regarded as a random sample from a population. The population from which multivariate data on p variables are drawn is modelled by the joint distribution of p random variables $X_1, \ldots, X_p$: collecting them in the random vector $\mathbf{x} = (X_1, \ldots, X_p)^\top$, its joint probability density function $f : \mathbb{R}^p \to [0, \infty)$ determines the distribution, written $\mathbf{x} = (X_1, \ldots, X_p)^\top \sim f$.

Example 1.31: distributions of random vectors (slides Example 2.1)

1. If $X_1, \ldots, X_p$ are independent with $X_i \sim \text{Unif}(a_i, b_i)$, the joint density is $f(x_1, \ldots, x_p) = \prod_{i=1}^p \frac{1}{b_i - a_i} \mathbf{1}\{x_i \in (a_i, b_i)\}$, where $b_i > a_i$ and $\mathbf{1}_{x \in (a,b)}$ is the indicator function: $\mathbf{x}$ is uniform on the hypercube $(a_1, b_1) \times \cdots \times (a_p, b_p)$.

2. If $X_1, \ldots, X_p$ are independent with $X_i \sim N(0, 1)$, the joint density is $f(x_1, \ldots, x_p) = \prod_i \frac{1}{\sqrt{2\pi}} e^{-x_i^2/2} = (2\pi)^{-p/2} e^{-(x_1^2 + \cdots + x_p^2)/2}$, the **standard multivariate normal distribution**.

3. If $(X_1, X_2)^\top$ is uniform on the triangle $0 < x_1 < x_2 < 1$, the joint density is $f(x_1, x_2) = 2 \cdot \mathbf{1}\{0 < x_1 < x_2 < 1\}$ (the triangle has area $1/2$, so the density is 2). Then $X_1 \mid X_2 = x_2 \sim \text{Unif}(0, x_2)$, $0 < x_2 < 1$: the conditional distribution of X_1 depends on the value of X_2, so the two components are not independent.

4. An example of a bivariate normal distribution with correlation ρ, $0 < |\rho| < 1$, has density $f(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1-\rho^2)}\}$, $(x_1, x_2) \in \mathbb{R}^2$. Each marginal is $X_1 \sim N(0, 1)$, $X_2 \sim N(0, 1)$, but the joint density does not factor into the product of the marginals, so the two components are not independent.

If n random vectors $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are a simple random sample from a population with density f, they are mutually independent and identically distributed (**i.i.d.**), written $\mathbf{x}_1, \ldots, \mathbf{x}_n \overset{\text{iid}}{\sim} f$.

Definition 1.32: moments of a random vector

For a random vector $\mathbf{X} = (X_1, \ldots, X_p)^\top$ (and a random vector $\mathbf{Y}$):

$$\boldsymbol{\mu} = \mathrm{E}(\mathbf{X}) = \begin{pmatrix} \mathrm{E}(X_1) \\ \vdots \\ \mathrm{E}(X_p) \end{pmatrix}, \qquad \mathrm{Cov}(\mathbf{X}) = \mathrm{E}\{(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top\} = \mathrm{E}(\mathbf{X}\mathbf{X}^\top) - \mathrm{E}(\mathbf{X})\mathrm{E}(\mathbf{X})^\top,$$

$$\mathrm{Cov}(\mathbf{X}, \mathbf{Y}) = \mathrm{E}\{(\mathbf{X} - \mathrm{E}\mathbf{X})(\mathbf{Y} - \mathrm{E}\mathbf{Y})^\top\} = \mathrm{Cov}(\mathbf{Y}, \mathbf{X})^\top.$$

For $\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{X}) \succ 0$ the Mahalanobis distance of $\mathbf{x}$ from $\boldsymbol{\mu}$ is $d_{\boldsymbol{\Sigma}}(\mathbf{x}, \boldsymbol{\mu}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}$. The **generalized variance** is $|\mathrm{Cov}(\mathbf{X})|$ and the **total variance** is $\mathrm{tr}\{\mathrm{Cov}(\mathbf{X})\}$; their sample versions are $|\mathbf{S}|$ and $\mathrm{tr}(\mathbf{S}) = \sum_j \lambda_j$.

Theorem 1.33: characterization of covariance matrices

$\boldsymbol{\Sigma}$ is the covariance matrix of some random vector if and only if $\boldsymbol{\Sigma}$ is symmetric and NND.

Proof. ($\Rightarrow$) [Supplement] $\mathrm{Cov}(\mathbf{X})^\top = \mathrm{Cov}(\mathbf{X})$, and for every $\mathbf{a}$, $\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a} = \mathrm{E}\{\mathbf{a}^\top(\mathbf{X}-\boldsymbol{\mu})\}^2 = \mathrm{Var}(\mathbf{a}^\top\mathbf{X}) \geq 0$.

($\Leftarrow$) As in the Remark on the existence of a Gaussian with covariance $\mathbf{S}$: $\mathbf{X} = \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$ with $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ has $\mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Sigma}$. ■

7.2 Mean, variance and correlation through variable vectors

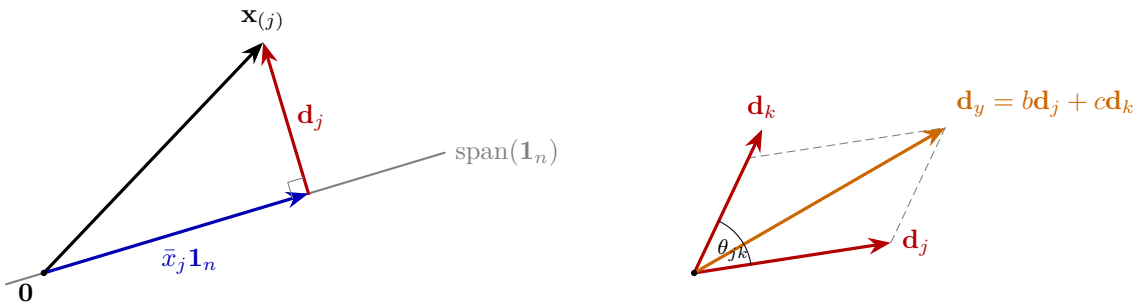
Using the variable vectors $\mathbf{x}_{(j)} \in \mathbb{R}^n$, that is, viewing the data as p vectors in n-dimensional space, gives a geometric reading of the statistics.

- **Mean**: the sample mean of variable j is $\bar{x}_j = \frac{1}{n}\sum_{i=1}^n x_{ij} = \frac{1}{n}\mathbf{1}_n^\top\mathbf{x}_{(j)} = (\mathbf{1}_n^\top\mathbf{1}_n)^{-1}\mathbf{1}_n^\top\mathbf{x}_{(j)}$, the coefficient of the orthogonal projection of $\mathbf{x}_{(j)}$ onto $\mathbf{1}_n$: $P_{\mathbf{1}_n}(\mathbf{x}_{(j)}) = \mathbf{1}_n(\mathbf{1}_n^\top\mathbf{1}_n)^{-1}\mathbf{1}_n^\top\mathbf{x}_{(j)} = \bar{x}_j\mathbf{1}_n$. The projection score of $\mathbf{x}_{(j)}$ onto $\mathrm{span}(\mathbf{1}_n)$ is $s_{\mathbf{1}}(\mathbf{x}_{(j)}) = \sqrt{n}\,\bar{x}_j$.
- **Deviation vector**: the deviation of $\mathbf{x}_{(j)}$ from its "mean vector" $\bar{x}_j\mathbf{1}_n$ is $\mathbf{d}_j = \mathbf{x}_{(j)} - \bar{x}_j\mathbf{1}_n = (x_{1j}-\bar{x}_j, \ldots, x_{nj}-\bar{x}_j)^\top$, and $\mathbf{x}_{(j)} = \bar{x}_j\mathbf{1}_n + \mathbf{d}_j$ is an orthogonal decomposition ($\mathbf{1}_n^\top\mathbf{d}_j = 0$). The deviation vectors carry the variability of the variables (standard deviation, variance) and the linear relations between pairs of variables (covariance, correlation).
- The sample standard deviation is proportional to the length of the deviation vector, $s_j = \frac{1}{\sqrt{n-1}}\|\mathbf{d}_j\|$; the sample variance is proportional to its squared length, $s_j^2 = \frac{1}{n-1}\|\mathbf{d}_j\|^2$; and the sample covariance is proportional to the inner product of two deviation vectors, $s_{jk} = \frac{1}{n-1}\mathbf{d}_j^\top\mathbf{d}_k = \frac{1}{n-1}\langle\mathbf{d}_j, \mathbf{d}_k\rangle$.
- **The sample correlation is the cosine of the angle between deviation vectors**: with the angle θ_{jk} between $\mathbf{d}_j$ and $\mathbf{d}_k$, $\mathbf{d}_j^\top\mathbf{d}_k = \|\mathbf{d}_j\|\|\mathbf{d}_k\|\cos\theta_{jk}$, so

$$r_{jk} = \frac{s_{jk}}{s_j s_k} = \frac{\mathbf{d}_j^\top\mathbf{d}_k}{\|\mathbf{d}_j\|\|\mathbf{d}_k\|} = \cos\theta_{jk}.$$

- The raw cross product splits accordingly: $\mathbf{x}_{(j)}^\top\mathbf{x}_{(k)} = n\bar{x}_j\bar{x}_k + \mathbf{d}_j^\top\mathbf{d}_k$, i.e. $\frac{n-1}{n}\,s_{jk} = \overline{x_jx_k} - \bar{x}_j\bar{x}_k$ with $\overline{x_jx_k} = \frac{1}{n}\sum_i x_{ij}x_{ik}$.

Geometric meaning: a longer deviation vector $\mathbf{d}_j$ means that the values of variable j lie farther from their mean, i.e. more variability. If $\mathbf{d}_j$ and $\mathbf{d}_k$ are parallel, $\mathbf{d}_j = c\mathbf{d}_k$ for some $c \neq 0$ and $\theta = 0$ (or π), and x_j, x_k are perfectly linearly related: $r_{jk} = +1$ for $\theta = 0$ (same direction) and $r_{jk} = -1$ for $\theta = \pi$ (opposite directions). Conversely, if x_j, x_k are perfectly linearly related, their deviation vectors are parallel. If $\mathbf{d}_j^\top\mathbf{d}_k = 0$, the deviation vectors are orthogonal and the *linear* correlation is $r_{jk} = 0$.



(a) $\mathbf{x}_{(j)} = \bar{x}_j\mathbf{1}_n + \mathbf{d}_j$ (b) $r_{jk} = \cos\theta_{jk}$, $s_j \propto \|\mathbf{d}_j\|$

Figure. *Variables as vectors in $\mathbb{R}^n$. (a) Projecting a variable vector on $\mathbf{1}_n$ gives its mean part; the orthogonal remainder is the deviation vector. (b) Lengths of deviation vectors are proportional to standard deviations, the cosine of the angle between them is the sample correlation, and a linear combination $y = a + bx_j + cx_k$ has deviation vector $b\mathbf{d}_j + c\mathbf{d}_k$.*

Theorem 1.34: unbiased estimation of the covariance matrix (slides Result 2.1)

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be a random sample from a joint distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Then $\bar{\mathbf{X}}$ is an unbiased estimator of $\boldsymbol{\mu}$,

$$\mathrm{E}(\bar{\mathbf{X}}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\bar{\mathbf{X}}) = \tfrac{1}{n}\boldsymbol{\Sigma},$$

(the population covariance divided by the sample size), and $\mathbf{S}_n$ is a biased estimator:

$$\mathrm{E}(\mathbf{S}_n) = \tfrac{n-1}{n}\boldsymbol{\Sigma} = \boldsymbol{\Sigma} - \tfrac{1}{n}\boldsymbol{\Sigma}, \qquad \mathrm{E}\Big(\tfrac{n}{n-1}\mathbf{S}_n\Big) = \mathrm{E}(\mathbf{S}) = \boldsymbol{\Sigma}.$$

Proof. Recall $\bar{\mathbf{X}} = (\mathbf{X}_1 + \mathbf{X}_2 + \cdots + \mathbf{X}_n)/n$. Repeated use of the properties of expectation gives $\mathrm{E}(\bar{\mathbf{X}}) = \mathrm{E}(\frac{1}{n}\mathbf{X}_1 + \cdots + \frac{1}{n}\mathbf{X}_n) = \frac{1}{n}\mathrm{E}(\mathbf{X}_1) + \cdots + \frac{1}{n}\mathrm{E}(\mathbf{X}_n) = \frac{1}{n}\boldsymbol{\mu} + \cdots + \frac{1}{n}\boldsymbol{\mu} = \boldsymbol{\mu}$. Next,

$$(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top = \Big(\frac{1}{n}\sum_{j=1}^{n}(\mathbf{X}_j - \boldsymbol{\mu})\Big)\Big(\frac{1}{n}\sum_{\ell=1}^{n}(\mathbf{X}_\ell - \boldsymbol{\mu})\Big)^\top = \frac{1}{n^2}\sum_{j=1}^{n}\sum_{\ell=1}^{n}(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_\ell - \boldsymbol{\mu})^\top,$$

so $\mathrm{Cov}(\bar{\mathbf{X}}) = \mathrm{E}\{(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top\} = \frac{1}{n^2}\sum_j\sum_\ell \mathrm{E}\{(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_\ell - \boldsymbol{\mu})^\top\}$. For $j \neq \ell$, $\mathrm{E}\{(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_\ell - \boldsymbol{\mu})^\top\} = \mathbf{0}$ by the independence of $\mathbf{X}_j$ and $\mathbf{X}_\ell$. Therefore $\mathrm{Cov}(\bar{\mathbf{X}}) = \frac{1}{n^2}(n\boldsymbol{\Sigma}) = \frac{1}{n}\boldsymbol{\Sigma}$, where $\boldsymbol{\Sigma}$ is the common population covariance matrix.

[Supplement] For $\mathbf{S}_n$: $\sum_j(\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})^\top = \sum_j(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_j - \boldsymbol{\mu})^\top - n(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top$ (expand $\mathbf{X}_j - \bar{\mathbf{X}} = (\mathbf{X}_j - \boldsymbol{\mu}) - (\bar{\mathbf{X}} - \boldsymbol{\mu})$ and use $\sum_j(\mathbf{X}_j - \boldsymbol{\mu}) = n(\bar{\mathbf{X}} - \boldsymbol{\mu})$). Its expectation is $n\boldsymbol{\Sigma} - n \cdot \frac{1}{n}\boldsymbol{\Sigma} = (n-1)\boldsymbol{\Sigma}$, so $\mathrm{E}(\mathbf{S}_n) = \frac{n-1}{n}\boldsymbol{\Sigma}$ and $\mathrm{E}(\mathbf{S}) = \boldsymbol{\Sigma}$. ■

7.3 Linear combinations of variables

When a new variable y is formed as a linear combination of variables, its mean, variance and covariance with other variables can be understood geometrically. For example, let the ith observation of a new variable be the linear combination of the first two variables

$$y_i = a + bx_{i1} + cx_{i2}.$$

The variable vector of y is the corresponding combination of variable vectors, $\mathbf{y} = (y_1, \ldots, y_n)^\top = a\mathbf{1}_n + b\mathbf{x}_{(1)} + c\mathbf{x}_{(2)}$. Recalling $\mathbf{x}_{(1)} = P_{\mathbf{1}_n}(\mathbf{x}_{(1)}) + \mathbf{d}_1$, its mean vector is

$$P_{\mathbf{1}_n}(\mathbf{y}) = a\mathbf{1}_n + bP_{\mathbf{1}_n}(\mathbf{x}_{(1)}) + cP_{\mathbf{1}_n}(\mathbf{x}_{(2)}) = (a + b\bar{x}_1 + c\bar{x}_2)\mathbf{1}_n,$$

and its deviation vector is $\mathbf{d}_y = b\mathbf{d}_1 + c\mathbf{d}_2$. The sample standard deviation of y is proportional to the length of $\mathbf{d}_y$, and with $\mathbf{d}_y^\top \mathbf{d}_y = (b\mathbf{d}_1 + c\mathbf{d}_2)^\top (b\mathbf{d}_1 + c\mathbf{d}_2)$ it can be written through the standard deviations and correlation of x_1, x_2:

$$\widehat{\mathrm{Var}}(y) = \frac{1}{n-1}\mathbf{d}_y^\top \mathbf{d}_y = \frac{b^2\|\mathbf{d}_1\|^2 + c^2\|\mathbf{d}_2\|^2 + 2bc\|\mathbf{d}_1\|\|\mathbf{d}_2\|\cos\theta}{n-1}.$$

$\sqrt{\widehat{\mathrm{Var}}(y)}$ is the sample standard deviation of the data $y_1, \ldots, y_n$. (The slides sketch $\mathbf{d}_y$ for $y = 2x_1 + x_2$ in the space of individuals.)

More generally, for constants a and $\mathbf{b} \in \mathbb{R}^p$ the new variable $z = a + \mathbf{b}^\top \mathbf{x} = a + \sum_{j=1}^p b_j x_j$ has variable vector $\mathbf{z} = (z_1, \ldots, z_n)^\top = a\mathbf{1}_n + \sum_j b_j \mathbf{x}_{(j)}$, with

$$P_{\mathbf{1}_n}(\mathbf{z}) = \Big(a + \sum_{j=1}^p b_j \bar{x}_j\Big)\mathbf{1}_n, \qquad \mathbf{d}_z = \sum_{j=1}^p b_j \mathbf{d}_j = [\mathbf{d}_1, \ldots, \mathbf{d}_p]\begin{pmatrix} b_1 \\ \vdots \\ b_p \end{pmatrix} = \mathbf{X}_C\mathbf{b}.$$

Theorem 1.35: sample variance and covariance of linear combinations

$$\widehat{\mathrm{Var}}(z) = \tfrac{1}{n-1}\mathbf{d}_z^\top \mathbf{d}_z = \tfrac{1}{n-1}\mathbf{b}^\top \mathbf{X}_C^\top \mathbf{X}_C \mathbf{b} = \mathbf{b}^\top \mathbf{S}\mathbf{b};$$

for two new variables $y = \mathbf{a}^\top \mathbf{x}$ and $z = \mathbf{b}^\top \mathbf{x}$ $(\mathbf{a}, \mathbf{b} \in \mathbb{R}^p)$, $\widehat{\mathrm{Cov}}(y, z) = \tfrac{1}{n-1}\mathbf{d}_y^\top \mathbf{d}_z = \tfrac{1}{n-1}\mathbf{a}^\top \mathbf{X}_C^\top \mathbf{X}_C \mathbf{b} = \mathbf{a}^\top \mathbf{S}\mathbf{b}$.

8 Dimension and Rank of the Data

Motivation. If $n \le p$, the sample covariance matrix is singular and cannot be inverted. The rank result here explains when classical multivariate procedures break down.

- **Linear independence**: a set of vectors $\{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ is linearly independent if no vector $\mathbf{x}_i$ is a linear combination of the others, i.e. $\sum_{i=1}^m c_i \mathbf{x}_i = \mathbf{0}$ implies $c_i = 0$ for all i.
- **Rank**: the rank of a matrix $\mathbf{X}$ is the maximal number of linearly independent columns (or rows). Column rank equals row rank for every matrix, and $\mathrm{rank}(\mathbf{X}) \le \min(n, p)$.
- The centered data matrix $\mathbf{X}_C = [\mathbf{x}_1 - \bar{\mathbf{x}}, \ldots, \mathbf{x}_n - \bar{\mathbf{x}}]^\top$ removes the mean in every dimension, adjusting the data to center $\mathbf{0}$. Its columns $\mathbf{d}_1, \ldots, \mathbf{d}_p$ lie in the subspace of $\mathbb{R}^n$ orthogonal to $\mathbf{1}_n$, whose span determines the rank: $\mathrm{rank}(\mathbf{X}_C) = \dim \mathrm{span}(\mathbf{d}_1, \ldots, \mathbf{d}_p) \le \min(n-1, p)$.
- $\mathrm{rank}(\mathbf{A}) = \mathrm{rank}(\mathbf{A}^\top) = \mathrm{rank}(\mathbf{A}^\top \mathbf{A})$. Hence $\mathrm{rank}(\mathbf{X}_C)$ is both its row and column rank, and $\mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{X}_C^\top \mathbf{X}_C) = \mathrm{rank}(\mathbf{X}_C)$.
- An $m \times k$ matrix $\mathbf{A}$ is **full-rank** if $\mathrm{rank}(\mathbf{A}) = \min(m, k)$ and **rank-deficient** if $\mathrm{rank}(\mathbf{A}) < \min(m, k)$.

If the sample covariance $\mathbf{S}$ is rank-deficient, $\mathrm{rank}(\mathbf{S}) = r$ is smaller than p: some variables are perfectly linearly related, so the variation of the data lies in an r-dimensional subspace of the p-dimensional data space and is zero in the remaining $p - r$ dimensions. (The slides show a 3-dimensional input cloud lying on a curved 2-dimensional sheet, mapped to a 2-dimensional embedded space.)

Theorem 1.36: rank-deficiency property (slides Proposition, homework)

For the NND sample covariance $\mathbf{S}$ of $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]^\top \in \mathbb{R}^{n\times p}$ the following are equivalent.

1. $\mathbf{S}$ is rank-deficient.
2. There is $\mathbf{a} \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ with $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0$ for all $i = 1, \dots, n$.
3. The projection scores onto some direction all coincide: $s_{\mathbf{a}}(\mathbf{x}_i) = s_{\mathbf{a}}(\mathbf{x}_j)$ for all $i, j = 1, \dots, n$.
4. The subspace $\operatorname{span}(\mathbf{d}_1, \dots, \mathbf{d}_p)$ spanned by the deviation vectors has dimension less than p.
5. At least one deviation vector $\mathbf{d}_j$ is a linear combination of the others.
6. At least one eigenvalue of $\mathbf{S}$ is 0.

Proof. (1)⇔(2): If $\mathbf{S}$ is rank-deficient, $\operatorname{rank}(\mathbf{S}) = \operatorname{rank}(\mathbf{X}_C^\top) = r < p$, so the column vectors $\{\mathbf{x}_i - \bar{\mathbf{x}} : i = 1, \dots, n\}$ of $\mathbf{X}_C^\top$ span a subspace of dimension r. Hence there is a vector $\mathbf{a} \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ orthogonal to that subspace, so $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0$ for all i. Conversely, if such an $\mathbf{a}$ exists, the vectors $\{\mathbf{x}_i - \bar{\mathbf{x}}\}$ lie in a subspace of dimension less than p, and $\mathbf{S}$ is rank-deficient.

(2)⇔(3): $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0$ for all i iff $\mathbf{a}^\top\mathbf{x}_i = \mathbf{a}^\top\bar{\mathbf{x}}$ for all i, iff all scores $s_{\mathbf{a}}(\mathbf{x}_i) = \mathbf{a}^\top\mathbf{x}_i/\|\mathbf{a}\|$ coincide (their common value is then the mean score).

(1)⇔(4): $\operatorname{rank}(\mathbf{S}) = \operatorname{rank}(\mathbf{X}_C) = \operatorname{rank}([\mathbf{d}_1, \dots, \mathbf{d}_p])$, so by the definition of rank (1)⇔(4).

(4)⇔(5): $\operatorname{span}(\mathbf{d}_1, \dots, \mathbf{d}_p)$ has dimension less than p iff $\{\mathbf{d}_1, \dots, \mathbf{d}_p\}$ is linearly dependent, so (4)⇔(5) by the definition of linear independence.

(2)⇒(6): For $\mathbf{a}$ as in (2), $\mathbf{X}_C\mathbf{a} = \mathbf{0}$, so $\mathbf{S}\mathbf{a} = \frac{1}{n-1}\mathbf{X}_C^\top\mathbf{X}_C\mathbf{a} = \mathbf{0} = 0 \cdot \mathbf{a}$: $\mathbf{a}$ is an eigenvector of $\mathbf{S}$ with eigenvalue 0.

(6)⇒(2): If an eigenvalue of $\mathbf{S}$ is 0 with eigenvector $\mathbf{u}$, then $\mathbf{S}\mathbf{u} = \mathbf{0}$, and $\mathbf{u}^\top\mathbf{S}\mathbf{u} = \frac{1}{n-1}\|\mathbf{X}_C\mathbf{u}\|^2 = 0$. The vector $\mathbf{X}_C\mathbf{u}$ has length 0, so all its components $\mathbf{u}^\top(\mathbf{x}_i - \bar{\mathbf{x}})$ are 0: $\mathbf{u}$ is a vector as in (2). ■

Statistical meaning. If $\mathbf{S}$ is full rank, the data have enough independent variation in every direction, and every component of the data contributes meaningfully to the analysis. If $\mathbf{S}$ is rank-deficient, the data give no information in some direction and the remaining directions are strongly correlated: compared with their size, the data provide insufficient information, there is little independent variation, and the reliability of statistical analysis drops. In particular $\mathbf{S}^{-1}$ does not exist, so the sphering $\mathbf{x} \mapsto \mathbf{S}^{-1/2}\mathbf{x}$ is undefined. In high dimensions ($p > n$) $\mathbf{S}$ is always rank-deficient:

$$\operatorname{rank}(\mathbf{S}) \le \min\{n - 1, p\} < p.$$

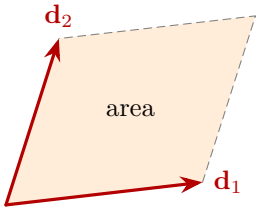
Remark 1.37: generalized variance as a squared volume [Supplement]

For $p = 2$, $|\mathbf{S}| = s_{11}s_{22} - s_{12}^2 = \frac{\|\mathbf{d}_1\|^2\|\mathbf{d}_2\|^2 - (\mathbf{d}_1^\top\mathbf{d}_2)^2}{(n-1)^2} = \frac{(\|\mathbf{d}_1\|\|\mathbf{d}_2\|\sin\theta_{12})^2}{(n-1)^2}$: the generalized variance is the squared area of the parallelogram spanned by the deviation vectors, divided by $(n-1)^2$. In general $|\mathbf{S}| = (n-1)^{-p}(\text{volume})^2$, where the volume is that of the parallelotope spanned by $\mathbf{d}_1, \dots, \mathbf{d}_p$.

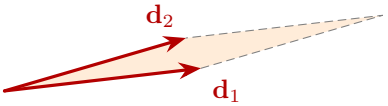
Proof for general p. Write $\mathbf{X}_C = [\mathbf{d}_1, \dots, \mathbf{d}_p] = \mathbf{Q}\mathbf{R}_0$ with $\mathbf{Q} \in \mathbb{R}^{n\times p}$ having orthonormal columns and $\mathbf{R}_0$ upper triangular (Gram–Schmidt; if the $\mathbf{d}_j$ are dependent, some diagonal entry of $\mathbf{R}_0$ is 0 and both sides below vanish). Then $|\mathbf{S}| = (n-1)^{-p}|\mathbf{X}_C^\top\mathbf{X}_C| = (n-1)^{-p}|\mathbf{R}_0^\top\mathbf{R}_0| = (n-1)^{-p}\prod_j(\mathbf{R}_0)_{jj}^2$, and $|(\mathbf{R}_0)_{jj}|$ is the distance from $\mathbf{d}_j$ to $\operatorname{span}(\mathbf{d}_1, \dots, \mathbf{d}_{j-1})$, the jth "height". The product of the heights is the volume.

So $|\mathbf{S}|$ is small when some deviation vector is nearly a combination of the others (strong

correlation), even if every variance is large, and $|\mathbf{S}| = 0$ exactly in the rank-deficient case of Theorem 1.36. Two data sets can share $\mathrm{tr}\mathbf{S}$ while having very different $|\mathbf{S}|$.



weak correlation: $|\mathbf{S}|$ large



strong correlation: $|\mathbf{S}|$ small

Figure. [Supplement] *For $p = 2$, $|\mathbf{S}| = (\text{area})^2/(n-1)^2$ with the parallelogram spanned by the deviation vectors. With equal lengths (equal variances), a small angle between $\mathbf{d}_1$ and $\mathbf{d}_2$ (correlation near ± 1) flattens the parallelogram and the generalized variance tends to 0.*

Chapter 2. Matrix, Calculus and Probability Tools

Matrix identities, eigenvalue inequalities, matrix calculus, optimization, convergence

This chapter collects the tools used in later proofs (MLE, Wishart, T^2, large-sample theory). It follows the Lab Session 3 handout "Preliminaries" (used for the proofs of Results 4.11–4.13), whose item numbers are quoted in the box titles, together with the Frobenius-norm part of Lab Session 2.

1 Basic Matrix Operations

Motivation. This chapter collects the linear-algebra and probability tools used throughout, so later proofs can cite them. We start with trace, determinant, norms and continuity.

Definition 2.1: basic facts (Lab 3, Recall 1.1)

(i) For $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{p\times q}$ and $\mathbf{B} = (b_{jk}) \in \mathbb{R}^{q\times r}$, $(\mathbf{AB})_{ik} = \sum_{j=1}^{q} a_{ij}b_{jk}$.

(ii) $(\mathbf{A}^\top)_{ij} = a_{ji}$; $(\mathbf{A}^\top)^\top = \mathbf{A}$, $(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$, and a 1×1 matrix (a scalar) equals its transpose.

(iii) $\mathrm{tr}(\mathbf{C}) = \sum_i c_{ii}$ for $\mathbf{C} \in \mathbb{R}^{p\times p}$, and the trace is linear: $\mathrm{tr}(a\mathbf{C} + b\mathbf{D}) = a\mathrm{tr}(\mathbf{C}) + b\mathrm{tr}(\mathbf{D})$.

(iv) $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$ for $p\times p$ matrices, so $|\mathbf{A}^{-1}| = 1/|\mathbf{A}|$ if $\mathbf{A}$ is invertible.

(v) (Leibniz) $|\mathbf{A}| = \sum_\sigma \mathrm{sgn}(\sigma)\prod_i a_{i\sigma(i)}$ over permutations σ of $\{1,\dots,p\}$, with $\mathrm{sgn}(\sigma) = (-1)^{N(\sigma)}$, $N(\sigma) = \#\{i < j : \sigma(i) > \sigma(j)\}$. Consequences: (a) $|\mathbf{A}|$ is a polynomial in the p^2 entries, so $\mathbf{A} \mapsto |\mathbf{A}|$ is continuous and infinitely differentiable, and $\{\mathbf{A} : |\mathbf{A}| \neq 0\}$ is open; (b) $|\mathbf{I}_p| = 1$, $|\mathrm{diag}(d_1,\dots,d_p)| = \prod_i d_i$, $|c\mathbf{A}| = c^p|\mathbf{A}|$.

(vi) (Adjugate) If $|\mathbf{A}| \neq 0$, $\mathbf{A}^{-1} = \mathrm{adj}(\mathbf{A})/|\mathbf{A}|$ with $\mathrm{adj}(\mathbf{A})_{ij} = (-1)^{i+j}|\mathbf{A}_{(j,i)}|$, where $\mathbf{A}_{(j,i)}$ deletes row j and column i. Each entry of $\mathbf{A}^{-1}$ is a polynomial divided by the nonzero polynomial $|\mathbf{A}|$, so $\mathbf{A} \mapsto \mathbf{A}^{-1}$ is continuous and infinitely differentiable on $\{|\mathbf{A}| \neq 0\}$ (Theorem 2.6).

(vii) (Cauchy–Schwarz for sums) $|\sum_k a_k b_k| \le (\sum_k a_k^2)^{1/2}(\sum_k b_k^2)^{1/2}$.

Definition 2.2: Frobenius norm (Lab 3, Definition 1.2; Lab 2)

For $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{p\times q}$, $\|\mathbf{A}\|_F = (\sum_{i,j} a_{ij}^2)^{1/2}$, and $\|\mathbf{x}\| = \|\mathbf{x}\|_F$ for vectors. Example: $\mathbf{M} = \begin{pmatrix} -0.1 & 0.3 \\ 0.2 & -0.2 \end{pmatrix}$ has $\|\mathbf{M}\|_F = \sqrt{0.18} \approx 0.424$. Estimation errors of the sample moments can be measured by $E_{\bar u} = \|\bar{\mathbf{u}} - \boldsymbol{\mu}\|_2$, $E_S = \|\mathbf{S} - \boldsymbol{\Sigma}\|_F$, $E_R = \|\mathbf{R} - \mathbf{R}_0\|_F$, where $\boldsymbol{\mu}, \boldsymbol{\Sigma}, \mathbf{R}_0$ are the population mean, covariance and correlation matrix.

Theorem 2.3: basic identities (Lab 3, Lemma 1.3)

For $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{y} \in \mathbb{R}^q$, $\mathbf{A} \in \mathbb{R}^{p\times q}$, $\mathbf{B} \in \mathbb{R}^{q\times p}$:

(i) (bilinear form) $\mathbf{x}^\top\mathbf{A}\mathbf{y} = \sum_{i,j} a_{ij}x_i y_j$;

(ii) (diagonal case) if $p = q$ and $\mathbf{A} = \mathrm{diag}(a_{11},\dots,a_{pp})$, $\mathbf{x}^\top\mathbf{A}\mathbf{y} = \sum_i a_{ii}x_i y_i$ and $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_i a_{ii}x_i^2$;

(iii) (trace) $\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$;

(iv) (Frobenius norm) $\|\mathbf{A}\|_F^2 = \operatorname{tr}(\mathbf{A}^\top \mathbf{A})$;

(v) (Euclidean norm) $\|\mathbf{x}\|^2 = \mathbf{x}^\top \mathbf{x}$.

Proof. (i) $(\mathbf{Ay})_i = \sum_j a_{ij} y_j$, so $\mathbf{x}^\top \mathbf{Ay} = \sum_i x_i (\mathbf{Ay})_i = \sum_{i,j} a_{ij} x_i y_j$.

(ii) If $\mathbf{A}$ is diagonal, $a_{ij} = 0$ for $i \neq j$, so only the terms $j = i$ remain in (i).

(iii) $\operatorname{tr}(\mathbf{AB}) = \sum_i \sum_j a_{ij} b_{ji} = \sum_j \sum_i b_{ji} a_{ij} = \operatorname{tr}(\mathbf{BA})$.

(iv) $(\mathbf{A}^\top \mathbf{A})_{jj} = \sum_i (\mathbf{A}^\top)_{ji} a_{ij} = \sum_i a_{ij}^2$; sum over j.

(v) Apply (iv) to the $p \times 1$ matrix $\mathbf{x}$: $\|\mathbf{x}\|^2 = \operatorname{tr}(\mathbf{x}^\top \mathbf{x}) = \mathbf{x}^\top \mathbf{x}$. ■

Theorem 2.4: first-order expansion of the determinant (Lab 3, Lemma 1.4)

For $\mathbf{N} \in \mathbb{R}^{p \times p}$, $|\mathbf{I}_p + t\mathbf{N}| = 1 + t \operatorname{tr}(\mathbf{N}) + O(t^2)$ as $t \to 0$.

Proof. By the Leibniz formula, $|\mathbf{I}_p + t\mathbf{N}| = \sum_\sigma \operatorname{sgn}(\sigma) \prod_i (\delta_{i\sigma(i)} + t n_{i\sigma(i)})$, where $\delta_{ii} = 1$ and $\delta_{ij} = 0$ for $i \neq j$. If σ is not the identity, $\sigma(i) \neq i$ for at least two indices, so its term is $O(t^2)$. The identity contributes $\prod_i (1 + t n_{ii}) = 1 + t \sum_i n_{ii} + O(t^2) = 1 + t \operatorname{tr}(\mathbf{N}) + O(t^2)$. ■

Definition 2.5: open sets and continuity (Lab 3, Definition 1.5)

Let V and W each be $\mathbb{R}^p$, $\mathbb{R}^{p \times q}$ or the space of $p \times p$ symmetric matrices, with distance $\|\mathbf{B} - \mathbf{A}\|_F$.

(i) $U \subset V$ is open iff for every $\mathbf{A} \in U$ there is $r > 0$ such that every $\mathbf{B} \in V$ with $\|\mathbf{B} - \mathbf{A}\|_F < r$ lies in U.

(ii) $f : U \to W$ is continuous at $\mathbf{A} \in U$ iff for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(\mathbf{B}) - f(\mathbf{A})\|_F < \epsilon$ for all $\mathbf{B} \in U$ with $\|\mathbf{B} - \mathbf{A}\|_F < \delta$; f is continuous on U if it is continuous at every point.

Theorem 2.6: entrywise continuity (Lab 3, Lemma 1.6)

With V, W as above, if every entry $F_{kl} : U \to \mathbb{R}$ of a map $F : U \to W$ is continuous at $\mathbf{A}$, then F is continuous at $\mathbf{A}$.

Proof. Let m be the number of entries of F. Given $\epsilon > 0$, choose $\delta > 0$ so that $|F_{kl}(\mathbf{B}) - F_{kl}(\mathbf{A})| < \epsilon/\sqrt{m}$ for all (k, l) whenever $\|\mathbf{B} - \mathbf{A}\|_F < \delta$. Then $\|F(\mathbf{B}) - F(\mathbf{A})\|_F^2 = \sum_{k,l} (F_{kl}(\mathbf{B}) - F_{kl}(\mathbf{A}))^2 < \epsilon^2$. ■

1.1 Block-matrix formulas

[Supplement] The following identities are not in the lab handout; they are used for conditional distributions, the Wishart theorem and likelihood ratio tests.

Theorem 2.7: block determinant and inverse [Supplement]

For $\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$ with $\mathbf{A}_{22}$ invertible,

$$|\mathbf{A}| = |\mathbf{A}_{22}| \, |\mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21}| = |\mathbf{A}_{11}| \, |\mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}| \quad (\mathbf{A}_{11} \text{ invertible}).$$

With the Schur complement $\mathbf{A}_{11\cdot 2} := \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}$,

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11\cdot 2}^{-1} & -\mathbf{A}_{11\cdot 2}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}_{11\cdot 2}^{-1} & \mathbf{A}_{22}^{-1} + \mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}_{11\cdot 2}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \end{pmatrix};$$

in particular $(\mathbf{A}^{-1})_{11} = \mathbf{A}_{11\cdot 2}^{-1}$.

Proof. Block Gaussian elimination:

$$\begin{pmatrix} \mathbf{I} & -\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \mathbf{A} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{I} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11\cdot 2} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22} \end{pmatrix}.$$

The outer factors are triangular with determinant 1, giving the determinant formula; inverting both sides gives $\mathbf{A}^{-1}$. ■

Theorem 2.8: matrix determinant lemma and Woodbury identity [Supplement]

$$|\mathbf{A} + \mathbf{u}\mathbf{v}^\top| = |\mathbf{A}|(1 + \mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u}), \qquad |\mathbf{I}_p + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top\mathbf{u},$$
$$(\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}\mathbf{A}^{-1}.$$

Proof. Compute $\begin{vmatrix} \mathbf{A} & -\mathbf{u} \\ \mathbf{v}^\top & 1 \end{vmatrix}$ with both forms of the block formula: $|\mathbf{A}|(1+\mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u}) = 1\cdot|\mathbf{A}+\mathbf{u}\mathbf{v}^\top|$. For Woodbury, multiply the right side by $\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V}$ and check that the product is $\mathbf{I}$. ■

2 Eigenvalues and Positive Definiteness

Motivation. Most multivariate statistics are functions of eigenvalues of covariance matrices; their basic properties are recorded here.

Theorem 2.9: spectral decomposition and consequences (Lab 3, Theorem 2.1, Lemmas 2.2 and 2.4)

Every symmetric $p \times p$ matrix can be written $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})\mathbf{U}^\top = \sum_i \lambda_i(\mathbf{A})\mathbf{u}_i\mathbf{u}_i^\top$ with $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ orthogonal ($\mathbf{U}^\top\mathbf{U} = \mathbf{U}\mathbf{U}^\top = \mathbf{I}_p$), real eigenvalues $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_p(\mathbf{A})$, $\mathbf{\Lambda}(\mathbf{A}) = \operatorname{diag}(\lambda_1(\mathbf{A}), \ldots, \lambda_p(\mathbf{A}))$ and $\mathbf{A}\mathbf{u}_i = \lambda_i(\mathbf{A})\mathbf{u}_i$. Throughout, $\lambda_i(\mathbf{A})$ denotes the ith largest eigenvalue. Moreover

$$|\mathbf{A}| = \prod_i \lambda_i(\mathbf{A}), \qquad \operatorname{tr}(\mathbf{A}) = \sum_i \lambda_i(\mathbf{A}), \qquad \mathbf{A} \succ 0 \iff \text{all } \lambda_i(\mathbf{A}) > 0,$$

where $\mathbf{A} \succ 0$ (positive definite, Lab 3 Definition 2.3) means that $\mathbf{A}$ is symmetric and $\mathbf{x}^\top\mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Proof. The decomposition is the spectral theorem for real symmetric matrices (any linear algebra textbook).

Determinant and trace: $|\mathbf{U}||\mathbf{U}^\top| = |\mathbf{U}\mathbf{U}^\top| = |\mathbf{I}_p| = 1$, so $|\mathbf{A}| = |\mathbf{U}||\mathbf{\Lambda}(\mathbf{A})||\mathbf{U}^\top| = |\mathbf{\Lambda}(\mathbf{A})| = \prod_i \lambda_i(\mathbf{A})$, and by Theorem 2.3(iii) $\operatorname{tr}(\mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top) = \operatorname{tr}(\mathbf{\Lambda}\mathbf{U}^\top\mathbf{U}) = \operatorname{tr}\mathbf{\Lambda}$.

Definiteness: with $\mathbf{z} = \mathbf{U}^\top\mathbf{x}$, $z_i = \mathbf{u}_i^\top\mathbf{x}$, Theorem 2.3(ii) gives $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \mathbf{z}^\top\mathbf{\Lambda}(\mathbf{A})\mathbf{z} = \sum_i \lambda_i(\mathbf{A})(\mathbf{u}_i^\top\mathbf{x})^2$. If all $\lambda_i(\mathbf{A}) > 0$ and $\mathbf{x} \neq \mathbf{0}$, some $\mathbf{u}_i^\top\mathbf{x} \neq 0$, so $\mathbf{x}^\top\mathbf{A}\mathbf{x} > 0$. Conversely, if $\mathbf{A} \succ 0$, $\lambda_i(\mathbf{A}) = \mathbf{u}_i^\top\mathbf{A}\mathbf{u}_i > 0$. ■

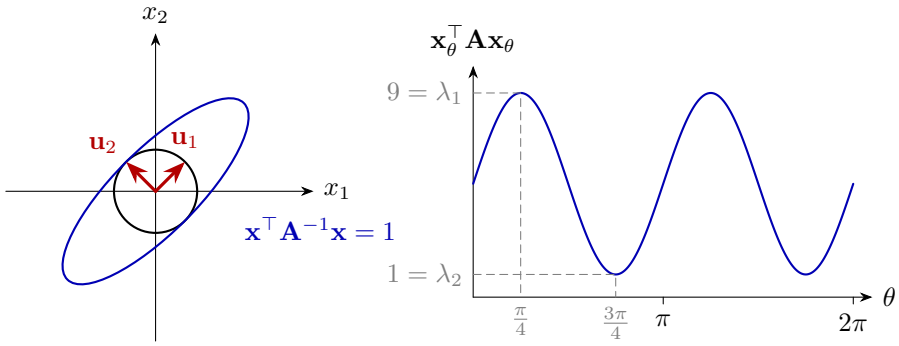
Theorem 2.10: smallest eigenvalue (Lab 3, Lemma 2.5)

For symmetric $p \times p$ matrices $\mathbf{A}, \mathbf{B}$:
(i) $\lambda_p(\mathbf{A}) = \min_{\|\mathbf{x}\|=1} \mathbf{x}^\top \mathbf{A}\mathbf{x}$;
(ii) $|\lambda_p(\mathbf{A}) - \lambda_p(\mathbf{B})| \le \|\mathbf{A} - \mathbf{B}\|_F$;
(iii) $\{\mathbf{A} \succ 0\}$ is convex and open in the space of symmetric matrices.

Proof. (i) With $\mathbf{z} = \mathbf{U}^\top \mathbf{x}$, $\|\mathbf{z}\|^2 = \mathbf{x}^\top \mathbf{U}\mathbf{U}^\top \mathbf{x} = \|\mathbf{x}\|^2 = 1$ and $\mathbf{x}^\top \mathbf{A}\mathbf{x} = \sum_i \lambda_i(\mathbf{A}) z_i^2 \ge \lambda_p(\mathbf{A})$, with equality at $\mathbf{x} = \mathbf{u}_p$.

(ii) For $\|\mathbf{x}\| = 1$, Theorem 2.3(i) and Cauchy–Schwarz give $|\mathbf{x}^\top \mathbf{A}\mathbf{x} - \mathbf{x}^\top \mathbf{B}\mathbf{x}| = |\sum_{ij}(a_{ij} - b_{ij}) x_i x_j| \le \|\mathbf{A} - \mathbf{B}\|_F (\sum_{ij} x_i^2 x_j^2)^{1/2} = \|\mathbf{A} - \mathbf{B}\|_F$. Taking the minimum over $\|\mathbf{x}\| = 1$ on both sides and using (i) gives the claim.

(iii) Convex: if $\mathbf{A}_0, \mathbf{A}_1 \succ 0$ and $s \in [0,1]$, $\mathbf{x}^\top\{(1-s)\mathbf{A}_0 + s\mathbf{A}_1\}\mathbf{x} > 0$ for $\mathbf{x} \ne \mathbf{0}$. Open: let $\mathbf{A} \succ 0$, so $\lambda_p(\mathbf{A}) > 0$. If $\mathbf{B}$ is symmetric with $\|\mathbf{B} - \mathbf{A}\|_F < \lambda_p(\mathbf{A})$, (ii) gives $\lambda_p(\mathbf{B}) \ge \lambda_p(\mathbf{A}) - \|\mathbf{A} - \mathbf{B}\|_F > 0$, so $\mathbf{B} \succ 0$; thus $r = \lambda_p(\mathbf{A})$ works in Definition 2.5(i). ∎



unit circle, eigenvectors of $\mathbf{A}$

Figure. *For* $\mathbf{A} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ *and* $\mathbf{x}_\theta = (\cos\theta, \sin\theta)^\top$, *the quadratic form on the unit circle is* $5 + 4\sin 2\theta$. *Its maximum* $\lambda_1 = 9$ *is attained at* $\pm\mathbf{u}_1$ *(*$\theta = \pi/4$*) and its minimum* $\lambda_2 = 1$ *at* $\pm\mathbf{u}_2$ *(*$\theta = 3\pi/4$*), as in Theorem 2.10(i) and its analogue for* λ_1.

Definition 2.11: symmetric square root (Lab 3, Definition 2.6)

For $\mathbf{A} \succ 0$ with $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})\mathbf{U}^\top$, $\mathbf{A}^{1/2} := \mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{1/2}\mathbf{U}^\top$ and $\mathbf{A}^{-1/2} := \mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1/2}\mathbf{U}^\top$, where $\mathbf{\Lambda}(\mathbf{A})^{\pm 1/2} = \operatorname{diag}(\lambda_1(\mathbf{A})^{\pm 1/2}, \dots, \lambda_p(\mathbf{A})^{\pm 1/2})$ is well defined because all $\lambda_i(\mathbf{A}) > 0$. One spectral decomposition of $\mathbf{A}$ is fixed throughout.

Theorem 2.12: properties of the square root (Lab 3, Lemma 2.7)

For $\mathbf{A} \succ 0$:
(i) $\mathbf{A}^{1/2}$ and $\mathbf{A}^{-1/2}$ are symmetric;
(ii) $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{A}$ and $\mathbf{A}^{-1/2}\mathbf{A}^{-1/2} = \mathbf{A}^{-1}$;
(iii) $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{I}_p$ and $\mathbf{A}^{1/2}\mathbf{A}^{-1/2} = \mathbf{A}^{-1/2}\mathbf{A}^{1/2} = \mathbf{I}_p$;
(iv) $\mathbf{A}$ is invertible, $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1}\mathbf{U}^\top$ is symmetric and $\mathbf{A}^{-1} \succ 0$, and $\lambda_i(\mathbf{A}^{-1}) = 1/\lambda_{p+1-i}(\mathbf{A})$ with eigenvector $\mathbf{u}_{p+1-i}$.

Proof. (i) Each matrix has the form $\mathbf{U}\mathbf{D}\mathbf{U}^\top$ with $\mathbf{D}$ diagonal, and $(\mathbf{U}\mathbf{D}\mathbf{U}^\top)^\top = \mathbf{U}\mathbf{D}\mathbf{U}^\top$.

(ii)–(iv) Since $\mathbf{U}^\top\mathbf{U} = \mathbf{U}\mathbf{U}^\top = \mathbf{I}_p$, $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})\mathbf{U}^\top = \mathbf{A}$, $\mathbf{A}^{-1/2}\mathbf{A}^{-1/2} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1}\mathbf{U}^\top$, $\{\mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1}\mathbf{U}^\top\}\mathbf{A} = \mathbf{A}\{\mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1}\mathbf{U}^\top\} = \mathbf{I}_p$, so $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1}\mathbf{U}^\top$, and $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{A}^{1/2}\mathbf{A}^{-1/2} = \mathbf{A}^{-1/2}\mathbf{A}^{1/2} = \mathbf{U}\mathbf{U}^\top = \mathbf{I}_p$. The diagonal entries of $\mathbf{\Lambda}(\mathbf{A})^{-1}$ are $1/\lambda_i(\mathbf{A}) > 0$; in

decreasing order they are $1/\lambda_p(\mathbf{A}) \geq \cdots \geq 1/\lambda_1(\mathbf{A})$, attached to $\mathbf{u}_p, \ldots, \mathbf{u}_1$. Hence $\lambda_i(\mathbf{A}^{-1}) = 1/\lambda_{p+1-i}(\mathbf{A})$, and $\mathbf{A}^{-1} \succ 0$ by Theorem 2.9. ■

3 Eigenvalue Inequalities

Motivation. PCA, canonical correlation and discriminant analysis all maximize a ratio of quadratic forms, and the normal MLE compares traces and determinants. These inequalities give a derivative-free way to optimize over covariance matrices; the trace inequality is used in the second proof of the MLE of $\boldsymbol{\Sigma}$ (Result 4.11).

Definition 2.13: Birkhoff's theorem and the rearrangement inequality (Lab 3, Recall 3.1)

(i) A $p \times p$ matrix $\mathbf{W} = (w_{ij})$ with $w_{ij} \geq 0$ and all row and column sums 1 (doubly stochastic) is a convex combination of permutation matrices.
(ii) If $a_1 \geq \cdots \geq a_p$ and $b_1 \geq \cdots \geq b_p$, then for every permutation σ, $\sum_i a_i b_{p+1-i} \leq \sum_i a_i b_{\sigma(i)} \leq \sum_i a_i b_i$.

Theorem 2.14: von Neumann's trace inequality (Lab 3, Theorem 3.2)

For symmetric $\mathbf{A} = \sum_i \lambda_i(\mathbf{A}) \mathbf{u}_i \mathbf{u}_i^\top$, $\mathbf{B} = \sum_j \lambda_j(\mathbf{B}) \mathbf{v}_j \mathbf{v}_j^\top$,

$$\sum_{i=1}^{p} \lambda_{p+1-i}(\mathbf{A}) \lambda_i(\mathbf{B}) \leq \operatorname{tr}(\mathbf{AB}) \leq \sum_{i=1}^{p} \lambda_i(\mathbf{A}) \lambda_i(\mathbf{B}).$$

The upper bound is attained when $\mathbf{v}_i = \mathbf{u}_i$ for all i and the lower bound when $\mathbf{v}_i = \mathbf{u}_{p+1-i}$ for all i (Part 1 slides, p. 176).

Proof.

Step 1 (reduction to a doubly stochastic matrix). By Theorem 2.3(iii), $\operatorname{tr}(\mathbf{AB}) = \sum_{i,j} \lambda_i(\mathbf{A}) \lambda_j(\mathbf{B}) w_{ij}$, $w_{ij} = (\mathbf{u}_i^\top \mathbf{v}_j)^2 \geq 0$. Since $\{\mathbf{u}_i\}, \{\mathbf{v}_j\}$ are orthonormal bases, $\sum_j w_{ij} = \|\mathbf{u}_i\|^2 = 1$ and $\sum_i w_{ij} = \|\mathbf{v}_j\|^2 = 1$: $\mathbf{W}$ is doubly stochastic.

Step 2 (extreme points). By Birkhoff $\mathbf{W}$ is a convex combination of permutation matrices; $\operatorname{tr}(\mathbf{AB})$ is linear in $\mathbf{W}$, so it lies between the minimum and maximum of $\sum_i \lambda_i(\mathbf{A}) \lambda_{\sigma(i)}(\mathbf{B})$ over permutations σ.

Step 3 (rearrangement). By the rearrangement inequality the maximum is at $\sigma(i) = i$ and the minimum at $\sigma(i) = p + 1 - i$.

Step 4 (equality). If $\mathbf{v}_i = \mathbf{u}_i$ (resp. $\mathbf{u}_{p+1-i}$) for all i, $\mathbf{W}$ is the identity (resp. reversal) permutation matrix and the bound is attained. ■

Theorem 2.15: an elementary inequality (Lab 3, Lemma 3.3)

$x - \log x \geq 1$ for $x > 0$, with equality iff $x = 1$.

Proof. $\phi(x) = x - \log x$ has $\phi'(x) = 1 - 1/x$, negative on $(0, 1)$ and positive on $(1, \infty)$. For $x \neq 1$ the mean value theorem gives $\phi(x) - \phi(1) = \phi'(\xi)(x - 1)$ for some ξ strictly between 1 and x, and $\phi'(\xi)(x - 1) > 0$. Hence $\phi(x) > \phi(1) = 1$. ■

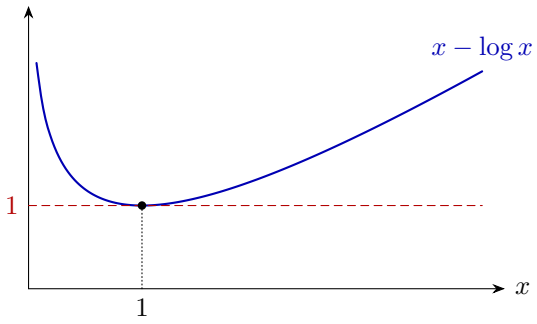


Figure. *The function $x - \log x$ decreases on $(0,1)$, increases on $(1,\infty)$ and touches its lower bound 1 only at $x = 1$. Applied to the eigenvalues of $\boldsymbol{\Sigma}^{-1}\mathbf{S}$ this inequality locates the normal MLE.*

4 Vector Calculus

Motivation. Maximum likelihood for the multivariate normal requires derivatives with respect to vectors and matrices, including $\log\det\boldsymbol{\Sigma}$. These rules are collected here.

Definition 2.16: calculus of one and several variables (Lab 3, Recall 4.1)

(i) If $g'' > 0$ on an interval, g' is strictly increasing and g is strictly convex there.
(ii) (Chain rule) If $\mathbf{y} = \mathbf{g}(\mathbf{x})$ is differentiable and f is differentiable, $\partial f(\mathbf{g}(\mathbf{x}))/\partial x_k = \sum_i (\partial f/\partial y_i)(\partial g_i/\partial x_k)$.
(iii) (Product rule) $\partial(uv)/\partial x_l = u\,\partial v/\partial x_l + v\,\partial u/\partial x_l$.

Definition 2.17: gradient, Jacobian, Hessian (Lab 3, Definition 4.2)

For differentiable $f : \mathbb{R}^p \to \mathbb{R}$, the gradient is the $p \times 1$ vector $\partial f/\partial \mathbf{x} = (\partial f/\partial x_1, \dots, \partial f/\partial x_p)^\top$, and $\partial f/\partial \mathbf{x}^\top := (\partial f/\partial \mathbf{x})^\top$. The Jacobian of a differentiable $\mathbf{g} : \mathbb{R}^p \to \mathbb{R}^q$ is the $q \times p$ matrix $\partial \mathbf{g}/\partial \mathbf{x}^\top$ with (i,j) entry $\partial g_i/\partial x_j$. For twice differentiable f the Hessian is the Jacobian of the gradient,

$$\frac{\partial^2 f}{\partial \mathbf{x}\partial \mathbf{x}^\top} := \frac{\partial}{\partial \mathbf{x}^\top}\Big(\frac{\partial f}{\partial \mathbf{x}}\Big), \qquad \Big(\frac{\partial^2 f}{\partial \mathbf{x}\partial \mathbf{x}^\top}\Big)_{ij} = \frac{\partial^2 f}{\partial x_j \partial x_i},$$

where $\partial^2 f/\partial x_j \partial x_i$ means differentiating first in x_i and then in x_j.

Theorem 2.18: symmetry of the Hessian, Schwarz's theorem (Lab 3, Theorem 4.3)

If the second partial derivatives of f are continuous, $\partial^2 f/\partial x_j \partial x_i = \partial^2 f/\partial x_i \partial x_j$ for all i, j; hence the Hessian is symmetric.

Proof. Fix $\mathbf{x}$ and $i \neq j$, and let $\Delta(h) = f(\mathbf{x} + h\mathbf{e}_i + h\mathbf{e}_j) - f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x} + h\mathbf{e}_j) + f(\mathbf{x})$.

Step 1. Applying the mean value theorem first in x_i and then in x_j, $\Delta(h) = h^2\,\partial^2 f(\boldsymbol{\xi}_h)/\partial x_j \partial x_i$ for some $\boldsymbol{\xi}_h$ with $\|\boldsymbol{\xi}_h - \mathbf{x}\| \le 2|h|$.

Step 2. Applying it in the other order, $\Delta(h) = h^2\,\partial^2 f(\boldsymbol{\eta}_h)/\partial x_i \partial x_j$ for some $\boldsymbol{\eta}_h$ with $\|\boldsymbol{\eta}_h - \mathbf{x}\| \le 2|h|$.

Step 3. Equate the two expressions, divide by h^2 and let $h \to 0$. By continuity of the second partials, $\partial^2 f(\mathbf{x})/\partial x_j \partial x_i = \partial^2 f(\mathbf{x})/\partial x_i \partial x_j$. ∎

Theorem 2.19: chain rule (Lab 3, Lemma 4.4)

Let $V \subset \mathbb{R}^p$, $U \subset \mathbb{R}^q$ be open and $\mathbf{g} : V \to U$, $f : U \to \mathbb{R}$ twice differentiable. Let $\mathbf{y} = \mathbf{g}(\mathbf{x})$ and $\mathbf{J} = \partial \mathbf{g}/\partial \mathbf{x}^\top$ $(q \times p)$. Then

(i) $\dfrac{\partial f(\mathbf{y})}{\partial \mathbf{x}} = \mathbf{J}^\top \dfrac{\partial f}{\partial \mathbf{y}}$;

(ii) $\dfrac{\partial^2 f(\mathbf{y})}{\partial \mathbf{x} \partial \mathbf{x}^\top} = \mathbf{J}^\top \dfrac{\partial^2 f}{\partial \mathbf{y} \partial \mathbf{y}^\top} \mathbf{J} + \sum_{i=1}^{q} \dfrac{\partial f}{\partial y_i} \dfrac{\partial^2 g_i}{\partial \mathbf{x} \partial \mathbf{x}^\top}$;

(iii) (affine map) if $\mathbf{g}(\mathbf{x}) = \mathbf{B}\mathbf{x} + \mathbf{b}$ with fixed $q \times p$ $\mathbf{B}$ and $\mathbf{b} \in \mathbb{R}^q$, then $\mathbf{J} = \mathbf{B}$, $\dfrac{\partial f(\mathbf{y})}{\partial \mathbf{x}} = \mathbf{B}^\top \dfrac{\partial f}{\partial \mathbf{y}}$ and $\dfrac{\partial^2 f(\mathbf{y})}{\partial \mathbf{x} \partial \mathbf{x}^\top} = \mathbf{B}^\top \dfrac{\partial^2 f}{\partial \mathbf{y} \partial \mathbf{y}^\top} \mathbf{B}$.

For $p = q = 1$, (i) and (ii) read $(f \circ g)' = f'(g) g'$ and $(f \circ g)'' = f''(g)(g')^2 + f'(g) g''$.

Proof. (i) Since $J_{ik} = \partial y_i / \partial x_k$, the chain rule gives $\partial f(\mathbf{y})/\partial x_k = \sum_{i=1}^{q} (\partial f/\partial y_i) J_{ik} = (\mathbf{J}^\top \partial f/\partial \mathbf{y})_k$.

(ii) Differentiate this in x_l. By the product rule and the chain rule applied to $\partial f/\partial y_i$,

$$\frac{\partial^2 f(\mathbf{y})}{\partial x_l \partial x_k} = \sum_{i,m} J_{ik} \frac{\partial^2 f}{\partial y_m \partial y_i} J_{ml} + \sum_{i=1}^{q} \frac{\partial f}{\partial y_i} \frac{\partial^2 g_i}{\partial x_l \partial x_k}.$$

The first sum is the (k,l) entry of $\mathbf{J}^\top (\partial^2 f/\partial \mathbf{y} \partial \mathbf{y}^\top) \mathbf{J}$ and the second is the (k,l) entry of $\sum_i (\partial f/\partial y_i) \partial^2 g_i / \partial \mathbf{x} \partial \mathbf{x}^\top$.

(iii) Here $g_i(\mathbf{x}) = \sum_k b_{ik} x_k + b_i$, so $J_{ik} = b_{ik}$ and every second partial of g_i is 0. ■

Theorem 2.20: gradient and Hessian of a quadratic form; Jacobians of basic maps (Lab 3, Proposition 4.5, Lemma 4.6)

For symmetric $\mathbf{A}$, $\dfrac{\partial\, \mathbf{x}^\top \mathbf{A} \mathbf{x}}{\partial \mathbf{x}} = 2\mathbf{A}\mathbf{x}$ and $\dfrac{\partial^2\, \mathbf{x}^\top \mathbf{A} \mathbf{x}}{\partial \mathbf{x} \partial \mathbf{x}^\top} = 2\mathbf{A}$. Also $\partial \mathbf{x}/\partial \mathbf{x}^\top = \mathbf{I}_p$ (the analogue of $dx/dx = 1$), $\partial(\mathbf{B}\mathbf{x}+\mathbf{b})/\partial \mathbf{x}^\top = \mathbf{B}$ for fixed $\mathbf{B} \in \mathbb{R}^{q \times p}$, $\mathbf{b} \in \mathbb{R}^q$, and, for fixed $\mathbf{x}$, $\partial(\mathbf{x} - \boldsymbol{\mu})/\partial \boldsymbol{\mu}^\top = -\mathbf{I}_p$.

Proof. By Theorem 2.3(i), $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$, so $\partial/\partial x_k = \sum_j a_{kj} x_j + \sum_i a_{ik} x_i = 2(\mathbf{A}\mathbf{x})_k$; differentiating again in x_l gives $2a_{kl}$, the (k,l) entry of $2\mathbf{A}$.

For the Jacobians: the ith component of $\mathbf{B}\mathbf{x} + \mathbf{b}$ is $\sum_k b_{ik} x_k + b_i$, whose partial in x_k is b_{ik}, the (i,k) entry of $\mathbf{B}$. Taking $\mathbf{B} = \mathbf{I}_p$, $\mathbf{b} = \mathbf{0}$ gives the identity map, and $\mathbf{B} = -\mathbf{I}_p$, $\mathbf{b} = \mathbf{x}$ (with $\boldsymbol{\mu}$ as the variable) gives the last claim. ■

5 Matrix Calculus

To optimize over covariance matrices we differentiate with respect to a matrix variable: matrix gradients give first-order conditions and directional derivatives give second-order conditions along symmetric directions. This is how the MLE of $\boldsymbol{\Sigma}$ is found in Result 4.11.

Definition 2.21: directional derivatives and matrix gradient (Lab 3, Definitions 5.1 and 5.3)

Let C be open in $\mathbb{R}^p$, $\mathbb{R}^{p\times q}$ or the space of $p\times p$ symmetric matrices, $f: C\to\mathbb{R}$, $\boldsymbol{\Theta}\in C$ and $\mathbf{H}$ a direction in the same space:

$$Df(\boldsymbol{\Theta})[\mathbf{H}] = \frac{d}{dt} f(\boldsymbol{\Theta}+t\mathbf{H})\Big|_{t=0}, \qquad D^2 f(\boldsymbol{\Theta})[\mathbf{H}] = \frac{d^2}{dt^2} f(\boldsymbol{\Theta}+t\mathbf{H})\Big|_{t=0}.$$

For f differentiable in the entries of $\mathbf{X}=(x_{ij})\in\mathbb{R}^{p\times q}$, the matrix gradient $\partial f/\partial\mathbf{X}$ is the $p\times q$ matrix with (i,j) entry $\partial f/\partial x_{ij}$.

Theorem 2.22: directional derivative and gradient (Lab 3, Lemma 5.2, Proposition 5.4)

(i) If $C\subset\mathbb{R}^p$ and f is twice differentiable, $Df(\boldsymbol{\mu})[\mathbf{h}] = \mathbf{h}^\top \dfrac{\partial f}{\partial \boldsymbol{\mu}}$ and $D^2 f(\boldsymbol{\mu})[\mathbf{h}] = \mathbf{h}^\top \dfrac{\partial^2 f}{\partial\boldsymbol{\mu}\partial\boldsymbol{\mu}^\top}\mathbf{h}$.
(ii) If f is differentiable in the pq entries of $\mathbf{X}\in\mathbb{R}^{p\times q}$, then for every $\mathbf{H}\in\mathbb{R}^{p\times q}$, $Df(\mathbf{X})[\mathbf{H}] = \operatorname{tr}\Big\{\Big(\dfrac{\partial f}{\partial \mathbf{X}}\Big)^\top \mathbf{H}\Big\} = \sum_{i,j}\dfrac{\partial f}{\partial x_{ij}} h_{ij}$. For $q=1$ this is (i).

Proof. (i) Let $g(t)=f(\boldsymbol{\mu}+t\mathbf{h})$. Theorem 2.19(iii) applied to the affine map $t\mapsto\boldsymbol{\mu}+t\mathbf{h}$ (with $\mathbf{B}=\mathbf{h}$, a $p\times 1$ matrix) gives $g'(t)=\mathbf{h}^\top\partial f(\boldsymbol{\mu}+t\mathbf{h})/\partial\boldsymbol{\mu}$ and $g''(t)=\mathbf{h}^\top\{\partial^2 f(\boldsymbol{\mu}+t\mathbf{h})/\partial\boldsymbol{\mu}\partial\boldsymbol{\mu}^\top\}\mathbf{h}$. Set $t=0$.

(ii) Each entry of $\mathbf{X}+t\mathbf{H}$ is $x_{ij}+th_{ij}$, so by the chain rule $\frac{d}{dt}f(\mathbf{X}+t\mathbf{H}) = \sum_{i,j}\frac{\partial f}{\partial x_{ij}}(\mathbf{X}+t\mathbf{H})h_{ij}$. At $t=0$ this is $\sum_{i,j} g_{ij}h_{ij}$ with $\mathbf{G}=(g_{ij})=\partial f/\partial\mathbf{X}$, and as in the proof of Theorem 2.3(iv), $\sum_{i,j} g_{ij}h_{ij} = \operatorname{tr}(\mathbf{G}^\top\mathbf{H})$. ■

Theorem 2.23: derivatives of basic matrix functions (Lab 3, Propositions 5.5 and 5.6)

For $\boldsymbol{\Sigma}\succ 0$, symmetric $\mathbf{H}$ and $p\times p$ $\mathbf{B}$:

$$D\log|\boldsymbol{\Sigma}|[\mathbf{H}] = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}), \qquad D^2\log|\boldsymbol{\Sigma}|[\mathbf{H}] = -\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\Sigma}^{-1}\mathbf{H}),$$

$$D\operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = \operatorname{tr}(\mathbf{H}\mathbf{B}), \qquad D^2\operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = 0,$$

$$\frac{\partial\log|\boldsymbol{\Sigma}|}{\partial\boldsymbol{\Sigma}} = \boldsymbol{\Sigma}^{-1}, \qquad \frac{\partial\operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})}{\partial\boldsymbol{\Sigma}} = \mathbf{B}^\top.$$

D and D^2 are linear: $D^k(a_1f_1+a_2f_2)(\boldsymbol{\Sigma})[\mathbf{H}] = a_1D^kf_1(\boldsymbol{\Sigma})[\mathbf{H}] + a_2D^kf_2(\boldsymbol{\Sigma})[\mathbf{H}]$, $k=1,2$.

Proof. Let $\mathbf{M}=\boldsymbol{\Sigma}^{-1/2}\mathbf{H}\boldsymbol{\Sigma}^{-1/2}$, symmetric by Theorem 2.12(i); write $\mathbf{M}=\mathbf{V}\boldsymbol{\Lambda}(\mathbf{M})\mathbf{V}^\top$.

Step 1. By Theorem 2.12, $\boldsymbol{\Sigma}+t\mathbf{H}=\boldsymbol{\Sigma}^{1/2}(\mathbf{I}_p+t\mathbf{M})\boldsymbol{\Sigma}^{1/2}$, and $\mathbf{I}_p+t\mathbf{M}=\mathbf{V}\{\mathbf{I}_p+t\boldsymbol{\Lambda}(\mathbf{M})\}\mathbf{V}^\top$ has eigenvalues $1+t\lambda_k(\mathbf{M})$. Hence for small $|t|$

$$\log|\boldsymbol{\Sigma}+t\mathbf{H}| = \log|\boldsymbol{\Sigma}| + \sum_{k=1}^{p}\log\{1+t\lambda_k(\mathbf{M})\}.$$

Step 2. The first and second derivatives at $t=0$ are $\sum_k\lambda_k(\mathbf{M})$ and $-\sum_k\lambda_k(\mathbf{M})^2$. Since $\mathbf{M}^2 = \mathbf{V}\boldsymbol{\Lambda}(\mathbf{M})^2\mathbf{V}^\top$, Theorems 2.9 and 2.3(iii) give $\sum_k\lambda_k(\mathbf{M}) = \operatorname{tr}\mathbf{M} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H})$ and $\sum_k\lambda_k(\mathbf{M})^2 = \operatorname{tr}(\mathbf{M}^2) = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\Sigma}^{-1}\mathbf{H})$.

Step 3. $\operatorname{tr}\{(\boldsymbol{\Sigma}+t\mathbf{H})\mathbf{B}\} = \operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B}) + t\operatorname{tr}(\mathbf{H}\mathbf{B})$ is linear in t. Linearity of D^k follows from that of d^k/dt^k.

Step 4 (gradients). $X \mapsto \log|\mathbf{X}|$ is differentiable in the entries on the open set $\{|\mathbf{X}| > 0\}$. For any $p \times p$ $\mathbf{H}$ let $\mathbf{N} = \boldsymbol{\Sigma}^{-1}\mathbf{H}$; then $|\boldsymbol{\Sigma} + t\mathbf{H}| = |\boldsymbol{\Sigma}||\mathbf{I}_p + t\mathbf{N}| = |\boldsymbol{\Sigma}|\{1 + t\operatorname{tr}(\mathbf{N}) + O(t^2)\}$ by Theorem 2.4, so $D\log|\boldsymbol{\Sigma}|[\mathbf{H}] = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}) = \operatorname{tr}\{(\boldsymbol{\Sigma}^{-1})^\top \mathbf{H}\}$, as $\boldsymbol{\Sigma}^{-1}$ is symmetric. Also $D\operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = \operatorname{tr}(\mathbf{H}\mathbf{B}) = \operatorname{tr}(\mathbf{B}\mathbf{H}) = \operatorname{tr}\{(\mathbf{B}^\top)^\top \mathbf{H}\}$. Comparing with Theorem 2.22(ii) for all $\mathbf{H}$ identifies the gradients. ■

Theorem 2.24: symmetric matrix variables (Lab 3, Corollary 5.7)

Let f be differentiable on an open subset of $\mathbb{R}^{p\times p}$ containing a symmetric $\mathbf{X}$, and suppose $\mathbf{G} = \partial f/\partial \mathbf{X}$ is symmetric. Then $Df(\mathbf{X})[\mathbf{H}] = 0$ for all symmetric $\mathbf{H}$ iff $\mathbf{G} = \mathbf{0}$.

Proof. If $\mathbf{G} = \mathbf{0}$, $Df(\mathbf{X})[\mathbf{H}] = \operatorname{tr}(\mathbf{G}^\top \mathbf{H}) = 0$ by Theorem 2.22(ii). Conversely, take $\mathbf{H} = \mathbf{G}$ and use Theorem 2.3(iv): $0 = Df(\mathbf{X})[\mathbf{G}] = \operatorname{tr}(\mathbf{G}^\top \mathbf{G}) = \|\mathbf{G}\|_F^2$, so $\mathbf{G} = \mathbf{0}$. ■

6 Optimization

Motivation. A critical point of a strictly convex function is its unique minimizer, some transformations of the objective do not change the optimizer, and a function of two parameters can be minimized one parameter at a time. Result 4.11 (the normal MLE) uses all three.

Theorem 2.25: convexity and unique minimizer (Lab 3, Proposition 6.1)

Let C be an open convex set in $\mathbb{R}^p$ or in the space of $p \times p$ symmetric matrices, and $f : C \to \mathbb{R}$ such that $t \mapsto f(\boldsymbol{\Theta} + t\mathbf{H})$ is twice differentiable for all $\boldsymbol{\Theta} \in C$ and all $\mathbf{H}$. Suppose $D^2 f(\boldsymbol{\Theta})[\mathbf{H}] > 0$ for every $\boldsymbol{\Theta} \in C$ and every $\mathbf{H} \neq \mathbf{0}$. Then
(i) f is strictly convex on C;
(ii) a point $\hat{\boldsymbol{\Theta}} \in C$ with $Df(\hat{\boldsymbol{\Theta}})[\mathbf{H}] = 0$ for all $\mathbf{H}$ is the unique global minimizer of f on C;
(iii) for $C = \mathbb{R}^p$ and f with continuous second partials, these conditions read $\partial^2 f/\partial\boldsymbol{\Theta}\partial\boldsymbol{\Theta}^\top \succ 0$ and $\partial f(\hat{\boldsymbol{\Theta}})/\partial\boldsymbol{\Theta} = \mathbf{0}$.

Proof. (i) Take $\boldsymbol{\Theta}_0 \neq \boldsymbol{\Theta}_1$ in C, let $\mathbf{H} = \boldsymbol{\Theta}_1 - \boldsymbol{\Theta}_0$ and $g(t) = f(\boldsymbol{\Theta}_0 + t\mathbf{H})$, $t \in [0,1]$; the segment lies in C by convexity. Then $g''(t) = D^2 f(\boldsymbol{\Theta}_0 + t\mathbf{H})[\mathbf{H}] > 0$, so g is strictly convex on every segment (Definition 2.16(i)): f is strictly convex.

(ii) Let $\boldsymbol{\Theta}_0 = \hat{\boldsymbol{\Theta}}$. Then $g'(0) = Df(\hat{\boldsymbol{\Theta}})[\mathbf{H}] = 0$ and g' is strictly increasing, so $g'(t) > 0$ for $t \in (0,1]$. By the mean value theorem $g(1) - g(0) = g'(\xi)$ for some $\xi \in (0,1)$, so $f(\boldsymbol{\Theta}_1) = g(1) > g(0) = f(\hat{\boldsymbol{\Theta}})$.

(iii) By Theorem 2.22(i), $D^2 f(\boldsymbol{\Theta})[\mathbf{H}] = \mathbf{H}^\top(\partial^2 f/\partial\boldsymbol{\Theta}\partial\boldsymbol{\Theta}^\top)\mathbf{H}$. The Hessian is symmetric (Theorem 2.18), so this is positive for all $\mathbf{H} \neq \mathbf{0}$ iff the Hessian is positive definite; and $Df(\hat{\boldsymbol{\Theta}})[\mathbf{H}] = \mathbf{H}^\top \partial f(\hat{\boldsymbol{\Theta}})/\partial\boldsymbol{\Theta} = 0$ for all $\mathbf{H}$ iff $\partial f(\hat{\boldsymbol{\Theta}})/\partial\boldsymbol{\Theta} = \mathbf{0}$. ■

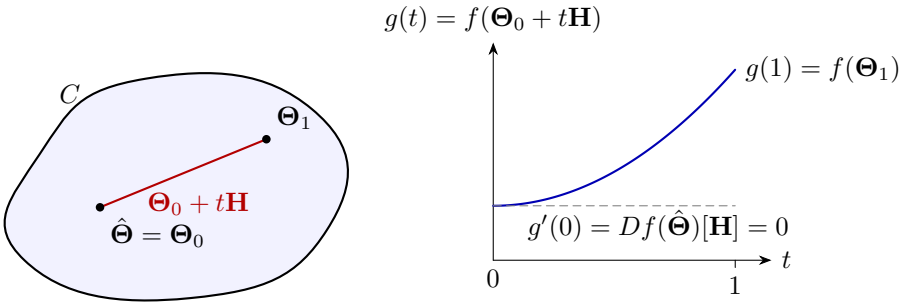


Figure. *The proof of Theorem 2.25: restricted to any segment from the critical point $\hat{\boldsymbol{\Theta}}$, the objective is a strictly convex function g with $g'(0) = 0$, so $g(1) > g(0)$. The open convex set C is, for example, the set of positive definite matrices.*

Theorem 2.26: transformations that keep the optimizer; profiling (Lab 3, Lemma 6.2, Proposition 6.3)

Let f be a function on a set Θ, $a > 0$ and c a constant not depending on θ.
(i) $\arg\min_\theta \{a f(\theta) + c\} = \arg\min_\theta f(\theta)$.
(ii) $\arg\max_\theta f(\theta) = \arg\min_\theta \{-f(\theta)\}$.
(iii) $\arg\max_\theta L(\theta) = \arg\max_\theta \log L(\theta)$ for $L > 0$.
(iv) (Change of variables) If $\phi : \Theta \to \Xi$ is a bijection, $\arg\min_\theta f(\theta) = \phi^{-1}(\arg\min_\eta f(\phi^{-1}(\eta)))$.
(v) (Profiling) Let $\ell(\theta, \eta)$ be a function of two arguments. Suppose that for each η, $\hat\theta(\eta)$ minimizes $\ell(\cdot, \eta)$, and define the profile function $p(\eta) = \ell(\hat\theta(\eta), \eta)$. If $\hat\eta$ minimizes p, then $(\hat\theta(\hat\eta), \hat\eta)$ minimizes ℓ jointly.

Proof. For any $\theta, \theta_1 \in \Theta$:

(i) $a f(\theta) + c \le a f(\theta_1) + c$ iff $f(\theta) \le f(\theta_1)$, since $a > 0$;

(ii) $f(\theta) \ge f(\theta_1)$ iff $-f(\theta) \le -f(\theta_1)$;

(iii) $L(\theta) \ge L(\theta_1)$ iff $\log L(\theta) \ge \log L(\theta_1)$, since log is strictly increasing;

(iv) with $\eta = \phi(\theta)$, $\eta_1 = \phi(\theta_1)$, $f(\theta) \le f(\theta_1)$ iff $f(\phi^{-1}(\eta)) \le f(\phi^{-1}(\eta_1))$, and η_1 ranges over all of Ξ as θ_1 ranges over Θ;

(v) for any (θ, η), $\ell(\theta, \eta) \ge \ell(\hat\theta(\eta), \eta) = p(\eta) \ge p(\hat\eta) = \ell(\hat\theta(\hat\eta), \hat\eta)$. ■

The MLE (Ch. 4) first profiles out $\boldsymbol{\mu}$ and then minimizes over $\boldsymbol{\Sigma}$.

7 Multivariate Normal Distribution: the Facts Used Here

Definition 2.27: multivariate normal distribution (Lab 3, Definition 7.1, Proposition 7.2)

For $\boldsymbol{\mu} \in \mathbb{R}^p$ and $\boldsymbol{\Sigma} \succ 0$, $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ iff $\mathbf{X}$ has density $f(\mathbf{x}) = |2\pi\boldsymbol{\Sigma}|^{-1/2} \exp\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\}$, $\mathbf{x} \in \mathbb{R}^p$ (Part 1 slides, p. 123). For i.i.d. $Z_1, \dots, Z_p \sim N(0,1)$, χ^2_p is the distribution of $Z_1^2 + \cdots + Z_p^2$. If $\mathbf{Z} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$, then (i) $\mathbf{t}^\top \mathbf{Z} \sim N(0, \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$ for every $\mathbf{t} \in \mathbb{R}^p$ ($N(0,0)$ is the point mass at 0), and (ii) $\mathbf{Z}^\top \boldsymbol{\Sigma}^{-1} \mathbf{Z} \sim \chi^2_p$ (Results 4.2 and 4.7; proofs in Ch. 3).

8 Probability and Convergence

Motivation. Results 4.12 and 4.13 describe the limits of the sample mean and the sample covariance matrix; this section collects the tools they need: convergence of random vectors and matrices, the CLT, Slutsky's lemma and the Cramér–Wold device.

Definition 2.28: standard results from mathematical statistics (Lab 3, Recall 8.1)

Definitions and moments.

(S0) For random vectors, $\mathbf{Y}_n \xrightarrow{p} \mathbf{c}$ means $P(\|\mathbf{Y}_n - \mathbf{c}\| > \epsilon) \to 0$ for every $\epsilon > 0$, and $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$ means $\mathrm{E}g(\mathbf{Y}_n) \to \mathrm{E}g(\mathbf{Y})$ for every bounded continuous g. $\mathrm{Cov}(\mathbf{Y}) = \mathrm{E}\{(\mathbf{Y} - \mathrm{E}\mathbf{Y})(\mathbf{Y} - \mathrm{E}\mathbf{Y})^\top\}$.

(S1) Markov: for a random variable $W \ge 0$ and $a > 0$, $P(W \ge a) \le \mathrm{E}(W)/a$.

(S2) $\mathrm{E}(\mathbf{AY}+\mathbf{b}) = \mathbf{A}\mathrm{E}\mathbf{Y}+\mathbf{b}$, $\mathrm{Cov}(\mathbf{AY}+\mathbf{b}) = \mathbf{A}\mathrm{Cov}(\mathbf{Y})\mathbf{A}^\top$; for independent $\mathbf{Y}_1,\dots,\mathbf{Y}_n$, $\mathrm{Cov}(\sum_j \mathbf{Y}_j) = \sum_j \mathrm{Cov}(\mathbf{Y}_j)$; $\mathrm{E}\|\mathbf{Y}-\mathrm{E}\mathbf{Y}\|^2 = \mathrm{tr}\mathrm{Cov}(\mathbf{Y})$; and $\mathrm{E}(V^2)=0$ implies $V=0$ almost surely.

Limit theorems.

(S3) Weak law of large numbers: if $Y_1, Y_2, \dots$ are i.i.d. with $\mathrm{E}|Y_1| < \infty$, $\frac{1}{n}\sum_{j=1}^n Y_j \xrightarrow{p} \mathrm{E}(Y_1)$.

(S4) Univariate CLT: if $Y_1, Y_2, \dots$ are i.i.d. with mean m and variance $v < \infty$, $\sqrt{n}(\bar{Y}_n - m) \xrightarrow{d} N(0, v)$.

Tools for limits.

(S5) Continuous mapping theorem (CMT): if $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$ and g is continuous, $g(\mathbf{Y}_n) \xrightarrow{d} g(\mathbf{Y})$; if $\mathbf{Y}_n \xrightarrow{p} \mathbf{c}$ and g is continuous at $\mathbf{c}$, $g(\mathbf{Y}_n) \xrightarrow{p} g(\mathbf{c})$ (g need only be continuous at the single point $\mathbf{c}$).

(S6) Slutsky: if $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$ and $\mathbf{W}_n \xrightarrow{p} \mathbf{c}$ for a constant $\mathbf{c}$, then $(\mathbf{Y}_n, \mathbf{W}_n) \xrightarrow{d} (\mathbf{Y}, \mathbf{c})$.

(S7) Joint convergence to constants: if $\mathbf{Y}_n^{(k)} \xrightarrow{p} \mathbf{c}_k$, $k = 1,\dots,m$, then $(\mathbf{Y}_n^{(1)},\dots,\mathbf{Y}_n^{(m)}) \xrightarrow{p} (\mathbf{c}_1,\dots,\mathbf{c}_m)$; a deterministic sequence $\mathbf{c}_n \to \mathbf{c}$ satisfies $\mathbf{c}_n \xrightarrow{p} \mathbf{c}$.

Theorem 2.29: Cauchy–Schwarz inequality for random variables (Lab 3, Lemma 8.2)

For random variables U, V with $\mathrm{E}(U^2) < \infty$ and $\mathrm{E}(V^2) < \infty$, $\mathrm{E}|UV| \le \{\mathrm{E}(U^2)\mathrm{E}(V^2)\}^{1/2}$.

Proof. Since $|UV| \le (U^2+V^2)/2$, $\mathrm{E}|UV| < \infty$. If $\mathrm{E}(V^2) = 0$, then $V = 0$ almost surely by (S2), and both sides are 0. Otherwise, for every $t \in \mathbb{R}$, $0 \le \mathrm{E}(|U| - t|V|)^2 = \mathrm{E}(U^2) - 2t\mathrm{E}|UV| + t^2\mathrm{E}(V^2)$. Taking $t = \mathrm{E}|UV|/\mathrm{E}(V^2)$ gives $0 \le \mathrm{E}(U^2) - \{\mathrm{E}|UV|\}^2/\mathrm{E}(V^2)$, which is the claim. ■

Theorem 2.30: vector Markov inequality and matrix convergence (Lab 3, Lemma 8.3, Definition 8.4, Lemma 8.5)

(i) For a random vector $\mathbf{Y}$ and $\epsilon > 0$, $P(\|\mathbf{Y}\| > \epsilon) \le \mathrm{E}\|\mathbf{Y}\|^2/\epsilon^2$.
(ii) For random $p \times q$ matrices $\mathbf{A}_n$ and a fixed $\mathbf{A}$, $\mathbf{A}_n \xrightarrow{p} \mathbf{A}$ means that every entry of $\mathbf{A}_n$ converges in probability to the corresponding entry of $\mathbf{A}$. Then $\mathbf{A}_n \xrightarrow{p} \mathbf{A}$ iff $\|\mathbf{A}_n - \mathbf{A}\|_F \xrightarrow{p} 0$ (Part 1 slides, p. 160).
(iii) Consequently $\mathbf{A}_n \xrightarrow{p} \mathbf{A}$ iff the vector of entries of $\mathbf{A}_n$ converges in probability, in the sense of (S0), to that of $\mathbf{A}$; hence (S5)–(S7) apply to random matrices.

Proof. (i) $\{\|\mathbf{Y}\| > \epsilon\} \subset \{\|\mathbf{Y}\|^2 \ge \epsilon^2\}$; apply (S1) to $W = \|\mathbf{Y}\|^2$ with $a = \epsilon^2$.

(ii) Suppose $\mathbf{A}_n \xrightarrow{p} \mathbf{A}$. Since $\|\mathbf{A}_n - \mathbf{A}\|_F^2 \le pq \max_{ij}(\mathbf{A}_n - \mathbf{A})_{ij}^2$, the event $\|\mathbf{A}_n - \mathbf{A}\|_F > \epsilon$ implies $|(\mathbf{A}_n - \mathbf{A})_{ij}| > \epsilon/\sqrt{pq}$ for some (i,j), so $P(\|\mathbf{A}_n - \mathbf{A}\|_F > \epsilon) \le \sum_{ij} P(|(\mathbf{A}_n - \mathbf{A})_{ij}| > \epsilon/\sqrt{pq}) \to 0$. Conversely $|(\mathbf{A}_n - \mathbf{A})_{ij}| \le \|\mathbf{A}_n - \mathbf{A}\|_F$ for every (i,j).

(iii) $\|\mathbf{A}_n - \mathbf{A}\|_F$ equals the Euclidean norm of the difference of the vectors of entries, so (iii) is (ii) restated with (S0). ■

Theorem 2.31: characteristic functions and the Cramér–Wold device (Lab 3, Theorem 8.6, Corollary 8.7)

For a random vector $\mathbf{Y} \in \mathbb{R}^k$ let $\varphi_{\mathbf{Y}}(\mathbf{t}) = \mathrm{E}\exp(i\mathbf{t}^\top \mathbf{Y})$, $\mathbf{t} \in \mathbb{R}^k$.
(i) (Uniqueness) $\varphi_{\mathbf{Y}}(\mathbf{t}) = \varphi_{\mathbf{W}}(\mathbf{t})$ for all $\mathbf{t}$ iff $\mathbf{Y}$ and $\mathbf{W}$ have the same distribution.
(ii) (Lévy's continuity theorem) $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$ iff $\varphi_{\mathbf{Y}_n}(\mathbf{t}) \to \varphi_{\mathbf{Y}}(\mathbf{t})$ for all $\mathbf{t} \in \mathbb{R}^k$.
These are proved in mathematical statistics (Part 1 slides, p. 125). **Cramér–Wold device**:
(iii) the distribution of $\mathbf{Y}$ is determined by the distributions of $\mathbf{t}^\top \mathbf{Y}$, $\mathbf{t} \in \mathbb{R}^k$: if $\mathbf{t}^\top \mathbf{Y}$ and $\mathbf{t}^\top \mathbf{W}$ have the same distribution for every $\mathbf{t}$, so do $\mathbf{Y}$ and $\mathbf{W}$;
(iv) $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$ iff $\mathbf{t}^\top \mathbf{Y}_n \xrightarrow{d} \mathbf{t}^\top \mathbf{Y}$ for every $\mathbf{t} \in \mathbb{R}^k$.

Proof of (iii) and (iv). For every $\mathbf{t}$, the definition gives the key identity $\varphi_{\mathbf{t}^\top \mathbf{Y}}(1) = \mathrm{E}e^{i\mathbf{t}^\top \mathbf{Y}} = \varphi_{\mathbf{Y}}(\mathbf{t})$.

(iii) If $\mathbf{t}^\top \mathbf{Y} \overset{d}{=} \mathbf{t}^\top \mathbf{W}$, then $\varphi_{\mathbf{Y}}(\mathbf{t}) = \varphi_{\mathbf{t}^\top \mathbf{Y}}(1) = \varphi_{\mathbf{t}^\top \mathbf{W}}(1) = \varphi_{\mathbf{W}}(\mathbf{t})$. This holds for all $\mathbf{t}$, so (i) gives the claim.

(iv) If $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$, then $\mathbf{t}^\top \mathbf{Y}_n \xrightarrow{d} \mathbf{t}^\top \mathbf{Y}$ by the CMT (S5). Conversely, if $\mathbf{t}^\top \mathbf{Y}_n \xrightarrow{d} \mathbf{t}^\top \mathbf{Y}$, (ii) applied to these scalars gives $\varphi_{\mathbf{t}^\top \mathbf{Y}_n}(s) \to \varphi_{\mathbf{t}^\top \mathbf{Y}}(s)$ for every $s \in \mathbb{R}$; at $s = 1$, $\varphi_{\mathbf{Y}_n}(\mathbf{t}) \to \varphi_{\mathbf{Y}}(\mathbf{t})$. Since $\mathbf{t}$ is arbitrary, (ii) gives $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$. ■

"All one-dimensional linear projections determine the multivariate distribution" underlies the general definition of the multivariate normal (Ch. 3) and the multivariate CLT.

9 Selected exercises

Motivation. Three further tools: the min–max characterization of every eigenvalue (which makes sample eigenvalues consistent), Hadamard's determinant inequality (which bounds $|\mathbf{R}|$ in independence tests), and the Kronecker product (the language for the covariance of a whole data matrix).

Exercise 2.1 (Courant–Fischer and Weyl)

[Supplement] Let $\mathbf{A}$ be symmetric $p \times p$ with eigenvalues $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_p(\mathbf{A})$ and orthonormal eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_p$.
(a) Show that for $k = 1, \ldots, p$,

$$\lambda_k(\mathbf{A}) = \max_{\mathcal{V}:\ \dim \mathcal{V} = k} \ \min_{\mathbf{x} \in \mathcal{V},\ \|\mathbf{x}\| = 1} \mathbf{x}^\top \mathbf{A} \mathbf{x}.$$

(b) For symmetric $\mathbf{A}, \mathbf{B}$ show $|\lambda_k(\mathbf{A}) - \lambda_k(\mathbf{B})| \leq \|\mathbf{A} - \mathbf{B}\|_F$ for every k, which extends Theorem 2.10(ii) to all eigenvalues.

Solution [Supplement].

(a)

Step 1 ($\geq$). Let $\mathcal{V}_k = \mathrm{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$. A unit vector $\mathbf{x} \in \mathcal{V}_k$ is $\mathbf{x} = \sum_{i \leq k} c_i \mathbf{u}_i$ with $\sum_i c_i^2 = 1$, so $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i \leq k} \lambda_i c_i^2 \geq \lambda_k$. Hence the minimum over $\mathcal{V}_k$ is at least λ_k (it equals λ_k at $\mathbf{x} = \mathbf{u}_k$), and the maximum over all $\mathcal{V}$ is at least λ_k.

Step 2 ($\leq$). Let $\mathcal{V}$ be any subspace with $\dim \mathcal{V} = k$ and $\mathcal{W} = \mathrm{span}(\mathbf{u}_k, \ldots, \mathbf{u}_p)$, $\dim \mathcal{W} = p - k + 1$. Since $\dim \mathcal{V} + \dim \mathcal{W} = p + 1 > p$, the intersection $\mathcal{V} \cap \mathcal{W}$ contains a unit vector $\mathbf{x} = \sum_{i \geq k} c_i \mathbf{u}_i$, and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i \geq k} \lambda_i c_i^2 \leq \lambda_k$. So the minimum over $\mathcal{V}$ is at most λ_k for every $\mathcal{V}$.

(b)

Step 1. Let $\epsilon = \|\mathbf{A}-\mathbf{B}\|_F$. As in the proof of Theorem 2.10(ii), $|\mathbf{x}^\top\mathbf{A}\mathbf{x} - \mathbf{x}^\top\mathbf{B}\mathbf{x}| \le \epsilon$ for every unit $\mathbf{x}$.

Step 2. Fix $\mathcal{V}$ with $\dim\mathcal{V} = k$. The unit sphere of $\mathcal{V}$ is compact, so $\mathbf{x}\mapsto\mathbf{x}^\top\mathbf{B}\mathbf{x}$ attains its minimum on it at some $\mathbf{x}_0$. Then

$$\min_{\mathbf{x}\in\mathcal{V},\|\mathbf{x}\|=1}\mathbf{x}^\top\mathbf{A}\mathbf{x} \le \mathbf{x}_0^\top\mathbf{A}\mathbf{x}_0 \le \mathbf{x}_0^\top\mathbf{B}\mathbf{x}_0 + \epsilon = \min_{\mathbf{x}\in\mathcal{V},\|\mathbf{x}\|=1}\mathbf{x}^\top\mathbf{B}\mathbf{x} + \epsilon.$$

Step 3. Taking the maximum over $\mathcal{V}$ and using (a) gives $\lambda_k(\mathbf{A}) \le \lambda_k(\mathbf{B}) + \epsilon$. Exchanging $\mathbf{A}$ and $\mathbf{B}$ gives $\lambda_k(\mathbf{B}) \le \lambda_k(\mathbf{A}) + \epsilon$.

Consequently each $\lambda_k(\cdot)$ is continuous, and $\mathbf{S}\xrightarrow{p}\boldsymbol{\Sigma}$ implies $\lambda_k(\mathbf{S})\xrightarrow{p}\lambda_k(\boldsymbol{\Sigma})$ for every k (sample PC variances are consistent). ■

Exercise 2.2 (Hadamard's inequality)

[Supplement] Let $\mathbf{A}\succ 0$ be $p\times p$. Show that $|\mathbf{A}| \le \prod_{i=1}^p a_{ii}$, with equality iff $\mathbf{A}$ is diagonal. Deduce that a correlation matrix $\mathbf{R}\succ 0$ satisfies $0 < |\mathbf{R}| \le 1$, with $|\mathbf{R}| = 1$ iff $\mathbf{R} = \mathbf{I}_p$.

Solution [Supplement].

Step 1 (reduction). $a_{ii} = \mathbf{e}_i^\top\mathbf{A}\mathbf{e}_i > 0$. Let $\mathbf{D} = \operatorname{diag}(a_{11},\dots,a_{pp})$ and $\mathbf{B} = \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$. Then $\mathbf{B}\succ 0$, $b_{ii} = 1$ and $|\mathbf{B}| = |\mathbf{A}|/\prod_i a_{ii}$, so it suffices to show $|\mathbf{B}| \le 1$ with equality iff $\mathbf{B} = \mathbf{I}_p$.

Step 2. Let $\mu_1,\dots,\mu_p > 0$ be the eigenvalues of $\mathbf{B}$. By Theorem 2.9, $\sum_i\mu_i = \operatorname{tr}\mathbf{B} = p$ and $\log|\mathbf{B}| = \sum_i\log\mu_i$. Theorem 2.15 gives $\log\mu_i \le \mu_i - 1$, so

$$\log|\mathbf{B}| \le \sum_{i=1}^p(\mu_i - 1) = \operatorname{tr}\mathbf{B} - p = 0.$$

Step 3 (equality). Equality holds iff $\mu_i = 1$ for all i, i.e. $\mathbf{B} = \mathbf{U}\mathbf{I}_p\mathbf{U}^\top = \mathbf{I}_p$, i.e. $\mathbf{A} = \mathbf{D}$ is diagonal.

For a correlation matrix $a_{ii} = 1$, and $|\mathbf{R}| > 0$ because $\mathbf{R}\succ 0$. ■

Exercise 2.3 (Kronecker products and vec)

[Supplement] For $\mathbf{A}\in\mathbb{R}^{m\times n}$ and $\mathbf{B}\in\mathbb{R}^{r\times s}$ let $\mathbf{A}\otimes\mathbf{B} = [a_{ij}\mathbf{B}]_{i,j}\in\mathbb{R}^{mr\times ns}$, and let $\operatorname{vec}(\mathbf{X})$ stack the columns of $\mathbf{X}$.
(a) Show $(\mathbf{A}\otimes\mathbf{B})(\mathbf{C}\otimes\mathbf{D}) = \mathbf{A}\mathbf{C}\otimes\mathbf{B}\mathbf{D}$ whenever the products are defined, $(\mathbf{A}\otimes\mathbf{B})^\top = \mathbf{A}^\top\otimes\mathbf{B}^\top$, and $(\mathbf{A}\otimes\mathbf{B})^{-1} = \mathbf{A}^{-1}\otimes\mathbf{B}^{-1}$ for invertible $\mathbf{A},\mathbf{B}$.
(b) For symmetric $\mathbf{A}$ $(m\times m)$ and $\mathbf{B}$ $(r\times r)$ with eigenpairs $(\lambda_i,\mathbf{u}_i)$ and $(\mu_j,\mathbf{v}_j)$, show that $\mathbf{A}\otimes\mathbf{B}$ has the orthonormal eigenvectors $\mathbf{u}_i\otimes\mathbf{v}_j$ with eigenvalues $\lambda_i\mu_j$; hence $|\mathbf{A}\otimes\mathbf{B}| = |\mathbf{A}|^r|\mathbf{B}|^m$ and $\operatorname{tr}(\mathbf{A}\otimes\mathbf{B}) = \operatorname{tr}\mathbf{A}\operatorname{tr}\mathbf{B}$.
(c) Show $\operatorname{vec}(\mathbf{A}\mathbf{X}\mathbf{B}) = (\mathbf{B}^\top\otimes\mathbf{A})\operatorname{vec}(\mathbf{X})$.
(d) If the $n\times p$ data matrix $\mathbf{X}$ has iid rows $N_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$, show $\operatorname{vec}(\mathbf{X})\sim N_{np}(\boldsymbol{\mu}\otimes\mathbf{1}_n,\ \boldsymbol{\Sigma}\otimes\mathbf{I}_n)$.

Solution [Supplement].

(a) The (i,k) block of the product is $\sum_j(a_{ij}\mathbf{B})(c_{jk}\mathbf{D}) = (\sum_j a_{ij}c_{jk})\mathbf{B}\mathbf{D} = (\mathbf{A}\mathbf{C})_{ik}\mathbf{B}\mathbf{D}$, the (i,k) block of $\mathbf{A}\mathbf{C}\otimes\mathbf{B}\mathbf{D}$. The (i,j) block of $(\mathbf{A}\otimes\mathbf{B})^\top$ is the transpose of the (j,i) block, $a_{ji}\mathbf{B}^\top$. Finally $(\mathbf{A}\otimes\mathbf{B})(\mathbf{A}^{-1}\otimes\mathbf{B}^{-1}) = \mathbf{I}_m\otimes\mathbf{I}_r = \mathbf{I}_{mr}$.

(b)

Step 1. Treating vectors as one-column matrices, (a) gives $(\mathbf{A}\otimes\mathbf{B})(\mathbf{u}_i\otimes\mathbf{v}_j) = \mathbf{A}\mathbf{u}_i\otimes\mathbf{B}\mathbf{v}_j = \lambda_i\mu_j\,\mathbf{u}_i\otimes\mathbf{v}_j$.

Step 2. $(\mathbf{u}_i \otimes \mathbf{v}_j)^\top (\mathbf{u}_k \otimes \mathbf{v}_l) = (\mathbf{u}_i^\top \mathbf{u}_k)(\mathbf{v}_j^\top \mathbf{v}_l) = \delta_{ik}\delta_{jl}$, so these mr vectors form an orthonormal eigenbasis of the symmetric matrix $\mathbf{A} \otimes \mathbf{B}$.

Step 3. By Theorem 2.9, $|\mathbf{A} \otimes \mathbf{B}| = \prod_{i,j} \lambda_i \mu_j = (\prod_i \lambda_i)^r (\prod_j \mu_j)^m = |\mathbf{A}|^r |\mathbf{B}|^m$ and $\operatorname{tr}(\mathbf{A} \otimes \mathbf{B}) = \sum_{i,j} \lambda_i \mu_j = \operatorname{tr}\mathbf{A} \operatorname{tr}\mathbf{B}$.

(c) Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have columns $\mathbf{x}_1, \dots, \mathbf{x}_p$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$. Column k of $\mathbf{AXB}$ is $\mathbf{AXb}_k = \sum_j b_{jk} \mathbf{A}\mathbf{x}_j = [b_{1k}\mathbf{A}, \dots, b_{pk}\mathbf{A}] \operatorname{vec}(\mathbf{X})$, and $[b_{1k}\mathbf{A}, \dots, b_{pk}\mathbf{A}]$ is the kth block row of $\mathbf{B}^\top \otimes \mathbf{A}$.

(d) Column j of $\mathbf{X}$ is $\mathbf{x}_{(j)} = (X_{1j}, \dots, X_{nj})^\top$ with mean $\mu_j \mathbf{1}_n$, so $\mathrm{E} \operatorname{vec}(\mathbf{X}) = \boldsymbol{\mu} \otimes \mathbf{1}_n$. For columns j, k, $\operatorname{Cov}(X_{ij}, X_{lk}) = \sigma_{jk}\delta_{il}$ by independence of the rows, so $\operatorname{Cov}(\mathbf{x}_{(j)}, \mathbf{x}_{(k)}) = \sigma_{jk}\mathbf{I}_n$, the (j,k) block of $\boldsymbol{\Sigma} \otimes \mathbf{I}_n$. Normality: the stacked rows $(\mathbf{X}_1^\top, \dots, \mathbf{X}_n^\top)^\top$ are jointly normal (independent normal blocks), and $\operatorname{vec}(\mathbf{X})$ is a permutation, hence a linear map, of this vector (Theorem 3.8). ■

Chapter 3. The Multivariate Normal Distribution

Definition and properties, conditional distributions, best linear prediction, multiple and partial correlation, quadratic forms

1 Definition

1.1 Non-singular case

The univariate normal density with mean μ and variance σ^2 is $f(x) = (2\pi\sigma^2)^{-1/2}\exp[-\frac{1}{2}(x-\mu)\sigma^{-2}(x-\mu)]$. About 68.3% of its mass lies within one standard deviation of μ and 95.4% within two. Replacing $(x-\mu)^2/\sigma^2$ by a squared Mahalanobis distance gives the multivariate version.

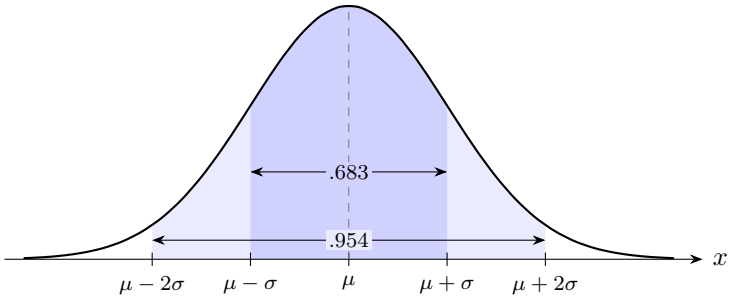


Figure. *The $N(\mu,\sigma^2)$ density. The dark band $\mu\pm\sigma$ carries probability* 0.683 *and the light band $\mu\pm2\sigma$ carries* 0.954.

Definition 3.1: multivariate normal (non-singular)

For $\boldsymbol{\mu}\in\mathbb{R}^p$ and $\boldsymbol{\Sigma}_{(p\times p)}\succ 0$, $\mathbf{X}\sim N_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$ means $\mathbf{X}$ has density

$$f(\mathbf{x}) = |2\pi\boldsymbol{\Sigma}|^{-1/2}\exp\Big[-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big], \qquad \mathbf{x}\in\mathbb{R}^p,$$

where $|2\pi\boldsymbol{\Sigma}| = (2\pi)^p|\boldsymbol{\Sigma}|$.

One checks that f is a valid joint density on $\mathbb{R}^p$: $f(\mathbf{x})>0$ for all $\mathbf{x}$, and $\int f(\mathbf{x})\,d\mathbf{x}=1$.

Proof that $\int f = 1$.

Step 1 (centering and diagonalizing). With $\boldsymbol{\Sigma} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$, $\mathbf{Q}$ orthogonal and $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1,\dots,\lambda_p)$, put $\mathbf{z} = \mathbf{Q}^\top(\mathbf{x}-\boldsymbol{\mu})$. Then $(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = \mathbf{z}^\top\boldsymbol{\Lambda}^{-1}\mathbf{z} = \sum_i z_i^2/\lambda_i$, $|\boldsymbol{\Sigma}| = \prod_i\lambda_i$, and the Jacobian is $|\mathbf{Q}|=\pm1$.

Step 2 (product of univariate integrals). The exponent separates, so

$$\int_{\mathbb{R}^p} f(\mathbf{x})\,d\mathbf{x} = \prod_{i=1}^{p}\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\lambda_i}}\exp\Big(-\frac{z_i^2}{2\lambda_i}\Big)dz_i = \prod_{i=1}^{p}1 = 1.$$

Equivalently, $\mathbf{Y} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{X}-\boldsymbol{\mu})$ has $d\mathbf{y}/d\mathbf{x} = |\boldsymbol{\Sigma}|^{-1/2}$, and

$$\int f\,d\mathbf{x} = \int\cdots\int(2\pi)^{-p/2}|\boldsymbol{\Sigma}|^{-1/2}e^{-\mathbf{y}^\top\mathbf{y}/2}|\boldsymbol{\Sigma}|^{1/2}d\mathbf{y} = \prod_{i=1}^{p}\int(2\pi)^{-1/2}e^{-y_i^2/2}dy_i = 1.$$

■

The substitution shows $\mathbf{Y} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{X}-\boldsymbol{\mu})\sim N_p(\mathbf{0},\mathbf{I}_p)$, i.e. $\mathbf{X}\overset{d}{=}\boldsymbol{\Sigma}^{1/2}\mathbf{Y}+\boldsymbol{\mu}$. Hence $\mathrm{E}\mathbf{X}=\boldsymbol{\mu}$, $\mathrm{Var}\mathbf{X}=\boldsymbol{\Sigma}$, and

$$U = (\mathbf{X}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) = \mathbf{Y}^\top\mathbf{Y} = \textstyle\sum_i Y_i^2\sim\chi^2(p).$$

1.2 General definition and Cramér–Wold

The characteristic function of a random vector $\mathbf{X}$ is $\varphi_{\mathbf{X}}(\mathbf{t}) = \mathrm{E}\exp(i\mathbf{t}^\top\mathbf{X})$, $\mathbf{t} \in \mathbb{R}^p$. It is $\mathbb{C}$-valued and always exists, and (Theorem 2.31): (1) $\varphi_{\mathbf{X}} = \varphi_{\mathbf{Y}}$ iff $\mathbf{X} \overset{d}{=} \mathbf{Y}$; (2) $\mathbf{X}_n \Rightarrow \mathbf{X}$ iff $\varphi_{\mathbf{X}_n}(\mathbf{t}) \to \varphi_{\mathbf{X}}(\mathbf{t})$ for all $\mathbf{t}$.

Remark 3.2: Cramér–Wold device (Corollary 2)

If $\mathbf{X}$ is a $p \times 1$ random vector, its distribution is uniquely determined by the distributions of the linear functions $\mathbf{t}^\top\mathbf{X}$, $\mathbf{t} \in \mathbb{R}^p$: the univariate laws of all linear combinations determine the multivariate joint law.

Proof. Let $F_{\mathbf{t}}$ be the distribution function of $\mathbf{t}^\top\mathbf{X}$. Then $\varphi_{\mathbf{X}}(\mathbf{t}) = \mathrm{E}[e^{i\mathbf{t}^\top\mathbf{X}}] = \int_{\mathbb{R}} e^{iu}\, dF_{\mathbf{t}}(u)$, so knowing $F_{\mathbf{t}}$ for every $\mathbf{t}$ determines $\varphi_{\mathbf{X}}$, which by (1) determines the distribution of $\mathbf{X}$. ■

Definition 3.3: multivariate normal (singular allowed)

$\mathbf{X} \in \mathbb{R}^p$ has a multivariate normal distribution iff $\mathbf{t}^\top\mathbf{X}$ is univariate normal for every $\mathbf{t} \in \mathbb{R}^p$ (including the point mass $N(c, 0)$).

This definition does not require $\mathrm{Cov}(\mathbf{X})$ to be non-singular: $\mathbf{X}$ is MVN iff *every* projection onto a one-dimensional subspace is normal. It is well defined because, by the Cramér–Wold device, the distributions of all $\mathbf{t}^\top\mathbf{X}$ determine that of $\mathbf{X}$.

Example 3.4: normal marginals do not imply joint normality (Example 4.8)

$X_1 \sim N(0,1)$ and $X_2 = -X_1$ if $-1 \le X_1 \le 1$, $X_2 = X_1$ otherwise.
(a) $X_2 \sim N(0,1)$.
(b) (X_1, X_2) is not bivariate normal.

Proof. (a) X_1 is standard normal and symmetric about zero, so the map $X_1 \mapsto -X_1$ on $\{-1 \le X_1 \le 1\}$ does not change the distribution. For $-1 \le X_1 \le 1$, $P[X_2 \le x_2 \mid -1 \le X_1 \le 1] = P[-X_1 \le x_2 \mid -1 \le X_1 \le 1] = P[X_1 \le x_2 \mid -1 \le X_1 \le 1]$ by symmetry, and for $|X_1| > 1$, $P[X_2 \le x_2 \mid |X_1| > 1] = P[X_1 \le x_2 \mid |X_1| > 1]$. Combining, $P[X_2 \le x_2] = P[X_1 \le x_2]$, so $X_2 \sim N(0,1)$.

(b) Consider the linear combination $X_1 - X_2$. For $-1 \le X_1 \le 1$ it equals $2X_1$, and otherwise $X_1 - X_2 = 0$. Since a linear combination of X_1 and X_2 is not normal, (X_1, X_2) cannot be jointly normal. (Concretely, $P(X_1 - X_2 = 0) = P(|X_1| > 1) \approx 0.317 \in (0,1)$, so $X_1 - X_2$ is neither a continuous normal nor a point mass.) ■

1.3 Characteristic function

Theorem 3.5: characteristic function of the MVN

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succeq 0$, then $\varphi_{\mathbf{X}}(\mathbf{t}) = \exp\left(i\mathbf{t}^\top\boldsymbol{\mu} - \tfrac{1}{2}\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t}\right)$. If $\boldsymbol{\Sigma} \succ 0$, the density above exists.

Proof. $\mathbf{t}^\top\mathbf{X} \sim N(\mathbf{t}^\top\boldsymbol{\mu}, \mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t})$ and $\varphi_{N(m,v)}(s) = \exp(ism - \tfrac{1}{2}s^2v)$, so $\varphi_{\mathbf{X}}(\mathbf{t}) = \varphi_{\mathbf{t}^\top\mathbf{X}}(1) = \exp(i\mathbf{t}^\top\boldsymbol{\mu} - \tfrac{1}{2}\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t})$. ■

[Supplement] Since the characteristic function depends only on $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the notation $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is justified under the general definition as well.

2 Geometry: Constant-Density Ellipsoids

Location parameter $\boldsymbol{\mu}$, shape parameter $\boldsymbol{\Sigma} \succ 0$. The contours of constant density are

$$E_c = \{\mathbf{x} \in \mathbb{R}^p : f(\mathbf{x}) = c_0\} = \{\mathbf{x} : (\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = c^2\},$$

where c is the Mahalanobis distance $d_M(\boldsymbol{\mu}, \mathbf{x})$. With $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\lambda_1 \geq \cdots \geq \lambda_p > 0$, E_c is an ellipsoid centered at $\boldsymbol{\mu}$ with principal axes along $\mathbf{u}_i$ and half-lengths proportional to $\sqrt{\lambda_i}$. If $\boldsymbol{\Sigma} = \mathbf{I}_p$ it is a sphere of radius c.

Example 3.6: bivariate normal

$N_2(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \mathbf{R}_{\pi/4}\mathrm{diag}(3,1)\mathbf{R}_{\pi/4}^\top$, $\mathbf{R}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$. Location: the origin $\boldsymbol{\mu} = \mathbf{0}$. Shape: elliptical contours with principal axes at $45°$ and $-45°$ and half-axis ratio $\sqrt{3} : 1$.

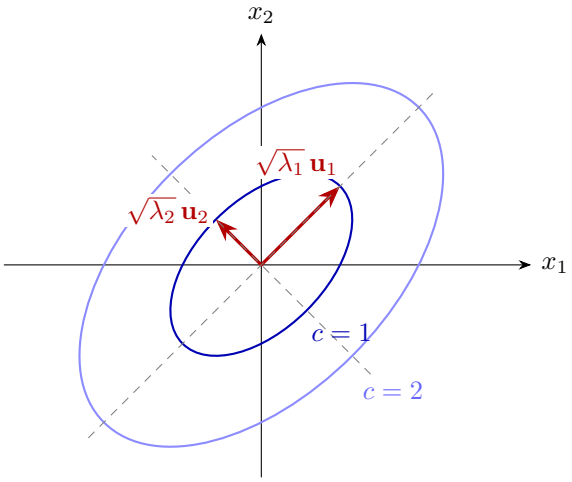


Figure. *Contours $(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = c^2$ of the bivariate normal in the example ($\lambda_1 = 3$, $\lambda_2 = 1$). The half-axes are $c\sqrt{\lambda_i}$ along the eigenvectors $\mathbf{u}_1 = (1,1)^\top/\sqrt{2}$ and $\mathbf{u}_2 = (-1,1)^\top/\sqrt{2}$. In higher dimensions ellipsoids play the same role.*

[Supplement] The sign and size of the correlation control the tilt and the eccentricity of the ellipses. For $\boldsymbol{\Sigma} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ the eigenvalues are $1 \pm \rho$ with eigenvectors $(1, \pm 1)^\top/\sqrt{2}$, so the contours are circles when $\rho = 0$ and become thinner along the $45°$ (or $-45°$) line as $|\rho| \to 1$.

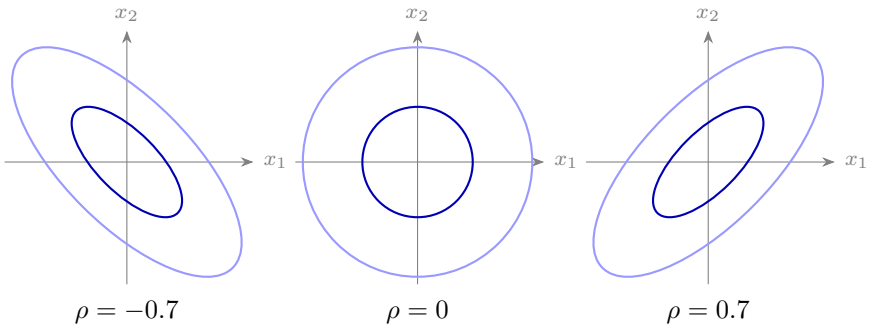


Figure. *[Supplement] Contours containing probability 0.5 (dark) and 0.95 (light) for unit variances and $\rho = -0.7$, 0, 0.7. Positive correlation stretches the ellipse along $x_2 = x_1$, negative correlation along $x_2 = -x_1$.*

3 Linear Transformations, Subsets, Independence, Conditionals

Motivation. The multivariate normal is the default model because it is closed under every operation we perform: linear maps, marginals and conditionals are again normal, and zero correlation means independence. The conditional distribution formula is the population version of linear regression.

Theorem 3.7: linear combinations (Result 4.2)

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then $\mathbf{a}^\top \mathbf{X} = a_1 X_1 + \cdots + a_p X_p \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$. Conversely, if $\mathbf{a}^\top \mathbf{X} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$ for every $\mathbf{a}$, then $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Proof. $\mathrm{E}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\mu}$ and $\mathrm{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$. Normality via the MGF: $M_{\mathbf{a}^\top \mathbf{X}}(t) = \mathrm{E} e^{t \mathbf{a}^\top \mathbf{X}} = M_{\mathbf{X}}(t\mathbf{a}) = \exp(t \mathbf{a}^\top \boldsymbol{\mu} + \tfrac{1}{2} t^2 \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$, the MGF of $N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$. The converse is the general definition. ∎

Theorem 3.8: linear transformations (Result 4.3)

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\mathbf{A} \in \mathbb{R}^{q \times p}$ and $\mathbf{d} \in \mathbb{R}^p$, then $\mathbf{A}\mathbf{X} \sim N_q(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top)$ and $\mathbf{X} + \mathbf{d} \sim N_p(\boldsymbol{\mu} + \mathbf{d}, \boldsymbol{\Sigma})$.

Proof. $\mathrm{E}(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$. Any $\mathbf{b}^\top(\mathbf{A}\mathbf{X}) = (\mathbf{A}^\top \mathbf{b})^\top \mathbf{X}$ is a linear combination of $\mathbf{X}$, normal by Theorem 3.7, so $\mathbf{A}\mathbf{X}$ is MVN. $\mathbf{a}^\top(\mathbf{X} + \mathbf{d}) = \mathbf{a}^\top \mathbf{X} + \mathbf{a}^\top \mathbf{d}$ only shifts the mean. ∎

Theorem 3.9: subsets (Result 4.4)

Partition $\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix}$, $\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}$, $\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}$. Then $\mathbf{X}_1 \sim N_q(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$; all subsets of $\mathbf{X}$ are normal.

Proof. Theorem 3.8 with $\mathbf{A} = [\mathbf{I}_{q \times q} \ \mathbf{0}_{q \times (p-q)}]$. Any subset can be relabelled as $\mathbf{X}_1$. ∎

Theorem 3.10: independence and covariance (Result 4.5)

(a) If $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2$, then $\mathrm{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \mathbf{0}$.

(b) If $\begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix} \sim N_{q_1+q_2}$, then $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2 \iff \boldsymbol{\Sigma}_{12} = \mathbf{0}$.

(c) If $\mathbf{X}_1 \sim N_{q_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$ and $\mathbf{X}_2 \sim N_{q_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$ are independent, then $(\mathbf{X}_1^\top, \mathbf{X}_2^\top)^\top \sim N_{q_1+q_2}\Big(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{22} \end{pmatrix} \Big)$.

Proof.

(a) [Supplement] $\mathrm{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \mathrm{E}[(\mathbf{X}_1 - \boldsymbol{\mu}_1)(\mathbf{X}_2 - \boldsymbol{\mu}_2)^\top]$, and by independence the expectation of each entry factorizes: $\mathrm{E}[(X_{1i} - \mu_{1i})(X_{2j} - \mu_{2j})] = \mathrm{E}(X_{1i} - \mu_{1i})\,\mathrm{E}(X_{2j} - \mu_{2j}) = 0$.

(b) The joint characteristic function is

$$\varphi_{\mathbf{X}}(\mathbf{t}_1, \mathbf{t}_2) = \exp\Big\{ i(\mathbf{t}_1^\top \boldsymbol{\mu}_1 + \mathbf{t}_2^\top \boldsymbol{\mu}_2) - \tfrac{1}{2} \begin{pmatrix} \mathbf{t}_1 \\ \mathbf{t}_2 \end{pmatrix}^\top \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{t}_1 \\ \mathbf{t}_2 \end{pmatrix} \Big\}.$$

If $\boldsymbol{\Sigma}_{12} = \mathbf{0}$ it factorizes as $\varphi_{\mathbf{X}_1}(\mathbf{t}_1)\varphi_{\mathbf{X}_2}(\mathbf{t}_2)$: independence. Conversely, independence forces the factorization, so the cross term $\mathbf{t}_1^\top \boldsymbol{\Sigma}_{12} \mathbf{t}_2$ must vanish for all $\mathbf{t}$, i.e. $\boldsymbol{\Sigma}_{12} = \mathbf{0}$.

(c) [Supplement] Independence gives $\varphi_{\mathbf{X}}(\mathbf{t}_1,\mathbf{t}_2)=\varphi_{\mathbf{X}_1}(\mathbf{t}_1)\varphi_{\mathbf{X}_2}(\mathbf{t}_2)$, which is the characteristic function above with $\boldsymbol{\Sigma}_{12}=\mathbf{0}$, i.e. that of the stated normal law. ■

Remark 3.11: "uncorrelated ⇒ independent" needs joint normality

[Supplement] In Example 3.4 each marginal is normal but the pair is not jointly normal, so (b) does not apply.

Theorem 3.12: conditional distribution (Result 4.6)

If $\boldsymbol{\Sigma}_{22}\succ 0$,

$$\mathbf{X}_1 \mid \mathbf{X}_2=\mathbf{x}_2 \;\sim\; N_r\big(\boldsymbol{\mu}_1+\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2-\boldsymbol{\mu}_2),\; \boldsymbol{\Sigma}_{11}-\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\big).$$

The conditional mean is linear in $\mathbf{x}_2$; the conditional covariance $\boldsymbol{\Sigma}_{11\cdot 2}:=\boldsymbol{\Sigma}_{11}-\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ does not depend on $\mathbf{x}_2$.

Proof. Define $\mathbf{X}_1^*=\mathbf{X}_1-\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$, $\mathbf{X}_2^*=\mathbf{X}_2$, i.e. $\mathbf{X}^*=\mathbf{A}\mathbf{X}$ with $\mathbf{A}=\begin{pmatrix}\mathbf{I}_r & -\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\\ \mathbf{0} & \mathbf{I}_s\end{pmatrix}$. By Theorem 3.8 $\mathbf{X}^*$ is MVN and

$$\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^{\top}=\begin{pmatrix}\boldsymbol{\Sigma}_{11}-\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{\Sigma}_{22}\end{pmatrix}.$$

$\operatorname{Cov}(\mathbf{X}_1^*,\mathbf{X}_2^*)=\mathbf{0}$, so $\mathbf{X}_1^*\perp\!\!\!\perp\mathbf{X}_2$ (Theorem 3.10(b)) and $\mathbf{X}_1^*\sim N_r(\boldsymbol{\mu}_1-\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\mu}_2,\boldsymbol{\Sigma}_{11\cdot 2})$. Since $\mathbf{X}_1=\mathbf{X}_1^*+\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$, conditioning on $\mathbf{X}_2=\mathbf{x}_2$ makes the second term constant without changing the law of $\mathbf{X}_1^*$: $\mathbf{X}_1\mid\mathbf{X}_2=\mathbf{x}_2\overset{d}{=}\mathbf{X}_1^*+\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{x}_2$. Therefore $\mathbf{X}_1\mid\mathbf{X}_2=\mathbf{x}_2\sim N_r(\boldsymbol{\mu}_1+\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2-\boldsymbol{\mu}_2),\ \boldsymbol{\Sigma}_{11}-\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21})$. ■

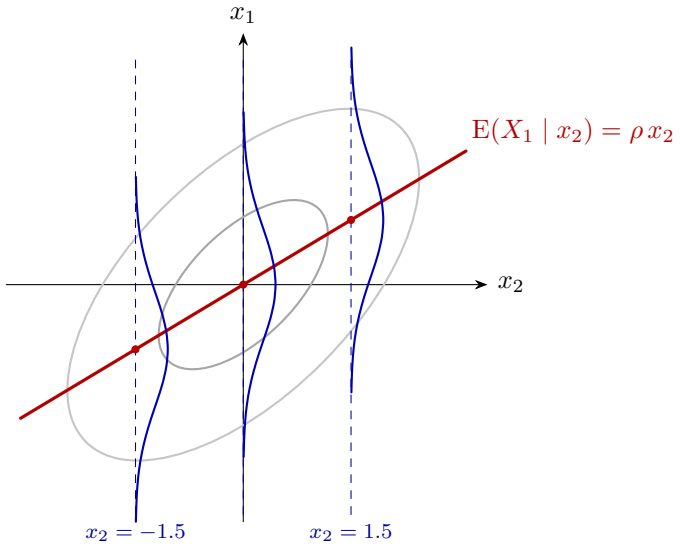


Figure. *Result 4.6 for a standardized pair with $\rho=0.6$. Each vertical slice $X_2=x_2$ of the joint density (gray contours) is a normal density (blue) centred on the regression line $\mu_1+\sigma_{12}\sigma_{22}^{-1}(x_2-\mu_2)=0.6\,x_2$, with the same variance $\sigma_{11\cdot 2}=1-\rho^2=0.64$ for every x_2.*

Remark 3.13: residuals, projections and partial variance

$\mathbf{X}_1^* = \mathbf{X}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$ is (up to centring) the residual $\mathbf{X}_1 - \mathrm{BLP}(\mathbf{X}_1 \mid \mathbf{X}_2)$ after projecting $\mathbf{X}_1$ onto the span of $\mathbf{X}_2$. The residual is uncorrelated with $\mathbf{X}_2$, and independent of it in the Gaussian case. $\sigma^2_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{21}^\top \boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}$ (more generally $\boldsymbol{\Sigma}_{11\cdot 2}$) is the **partial variance** of $\mathbf{X}_1$ after removing the effect of $\mathbf{X}_2$, i.e. $\mathrm{Var}(\mathbf{X}_1^*)$.

4 Best Linear Prediction, Multiple and Partial Correlation

Motivation. How well can Y be predicted linearly from $\mathbf{X}$, and how correlated are two variables once a third is held fixed? Both answers come from the conditional covariance of the normal distribution.

Let $\mathbf{X} \in \mathbb{R}^q$ be the predictor and $Y \in \mathbb{R}$ the response, $\mathbf{U} = (\mathbf{X}^\top, Y)^\top \in \mathbb{R}^p$ $(p = q+1)$,

$$\boldsymbol{\mu} = \mathrm{E}(\mathbf{U}) = \begin{pmatrix} \boldsymbol{\mu}_X \\ \mu_Y \end{pmatrix}, \qquad \boldsymbol{\Sigma} = \mathrm{Var}(\mathbf{U}) = \begin{pmatrix} \boldsymbol{\Sigma}_{XX} & \boldsymbol{\Sigma}_{XY} \\ \boldsymbol{\Sigma}_{YX} & \sigma_Y^2 \end{pmatrix},$$

with sample counterparts $\bar{\mathbf{u}} = (\bar{\mathbf{x}}^\top, \bar{y})^\top$ and $\mathbf{S} = \begin{pmatrix} \mathbf{S}_{XX} & \mathbf{S}_{XY} \\ \mathbf{S}_{YX} & s_Y^2 \end{pmatrix}$.

4.1 Best linear predictor

Theorem 3.14: best linear predictor

$\mathrm{BLP}(Y \mid \mathbf{X}) := \mathbf{A}^*\mathbf{X} + b^*$, $(\mathbf{A}^*, b^*) = \arg\min_{\mathbf{A}\in\mathbb{R}^{1\times q},\, b\in\mathbb{R}} \mathrm{E}\{Y - (\mathbf{A}\mathbf{X} + b)\}^2$. With finite second moments and $\boldsymbol{\Sigma}_{XX} \succ 0$,

$$\mathbf{A}^* = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}, \qquad b^* = \mu_Y - \mathbf{A}^*\boldsymbol{\mu}_X, \qquad \mathrm{BLP}(Y \mid \mathbf{X}) = \mu_Y + \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}(\mathbf{X} - \boldsymbol{\mu}_X).$$

Proof. Let $W := Y - \mu_Y - \mathbf{A}^*(\mathbf{X} - \boldsymbol{\mu}_X)$; then $\mathrm{E}W = 0$ and $\mathrm{Cov}(W, \mathbf{X}) = \boldsymbol{\Sigma}_{YX} - \mathbf{A}^*\boldsymbol{\Sigma}_{XX} = \mathbf{0}$. For any $(\mathbf{A}, b)$,

$$Y - \mathbf{A}\mathbf{X} - b = W + (\mathbf{A}^* - \mathbf{A})(\mathbf{X} - \boldsymbol{\mu}_X) + \{\mu_Y - \mathbf{A}\boldsymbol{\mu}_X - b\},$$

three mutually uncorrelated terms (the last is constant), so

$$\mathrm{E}(Y - \mathbf{A}\mathbf{X} - b)^2 = \mathrm{E}W^2 + (\mathbf{A}^* - \mathbf{A})\boldsymbol{\Sigma}_{XX}(\mathbf{A}^* - \mathbf{A})^\top + (\mu_Y - \mathbf{A}\boldsymbol{\mu}_X - b)^2 \geq \mathrm{E}W^2,$$

with equality iff $\mathbf{A} = \mathbf{A}^*$ and $b = \mu_Y - \mathbf{A}^*\boldsymbol{\mu}_X$. ■

Remark 3.15: properties of the BLP

$\mathrm{E}[\mathrm{BLP}(Y \mid \mathbf{X})] = \mu_Y$, $\mathrm{Var}[\mathrm{BLP}(Y \mid \mathbf{X})] = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$; the residual $W = Y - \mathrm{BLP}(Y \mid \mathbf{X})$ has $\mathrm{E}W = 0$, $\mathrm{Var}W = \sigma_Y^2 - \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ and $\mathrm{Cov}(W, \mathrm{BLP}) = 0$. Under joint normality $\mathrm{BLP}(Y \mid \mathbf{X}) = \mathrm{E}(Y \mid \mathbf{X})$ (Theorem 3.12). In block notation $\mathrm{E}[\mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_1)] = \boldsymbol{\mu}_2$, $\mathrm{Var}[\mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_1)] = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}$.

4.2 Multiple correlation

Definition 3.16: multiple correlation of Y with $\mathbf{X}$

$$\rho_{\text{mult}} = \text{mcorr}(\mathbf{X}, Y) := \max_{\mathbf{a}\in\mathbb{R}^q\setminus\{\mathbf{0}\}} \text{Corr}(Y, \mathbf{a}^\top\mathbf{X}), \qquad \rho^2_{\text{mult}} = \frac{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}}{\sigma_Y^2}.$$

With $\hat{Y} = \text{BLP}(Y \mid \mathbf{X})$, $\rho_{\text{mult}} = \text{Corr}(Y, \hat{Y})$.

Proof. $\text{Corr}(Y, \mathbf{a}^\top\mathbf{X} + b) = \dfrac{\mathbf{a}^\top\boldsymbol{\Sigma}_{XY}}{\sigma_Y\sqrt{\mathbf{a}^\top\boldsymbol{\Sigma}_{XX}\mathbf{a}}}$. With $\mathbf{b} = \boldsymbol{\Sigma}_{XX}^{1/2}\mathbf{a}$, Cauchy–Schwarz gives

$$\mathbf{a}^\top\boldsymbol{\Sigma}_{XY} = \mathbf{b}^\top\boldsymbol{\Sigma}_{XX}^{-1/2}\boldsymbol{\Sigma}_{XY} \le \sqrt{\mathbf{a}^\top\boldsymbol{\Sigma}_{XX}\mathbf{a}}\,\sqrt{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}},$$

so Corr $\le \sqrt{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}}/\sigma_Y$, with equality iff $\mathbf{a} \propto \boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ (the BLP direction). Then $\text{Cov}(Y, \hat{Y}) = \text{Var}(\hat{Y}) = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ and $\text{Corr}(Y, \hat{Y}) = \sqrt{\text{Var}\hat{Y}}/\sigma_Y$. ■

For two vectors, $\rho(\mathbf{X}_1, \mathbf{X}_2) := \max \text{Corr}(\mathbf{a}^\top\mathbf{X}_1, \mathbf{b}^\top\mathbf{X}_2)$ is the largest canonical correlation (Ch. 8).

4.3 Partial covariance and partial correlation

Partition $\mathbf{X} = (X, \mathbf{Z}^\top)^\top$ with X the predictor of interest and $\mathbf{Z} \in \mathbb{R}^{q-1}$ the others ($\text{Var}\mathbf{Z} \succ 0$), and let $\tilde{X} = X - \text{BLP}(X \mid \mathbf{Z})$, $\tilde{Y} = Y - \text{BLP}(Y \mid \mathbf{Z})$.

Definition 3.17: partial covariance and partial correlation

$$\sigma_{YX\cdot Z} := \text{Cov}(\tilde{Y}, \tilde{X}) = \text{Cov}(Y, X) - \text{Cov}(Y, \mathbf{Z})\text{Var}(\mathbf{Z})^{-1}\text{Cov}(\mathbf{Z}, X),$$

$$\rho_{YX\cdot Z} := \frac{\sigma_{YX\cdot Z}}{\sqrt{\text{Var}(\tilde{Y})\text{Var}(\tilde{X})}} = \text{Corr}(\tilde{Y}, \tilde{X}).$$

In block notation: $\text{pcorr}(\mathbf{X}_1, \mathbf{X}_2; \mathbf{X}_3) = \text{Corr}(\mathbf{X}_1 - \text{BLP}(\mathbf{X}_1 \mid \mathbf{X}_3), \mathbf{X}_2 - \text{BLP}(\mathbf{X}_2 \mid \mathbf{X}_3)) = \text{Corr}(\mathbf{X}_1^*, \mathbf{X}_2^*)$ and the partial covariance matrix $\text{pVar}(\mathbf{X}_1; \mathbf{X}_2) = \text{Var}(\mathbf{X}_1 - \text{BLP}(\mathbf{X}_1 \mid \mathbf{X}_2)) = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$.

Partial covariance formula [Supplement]. $\tilde{Y} = Y - \mu_Y - \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}(\mathbf{Z} - \boldsymbol{\mu}_Z)$ and similarly for $\tilde{X}$. Expanding, $\text{Cov}(\tilde{Y}, \tilde{X}) = \sigma_{YX} - 2\boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX} + \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX} = \sigma_{YX} - \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX}$. ■

[Supplement] So the partial covariance matrix of (Y, X) is a Schur complement, and under joint normality it equals the conditional covariance given $\mathbf{Z}$.

Theorem 3.18: Frisch–Waugh–Lovell theorem

Let $\hat{\beta}_X$ be the coefficient of X in the OLS regression of Y on $(X, \mathbf{Z})$, and let $\tilde{y}_i, \tilde{x}_i$ be the residuals from the OLS regressions of Y and X on $\mathbf{Z}$. Then

$$\hat{\beta}_X = \frac{\sum_i \tilde{x}_i\tilde{y}_i}{\sum_i \tilde{x}_i^2},$$

the slope of the residual-on-residual regression (all regressions with intercept; full-column-rank design).

Proof [Supplement]. Let $\mathbf{M}_Z = \mathbf{I} - \mathbf{P}_{[\mathbf{1},\mathbf{Z}]}$. The normal equations give $\mathbf{y} = \hat{\beta}_X \mathbf{x} + [\mathbf{1}, \mathbf{Z}]\hat{\boldsymbol{\gamma}} + \mathbf{e}$ with $\mathbf{e} \perp \mathbf{x}, \mathbf{1}, \mathbf{Z}$. Multiplying by $\mathbf{M}_Z$ ($\mathbf{M}_Z[\mathbf{1}, \mathbf{Z}] = \mathbf{0}$, $\mathbf{M}_Z\mathbf{e} = \mathbf{e}$) gives $\tilde{\mathbf{y}} = \hat{\beta}_X \tilde{\mathbf{x}} + \mathbf{e}$. Multiply by $\tilde{\mathbf{x}}^\top$: $\tilde{\mathbf{x}}^\top \mathbf{e} = \mathbf{x}^\top \mathbf{M}_Z \mathbf{e} = \mathbf{x}^\top \mathbf{e} = 0$, so $\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} = \hat{\beta}_X \tilde{\mathbf{x}}^\top \tilde{\mathbf{x}}$. ■

Example 3.19: mtcars — marginal versus partial correlation

Y = mpg, X = displacement, $\mathbf{Z}$ = (horsepower, weight). The marginal correlation is $\hat{\rho}_{YX} \approx -0.805$, but after adjusting for horsepower and weight $\hat{\rho}_{YX \cdot Z} \approx -0.044$: little linear association remains. The displacement–mpg relation runs mostly through horsepower and weight.

5 Quadratic Forms in Normal Random Vectors

Motivation. Test statistics are typically quadratic forms such as $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu})$. Their χ^2 distributions are the basis of all tests in later chapters.

For $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and symmetric $\mathbf{A}$, a quadratic form is $Y = \mathbf{X}^\top \mathbf{A} \mathbf{X} = \sum_{j,k} X_j a_{jk} X_k$. We want its distribution and conditions for independence. Special case: $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, $\mathbf{A} = \mathbf{I}_p$ gives $Y = \sum X_i^2 \sim \chi^2(p)$.

Remark 3.20: matrix algebra recall

$\mathbf{A}$ is idempotent if $\mathbf{A}^2 = \mathbf{A}$. If $\mathbf{A}$ is symmetric, $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$; if it is also idempotent, its eigenvalues are 0 or 1 and $\operatorname{rank}(\mathbf{A}) = \#\{\text{nonzero eigenvalues}\} = \operatorname{tr}(\mathbf{A})$.

Theorem 3.21: chi-square quadratic forms, $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ (Result 4.7, Case 001)

Let $\mathbf{X} \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I})$ and $\mathbf{A}$ symmetric. Then $Y = \dfrac{\mathbf{X}^\top \mathbf{A} \mathbf{X}}{\sigma^2} \sim \chi^2(m)$ iff $\mathbf{A}$ is idempotent of rank $m \le p$.

Proof. ($\Leftarrow$) $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda} = \operatorname{diag}(1, \ldots, 1, 0, \ldots, 0)$ (m ones). Let $\mathbf{z} = \frac{1}{\sigma}\boldsymbol{\Gamma}^\top \mathbf{X} \sim N(\mathbf{0}, \boldsymbol{\Gamma}^\top \boldsymbol{\Gamma}) = N(\mathbf{0}, \mathbf{I}_p)$; then $Y = \mathbf{z}^\top \boldsymbol{\Lambda} \mathbf{z} = z_1^2 + \cdots + z_m^2 \sim \chi^2(m)$.

($\Rightarrow$) [Supplement] In general $Y = \sum_i \lambda_i z_i^2$ has MGF $\prod_i (1 - 2t\lambda_i)^{-1/2}$; matching $(1 - 2t)^{-m/2}$ near $t = 0$ forces the nonzero λ_i to be m ones. ■

Simple example: $\mathbf{A} = \operatorname{diag}(1, 0, \ldots, 0)$ gives $\mathbf{X}^\top \mathbf{A} \mathbf{X}/\sigma^2 = X_1^2/\sigma^2 \sim \chi_1^2$.

Theorem 3.22: chi-square quadratic forms, general $\boldsymbol{\Sigma}$ (Result 4.7, Case 002)

Let $\mathbf{X} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$ and $\mathbf{A}$ symmetric. Then $Y = \mathbf{X}^\top \mathbf{A} \mathbf{X} \sim \chi^2(m)$ iff either (1) $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent of rank m, or (2) $\boldsymbol{\Sigma}\mathbf{A}$ is idempotent of rank m.

Proof. By the representation theorem $\mathbf{X} = \boldsymbol{\Sigma}^{1/2} \mathbf{z}$, $\mathbf{z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, so $Y = \mathbf{z}^\top \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2} \mathbf{z}$. By Theorem 3.21, $Y \sim \chi_m^2$ iff $\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$ is (symmetric) idempotent of rank m. Now

$$(\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2})^2 = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2} \iff \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2} \iff \mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A},$$

(multiply by $\boldsymbol{\Sigma}^{-1/2}$ on both sides), and $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A} \iff (\mathbf{A}\boldsymbol{\Sigma})^2 = \mathbf{A}\boldsymbol{\Sigma}$ (multiply on the right by $\boldsymbol{\Sigma}$ or $\boldsymbol{\Sigma}^{-1}$). Ranks: $\operatorname{rank}(\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}) = \operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}\boldsymbol{\Sigma})$. Since $\boldsymbol{\Sigma}\mathbf{A} = (\mathbf{A}\boldsymbol{\Sigma})^\top$, the same holds for $\boldsymbol{\Sigma}\mathbf{A}$. ■

Example 3.23: distribution of the Mahalanobis distance (Example 4)

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$, then $(\mathbf{X}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) \sim \chi^2(p)$ ($\mathbf{A} = \boldsymbol{\Sigma}^{-1}$ makes $\mathbf{A}\boldsymbol{\Sigma} = \mathbf{I}_p$ idempotent of rank p; apply Case 002 to $\mathbf{X}-\boldsymbol{\mu} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$). Hence $\{\mathbf{x} : (\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \le \chi^2_p(\alpha)\}$ is an ellipsoid of probability $1-\alpha$.

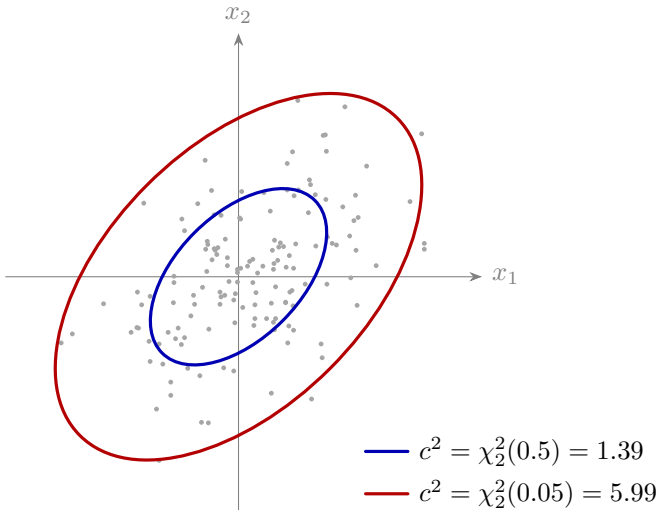


Figure. *Example 4 with* $p = 2$, $\boldsymbol{\mu} = \mathbf{0}$, $\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$*: the ellipses* $(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \le \chi^2_2(\alpha)$ *for* $\alpha = 0.5$ *and* 0.05 *have probability* 0.5 *and* 0.95*. In* 150 *simulated draws,* 54% *and* 96% *fall inside (one draw lies outside the frame).*

Theorem 3.24: independence of linear and quadratic forms (Theorem 9)

Let $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I})$, $\mathbf{A}$ a $p \times p$ symmetric matrix and $\mathbf{B}$ a $k \times p$ matrix. If $\mathbf{BA} = \mathbf{0}$, then $\mathbf{BX}$ and $\mathbf{X}^\top \mathbf{A}\mathbf{X}$ are independent.

Proof. If $\mathbf{A}$ is full rank, $\mathbf{BA} = \mathbf{0}$ forces $\mathbf{B} = \mathbf{0}$: trivial. If $\operatorname{rank}(\mathbf{A}) = m < p$, write $\mathbf{A} = \boldsymbol{\Gamma} \begin{pmatrix} \boldsymbol{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda}_m = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$, all $\lambda_i \neq 0$. From $\mathbf{BA} = \mathbf{0}$, $\mathbf{B}\boldsymbol{\Gamma} \begin{pmatrix} \boldsymbol{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} = \mathbf{0}$. Writing $\mathbf{B}\boldsymbol{\Gamma} = [\mathbf{C}_1, \mathbf{C}_2]$ gives $\mathbf{C}_1 \boldsymbol{\Lambda}_m = \mathbf{0}$, so $\mathbf{C}_1 = \mathbf{0}$ and $\mathbf{B}\boldsymbol{\Gamma} = [\mathbf{0}, \mathbf{C}_2]$. Let $\mathbf{z} = \begin{pmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \end{pmatrix} = \boldsymbol{\Gamma}^\top \mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I})$, so $\mathbf{z}_1 \perp\!\!\!\perp \mathbf{z}_2$. Then

$$\mathbf{BX} = \mathbf{B}\boldsymbol{\Gamma}\mathbf{z} = \mathbf{C}_2 \mathbf{z}_2, \qquad \mathbf{X}^\top \mathbf{A}\mathbf{X} = \mathbf{z}^\top \begin{pmatrix} \boldsymbol{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \mathbf{z} = \mathbf{z}_1^\top \boldsymbol{\Lambda}_m \mathbf{z}_1,$$

functions of $\mathbf{z}_2$ and $\mathbf{z}_1$ respectively, hence independent. ■

Example 3.25: $\bar{X} \perp\!\!\!\perp S^2$

For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ and $\mathbf{X} = (X_1, \ldots, X_n)^\top$, $\bar{X}_n = \frac{1}{n}\mathbf{1}_n^\top \mathbf{X}$ and $S_n^2 = \frac{1}{n-1}\mathbf{X}^\top(\mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top)\mathbf{X}$. With $\mathbf{B} = \frac{1}{n}\mathbf{1}_n^\top$ and $\mathbf{A} = \frac{1}{n-1}\mathbf{H}_n$, $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$, we have $\mathbf{BA} = \mathbf{0}$. With $\mathbf{Z} = (\mathbf{X} - \mu\mathbf{1}_n)/\sigma \sim N_n(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{H}_n\mathbf{1}_n = \mathbf{0}$, $\bar{X} = \mu + \sigma\mathbf{B}\mathbf{Z}$ and $S^2 = \sigma^2 \mathbf{Z}^\top \mathbf{A}\mathbf{Z}$, so $\bar{X} \perp\!\!\!\perp S^2$ by Theorem 3.24. The multivariate version is in Ch. 4.

Remark 3.26: same argument, two quadratic forms

[Supplement] If $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and symmetric $\mathbf{A}, \mathbf{B}$ satisfy $\mathbf{AB} = \mathbf{0}$, then $\mathbf{X}^\top \mathbf{AX} \perp\!\!\!\perp \mathbf{X}^\top \mathbf{BX}$. Indeed $\mathbf{X}^\top \mathbf{BX} = (\mathbf{BX})^\top \mathbf{B}^+ (\mathbf{BX})$ with $\mathbf{B}^+$ the Moore–Penrose inverse ($\mathbf{BB}^+\mathbf{B} = \mathbf{B}$), so it is a function of $\mathbf{BX}$, and $\mathbf{BA} = (\mathbf{AB})^\top = \mathbf{0}$; Theorem 3.24 applies.

6 Selected exercises

Motivation. These exercises sharpen two warnings and add one tool: zero correlation with normal marginals still allows dependence, the normal density factorizes into a marginal and a conditional part, and quadratic forms in normal vectors have explicit means and variances.

Exercise 3.1 (Uncorrelated but dependent; J&W 4.9)

Let $X_1 \sim N(0,1)$, $c > 0$, and $X_2 = -X_1$ if $|X_1| \le c$, $X_2 = X_1$ otherwise.
(a) Show that $X_2 \sim N(0,1)$ for every c.
(b) Show that c can be chosen so that $\mathrm{Cov}(X_1, X_2) = 0$.
(c) For this c, show that X_1 and X_2 are not independent, and conclude that (X_1, X_2) is not bivariate normal.

Solution [Supplement].

(a) For a Borel set B, $P(X_2 \in B) = P(X_1 \in B, |X_1| > c) + P(-X_1 \in B, |X_1| \le c)$. Since $-X_1 \overset{d}{=} X_1$ and $\{|X_1| \le c\} = \{|-X_1| \le c\}$, the second term equals $P(X_1 \in B, |X_1| \le c)$, so $P(X_2 \in B) = P(X_1 \in B)$ (Example 3.4 is the case $c = 1$).

(b)

Step 1. With $g(c) = \mathrm{E}[X_1^2 \mathbf{1}\{|X_1| \le c\}] = \int_{-c}^{c} x^2 \phi(x)\, dx$,

$$\mathrm{Cov}(X_1, X_2) = \mathrm{E}(X_1 X_2) = \mathrm{E}[X_1^2 \mathbf{1}\{|X_1| > c\}] - \mathrm{E}[X_1^2 \mathbf{1}\{|X_1| \le c\}] = 1 - 2g(c).$$

Step 2. g is continuous, $g(c) \to 0$ as $c \downarrow 0$, and $g(c) \to \mathrm{E}X_1^2 = 1$ as $c \to \infty$ (monotone convergence). So $h(c) = 1 - 2g(c)$ is continuous with limits 1 and -1, and by the intermediate value theorem $h(c^*) = 0$ for some $c^* > 0$ (numerically $c^* \approx 1.54$).

(c)

Step 1. $|X_2| = |X_1|$, so $X_2^2 = X_1^2$. If $X_1 \perp\!\!\!\perp X_2$, then $X_1^2 \perp\!\!\!\perp X_2^2$ and $\mathrm{Var}(X_1^2) = \mathrm{Cov}(X_1^2, X_2^2) = 0$, contradicting $\mathrm{Var}(X_1^2) = \mathrm{E}X_1^4 - 1 = 2$.

Step 2. If (X_1, X_2) were bivariate normal, $\mathrm{Cov}(X_1, X_2) = 0$ would imply independence by Theorem 3.10. Hence it is not jointly normal, although both marginals are $N(0,1)$. ■

Exercise 3.2 (Factorizing the normal density; J&W 4.13)

Let $\mathbf{X} = (\mathbf{X}_1^\top, \mathbf{X}_2^\top)^\top \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$ partitioned as in Theorem 3.12, and $\boldsymbol{m}(\mathbf{x}_2) = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2)$.
(a) Show $|\boldsymbol{\Sigma}| = |\boldsymbol{\Sigma}_{22}|\,|\boldsymbol{\Sigma}_{11\cdot 2}|$.
(b) Show

$$(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = \{\mathbf{x}_1 - \boldsymbol{m}(\mathbf{x}_2)\}^\top \boldsymbol{\Sigma}_{11\cdot 2}^{-1} \{\mathbf{x}_1 - \boldsymbol{m}(\mathbf{x}_2)\} + (\mathbf{x}_2 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{x}_2 - \boldsymbol{\mu}_2).$$

(c) Using only the density, deduce the marginal law of $\mathbf{X}_2$ and the conditional law of $\mathbf{X}_1$ given $\mathbf{X}_2 = \mathbf{x}_2$.

Solution [Supplement].

(a) This is the first determinant formula of Theorem 2.7 with $\mathbf{A} = \boldsymbol{\Sigma}$.

(b)

Step 1. Let $\mathbf{E} = \begin{pmatrix} \mathbf{I} & -\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$. Since $\boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^\top$ and $\boldsymbol{\Sigma}_{22}$ is symmetric, the elimination identity in the proof of Theorem 2.7 reads $\mathbf{E}\boldsymbol{\Sigma}\mathbf{E}^\top = \mathbf{D} := \operatorname{diag}(\boldsymbol{\Sigma}_{11\cdot 2}, \boldsymbol{\Sigma}_{22})$.

Step 2. Both $\mathbf{E}$ and $\mathbf{D}$ are invertible, so $\boldsymbol{\Sigma}^{-1} = \mathbf{E}^\top \mathbf{D}^{-1}\mathbf{E}$ and the quadratic form equals $\{\mathbf{E}(\mathbf{x} - \boldsymbol{\mu})\}^\top \mathbf{D}^{-1}\{\mathbf{E}(\mathbf{x} - \boldsymbol{\mu})\}$.

Step 3. $\mathbf{E}(\mathbf{x} - \boldsymbol{\mu}) = (\mathbf{x}_1 - \boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2),\ \mathbf{x}_2 - \boldsymbol{\mu}_2) = (\mathbf{x}_1 - \boldsymbol{m}(\mathbf{x}_2),\ \mathbf{x}_2 - \boldsymbol{\mu}_2)$, and $\mathbf{D}^{-1}$ is block diagonal, which gives the identity.

(c)

Step 1. $\boldsymbol{\Sigma}_{11\cdot 2} = \{(\boldsymbol{\Sigma}^{-1})_{11}\}^{-1}$ is the inverse of a principal block of a positive definite matrix, hence positive definite. With $r = \dim \mathbf{X}_1$, $s = \dim \mathbf{X}_2$, (a) and (b) give

$$f(\mathbf{x}_1, \mathbf{x}_2) = g(\mathbf{x}_1 \mid \mathbf{x}_2)\, h(\mathbf{x}_2),$$

where $g(\cdot \mid \mathbf{x}_2)$ is the $N_r(\boldsymbol{m}(\mathbf{x}_2), \boldsymbol{\Sigma}_{11\cdot 2})$ density and h the $N_s(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$ density (the constants split as $(2\pi)^{-p/2} = (2\pi)^{-r/2}(2\pi)^{-s/2}$).

Step 2. For each $\mathbf{x}_2$, $\int g(\mathbf{x}_1 \mid \mathbf{x}_2)\, d\mathbf{x}_1 = 1$, so the marginal density of $\mathbf{X}_2$ is $\int f\, d\mathbf{x}_1 = h(\mathbf{x}_2)$: $\mathbf{X}_2 \sim N_s(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$.

Step 3. The conditional density is $f(\mathbf{x}_1, \mathbf{x}_2)/h(\mathbf{x}_2) = g(\mathbf{x}_1 \mid \mathbf{x}_2)$, which is Theorem 3.12 again, now obtained from the density. ■

Exercise 3.3 (Mean and variance of a normal quadratic form)

Let $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ ($\boldsymbol{\Sigma} \succeq 0$) and $\mathbf{A}$ symmetric. Show

$$\mathrm{E}(\mathbf{X}^\top \mathbf{A}\mathbf{X}) = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}, \qquad \operatorname{Var}(\mathbf{X}^\top \mathbf{A}\mathbf{X}) = 2\operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}) + 4\boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\mu}.$$

Solution [Supplement].

Step 1 (mean). $\mathrm{E}(\mathbf{X}^\top \mathbf{A}\mathbf{X}) = \mathrm{E}\operatorname{tr}(\mathbf{A}\mathbf{X}\mathbf{X}^\top) = \operatorname{tr}\{\mathbf{A}(\boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top)\} = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}$ (normality is not needed here).

Step 2 (representation). Let $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\boldsymbol{\Sigma}^{1/2}$ the symmetric square root. By Theorem 3.8, $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$ is normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and a normal law is determined by its mean and covariance (Theorem 3.5), so $\mathbf{X} \overset{d}{=} \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$. Let $\mathbf{B} = \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}^\top$ (spectral decomposition), $\mathbf{W} = \mathbf{P}^\top \mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\mathbf{b} = \mathbf{P}^\top \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\mu}$. Then

$$\mathbf{X}^\top \mathbf{A}\mathbf{X} \overset{d}{=} \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu} + 2\mathbf{b}^\top \mathbf{W} + \mathbf{W}^\top \boldsymbol{\Lambda}\mathbf{W} = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu} + \sum_{i=1}^{p} (2b_i W_i + \lambda_i W_i^2).$$

Step 3 (variance). The summands are independent. With $\mathrm{E}W_i^3 = 0$ and $\operatorname{Var}(W_i^2) = 3 - 1 = 2$,

$$\operatorname{Var}(2b_i W_i + \lambda_i W_i^2) = 4b_i^2 + 2\lambda_i^2 + 4b_i\lambda_i \operatorname{Cov}(W_i, W_i^2) = 4b_i^2 + 2\lambda_i^2.$$

Step 4. $\sum_i \lambda_i^2 = \operatorname{tr}(\mathbf{B}^2) = \operatorname{tr}(\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}^{1/2}) = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma})$ and $\sum_i b_i^2 = \mathbf{b}^\top \mathbf{b} = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\Sigma}^{1/2}\mathbf{P}\mathbf{P}^\top \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\mu}$, which gives the variance formula.

Check: $\boldsymbol{\mu} = \mathbf{0}$ and $\mathbf{A} = \boldsymbol{\Sigma}^{-1}$ give mean p and variance $2p$, those of χ_p^2 (Example 3.23). ■

Chapter 4. Sampling Distributions and Large-Sample Theory

The Wishart distribution, distributions of $\bar{\mathbf{X}}$ and $\mathbf{S}$, MLE, consistency, multivariate CLT

1 The Wishart Distribution

Motivation. The sample covariance matrix is random, and inference about $\boldsymbol{\Sigma}$ needs its distribution. The Wishart distribution is the matrix analogue of χ^2: a sum of outer products of independent normal vectors.

Definition 4.1: Wishart distribution

- The Wishart distribution is a family of distributions for symmetric positive definite matrices.
- The distribution of the $p \times p$ random matrix $\mathbf{M} = \mathbf{X}^\top \mathbf{X} = \sum_{i=1}^{n} \mathbf{X}_i \mathbf{X}_i^\top$ is a Wishart distribution.
- With m degrees of freedom,

$$W_p(m \mid \boldsymbol{\Sigma}) = \text{Wishart with } m \text{ d.f.} = \text{distribution of } \sum_{j=1}^{m} \mathbf{Z}_j \mathbf{Z}_j^\top, \qquad \mathbf{Z}_j \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

We also write $W_p(m, \boldsymbol{\Sigma})$.

- It is the multivariate extension of χ^2: if $\mathbf{M} \sim W_1(n, \sigma^2)$ then $\mathbf{M}/\sigma^2 \sim \chi^2_n$. E.g. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(0, \sigma^2)$ gives $\mathbf{M} = \sum X_i^2$, $\mathbf{M}/\sigma^2 \sim \chi^2_n$.
- $W_p(n, \mathbf{I})$ is the *standard Wishart* distribution.
- If $\boldsymbol{\mu} = \mathbf{0}$ is known and $\mathbf{S} = \frac{1}{n}\mathbf{X}^\top \mathbf{X} = \frac{1}{n}\sum \mathbf{X}_i \mathbf{X}_i^\top$, then $n\mathbf{S} \sim W_p(n, \boldsymbol{\Sigma})$.
- [Supplement] If $m \geq p$ and $\boldsymbol{\Sigma} \succ 0$, $\mathbf{M}$ is PD with probability 1.

Theorem 4.2: properties of the Wishart distribution (Proposition 1)

(i) For $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$ and $\mathbf{B}_{(p\times m)}$, $\mathbf{B}^\top \mathbf{M} \mathbf{B} \sim W_m(n, \mathbf{B}^\top \boldsymbol{\Sigma} \mathbf{B})$.

(ii) For $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$, $\boldsymbol{\Sigma}^{-1/2} \mathbf{M} \boldsymbol{\Sigma}^{-1/2} \sim W_p(n, \mathbf{I}_p)$.

(iii) If $\mathbf{M}_i \sim W_p(n_i, \boldsymbol{\Sigma})$ $(i = 1, \ldots, k)$ are independent, $\sum_i \mathbf{M}_i \sim W_p(n, \boldsymbol{\Sigma})$ with $n = n_1 + \cdots + n_k$.

(iv) For $\mathbf{M}_n \sim W_p(n, \boldsymbol{\Sigma})$, $\mathrm{E}(\mathbf{M}_n) = n\boldsymbol{\Sigma}$ and $\mathrm{Var}((\mathbf{M}_n)_{ij}) = n(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})$. Hence $\hat{\boldsymbol{\Sigma}}_n = \mathbf{M}_n/n$ has $\mathrm{E}(\hat{\boldsymbol{\Sigma}}_n) = \boldsymbol{\Sigma}$ and $\mathrm{Var}[(\hat{\boldsymbol{\Sigma}}_n)_{ij}] = (\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})/n$.

(v) The law of large numbers and the Cramér–Wold device give $\mathbf{M}_n/n \to \boldsymbol{\Sigma}$ in probability as $n \to \infty$.

Proof. (i) Write $\mathbf{M} = \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^\top$ with $\mathbf{x}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$. Then $\mathbf{B}^\top \mathbf{M} \mathbf{B} = \sum_i \mathbf{B}^\top \mathbf{x}_i (\mathbf{B}^\top \mathbf{x}_i)^\top$ and $\mathbf{B}^\top \mathbf{x}_i \overset{\text{iid}}{\sim} N_m(\mathbf{0}, \mathbf{B}^\top \boldsymbol{\Sigma} \mathbf{B})$.

(ii) Special case of (i) with $\mathbf{B} = \boldsymbol{\Sigma}^{-1/2}$.

(iii) With $\mathbf{M}_i = \sum_j \mathbf{x}_{ij}\mathbf{x}_{ij}^\top$, $\sum_i \mathbf{M}_i = \sum_i \sum_j \mathbf{x}_{ij}\mathbf{x}_{ij}^\top \sim W_p(\sum n_i, \boldsymbol{\Sigma})$.

(iv) $\mathrm{E}[\mathbf{M}_n] = \mathrm{E}(\sum_k \mathbf{x}_k\mathbf{x}_k^\top) = \sum_k \mathrm{Var}(\mathbf{x}_k) = n\boldsymbol{\Sigma}$. Terms with different k are independent, so $\mathrm{Var}((\mathbf{M}_n)_{ij}) = n\{\mathrm{E}(X_{ki}^2X_{kj}^2) - \Sigma_{ij}^2\}$. Using $\mathrm{E}(X_{ki} \mid X_{kj}) = \frac{\Sigma_{ij}}{\Sigma_{jj}}X_{kj}$ and $X_{ki} - \mathrm{E}(X_{ki} \mid X_{kj}) \perp\!\!\!\perp X_{kj}$,

$$\mathrm{E}[(X_{ki}X_{kj})^2] = \mathrm{E}[X_{kj}^2\mathrm{E}(X_{ki}^2 \mid X_{kj})] = \frac{\Sigma_{ij}^2}{\Sigma_{jj}^2}3\Sigma_{jj}^2 + \Big(\Sigma_{ii} - \frac{\Sigma_{ij}^2}{\Sigma_{jj}}\Big)\Sigma_{jj} = 2\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj},$$

so $\mathrm{Var} = n(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})$.

(v) By Cramér–Wold it suffices that $\sum_{ij} a_{ij}M_{ij}/n \xrightarrow{p} \sum_{ij} a_{ij}\Sigma_{ij}$ for all a_{ij}. Since $\mathbf{M}_n/n = \frac{1}{n}\sum_k \mathbf{x}_k\mathbf{x}_k^\top$ is an average of iid random matrices, the LLN gives $\sum_{ij} a_{ij}\frac{M_{ij}}{n} \xrightarrow{p} \sum_{ij} a_{ij}\Sigma_{ij}$. ■

2 Wishart Quadratic Forms and the Distribution of S

Motivation. Projections of a Wishart matrix are again Wishart or χ^2, which yields the distribution of $\mathbf{S}$ and its independence from $\bar{\mathbf{X}}$, just as in the univariate case.

Theorem 4.3: Wishart quadratic forms (Lemma 5)

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be iid $N_p(\mathbf{0}, \boldsymbol{\Sigma})$, $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n)^\top$, and $\mathbf{A}$ an $n \times n$ real symmetric matrix. Then

$$\mathbf{X}^\top\mathbf{A}\mathbf{X} \sim W_p(r, \boldsymbol{\Sigma}) \iff \mathbf{A} \text{ is idempotent with } \mathrm{rank}(\mathbf{A}) = r.$$

Partial proof ($\Leftarrow$). Since $\mathbf{A}$ is symmetric idempotent of rank r, $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda} = \mathrm{diag}(1, \dots, 1, 0, \dots, 0)$. Write $\mathbf{A} = \mathbf{P}\begin{pmatrix}\mathbf{I}_r & \mathbf{0}\\ \mathbf{0} & \mathbf{0}\end{pmatrix}\mathbf{P}^\top$ for an $n \times n$ orthogonal $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_n]$, so $\mathbf{A} = \sum_{j=1}^r \mathbf{p}_j\mathbf{p}_j^\top$ and

$$\mathbf{X}^\top\mathbf{A}\mathbf{X} = \sum_{j=1}^r \mathbf{X}^\top\mathbf{p}_j\mathbf{p}_j^\top\mathbf{X}.$$

Now $\mathbf{X}^\top\mathbf{p}_j = \sum_{i=1}^n p_{ij}\mathbf{X}_i \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, since

$$\mathrm{Cov}\Big(\sum_i p_{ij}\mathbf{X}_i\Big) = \boldsymbol{\Sigma}\cdot\mathbf{p}_j^\top\mathbf{p}_j = \boldsymbol{\Sigma}, \qquad \mathrm{Cov}\Big(\sum_i p_{ij}\mathbf{X}_i, \sum_i p_{ik}\mathbf{X}_i\Big) = \mathbf{p}_j^\top\mathbf{p}_k\boldsymbol{\Sigma} = \mathbf{0}\ (j \neq k);$$

being jointly normal, $\mathbf{z}_j = \mathbf{X}^\top\mathbf{p}_j$ are iid $N_p(\mathbf{0}, \boldsymbol{\Sigma})$. Hence $\mathbf{X}^\top\mathbf{A}\mathbf{X} = \sum_{j=1}^r \mathbf{z}_j\mathbf{z}_j^\top \sim W_p(r, \boldsymbol{\Sigma})$. ([Supplement] $\Rightarrow$ reduces to Theorem 3.21 with $p = 1$: $\mathbf{a}^\top\mathbf{X}^\top\mathbf{A}\mathbf{X}\mathbf{a} \sim (\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a})\chi_r^2$ for every $\mathbf{a}$.) ■

Theorem 4.4: distribution of the sample covariance matrix; $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$

If $\mathbf{X}_1, \dots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then

$$\bar{\mathbf{X}}_n \sim N_p(\boldsymbol{\mu}, \tfrac{1}{n}\boldsymbol{\Sigma}), \qquad (n-1)\mathbf{S} = \mathbf{X}^\top\Big(\mathbf{I}_n - \tfrac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\Big)\mathbf{X} \sim W_p(n-1, \boldsymbol{\Sigma}), \qquad \bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}.$$

Proof.

Step 1 ($\bar{\mathbf{X}}$). $\bar{\mathbf{X}} = \frac{1}{n}\mathbf{X}^\top\mathbf{1}_n$ is a linear transformation of the jointly normal $(\mathbf{X}_1^\top, \dots, \mathbf{X}_n^\top)^\top$ with mean $\boldsymbol{\mu}$ and covariance $\frac{1}{n^2}\sum_i \boldsymbol{\Sigma} = \frac{1}{n}\boldsymbol{\Sigma}$, so $\bar{\mathbf{X}} \sim N_p(\boldsymbol{\mu}, \frac{1}{n}\boldsymbol{\Sigma})$ by Theorem 3.8.

Step 2 ($\mathbf{S}$). $\mathbf{S} = \frac{1}{n-1}\mathbf{X}_C^\top\mathbf{X}_C = \frac{1}{n-1}\mathbf{X}^\top(\mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top)\mathbf{X}$ with $\mathbf{X}_C = \mathbf{H}_n\mathbf{X}$. The matrix $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$ is symmetric idempotent of rank $n-1$. Since it annihilates $\mathbf{1}_n$, $\mathbf{X}^\top\mathbf{H}_n\mathbf{X} = \mathbf{Y}^\top\mathbf{H}_n\mathbf{Y}$ with $\mathbf{Y} = \mathbf{X} - \mathbf{1}_n\boldsymbol{\mu}^\top$, whose rows are iid $N_p(\mathbf{0}, \boldsymbol{\Sigma})$, and Theorem 4.3 gives $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$.

Step 3 (independence) [Supplement]. Choose an orthogonal $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_n]$ with $\mathbf{p}_1 = \mathbf{1}_n/\sqrt{n}$ and let $\mathbf{z}_j = (\mathbf{X} - \mathbf{1}_n\boldsymbol{\mu}^\top)^\top \mathbf{p}_j$. As above $\mathbf{z}_1, \dots, \mathbf{z}_n \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$, $\mathbf{z}_1 = \sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu})$, and since $\mathbf{H}_n = \sum_{j=2}^n \mathbf{p}_j\mathbf{p}_j^\top$, $(n-1)\mathbf{S} = \sum_{j=2}^n \mathbf{z}_j\mathbf{z}_j^\top$. The first depends only on $\mathbf{z}_1$, the second only on $\mathbf{z}_2, \dots, \mathbf{z}_n$. ∎

3 A Theorem on Inverse Wishart Ratios

Motivation. Hotelling's T^2 involves $\mathbf{S}^{-1}$. This theorem reduces quadratic forms in an inverse Wishart matrix to ratios of χ^2 variables, which is the key step toward the F distribution of T^2.

Theorem 4.5: ratios involving an inverse Wishart matrix (Theorem 3)

Suppose $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$, $\mathbf{a} \in \mathbb{R}^p$, and $n > p - 1$. Then

(i) $(\boldsymbol{\Sigma}^{-1})_{11}/(\mathbf{M}^{-1})_{11} \sim \chi^2_{n-p+1}$;

(ii) for any constant $\mathbf{a} \neq \mathbf{0}$, $\dfrac{\mathbf{a}^\top\boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top\mathbf{M}^{-1}\mathbf{a}} \sim \chi^2_{n-p+1}$;

(iii) if $\mathbf{y} \in \mathbb{R}^p$ is random and independent of $\mathbf{M}$ (with $P(\mathbf{y} = \mathbf{0}) = 0$), then $\dfrac{\mathbf{y}^\top\boldsymbol{\Sigma}^{-1}\mathbf{y}}{\mathbf{y}^\top\mathbf{M}^{-1}\mathbf{y}} \sim \chi^2_{n-p+1}$ and is independent of $\mathbf{y}$.

Sketch. (i) is a special case of (ii), but we prove (i) first and generalize it to obtain (ii); (ii) implies (iii), since the conditional distribution of $\mathbf{y}^\top\boldsymbol{\Sigma}^{-1}\mathbf{y}/\mathbf{y}^\top\mathbf{M}^{-1}\mathbf{y}$ given $\mathbf{y} = \mathbf{a}$ is χ^2_{n-p+1} for almost all $\mathbf{a}$ (i.e. $\mathbf{a} \neq \mathbf{0}$) by (ii).

Proof. (i) Write

$$\mathbf{M} = \begin{pmatrix} m_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \boldsymbol{\sigma}_{12} \\ \boldsymbol{\sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}.$$

By the block-inverse formula,

$$\frac{(\boldsymbol{\Sigma}^{-1})_{11}}{(\mathbf{M}^{-1})_{11}} = \frac{m_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21}}{\sigma_{11} - \boldsymbol{\sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}}.$$

Write $\mathbf{M} = \mathbf{X}^\top\mathbf{X} = \sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^\top$ with $\mathbf{x}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\mathbf{X} = [\mathbf{x}_1, \mathbf{X}_2]$ ($\mathbf{x}_1$: first column). Then

$$m_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21} = \mathbf{x}_1^\top\mathbf{x}_1 - \mathbf{x}_1^\top\mathbf{X}_2(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{x}_1 = \mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1,$$

$$\mathbf{A} = \mathbf{A}(\mathbf{X}_2) = \mathbf{I}_n - \mathbf{X}_2(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top.$$

Given $\mathbf{X}_2$, $\mathbf{x}_1 \mid \mathbf{X}_2 \sim N_n(\mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21},\ \sigma_{11\cdot 2}\mathbf{I}_n)$, $\sigma_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}$. Since $\mathbf{A}\mathbf{X}_2 = \mathbf{0}$, $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 = (\mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21})^\top\mathbf{A}(\mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21})$. For any given $\mathbf{X}_2$, $\mathbf{A}$ is symmetric idempotent with $\operatorname{rank}(\mathbf{A}) = \operatorname{tr}(\mathbf{A}) = n - \operatorname{tr}\{(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{X}_2\} = n - (p-1)$, so Theorem 3.21 gives

$$\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 \mid \mathbf{X}_2 \sim \sigma_{11\cdot 2}\,\chi^2_{n-p+1}.$$

This does not depend on $\mathbf{X}_2$, so marginally $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 \sim \sigma_{11\cdot 2}\chi^2_{n-p+1}$, independent of $\mathbf{X}_2$. Hence the ratio is $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1/\sigma_{11\cdot 2} \sim \chi^2_{n-p+1}$.

(i)⇒(ii) Given $\mathbf{a}$, take a $p \times p$ orthogonal $\mathbf{P}$ whose first column is $\mathbf{a}/\|\mathbf{a}\|$, i.e. $\mathbf{a}^\top\mathbf{P} = \mathbf{e}_1^\top$ (w.l.o.g. $\|\mathbf{a}\| = 1$). Then

$$\frac{\mathbf{a}^\top\boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top\mathbf{M}^{-1}\mathbf{a}} = \frac{\mathbf{a}^\top\mathbf{P}\mathbf{P}^\top\boldsymbol{\Sigma}^{-1}\mathbf{P}\mathbf{P}^\top\mathbf{a}}{\mathbf{a}^\top\mathbf{P}\mathbf{P}^\top\mathbf{M}^{-1}\mathbf{P}\mathbf{P}^\top\mathbf{a}} = \frac{\mathbf{e}_1^\top\boldsymbol{\Sigma}_{\mathbf{P}}^{-1}\mathbf{e}_1}{\mathbf{e}_1^\top\mathbf{M}_{\mathbf{P}}^{-1}\mathbf{e}_1}, \quad \boldsymbol{\Sigma}_{\mathbf{P}}^{-1} = \mathbf{P}^\top\boldsymbol{\Sigma}^{-1}\mathbf{P},\ \mathbf{M}_{\mathbf{P}}^{-1} = \mathbf{P}^\top\mathbf{M}^{-1}\mathbf{P}.$$

By Theorem 4.2, $\mathbf{M_P} = \mathbf{P}^\top \mathbf{M}\mathbf{P} \sim W_p(n, \mathbf{\Sigma_P})$, so (i) applies: $(\mathbf{\Sigma_P}^{-1})_{11}/(\mathbf{M_P}^{-1})_{11} \sim \chi^2_{n-p+1}$.

(ii)⇒(iii) Conditioning on $\mathbf{y} = \mathbf{a}$ leaves the law of $\mathbf{M}$ unchanged (independence), so by (ii) the conditional distribution is χ^2_{n-p+1} for almost all $\mathbf{a} \neq \mathbf{0}$. It does not depend on $\mathbf{a}$, so it is also the marginal law and is independent of $\mathbf{y}$. ■

Remark 4.6: residuals and projections in multivariate analysis

1. The residual of $\mathbf{x}_1$ after projecting onto the span of $\mathbf{X}_2$ is $\mathbf{x}_1^* = \mathbf{x}_1 - \mathbf{X}_2\mathbf{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} = \mathbf{x}_1 - \mathrm{BLP}(\mathbf{x}_1 \mid \mathbf{X}_2)$.
2. $\mathbf{x}_1^*$ is orthogonal to (uncorrelated with) $\mathbf{X}_2$; being Gaussian, they are independent.
3. The partial variance $\sigma_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{21}^\top \mathbf{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} = \mathrm{Var}(\mathbf{x}_1^*)$ is the variance left after removing $\mathbf{X}_2$.

The residual sum of squares has $n - (p-1)$ degrees of freedom. This theorem is the key to the F distribution of Hotelling's T^2 (Ch. 5).

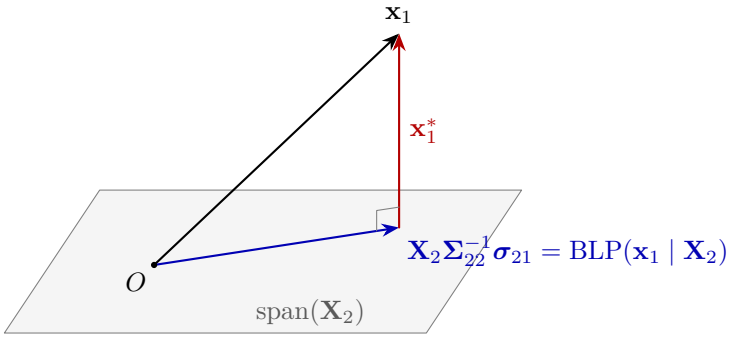


Figure. *Residual and projection. The residual $\mathbf{x}_1^*$ is what is left of $\mathbf{x}_1$ after projecting it onto the span of $\mathbf{X}_2$; it is orthogonal to that span, and $\sigma_{11\cdot 2} = \mathrm{Var}(\mathbf{x}_1^*)$ measures its size.*

4 Maximum Likelihood Estimation

Motivation. Under normality the maximum likelihood estimates of $\boldsymbol{\mu}$ and $\mathbf{\Sigma}$ are the sample mean and the (biased) sample covariance. The proof uses a trace inequality rather than differentiation.

Theorem 4.7: MLE of μ and Σ (Result 4.11)

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be iid $N_p(\boldsymbol{\mu}, \mathbf{\Sigma})$, $n > p$. The MLEs are

$$\hat{\boldsymbol{\mu}}_{\mathrm{MLE}} = \bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i, \qquad \hat{\mathbf{\Sigma}}_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top =: \mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}.$$

As random quantities, $\bar{\mathbf{X}}$ and $\frac{1}{n}\sum_j (\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})^\top$ are the maximum likelihood *estimators*; their observed values are the maximum likelihood *estimates*.

The MLE jointly minimizes the negative log-likelihood

$$\begin{aligned}\ell(\boldsymbol{\mu}, \mathbf{\Sigma} \mid \mathbf{x}^n) &:= -\log L(\boldsymbol{\mu}, \mathbf{\Sigma}) = \frac{n}{2}\log|\mathbf{\Sigma}| + \frac{1}{2}\sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\mu}) + c \\ &= \frac{n}{2}\log|\mathbf{\Sigma}| + \frac{n}{2}(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) + \frac{1}{2}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{\Sigma}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) + c.\end{aligned}$$

(Expand $\mathbf{x}_i - \boldsymbol{\mu} = (\mathbf{x}_i - \bar{\mathbf{x}}) + (\bar{\mathbf{x}} - \boldsymbol{\mu})$; the cross term vanishes.)

Step 1 (profile out $\boldsymbol{\mu}$). ℓ depends on $\boldsymbol{\mu}$ only through $(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \geq 0$, minimized by $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ for every $\boldsymbol{\Sigma}$. With $\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) = \operatorname{tr}\{\boldsymbol{\Sigma}^{-1} \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top\} = n\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$,

$$2\ell(\hat{\boldsymbol{\mu}}_{\text{MLE}}, \boldsymbol{\Sigma}) = n \log |\boldsymbol{\Sigma}| + n\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) + c,$$

so it remains to minimize $f(\boldsymbol{\Sigma}) = \log |\boldsymbol{\Sigma}| + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$ over $\boldsymbol{\Sigma} \succ 0$.

First proof: von Neumann's inequality. Decompose $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$ and $\mathbf{S}_0 = \mathbf{V}\mathbf{D}\mathbf{V}^\top$ (λ_i, d_i in decreasing order; $d_i > 0$ a.s. when $n > p$). The eigenvalues of $\boldsymbol{\Sigma}^{-1}$ in increasing order are $1/\lambda_1 \leq \cdots \leq 1/\lambda_p$, so von Neumann's lower bound gives

$$\log |\boldsymbol{\Sigma}| + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) \geq \log(\lambda_1 \cdots \lambda_p) + \sum_{i=1}^{p} \frac{d_i}{\lambda_i} = \sum_{i=1}^{p} \Big(\log \lambda_i + \frac{d_i}{\lambda_i} \Big),$$

with equality iff $\mathbf{U} = \mathbf{V}$. Let $a_i = d_i/\lambda_i$. Then

$$\sum_{i=1}^{p} \Big(\log \lambda_i + \frac{d_i}{\lambda_i} \Big) = \sum_{i=1}^{p} (\log d_i + a_i - \log a_i) = \log |\mathbf{S}_0| + \sum_{i=1}^{p} (a_i - \log a_i) \geq \log |\mathbf{S}_0| + p,$$

since $x - \log x \geq 1$. The bound is attained iff (1) $\mathbf{U} = \mathbf{V}$ and (2) $a_i = 1$ for all i, i.e. $\boldsymbol{\Lambda} = \mathbf{D}$: $\hat{\boldsymbol{\Sigma}}_{\text{MLE}} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top = \mathbf{V}\mathbf{D}\mathbf{V}^\top = \mathbf{S}_0$. ■

Second proof: matrix calculus. Reparametrise by the precision matrix $\mathbf{K} = \boldsymbol{\Sigma}^{-1}$ ($\boldsymbol{\Sigma} \mapsto \boldsymbol{\Sigma}^{-1}$ is a bijection of the PD cone, so by the change-of-variables part of Theorem 2.26 the optimizers correspond). $g(\mathbf{K}) = -\log |\mathbf{K}| + \operatorname{tr}(\mathbf{K}\mathbf{S}_0)$ is defined on the open convex set $\{\mathbf{K} \succ 0\}$, and for symmetric $\mathbf{H} \neq \mathbf{0}$

$$D^2 g(\mathbf{K})[\mathbf{H}] = \operatorname{tr}(\mathbf{K}^{-1}\mathbf{H}\mathbf{K}^{-1}\mathbf{H}) = \|\mathbf{K}^{-1/2}\mathbf{H}\mathbf{K}^{-1/2}\|_F^2 > 0,$$

so g is strictly convex (Theorem 2.25). The first-order condition

$$Dg(\mathbf{K})[\mathbf{H}] = -\operatorname{tr}(\mathbf{K}^{-1}\mathbf{H}) + \operatorname{tr}(\mathbf{S}_0\mathbf{H}) = \operatorname{tr}\{(\mathbf{S}_0 - \mathbf{K}^{-1})\mathbf{H}\} = 0 \quad \forall \mathbf{H}$$

and Theorem 2.24 give $\mathbf{K}^{-1} = \mathbf{S}_0$, the unique minimizer. By profiling (Theorem 2.26), $(\bar{\mathbf{x}}, \mathbf{S}_0)$ is the joint minimizer. ■

Remark 4.8: invariance and the maximized likelihood

[Supplement] A function of the MLE is the MLE of the function: e.g. the eigenvalues of $\mathbf{S}_0$ estimate those of $\boldsymbol{\Sigma}$, and r_{ij} is the MLE of ρ_{ij}. The maximum is $L(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}) = (2\pi)^{-np/2} |\mathbf{S}_0|^{-n/2} e^{-np/2}$, which depends on the data only through the generalized variance $|\mathbf{S}_0|$; this is used for likelihood ratio tests (Ch. 5).

5 Large-Sample Behavior of $\bar{\mathbf{X}}$ and $\mathbf{S}$

Motivation. Without normality, the CLT still makes $\bar{\mathbf{X}}$ approximately normal and $\mathbf{S}$ consistent, so normal-theory procedures remain valid for large samples.

Theorem 4.9: law of large numbers (Result 4.12)

Let $\mathbf{X}_1, \dots, \mathbf{X}_n \in \mathbb{R}^p$ be iid with $\mathrm{E}(\mathbf{X}_j) = \boldsymbol{\mu}$ and finite $\mathrm{Cov}(\mathbf{X}_j) = \boldsymbol{\Sigma}$. Then, as $n \to \infty$,

$$\text{(a) } \bar{\mathbf{X}}_n \xrightarrow{p} \boldsymbol{\mu}, \qquad \text{(b) } \mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}, \qquad \text{(c) } \mathbf{S}_0 \xrightarrow{p} \boldsymbol{\Sigma}.$$

Moreover $\bar{\mathbf{X}}_n$ and $\mathbf{S}$ are unbiased.

Proof. (a) By Markov's inequality,

$$P(\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\| > \epsilon) \le \frac{1}{\epsilon^2}\mathrm{E}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|^2 = \frac{1}{\epsilon^2}\mathrm{E}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) = \frac{1}{\epsilon^2}\mathrm{E}[\mathrm{tr}\{(\bar{\mathbf{X}}_n - \boldsymbol{\mu})(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top\}]$$
$$= \frac{1}{\epsilon^2}\mathrm{tr}(\mathrm{Var}(\bar{\mathbf{X}}_n)) = \frac{\mathrm{tr}(\boldsymbol{\Sigma})}{n\epsilon^2} \xrightarrow[n\to\infty]{} 0.$$

(b) Adding and subtracting $\boldsymbol{\mu}$ inside each factor and using $\sum_i(\mathbf{X}_i - \boldsymbol{\mu}) = n(\bar{\mathbf{X}} - \boldsymbol{\mu})$,

$$(n-1)\mathbf{S} = \sum_{i=1}^n(\mathbf{X}_i - \bar{\mathbf{X}}_n)(\mathbf{X}_i - \bar{\mathbf{X}}_n)^\top = \sum_{i=1}^n(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top - n(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top.$$

Hence

$$\frac{n-1}{n}\mathbf{S} = \frac{1}{n}\sum_{i=1}^n(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top - (\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \xrightarrow{p} \boldsymbol{\Sigma} - \mathbf{0} = \boldsymbol{\Sigma}.$$

The first term is an average of iid matrices with mean $\mathrm{E}(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top = \boldsymbol{\Sigma}$, so it converges to $\boldsymbol{\Sigma}$ by the weak LLN applied entrywise (matrix convergence is entrywise, Theorem 2.30). The second term tends to $\mathbf{0}$ by (a) and the CMT ($\mathbf{x} \mapsto \mathbf{x}\mathbf{x}^\top$ is continuous). Since $n/(n-1) \to 1$, $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$.

(c) $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$, which was shown to converge to $\boldsymbol{\Sigma}$ in (b).

Unbiasedness. $\mathrm{E}\bar{\mathbf{X}}_n = \boldsymbol{\mu}$. Under normality $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$, so $\mathrm{E}\{(n-1)\mathbf{S}\} = (n-1)\boldsymbol{\Sigma}$ by Theorem 4.2(iv), i.e. $\mathrm{E}\mathbf{S} = \boldsymbol{\Sigma}$. [Supplement] Without normality, taking expectations in the identity of (b) gives the same, $\mathrm{E}\{(n-1)\mathbf{S}\} = n\boldsymbol{\Sigma} - n\mathrm{Var}(\bar{\mathbf{X}}_n) = (n-1)\boldsymbol{\Sigma}$; under normality Theorem 4.2(iv) also gives the rate $\mathrm{E}\|\mathbf{S} - \boldsymbol{\Sigma}\|_F^2 = \sum_{ij}(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})/(n-1) \to 0$. ■

Remark 4.10: rate of the estimation error

With the Frobenius norm $\|\mathbf{A}\|_F = (\sum_{i,j} a_{ij}^2)^{1/2}$, the estimation errors are $E_{\bar{u}} = \|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2$, $E_S = \|\mathbf{S} - \boldsymbol{\Sigma}\|_F$ and $E_R = \|\mathbf{R} - \mathbf{R}_0\|_F$ ($\mathbf{R}_0$ the population correlation matrix). [Supplement] $\mathrm{E}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2^2 = \mathrm{tr}\boldsymbol{\Sigma}/n$, so $E_{\bar{u}} = \|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2$ decays at rate $n^{-1/2}$; so does $\|\mathbf{S} - \boldsymbol{\Sigma}\|_F$ under normality. On a log–log plot of $E_{\bar{u}}, E_S, E_R$ against n the slope is about $-1/2$.

Theorem 4.11: multivariate central limit theorem (Result 4.13(i))

Let $\mathbf{X}_1, \mathbf{X}_2, \dots \in \mathbb{R}^p$ be iid with $\mathrm{E}(\mathbf{X}_i) = \boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{X}_i) = \boldsymbol{\Sigma}$. Then

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n(\mathbf{X}_i - \boldsymbol{\mu}) \Rightarrow N_p(\mathbf{0}, \boldsymbol{\Sigma}), \qquad \text{equivalently} \qquad \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \Rightarrow N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

Proof. By Cramér–Wold it suffices to show $\mathbf{t}^\top\sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \Rightarrow \mathbf{t}^\top\mathbf{Y}$, $\mathbf{Y} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, for every $\mathbf{t}$. $W_i = \mathbf{t}^\top(\mathbf{X}_i - \boldsymbol{\mu})$ are iid scalars with mean 0 and variance $\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t}$, so the univariate CLT gives $\sqrt{n}\bar{W}_n \Rightarrow N(0, \mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t})$, the law of $\mathbf{t}^\top\mathbf{Y}$ (if $\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t} = 0$, $W_i = 0$ a.s. and the claim is trivial). ■

Theorem 4.12: large-sample chi-square (Result 4.13(ii))

Under the same assumptions with $\boldsymbol{\Sigma} \succ 0$, $P(\mathbf{S} \text{ invertible}) \to 1$ and $T_n := n(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top r(\mathbf{S})(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \Rightarrow \chi^2_p$, where $r(\mathbf{A}) := \mathbf{A}^{-1}$ if $|\mathbf{A}| \neq 0$ and $r(\mathbf{A}) := \mathbf{I}_p$ otherwise; thus $T_n = n(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})$ whenever $\mathbf{S}$ is invertible.

Proof. Let $\mathbf{Z}_n := \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})$ and $\mathbf{Z} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$.

(i) $\mathbf{Z}_n \Rightarrow \mathbf{Z}$. For every $\mathbf{t} \in \mathbb{R}^p$, $\mathbf{t}^\top \mathbf{Z}_n = \sqrt{n}(\mathbf{t}^\top \bar{\mathbf{X}}_n - \mathbf{t}^\top \boldsymbol{\mu}) \Rightarrow N(0, \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$ by the univariate CLT, so $\mathbf{Z}_n \Rightarrow \mathbf{Z}$ by Cramér–Wold (this is Theorem 4.11).

(ii) Handling a possibly singular $\mathbf{S}$. Let $A_n := \{\mathbf{S} \text{ is invertible}\}$. Since $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$ (Theorem 4.9(b)) and $\mathbf{A} \mapsto |\mathbf{A}|$ is continuous, $|\mathbf{S}| \xrightarrow{p} |\boldsymbol{\Sigma}| > 0$, hence

$$P(A_n) \geq P(|\mathbf{S}| \geq \tfrac{1}{2}|\boldsymbol{\Sigma}|) \longrightarrow 1.$$

Define $r(\mathbf{A}) = \mathbf{A}^{-1}$ if $|\mathbf{A}| \neq 0$ and $r(\mathbf{A}) = \mathbf{I}_p$ otherwise. By the definition of r,

$$T_n = n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu})\, I_{A_n} + n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top (\bar{\mathbf{X}} - \boldsymbol{\mu})\, I_{A_n^c} = \mathbf{Z}_n^\top r(\mathbf{S}) \mathbf{Z}_n,$$

and T_n coincides with the statistic of the theorem on A_n, an event of probability tending to one.

(iii) $r(\mathbf{S}) \xrightarrow{p} \boldsymbol{\Sigma}^{-1}$. Since $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$, by the CMT it suffices that r is continuous at the point $\boldsymbol{\Sigma}$. Since $|\boldsymbol{\Sigma}| > 0$ and $\{\mathbf{A} : |\mathbf{A}| \neq 0\}$ is open, $r(\mathbf{A}) = \mathbf{A}^{-1} = \operatorname{adj}(\mathbf{A})/|\mathbf{A}|$ on a neighborhood of $\boldsymbol{\Sigma}$ (Section 1 of Ch. 2), whose entries are polynomials divided by a nonvanishing polynomial; so r is continuous at $\boldsymbol{\Sigma}$ and $r(\mathbf{S}) \xrightarrow{p} \boldsymbol{\Sigma}^{-1}$.

(iv) Conclusion. Let $Q(\mathbf{z}, \mathbf{A}) = \mathbf{z}^\top \mathbf{A} \mathbf{z}$, so $T_n = Q(\mathbf{Z}_n, r(\mathbf{S}))$. By Slutsky, $(\mathbf{Z}_n, r(\mathbf{S})) \Rightarrow (\mathbf{Z}, \boldsymbol{\Sigma}^{-1})$, and Q is continuous, so by the CMT

$$T_n = Q(\mathbf{Z}_n, r(\mathbf{S})) \Rightarrow Q(\mathbf{Z}, \boldsymbol{\Sigma}^{-1}) = \mathbf{Z}^\top \boldsymbol{\Sigma}^{-1} \mathbf{Z} \sim \chi^2_p$$

(Example 3.23). ■

This is the basis of large-sample tests and confidence regions for the mean without normality (Theorems 5.16 and 5.17).

Theorem 4.13: asymptotic normality of $h(\bar{\mathbf{X}}_n)$ (delta method)

For $h : \mathbb{R}^p \to \mathbb{R}$ differentiable at $\boldsymbol{\mu}$ with gradient $\nabla h(\boldsymbol{\mu})$,

$$\sqrt{n}\{h(\bar{\mathbf{X}}_n) - h(\boldsymbol{\mu})\} \Rightarrow N_p\big(0,\ (\nabla h(\boldsymbol{\mu}))^\top \boldsymbol{\Sigma} (\nabla h(\boldsymbol{\mu}))\big).$$

Proof. Taylor: $h(\bar{\mathbf{X}}_n) = h(\boldsymbol{\mu}) + (\nabla h(\boldsymbol{\mu}))^\top (\bar{\mathbf{X}}_n - \boldsymbol{\mu}) + O_p(\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2^2)$. Multiply by $\sqrt{n}$: $\sqrt{n}(h(\bar{\mathbf{X}}_n) - h(\boldsymbol{\mu})) = (\nabla h(\boldsymbol{\mu}))^\top \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) + O_p(\sqrt{n}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|^2)$, and the remainder is $o_p(1)$. Slutsky gives $\Rightarrow (\nabla h(\boldsymbol{\mu}))^\top N_p(\mathbf{0}, \boldsymbol{\Sigma})$. ■

6 Assessing the Assumption of Normality

Motivation. Most procedures in later chapters assume $\mathbf{X}_j \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Normality itself cannot be verified, but its consequences can be checked one at a time: normal marginals, elliptical pairs and χ^2_p distances.

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then

- every $\mathbf{a}^\top \mathbf{X}$ is normal (Result 4.2), so each marginal X_i is normal;

- the density contours are the ellipsoids $(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = c^2$, so each pair (X_i, X_k) has elliptical contours;
- the squared distance $(\mathbf{X}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) \sim \chi^2_p$ (Result 4.7): very large distances are rare.

These are necessary conditions only: a violation rules out normality, but passing them does not prove it. The checks are: (1) normal marginals: proportions, Q–Q plot, r_Q; (2) elliptical pairs: the 50% contour; (3) overall fit and outliers: the chi-square plot.

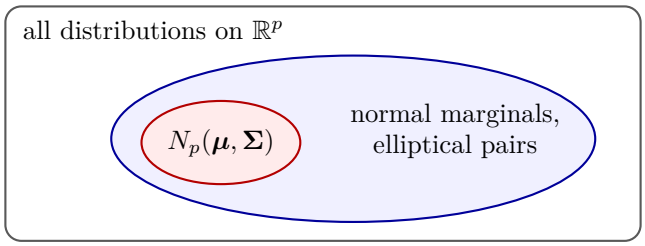


Figure. *The checks below test necessary conditions. Data from a non-normal law in the blue region pass the marginal and pairwise checks, so passing them supports, but does not establish, normality.*

6.1 Normal marginals

Proportion check. For variable i and $k = 1, 2$ let $p_{ik} := P(|X_i - \mu_i| < k\sqrt{\sigma_{ii}})$. If $X_i \sim N(\mu_i, \sigma_{ii})$, then $p_{i1} = P(|Z| < 1) = 0.683$ and $p_{i2} = P(|Z| < 2) = 0.954$ for $Z \sim N(0,1)$. Replace μ_i, σ_{ii} by $\bar{x}_i, s_{ii}$ and count:

$$\hat{p}_{i1} := \frac{\#\{j : |x_{ji} - \bar{x}_i| < \sqrt{s_{ii}}\}}{n}, \qquad \hat{p}_{i2} := \frac{\#\{j : |x_{ji} - \bar{x}_i| < 2\sqrt{s_{ii}}\}}{n}.$$

Flag non-normality if $\hat{p}_{ik}$ is more than three standard errors $\sqrt{p_{ik}(1-p_{ik})/n}$ away from p_{ik}:

$$|\hat{p}_{i1} - 0.683| > 3\sqrt{\frac{(0.683)(0.317)}{n}} = \frac{1.396}{\sqrt{n}} \quad \text{or} \quad |\hat{p}_{i2} - 0.954| > 3\sqrt{\frac{(0.954)(0.046)}{n}} = \frac{0.628}{\sqrt{n}}.$$

Proportions that are too small suggest tails heavier than normal.

Normal Q–Q plot. If $X \sim N(\mu, \sigma^2)$, its u-quantile is $\mu + \sigma\Phi^{-1}(u)$, linear in $\Phi^{-1}(u)$. The order statistic $x_{(j)}$ estimates the $\frac{j-1/2}{n}$-quantile of X, so with the standard normal quantiles

$$q_{(j)} := \Phi^{-1}\Big(\frac{j - \frac{1}{2}}{n}\Big), \qquad j = 1, \ldots, n,$$

the steps are: (1) order $x_{(1)} \le \cdots \le x_{(n)}$ and compute $q_{(j)}$; (2) plot the pairs $(q_{(j)}, x_{(j)})$; (3) under normality $x_{(j)} \approx \mu + \sigma q_{(j)}$, so check straightness.

Correlation coefficient test. Under normality the Q–Q points lie near a line, so their correlation

$$r_Q := \frac{\sum_{j=1}^n (x_{(j)} - \bar{x})(q_{(j)} - \bar{q})}{\sqrt{\sum_{j=1}^n (x_{(j)} - \bar{x})^2}\sqrt{\sum_{j=1}^n (q_{(j)} - \bar{q})^2}}$$

is near 1. Reject normality at level α if r_Q is below the critical point (Table 4.2 of J&W; excerpt below).

n	$\alpha = .01$	.05	.10
10	.8801	.9198	.9351
20	.9269	.9508	.9604
30	.9479	.9652	.9715
50	.9671	.9768	.9809
100	.9822	.9873	.9895

Example 4.14: Q–Q plot and r_Q (Examples 4.9 and 4.11)

The $n = 10$ ordered observations $-1.00, -0.10, 0.16, 0.41, 0.62, 0.80, 1.26, 1.54, 1.71, 2.30$ have $\bar{x} = 0.770$, $\sqrt{s_{11}} = 0.970$, and $q_{(j)} = \Phi^{-1}((j - \frac{1}{2})/10)$ runs from -1.645 to 1.645. The points lie close to the line $\bar{x} + \sqrt{s_{11}}\, q = 0.770 + 0.970\, q$ (Figure below), and

$$r_Q = \frac{8.584}{\sqrt{8.472}\,\sqrt{8.795}} = 0.994 > 0.9351,$$

so normality is not rejected at $\alpha = 0.10$.

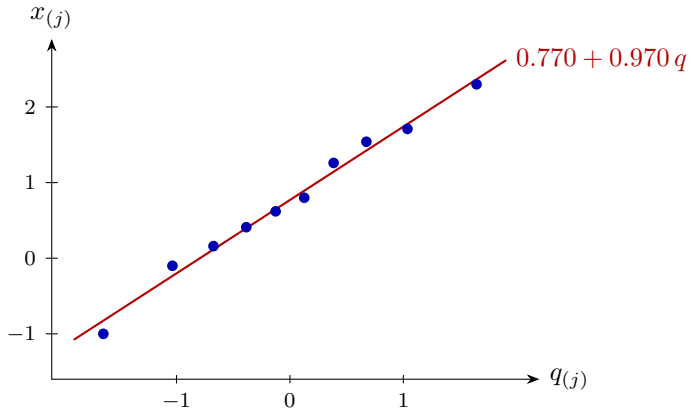


Figure. *Normal Q–Q plot for Example 4.9 ($n = 10$): the ordered data against $q_{(j)} = \Phi^{-1}((j - \frac{1}{2})/n)$. The points are close to the line $\bar{x} + \sqrt{s_{11}}\, q$; $r_Q = 0.994$.*

6.2 Elliptical pairs: the 50% contour

For $p = 2$, $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_2$ by Result 4.7, so the ellipse $\{\mathbf{x} : (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \le \chi^2_2(0.5)\}$, $\chi^2_2(0.5) = 1.39$, has probability 0.5. Replacing $\boldsymbol{\mu}, \boldsymbol{\Sigma}$ by $\bar{\mathbf{x}}, \mathbf{S}$, the estimated 50% ellipse $\{\mathbf{x} : (\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x} - \bar{\mathbf{x}}) \le \chi^2_2(0.5)\}$ should contain about half of the data.

Example 4.15: the 50% contour (Example 4.12)

For the sales–profits data ($n = 10$) the squared Mahalanobis distances are $1.61, 0.30, 0.62, 1.79, 1.30, 4.38, 1.64, 3.53, 1.71, 1.16$. Only 4 of 10 are ≤ 1.39: proportion 0.40 versus 0.50, and $n = 10$ is too small to reject normality.

6.3 Overall fit and outliers: the chi-square plot

$D^2 := (\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_p$ has u-quantile $q_{c,p}(u)$. Replacing $\boldsymbol{\mu}, \boldsymbol{\Sigma}$ by $\bar{\mathbf{x}}, \mathbf{S}$, the ordered squared distances

$$d^2_j := (\mathbf{x}_j - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x}_j - \bar{\mathbf{x}}), \qquad j = 1, \dots, n,$$

estimate the $\frac{j-1/2}{n}$-quantiles of D^2. (1) Order $d^2_{(1)} \le \cdots \le d^2_{(n)}$ and plot the pairs $(q_{c,p}((j - \frac{1}{2})/n),\, d^2_{(j)})$. (2) Compare with the line of slope 1 through the origin.

- Curvature indicates non-normality; a few points far above the line indicate outliers; about half of the d^2_j should be $\le q_{c,p}(0.5)$.
- The d^2_j are neither independent nor exactly χ^2_p, so use the plot as a guide when n and $n - p$ are at least 25–30.

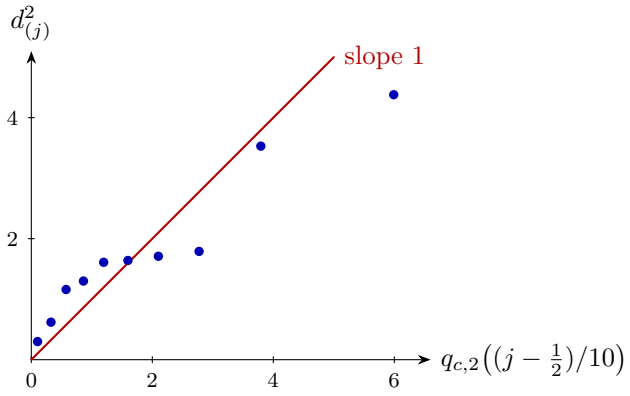


Figure. *Chi-square plot for the sales–profits data (Example 4.13, $p = 2$, $n = 10$), using the distances of Example 4.12. The points follow the 45° line reasonably well, so normality is hard to reject.*

Example 4.16: chi-square plots (Examples 4.13 and 4.14)

(a) Sales–profits ($p = 2$, $n = 10$): the plot above is reasonably straight; it is hard to reject normality.
(b) Board stiffness ($p = 4$, $n = 30$): 17 of the 30 values d_j^2 are $\le q_{c,4}(0.5) = 3.36$ (about half), yet the two largest $d^2_{(j)}$ (boards 9 and 16) break away from the line and the upper end curves, so these boards merit attention.

With small n only gross departures are detected. With large n the tests nearly always reject, but inference on $\boldsymbol{\mu}$ is robust (CLT, Result 4.13).

7 Summary: Distributional Facts Used Later

[Supplement] The facts of Chapters 3 and 4 that the inference chapters rely on:

Fact	Source
$\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$	Theorem 3.8
$(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$, $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$	Theorem 4.3
$(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_p$	Theorem 3.22, Example 3.23
$\mathbf{a}^\top \boldsymbol{\Sigma}^{-1} \mathbf{a} / \mathbf{a}^\top \mathbf{M}^{-1} \mathbf{a} \sim \chi^2_{m-p+1}$, also for independent random $\mathbf{y}$	Theorem 4.5
$n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu}) \Rightarrow \chi^2_p$ (no normality)	Theorem 4.11 + Slutsky

8 Selected exercises

Motivation. Two classical consequences of this chapter: the exact law of the generalized variance $|\mathbf{S}|$, obtained from the argument behind Theorem 4.5, and the large-sample law of the sample correlation, obtained from the CLT and the delta method.

Exercise 4.1 (The generalized variance)

Let $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$ and $n \ge p$.
(a) Show that $|\mathbf{M}|/|\boldsymbol{\Sigma}| \overset{d}{=} \prod_{i=1}^{p} V_i$ with $V_1, \dots, V_p$ independent and $V_i \sim \chi^2_{n-i+1}$.
(b) For a sample of size $n > p$ from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, deduce $\mathrm{E}|\mathbf{S}| = |\boldsymbol{\Sigma}| \prod_{i=1}^{p} \dfrac{n-i}{n-1}$, so $|\mathbf{S}|$ underestimates $|\boldsymbol{\Sigma}|$ on average when $p \ge 2$.

Solution [Supplement].

(a) Induction on p. For $p = 1$, $M/\sigma \sim \chi^2_n$ by definition.

Step 1 (set-up). Let $p \geq 2$ and write $\mathbf{M} = \mathbf{X}^\top \mathbf{X}$, where the $n \times p$ matrix $\mathbf{X} = [\mathbf{x}_1, \mathbf{X}_2]$ has iid rows $N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\mathbf{x}_1$ is its first column. The rows of $\mathbf{X}_2$ are iid $N_{p-1}(\mathbf{0}, \boldsymbol{\Sigma}_{22})$, so $\mathbf{M}_{22} = \mathbf{X}_2^\top \mathbf{X}_2 \sim W_{p-1}(n, \boldsymbol{\Sigma}_{22})$, and $\mathbf{M}_{22} \succ 0$ a.s. since $n \geq p - 1$.

Step 2 (block determinants). By Theorem 2.7, $|\mathbf{M}| = |\mathbf{M}_{22}|\, m_{11\cdot 2}$ and $|\boldsymbol{\Sigma}| = |\boldsymbol{\Sigma}_{22}|\, \sigma_{11\cdot 2}$, so

$$\frac{|\mathbf{M}|}{|\boldsymbol{\Sigma}|} = \frac{|\mathbf{M}_{22}|}{|\boldsymbol{\Sigma}_{22}|} \cdot \frac{m_{11\cdot 2}}{\sigma_{11\cdot 2}}.$$

Step 3 (the last factor). The proof of Theorem 4.5(i) shows that, conditionally on $\mathbf{X}_2$, $U := m_{11\cdot 2}/\sigma_{11\cdot 2} \sim \chi^2_{n-p+1}$. This conditional law does not depend on $\mathbf{X}_2$, so $U \sim \chi^2_{n-p+1}$ and U is independent of $\mathbf{X}_2$, hence of $|\mathbf{M}_{22}|/|\boldsymbol{\Sigma}_{22}|$.

Step 4 (induction). By the induction hypothesis for $\mathbf{M}_{22}$ (dimension $p-1$, $n \geq p-1$), $|\mathbf{M}_{22}|/|\boldsymbol{\Sigma}_{22}| \overset{d}{=} \prod_{i=1}^{p-1} V_i$ with independent $V_i \sim \chi^2_{n-i+1}$. Put $V_p = U \sim \chi^2_{n-p+1}$; by Step 3 it is independent of $(V_1, \ldots, V_{p-1})$.

(b) By Theorem 4.4, $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$ with $n - 1 \geq p$, so by (a) $(n-1)^p |\mathbf{S}|/|\boldsymbol{\Sigma}|$ is a product of independent χ^2_{n-i} variables, $i = 1, \ldots, p$. Independence gives $\mathrm{E}\{(n-1)^p |\mathbf{S}|/|\boldsymbol{\Sigma}|\} = \prod_{i=1}^{p}(n-i)$. For $p \geq 2$ the factor $(n-2)/(n-1)$ is < 1 and the others are ≤ 1. ∎

Exercise 4.2 (Asymptotics of the sample correlation)

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be iid $N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\rho = \sigma_{12}/\sqrt{\sigma_{11}\sigma_{22}}$, $|\rho| < 1$, and $r = s_{12}/\sqrt{s_{11}s_{22}}$.
(a) Show $\sqrt{n}(r - \rho) \Rightarrow N(0, (1-\rho^2)^2)$.
(b) Show that Fisher's $z = \frac{1}{2}\log\frac{1+r}{1-r}$ satisfies $\sqrt{n}(z - \zeta) \Rightarrow N(0,1)$, $\zeta = \frac{1}{2}\log\frac{1+\rho}{1-\rho}$, and give an approximate $1-\alpha$ confidence interval for ρ.

Solution [Supplement].

(a)

Step 1 (reduction). r is unchanged by $X_{ij} \mapsto (X_{ij} - \mu_j)/\sqrt{\sigma_{jj}}$, so assume $\boldsymbol{\mu} = \mathbf{0}$, $\sigma_{11} = \sigma_{22} = 1$, $\sigma_{12} = \rho$.

Step 2 (moments about 0). Let $\mathbf{Y}_i = (X_{i1}^2, X_{i2}^2, X_{i1}X_{i2})^\top$, iid with mean $\boldsymbol{m} = (1, 1, \rho)^\top$, and

$$\mathbf{W}_n = \bar{\mathbf{Y}}_n - (\bar{X}_1^2,\ \bar{X}_2^2,\ \bar{X}_1\bar{X}_2)^\top = \tfrac{n-1}{n}(s_{11}, s_{22}, s_{12})^\top.$$

Since $\sqrt{n}\,\bar{X}_j\bar{X}_k = (\sqrt{n}\bar{X}_j)\bar{X}_k = O_p(1)\,o_p(1) \xrightarrow{p} 0$ (CLT and LLN), Theorem 4.11 and Slutsky give $\sqrt{n}(\mathbf{W}_n - \boldsymbol{m}) \Rightarrow N_3(\mathbf{0}, \boldsymbol{\Gamma})$ with $\boldsymbol{\Gamma} = \mathrm{Cov}(\mathbf{Y}_1)$. The factor $\frac{n-1}{n}$ cancels in r, so $r = g(\mathbf{W}_n)$ with $g(a,b,c) = c/\sqrt{ab}$.

Step 3 ($\boldsymbol{\Gamma}$). Write $X_2 = \rho X_1 + \sqrt{1-\rho^2}\,E$ with $E \sim N(0,1)$ independent of X_1 (Theorem 3.12). Using $\mathrm{E}X_1^4 = 3$ and $\mathrm{E}X_1^3 = 0$,

$$\mathrm{E}X_1^2X_2^2 = 3\rho^2 + (1-\rho^2) = 1 + 2\rho^2, \qquad \mathrm{E}X_1^3X_2 = 3\rho, \qquad \mathrm{E}X_1X_2^3 = 3\rho,$$

and subtracting the products of the means,

$$\boldsymbol{\Gamma} = \begin{pmatrix} 2 & 2\rho^2 & 2\rho \\ 2\rho^2 & 2 & 2\rho \\ 2\rho & 2\rho & 1+\rho^2 \end{pmatrix}.$$

Step 4 (delta method). g is continuously differentiable near $\boldsymbol{m}$ with $\nabla g(\boldsymbol{m}) = (-\rho/2, -\rho/2, 1)^\top$. The proof of Theorem 4.13 uses only $\sqrt{n}(\mathbf{W}_n - \boldsymbol{m}) \Rightarrow N_3(\mathbf{0}, \boldsymbol{\Gamma})$ and a Taylor expansion, so $\sqrt{n}(r -$

$\rho) \Rightarrow N(0, \nabla g^\top \mathbf{\Gamma} \nabla g)$ with

$$\begin{aligned}\nabla g^\top \mathbf{\Gamma} \nabla g &= \frac{\rho^2}{4}(\Gamma_{11} + \Gamma_{22} + 2\Gamma_{12}) + \Gamma_{33} - \rho(\Gamma_{13} + \Gamma_{23}) \\ &= \rho^2(1+\rho^2) + 1 + \rho^2 - 4\rho^2 = (1-\rho^2)^2.\end{aligned}$$

(b) $h(x) = \frac{1}{2}\log\frac{1+x}{1-x}$ has $h'(\rho) = 1/(1-\rho^2)$. The one-dimensional delta method applied to (a) gives $\sqrt{n}(z-\zeta) \Rightarrow N\big(0, (1-\rho^2)^2/(1-\rho^2)^2\big) = N(0,1)$: the transformation stabilizes the variance. Hence $P(|z-\zeta| \le z_{\alpha/2}/\sqrt{n}) \to 1-\alpha$, and since $\tanh = h^{-1}$ is increasing,

$$\big[\tanh(z - z_{\alpha/2}/\sqrt{n}),\ \tanh(z + z_{\alpha/2}/\sqrt{n})\big]$$

is an approximate $1-\alpha$ interval for ρ (in practice $\sqrt{n-3}$ is used, which has the same limit). ■

Chapter 5. Inference about a Mean Vector

Hotelling's T^2, likelihood ratio tests, confidence regions and simultaneous intervals, large-sample inference

1 Starting Point: the Univariate t and Known Σ

In one dimension $H_0 : \mu = \mu_0$ is tested with $t = \sqrt{n}(\bar{X} - \mu_0)/s$, and

$$t_{n-1}^2 = \sqrt{n}(\bar{X} - \mu_0)\,(s^2)^{-1}\sqrt{n}(\bar{X} - \mu_0) = \begin{pmatrix}\text{normal}\\ \text{r.v.}\end{pmatrix}\begin{pmatrix}\text{(scaled) chi-square}\\ \text{r.v./d.f.}\end{pmatrix}^{-1}\begin{pmatrix}\text{normal}\\ \text{r.v.}\end{pmatrix}.$$

If $\boldsymbol{\Sigma}$ is known, $\sqrt{n}\boldsymbol{\Sigma}^{-1/2}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \mathbf{I}_p)$, so

$$n(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \sim \chi_p^2, \qquad C_\alpha = \{\boldsymbol{\mu} \in \mathbb{R}^p : (\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top (\tfrac{1}{n}\boldsymbol{\Sigma})^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \le \chi_{p,\alpha}^2\}.$$

Replacing $\boldsymbol{\Sigma}$ by $\mathbf{S}$ gives Hotelling's T^2.

2 Hotelling's T^2 Statistic

Motivation. Testing $\boldsymbol{\mu} = \boldsymbol{\mu}_0$ with p variables by p separate t-tests ignores correlation and inflates error. Hotelling's T^2 is the multivariate t-statistic: it combines all coordinates using $\mathbf{S}^{-1}$ and has an exact F distribution.

Definition 5.1: Hotelling's T^2

Suppose $\mathbf{X}$ and $\mathbf{S}$ are independent with $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $m\mathbf{S} \sim W_p(m, \boldsymbol{\Sigma})$. Then

$$T_p^2(m) = (\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\mathbf{X} - \boldsymbol{\mu})$$

is Hotelling's T^2 statistic. It plays the role of Student's t when $\boldsymbol{\Sigma}$ is *unknown*:

$$T_p^2(n-1) = \begin{pmatrix}\text{multivariate normal}\\ \text{random vector}\end{pmatrix}^\top \begin{pmatrix}\text{Wishart random}\\ \text{matrix/d.f.}\end{pmatrix}^{-1} \begin{pmatrix}\text{multivariate normal}\\ \text{random vector}\end{pmatrix}$$

$$= N_p(\mathbf{0}, \boldsymbol{\Sigma})^\top \left[\tfrac{1}{n-1} W_p(n-1, \boldsymbol{\Sigma})\right]^{-1} N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

Theorem 5.2 (slides: Theorem 7): T^2 and the F distribution

If $m > p - 1$, then $\dfrac{m - p + 1}{mp} T_p^2(m) \sim F_{p, m-p+1}$.

Proof. Multiplying and dividing by $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})$,

$$\frac{m-p+1}{mp} T_p^2(m) = \frac{(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})/p}{\dfrac{(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})}{(\mathbf{X} - \boldsymbol{\mu})^\top (m\mathbf{S})^{-1}(\mathbf{X} - \boldsymbol{\mu})} \Big/ (m - p + 1)} = \frac{Y/p}{W/(m-p+1)},$$

where $Y = (\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi_p^2$ (Example 3.23) and $W = \dfrac{\mathbf{a}^\top \boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top (m\mathbf{S})^{-1}\mathbf{a}}$ with the random vector $\mathbf{a} = \mathbf{X} - \boldsymbol{\mu}$ independent of $m\mathbf{S} \sim W_p(m, \boldsymbol{\Sigma})$; by Theorem 4.5(iii) $W \sim \chi_{m-p+1}^2$ and is independent of $\mathbf{a}$, hence of Y. By definition the ratio is $F_{p, m-p+1}$. ∎

Theorem 5.3: one-sample T^2

If $\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\bar{\mathbf{X}} = \frac{1}{n}\sum \mathbf{X}_j$, $\mathbf{S} = \frac{1}{n-1}\sum(\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})^\top$, then $\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$ independently, so

$$n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim T^2_p(n-1), \qquad n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim \frac{(n-1)p}{n-p} F_{p,n-p}.$$

($m = n-1$ gives $\frac{m-p+1}{mp} = \frac{n-p}{(n-1)p}$.) Hence $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ is rejected at level α if

$$T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) > \frac{(n-1)p}{n-p} F_{p,n-p}(\alpha),$$

with p-value $P(F_{p,n-p} \geq \frac{n-p}{(n-1)p} T^2_{\text{obs}})$.

Theorem 5.4: invariance of T^2 under linear transformations

The T^2 statistic is unchanged by changes of units of the form $\mathbf{Y}_{(p\times 1)} = \mathbf{C}_{(p\times p)}\mathbf{X}_{(p\times 1)} + \mathbf{d}_{(p\times 1)}$ with $\mathbf{C}$ non-singular: $\bar{\mathbf{y}} = \mathbf{C}\bar{\mathbf{x}} + \mathbf{d}$, $\mathbf{S}_y = \mathbf{C}\mathbf{S}\mathbf{C}^\top$, $\boldsymbol{\mu}_Y = \mathbf{C}\boldsymbol{\mu}_X + \mathbf{d}$, and

$$n(\bar{\mathbf{y}} - \boldsymbol{\mu}_Y)^\top \mathbf{S}_y^{-1}(\bar{\mathbf{y}} - \boldsymbol{\mu}_Y) = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_x)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x) \sim T^2_p(n-1).$$

Proof. $n(\mathbf{C}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x))^\top(\mathbf{C}\mathbf{S}\mathbf{C}^\top)^{-1}\mathbf{C}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x) = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_x)^\top \mathbf{C}^\top \mathbf{C}^{-\top}\mathbf{S}^{-1}\mathbf{C}^{-1}\mathbf{C}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x)$. ■

Example 5.5 (Example 5.2): testing a mean vector with T^2 (sweat data)

Perspiration from $n = 20$ healthy females: X_1 sweat rate, X_2 sodium, X_3 potassium. Test $H_0 : \boldsymbol{\mu}^\top = [4, 50, 10]$ at $\alpha = 0.10$.

$$\bar{\mathbf{x}} = \begin{pmatrix} 4.640 \\ 45.400 \\ 9.965 \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 2.879 & 10.010 & -1.810 \\ 10.010 & 199.788 & -5.640 \\ -1.810 & -5.640 & 3.628 \end{pmatrix}, \quad \mathbf{S}^{-1} = \begin{pmatrix} .586 & -.022 & .258 \\ -.022 & .006 & -.002 \\ .258 & -.002 & .402 \end{pmatrix}.$$

$\bar{\mathbf{x}} - \boldsymbol{\mu}_0 = (.640, -4.600, -.035)^\top$ and $\mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) = (.467, -.042, .160)^\top$, so $T^2 = 20(.640 \cdot .467 + 4.600 \cdot .042 - .035 \cdot .160) = 9.74$. Critical value $\frac{(n-1)p}{n-p} F_{p,n-p}(.10) = \frac{19(3)}{17} F_{3,17}(.10) = 3.353(2.44) = 8.18$. Since $9.74 > 8.18$, reject H_0 at the 10% level.

3 Likelihood Ratio Tests

3.1 $\boldsymbol{\Sigma}$ unknown

Under $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ the likelihood is

$$L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{np/2}|\boldsymbol{\Sigma}|^{n/2}} \exp\Big(-\frac{1}{2}\sum_{j=1}^{n}(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}_j - \boldsymbol{\mu}_0)\Big),$$

and the exponent equals $-\frac{1}{2}\text{tr}[\boldsymbol{\Sigma}^{-1}\sum_j(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top]$. Maximizing over $\boldsymbol{\Sigma}$ as in the proof of Theorem 4.7 (with $\sum_j(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top$ in place of $\sum_j(\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top$),

$$\max_{\boldsymbol{\Sigma}} L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{np/2}|\hat{\boldsymbol{\Sigma}}_0|^{n/2}} e^{-np/2}, \qquad \hat{\boldsymbol{\Sigma}}_0 = \frac{1}{n}\sum_{j=1}^{n}(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top.$$

The unrestricted maximum has the same form with $\hat{\boldsymbol{\Sigma}} = \frac{1}{n}\sum(\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top$, so the *likelihood ratio statistic* is

$$\Lambda = \frac{\max_{\boldsymbol{\Sigma}} L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})}{\max_{\boldsymbol{\mu},\boldsymbol{\Sigma}} L(\boldsymbol{\mu}, \boldsymbol{\Sigma})} = \Big(\frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|}\Big)^{n/2}, \qquad \Lambda^{2/n} = \frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|} = \frac{|\sum_j (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top|}{|\sum_j (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top|}.$$

To determine whether $\boldsymbol{\mu}_0$ is a plausible value of $\boldsymbol{\mu}$, the maximum of $L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})$ is compared with the unrestricted maximum of $L(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The equivalent statistic $\Lambda^{2/n} = |\hat{\boldsymbol{\Sigma}}|/|\hat{\boldsymbol{\Sigma}}_0|$ is *Wilks' lambda*. If the observed value of the likelihood ratio is too small, $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ is unlikely to be true and is rejected: reject H_0 if $\Lambda < c_\alpha$, the lower 100αth percentile of its distribution; fortunately the distribution of $\Lambda^{2/n}$ is available.

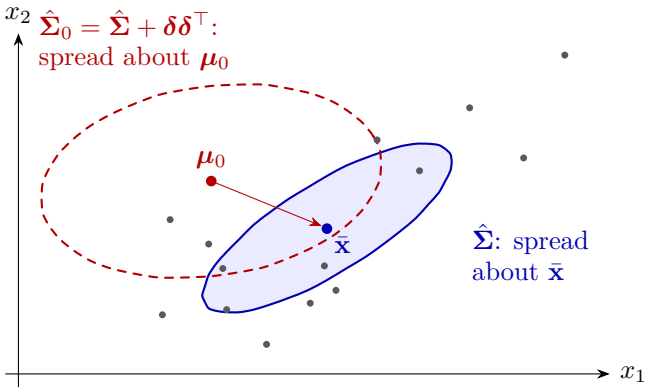


Figure. *A sample of $n = 14$ points with the contours $\{\mathbf{x} : (\mathbf{x} - \mathbf{c})^\top \mathbf{A}^{-1}(\mathbf{x} - \mathbf{c}) = 1\}$ of $\hat{\boldsymbol{\Sigma}}$ about $\bar{\mathbf{x}}$ (solid, shaded) and of $\hat{\boldsymbol{\Sigma}}_0$ about $\boldsymbol{\mu}_0$ (dashed). The area of each ellipse is proportional to the square root of the generalized variance, so Wilks' lambda $\Lambda^{2/n} = |\hat{\boldsymbol{\Sigma}}|/|\hat{\boldsymbol{\Sigma}}_0|$ is the squared ratio of the areas (here 0.16). The further $\boldsymbol{\mu}_0$ lies from $\bar{\mathbf{x}}$, measured by $\boldsymbol{\delta} = \bar{\mathbf{x}} - \boldsymbol{\mu}_0$, the larger $\hat{\boldsymbol{\Sigma}}_0$ and the smaller Λ.*

Theorem 5.6 (Result 5.1): likelihood ratio test and T^2

For a random sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the T^2 test is equivalent to the likelihood ratio test of $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ versus $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$ because

$$\Lambda^{2/n} = \Big(1 + \frac{T^2}{n-1}\Big)^{-1}.$$

H_0 is rejected for small $\Lambda^{2/n}$, equivalently large T^2.

First proof: block determinant. Let $\mathbf{Q} = \sum_j (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top = (n-1)\mathbf{S}$, $\boldsymbol{\delta} = \bar{\mathbf{x}} - \boldsymbol{\mu}_0$, and consider the $(p+1) \times (p+1)$ matrix

$$\mathbf{A} = \begin{pmatrix} \mathbf{Q} & \sqrt{n}\,\boldsymbol{\delta} \\ \sqrt{n}\,\boldsymbol{\delta}^\top & -1 \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}.$$

From $|\mathbf{A}| = |\mathbf{A}_{22}||\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}| = |\mathbf{A}_{11}||\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}|$,

$$(-1)|\mathbf{Q} + n\boldsymbol{\delta}\boldsymbol{\delta}^\top| = |\mathbf{Q}|(-1 - n\boldsymbol{\delta}^\top \mathbf{Q}^{-1}\boldsymbol{\delta}) = -|\mathbf{Q}|\Big(1 + \frac{T^2}{n-1}\Big).$$

Also $\sum_j (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top = \sum_j (\mathbf{x}_j - \bar{\mathbf{x}} + \bar{\mathbf{x}} - \boldsymbol{\mu}_0)(\cdot)^\top = \mathbf{Q} + n\boldsymbol{\delta}\boldsymbol{\delta}^\top$. Hence $|n\hat{\boldsymbol{\Sigma}}_0| = |n\hat{\boldsymbol{\Sigma}}|(1 + \frac{T^2}{n-1})$ and $\Lambda^{2/n} = |\hat{\boldsymbol{\Sigma}}|/|\hat{\boldsymbol{\Sigma}}_0| = (1 + \frac{T^2}{n-1})^{-1}$. ■

Second proof: determinant lemma. With $\boldsymbol{\delta} = \bar{\mathbf{x}} - \boldsymbol{\mu}_0$ and $\mathbf{S}_0 = \hat{\boldsymbol{\Sigma}}$, $\hat{\boldsymbol{\Sigma}}_0 = \mathbf{S}_0 + \boldsymbol{\delta}\boldsymbol{\delta}^\top$ and $|\mathbf{S}_0 + \boldsymbol{\delta}\boldsymbol{\delta}^\top| = |\mathbf{S}_0^{1/2}||\mathbf{I}_p + \mathbf{S}_0^{-1/2}\boldsymbol{\delta}\boldsymbol{\delta}^\top\mathbf{S}_0^{-1/2}||\mathbf{S}_0^{1/2}| = |\mathbf{S}_0|(1 + \boldsymbol{\delta}^\top\mathbf{S}_0^{-1}\boldsymbol{\delta})$ using $|\mathbf{I} + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top\mathbf{u}$. Since $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$, $\boldsymbol{\delta}^\top\mathbf{S}_0^{-1}\boldsymbol{\delta} = \frac{n}{n-1}\boldsymbol{\delta}^\top\mathbf{S}^{-1}\boldsymbol{\delta} = \frac{T^2}{n-1} =: \frac{T_0^2}{n-1}$. ■

Remark 5.7: approximation of $W = -2\log\Lambda$

$W = -2\log(L_0/L_1) = n\log\left(1 + \frac{T^2}{n-1}\right)$. Under H_0, $T^2 = O_p(1)$, so $W \approx \frac{n}{n-1}T^2 \approx T^2$, and for large n, $\frac{np}{n-p}F_{p,n-p} \approx \chi^2_p$: $W \dot{\sim} \chi^2_p$, in agreement with general likelihood ratio theory. Exact test: reject if $T_0^2 > \frac{(n-1)p}{n-p}F_{p,n-p,\alpha}$; approximate test: reject if $W > \chi^2_{p,\alpha}$.

3.2 $\boldsymbol{\Sigma}$ known

With $L_0 = L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})$ and $L_1 = L(\bar{\mathbf{x}}, \boldsymbol{\Sigma})$, $W = -2\log L_0 - (-2\log L_1) = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$, which is exactly χ^2_p under H_0.

3.3 Testing a covariance matrix: $H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$, $\boldsymbol{\mu}$ unknown

[Supplement] The slides test only mean vectors; for completeness the likelihood ratio test for a covariance matrix is added here.

Theorem 5.8: likelihood ratio test for $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$

[Supplement] Let $\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu}$ unknown, and $\boldsymbol{\Sigma}_0 \succ 0$ given. For $H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$ versus $H_1 : \boldsymbol{\Sigma} \neq \boldsymbol{\Sigma}_0$,

$$W = -2\log\Lambda = -n\log|\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0| - np + n\,\mathrm{tr}(\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0) \;\dot{\sim}\; \chi^2_\nu, \qquad \nu = \frac{p(p+1)}{2},$$

where $\mathbf{S}_0 = \frac{1}{n}\sum_i(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$. Equivalently, with $l_1, \ldots, l_p$ the eigenvalues of $\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0$,

$$W = n\sum_{i=1}^{p}(l_i - \log l_i - 1) \;\geq 0.$$

Reject H_0 if $W > \chi^2_\nu(\alpha)$.

Proof [Supplement]. Step 1 (the statistic). By Theorem 4.7, $-2\log L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = n\log|\boldsymbol{\Sigma}| + n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) + n\mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) + c$. Under H_1 the maximum is at $(\bar{\mathbf{x}}, \mathbf{S}_0)$: $-2\log L_1 = n\log|\mathbf{S}_0| + np + c$. Under H_0 only $\boldsymbol{\mu}$ is free and its maximizer is $\bar{\mathbf{x}}$ whatever $\boldsymbol{\Sigma}_0$ is: $-2\log L_0 = n\log|\boldsymbol{\Sigma}_0| + n\mathrm{tr}(\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0) + c$. Hence

$$\begin{aligned} W &= -2\log L_0 + 2\log L_1 = n\log|\boldsymbol{\Sigma}_0| - n\log|\mathbf{S}_0| + n\mathrm{tr}(\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0) - np \\ &= -n\log|\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0| - np + n\mathrm{tr}(\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0). \end{aligned}$$

The eigenvalues l_i of $\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0$ equal those of the PD matrix $\boldsymbol{\Sigma}_0^{-1/2}\mathbf{S}_0\boldsymbol{\Sigma}_0^{-1/2}$, so $l_i > 0$, $|\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0| = \prod l_i$, $\mathrm{tr}(\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0) = \sum l_i$, giving $W = n\sum(l_i - \log l_i - 1) \geq 0$ by Theorem 2.15, with $W = 0$ iff $\mathbf{S}_0 = \boldsymbol{\Sigma}_0$.

Step 2 (reduction to $\boldsymbol{\Sigma}_0 = \mathbf{I}$). Put $\mathbf{Y}_i = \boldsymbol{\Sigma}_0^{-1/2}\mathbf{X}_i$. Then $\mathbf{S}_0^{(Y)} = \boldsymbol{\Sigma}_0^{-1/2}\mathbf{S}_0\boldsymbol{\Sigma}_0^{-1/2}$ has the same eigenvalues l_i, so W is unchanged, and under H_0, $\mathbf{Y}_i \overset{\text{iid}}{\sim} N_p(\boldsymbol{\Sigma}_0^{-1/2}\boldsymbol{\mu}, \mathbf{I}_p)$. So assume $\boldsymbol{\Sigma}_0 = \mathbf{I}$ and write $\mathbf{S}_0$ for the MLE.

Step 3 (quadratic approximation). Under H_0, $\mathbf{S}_0 \xrightarrow{p} \mathbf{I}$, so all $l_i \xrightarrow{p} 1$. Taylor expansion of $g(l) = l - \log l - 1$ at $l = 1$: $g(1) = g'(1) = 0$, $g''(1) = 1$, so $g(l) = \frac{1}{2}(l-1)^2 + O(|l-1|^3)$ and

$$W = \frac{n}{2}\sum_i(l_i - 1)^2 + n\,O_p\Big(\max_i|l_i - 1|^3\Big) = \frac{n}{2}\|\mathbf{S}_0 - \mathbf{I}\|_F^2 + o_p(1),$$

using $\sum_i(l_i-1)^2 = \mathrm{tr}\{(\mathbf{S}_0 - \mathbf{I})^2\} = \|\mathbf{S}_0 - \mathbf{I}\|_F^2$ and $\sqrt{n}(\mathbf{S}_0 - \mathbf{I}) = O_p(1)$ (Step 4), so that $n\max|l_i - 1|^3 = O_p(n^{-1/2})$.

Step 4 (limit of $\sqrt{n}(\mathbf{S}_0 - \mathbf{I})$). Write $\mathbf{S}_0 = \frac{1}{n}\sum_i \mathbf{Z}_i\mathbf{Z}_i^\top - (\bar{\mathbf{Z}})(\bar{\mathbf{Z}})^\top$ with $\mathbf{Z}_i = \mathbf{Y}_i - \mathrm{E}\mathbf{Y}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \mathbf{I})$; the second term is $O_p(1/n)$. For the vector of distinct entries $\{Z_jZ_k\}_{j\le k}$, $\mathrm{E}(Z_jZ_k) = \delta_{jk}$ and

$$\mathrm{Cov}(Z_jZ_k, Z_lZ_m) = \mathrm{E}(Z_jZ_kZ_lZ_m) - \delta_{jk}\delta_{lm} = \delta_{jl}\delta_{km} + \delta_{jm}\delta_{kl},$$

so $\mathrm{Var}(Z_j^2) = 2$, $\mathrm{Var}(Z_jZ_k) = 1$ $(j < k)$, and different pairs $\{j,k\} \neq \{l,m\}$ are uncorrelated (consistent with Theorem 4.2(iv): $\Sigma_{jk}^2 + \Sigma_{jj}\Sigma_{kk}$). By the multivariate CLT,

$$\sqrt{n}(s_{jj} - 1) \Rightarrow N(0,2), \qquad \sqrt{n}\, s_{jk} \Rightarrow N(0,1)\ (j < k),$$

jointly, with a diagonal limiting covariance, i.e. all $p(p+1)/2$ limits are independent.

Step 5 (chi-square limit). By the CMT,

$$\frac{n}{2}\|\mathbf{S}_0 - \mathbf{I}\|_F^2 = \sum_{j=1}^{p} \frac{\{\sqrt{n}(s_{jj} - 1)\}^2}{2} + \sum_{j<k}\{\sqrt{n}\, s_{jk}\}^2$$

$$\Rightarrow \sum_{j=1}^{p} \frac{(\sqrt{2}Z_{jj})^2}{2} + \sum_{j<k} Z_{jk}^2 = \chi^2_{p+p(p-1)/2} = \chi^2_{p(p+1)/2},$$

with independent $Z_{jk} \sim N(0,1)$. By Slutsky, $W \Rightarrow \chi^2_{p(p+1)/2}$. The degrees of freedom $p(p+1)/2$ are the number of free parameters in a symmetric $\mathbf{\Sigma}$ fixed by H_0, as general likelihood ratio theory predicts. ■

Remark 5.9: special cases

[Supplement]

- $\mathbf{\Sigma}_0 = \sigma_0^2\mathbf{I}$: $W = n\{\mathrm{tr}(\mathbf{S}_0)/\sigma_0^2 - \log|\mathbf{S}_0/\sigma_0^2| - p\}$.
- $p = 1$: $W = n\{s_0^2/\sigma_0^2 - \log(s_0^2/\sigma_0^2) - 1\}$, an asymptotic version of the χ^2 test for a variance; the exact test uses $ns_0^2/\sigma_0^2 \sim \chi^2_{n-1}$.
- Bartlett-type corrections replace n by $n-1$ (using $\mathbf{S}$ instead of $\mathbf{S}_0$) and an additional factor to improve the χ^2 approximation.

3.4 Summary: likelihood ratio tests under normality

[Supplement] The four likelihood ratio tests met so far (rows (i), (ii) from the lectures; (iii) above; (iv) below, anticipating Ch. 6):

Hypothesis	Statistic and distribution
(i) $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$, $\mathbf{\Sigma}$ known	$W = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top\mathbf{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) \sim \chi_p^2$ (exact)
(ii) $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$, $\mathbf{\Sigma}$ unknown	$W = n\log(1 + T^2/(n-1))$; exact: $T^2 \sim \frac{(n-1)p}{n-p}F_{p,n-p}$
(iii) $H_0 : \mathbf{\Sigma} = \mathbf{\Sigma}_0$, $\boldsymbol{\mu}$ unknown	$W = -n\log\lvert\mathbf{\Sigma}_0^{-1}\mathbf{S}_0\rvert - np + n\mathrm{tr}(\mathbf{\Sigma}_0^{-1}\mathbf{S}_0) \overset{\cdot}{\sim} \chi^2_{p(p+1)/2}$
(iv) $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$, common $\mathbf{\Sigma}$ unknown	$W = n\log\left(1 + \frac{T^2}{n_1+n_2-2}\right)$; exact: $\frac{n_1+n_2-p-1}{(n_1+n_2-2)p}T^2 \sim F_{p,n_1+n_2-p-1}$

Theorem 5.10: two-sample likelihood ratio test

[Supplement] Let $\mathbf{X}_{11},\dots,\mathbf{X}_{1n_1} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_1,\boldsymbol{\Sigma})$ and $\mathbf{X}_{21},\dots,\mathbf{X}_{2n_2} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_2,\boldsymbol{\Sigma})$ be independent, $n = n_1 + n_2$. For $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ versus $H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$ with $\boldsymbol{\Sigma}$ unknown,

$$\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 \sim N_p\Big(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2, \frac{n_1+n_2}{n_1 n_2}\boldsymbol{\Sigma}\Big), \qquad (n_1+n_2-2)\mathbf{S}_p \sim W_p(n_1+n_2-2,\boldsymbol{\Sigma}),$$

$$T^2(n_1+n_2-2) = \frac{n_1 n_2}{n_1+n_2}(\bar{\mathbf{X}}_1-\bar{\mathbf{X}}_2)^\top \mathbf{S}_p^{-1}(\bar{\mathbf{X}}_1-\bar{\mathbf{X}}_2),$$

$$\frac{n_1+n_2-p-1}{(n_1+n_2-2)p}T^2(n_1+n_2-2) \sim F_{p,n_1+n_2-p-1},$$

and the likelihood ratio satisfies $\Lambda^{2/n} = \big(1+\frac{T^2}{n_1+n_2-2}\big)^{-1}$, so the LRT is the T^2 test.

Proof [Supplement]. The distributional facts are Theorem 6.5. Under H_1 the MLEs are $\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2$ and $\hat{\boldsymbol{\Sigma}}_1 = \mathbf{W}/n$, $\mathbf{W} = (n_1+n_2-2)\mathbf{S}_p$; under H_0 they are the pooled mean $\bar{\mathbf{x}}$ and $\hat{\boldsymbol{\Sigma}}_0 = (\mathbf{W}+\mathbf{B})/n$ with $\mathbf{B} = \frac{n_1 n_2}{n}\mathbf{d}\mathbf{d}^\top$, $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$ (the $g=2$ case of Ch. 6). As in Theorem 5.6, $\Lambda^{2/n} = |\hat{\boldsymbol{\Sigma}}_1|/|\hat{\boldsymbol{\Sigma}}_0| = |\mathbf{W}|/|\mathbf{W}+\mathbf{B}|$, and the determinant lemma gives $|\mathbf{W}+\mathbf{B}| = |\mathbf{W}|(1+\frac{n_1 n_2}{n}\mathbf{d}^\top\mathbf{W}^{-1}\mathbf{d}) = |\mathbf{W}|(1+\frac{T^2}{n_1+n_2-2})$. ■

[Supplement] *Note.* The F constant is $\frac{n_1+n_2-p-1}{(n_1+n_2-2)p}$, from Theorem 5.2 with $m = n_1+n_2-2$.

4 Confidence Regions

Motivation. Inverting the T^2 test gives a confidence ellipsoid for $\boldsymbol{\mu}$, whose axes are the eigenvectors of $\mathbf{S}$.

Theorem 5.11: $100(1-\alpha)\%$ confidence region for $\boldsymbol{\mu}$

$$R(\mathbf{X}) = \Big\{\boldsymbol{\mu}\in\mathbb{R}^p : \frac{n}{n-1}\frac{n-p}{p}(\bar{\mathbf{x}}-\boldsymbol{\mu})^\top\mathbf{S}_0^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) \le F_{p,n-p,\alpha}\Big\}$$

$$= \Big\{\boldsymbol{\mu} : n(\bar{\mathbf{x}}-\boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) \le \frac{(n-1)p}{n-p}F_{p,n-p}(\alpha)\Big\}.$$

[Supplement] With $\mathbf{S} = \sum\lambda_i\mathbf{e}_i\mathbf{e}_i^\top$, R is an ellipsoid centered at $\bar{\mathbf{x}}$ with axes along $\mathbf{e}_i$ and half-lengths $\sqrt{\lambda_i}\sqrt{\frac{(n-1)p}{n(n-p)}F_{p,n-p}(\alpha)}$: writing $\boldsymbol{\mu}-\bar{\mathbf{x}} = \sum_i y_i\mathbf{e}_i$, the defining inequality becomes $\sum_i y_i^2/\lambda_i \le \frac{(n-1)p}{n(n-p)}F_{p,n-p}(\alpha)$. Checking whether $\boldsymbol{\mu}_0 \in R$ is the T^2 test. For $p \ge 3$ the ellipsoid cannot be drawn, so componentwise intervals are needed.

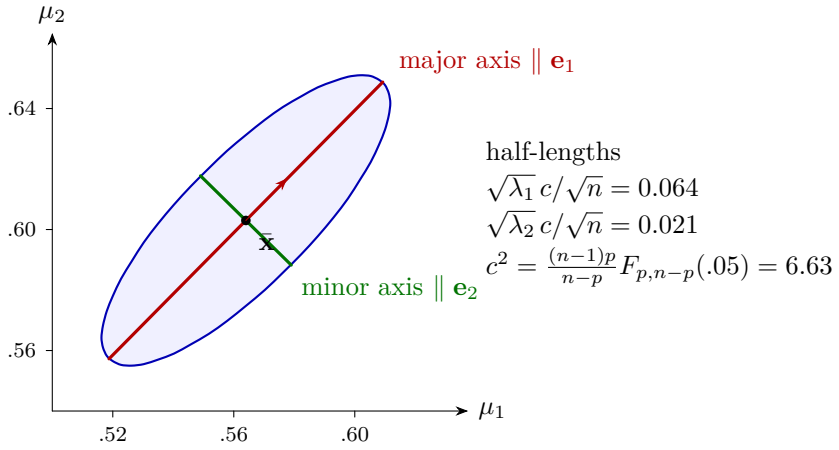


Figure. *[Supplement] The 95% confidence ellipse for $\boldsymbol{\mu} = (\mu_1, \mu_2)^\top$ in the microwave radiation data (Example 5.13; $n = 42$, $\mathbf{S}$ with $s_{12} = 0.0117$). Its axes point along the eigenvectors $\mathbf{e}_1 \approx (0.70, 0.71)^\top$, $\mathbf{e}_2 \approx (-0.71, 0.70)^\top$ of $\mathbf{S}$ ($\lambda_1 = 0.0262$, $\lambda_2 = 0.0028$); the strong positive correlation makes the ellipse long and thin.*

5 Simultaneous Confidence Intervals

5.1 A single linear combination

For fixed $\mathbf{a}$, $\mathbf{a}^\top(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim N_1(0, \frac{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}{n})$ and $(n-1)\mathbf{a}^\top\mathbf{S}\mathbf{a} \sim W_1(n-1, \mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}) \iff (n-1)\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}} \sim \chi^2_{n-1}$, independent, so

$$t(\mathbf{a}) = \frac{\sqrt{n}\,\mathbf{a}^\top(\bar{\mathbf{X}} - \boldsymbol{\mu})/\sqrt{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}}{\sqrt{(n-1)\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}/(n-1)}} = \frac{\mathbf{a}^\top(\bar{\mathbf{X}} - \boldsymbol{\mu})}{\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}} \sim t_{n-1}, \qquad C_\alpha = \Big\{\theta : \frac{|\mathbf{a}^\top\bar{\mathbf{x}} - \theta|}{\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}} \le t_{n-1,\alpha/2}\Big\}.$$

Using these intervals for several $\mathbf{a}$ at once lowers the overall confidence below $1-\alpha$.

5.2 T^2 simultaneous intervals

Theorem 5.12 (Result 5.3): simultaneous confidence intervals for $\mathbf{a}^\top\boldsymbol{\mu}$

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be a random sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$. Then, *simultaneously for all* $\mathbf{a}$, the interval

$$\Big(\mathbf{a}^\top\bar{\mathbf{X}} - \sqrt{\frac{p(n-1)}{n(n-p)}F_{p,n-p}(\alpha)\mathbf{a}^\top\mathbf{S}\mathbf{a}},\ \ \mathbf{a}^\top\bar{\mathbf{X}} + \sqrt{\frac{p(n-1)}{n(n-p)}F_{p,n-p}(\alpha)\mathbf{a}^\top\mathbf{S}\mathbf{a}}\Big)$$

contains $\mathbf{a}^\top\boldsymbol{\mu}$ with probability $1-\alpha$.

Proof. Simple algebra gives, for every $\mathbf{a}$,

$$\mathbf{a}^\top\bar{\mathbf{x}} - c\sqrt{\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{n}} \le \mathbf{a}^\top\boldsymbol{\mu} \le \mathbf{a}^\top\bar{\mathbf{x}} + c\sqrt{\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{n}} \iff t(\mathbf{a})^2 = \frac{n(\mathbf{a}^\top\bar{\mathbf{x}} - \mathbf{a}^\top\boldsymbol{\mu})^2}{\mathbf{a}^\top\mathbf{S}\mathbf{a}} \le c^2.$$

The key is $\max_{\mathbf{a}} t(\mathbf{a})^2 = T^2$: with $\mathbf{b} = \mathbf{S}^{1/2}\mathbf{a}$ and $\mathbf{v} = \sqrt{n}\mathbf{S}^{-1/2}(\bar{\mathbf{x}} - \boldsymbol{\mu})$,

$$\max_{\mathbf{a}} t(\mathbf{a})^2 = \max_{\mathbf{b}} \frac{(\mathbf{b}^\top\mathbf{v})^2}{\mathbf{b}^\top\mathbf{b}} \le \mathbf{v}^\top\mathbf{v} = n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) = T^2(n-1)$$

by Cauchy–Schwarz, with equality at $\mathbf{a} \propto \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu})$. By Theorem 5.2, choosing $c^2 = \frac{p(n-1)}{n-p} F_{p,n-p}(\alpha)$ gives $P(\bigcap_{\mathbf{a}} \{t(\mathbf{a})^2 \le c^2\}) = P(\max_{\mathbf{a}} t(\mathbf{a})^2 \le c^2) = P(T^2 \le c^2) = 1 - \alpha$. ■

- These are called T^2-intervals, since the coverage is determined by the distribution of T^2. With $\mathbf{a} = \mathbf{e}_j$: $\bar{x}_j \pm \sqrt{\frac{p(n-1)}{n-p} F_{p,n-p}(\alpha)} \sqrt{\frac{s_{jj}}{n}}$, holding simultaneously with confidence $1 - \alpha$.
- With $\mathbf{a}^\top = [0, \dots, 0, a_j, 0, \dots, 0, a_k, 0, \dots]$, $a_j = 1$, $a_k = -1$: $\mathbf{a}^\top \mathbf{S} \mathbf{a} = s_{jj} - 2s_{jk} + s_{kk}$, giving intervals for $\mu_j - \mu_k$.
- The coefficient $1 - \alpha$ holds for any choice of $\mathbf{a}$, so these intervals are ideal for "data snooping".
- The rectangle formed by the componentwise T^2 intervals is the shadow of the confidence ellipsoid on the coordinate axes.

Example 5.13 (Example 5.4): simultaneous intervals (microwave radiation data)

$n = 42$, $p = 2$, $\bar{\mathbf{x}} = (0.564, 0.603)^\top$, $s_{11} = 0.0144$, $s_{22} = 0.0146$ (and $s_{12} = 0.0117$, from the textbook), $F_{2,40}(.05) = 3.23$. The 95% simultaneous T^2 intervals are

$$0.564 \pm \sqrt{\tfrac{2(41)}{40} 3.23} \sqrt{\tfrac{0.0144}{42}} = (0.516, 0.612), \qquad 0.603 \pm \sqrt{\tfrac{2(41)}{40} 3.23} \sqrt{\tfrac{0.0146}{42}} = (0.555, 0.651).$$

5.3 The Bonferroni method

When attention is restricted to a small number m of pre-specified linear combinations, shorter (more precise) simultaneous intervals are possible.

Theorem 5.14: Bonferroni inequality

Let C_i be a confidence statement about $\mathbf{a}_i^\top \boldsymbol{\mu}$ with $P(C_i \text{ true}) = 1 - \alpha_i$. Then

$$P[\text{all } C_i \text{ true}] = 1 - P[\text{at least one } C_i \text{ false}] \ge 1 - \sum_{i=1}^{m} P(C_i \text{ false}) = 1 - (\alpha_1 + \cdots + \alpha_m).$$

With $\alpha_i = \alpha/m$, the individual intervals $\bar{x}_i \pm t_{n-1}(\frac{\alpha}{2m}) \sqrt{s_{ii}/n}$, $i = 1, \dots, m$, hold simultaneously with confidence at least $1 - \alpha$. (Equivalently, with $C_i(\alpha) = \{\mu_i : |\bar{x}_i - \mu_i| / (s_i/\sqrt{n}) < t_{n-1,\alpha/2}\}$; setting $\mu_i \in C_i(\alpha/m)$ gives $P(\mu_i \in C_i(\alpha/m),\ i = 1, \dots, m) \ge 1 - \sum_i P(\mu_i \notin C_i(\alpha/m)) = 1 - \alpha$.)

Remark 5.15: comparison of Bonferroni and T^2 interval lengths

Both have the form $\mathbf{a}^\top \bar{\mathbf{x}} \pm (\text{critical value}) \sqrt{\mathbf{a}^\top \mathbf{S} \mathbf{a}/n}$, so with $\alpha_i = \alpha/m$

$$\frac{\text{length of Bonferroni interval}}{\text{length of } T^2 \text{ interval}} = \frac{t_{n-1}(\alpha/2m)}{\sqrt{\frac{p(n-1)}{n-p} F_{p,n-p}(\alpha)}},$$

which does not depend on $\bar{\mathbf{X}}$ or $\mathbf{S}$. For $1 - \alpha = .95$, $\alpha_i = .05/m$, $m = p$:

n	$m=2$	$m=4$	$m=10$
15	.88	.69	.29
25	.90	.75	.48
50	.91	.78	.58
100	.91	.80	.62
∞	.91	.81	.66

For a small number of specified functions $\mathbf{a}^\top\boldsymbol{\mu}$ the Bonferroni intervals are shorter; being easy to apply, they are used often. In Example 5.13 the Bonferroni intervals are $(0.521, 0.607)$ and $(0.560, 0.646)$, inside the T^2 intervals.

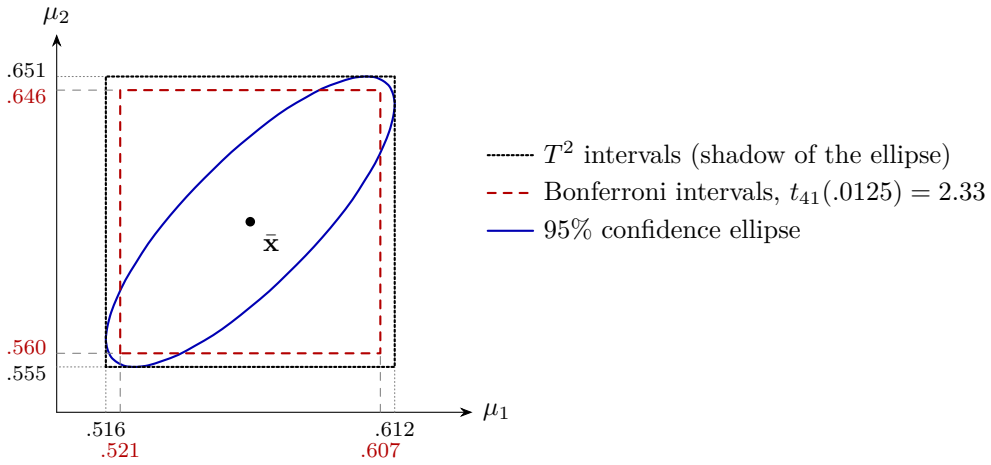


Figure. *The 95% T^2 and 95% Bonferroni simultaneous intervals for the component means of the microwave radiation data. The T^2 intervals are exactly the projections (shadows) of the confidence ellipse on the axes; the Bonferroni rectangle is slightly smaller, and the tips of the ellipse stick out of it at the lower-left and upper-right corners.*

6 Large-Sample Inference about a Mean Vector

Motivation. When normality is doubtful but n is large, T^2 is approximately χ^2_p, and the same procedures apply with χ^2 critical values.

When the sample size is large, tests and confidence regions for $\boldsymbol{\mu}$ can be constructed *without* the assumption of a normal population.

Theorem 5.16 (Result 5.4): hypothesis testing for $\boldsymbol{\mu}$ with large $n-p$

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be a random sample from a population with mean $\boldsymbol{\mu}$ and positive definite covariance $\boldsymbol{\Sigma}$ (not necessarily normal). When $n-p$ is large, $H_0: \boldsymbol{\mu} = \boldsymbol{\mu}_0$ is rejected in favor of $H_1: \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$, at level approximately α, if

$$n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) > \chi^2_p(\alpha),$$

where $\chi^2_p(\alpha)$ is the upper (100α)th percentile of the chi-square distribution with p degrees of freedom.

Proof. [Supplement] By Theorem 4.12, under H_0 the statistic $T_n = n(\bar{\mathbf{X}} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}_0)$ converges in distribution to χ^2_p. Since the χ^2_p distribution function is continuous, $P_{H_0}(T_n > \chi^2_p(\alpha)) \to P(\chi^2_p > \chi^2_p(\alpha)) = \alpha$. ■

Comparison with normal theory. The test statistics have the same structure; only the critical values differ. The similarity follows from the fact that, for n large relative to p (large $n-p$),

$$\frac{(n-1)p\,F_{p,n-p}(\alpha)}{n-p} \approx \chi^2_p(\alpha).$$

[Supplement] Indeed $F_{p,\nu} = (\chi^2_p/p)/(\chi^2_\nu/\nu)$ and $\chi^2_\nu/\nu \xrightarrow{p} 1$ as $\nu \to \infty$, so $pF_{p,\nu} \Rightarrow \chi^2_p$, while $(n-1)/(n-p) \to 1$.

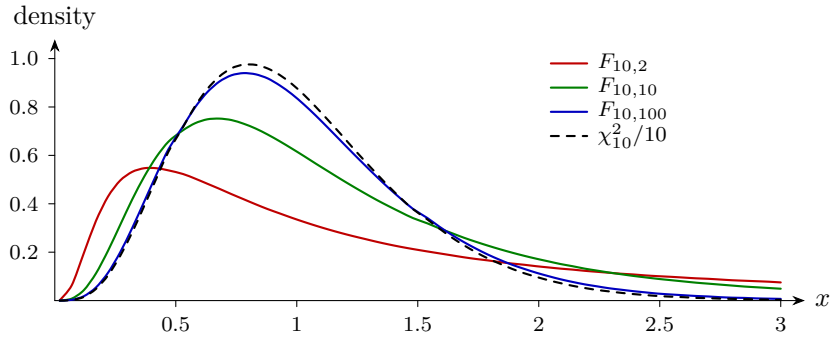


Figure. *Densities of $F_{10,\nu}$ for $\nu = 2, 10, 100$ against the density of $\chi^2_{10}/10$ (dashed). As the denominator degrees of freedom grow, $F_{p,\nu}$ approaches χ^2_p/p, which is why the normal-theory critical value $\frac{(n-1)p}{n-p}F_{p,n-p}(\alpha)$ and the large-sample one $\chi^2_p(\alpha)$ nearly coincide when $n-p$ is large.*

Theorem 5.17 (Result 5.5): large-sample simultaneous intervals

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be a random sample from a population with mean $\boldsymbol{\mu}$ and positive definite covariance $\boldsymbol{\Sigma}$. If $n-p$ is large, the interval $\mathbf{a}^\top\bar{\mathbf{X}} \pm \sqrt{\chi^2_p(\alpha)}\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}$ contains $\mathbf{a}^\top\boldsymbol{\mu}$, for every $\mathbf{a}$, with probability approximately $1-\alpha$. Consequently the $100(1-\alpha)\%$ simultaneous statements

$$\bar{x}_1 \pm \sqrt{\chi^2_p(\alpha)}\sqrt{\frac{s_{11}}{n}} \text{ contains } \mu_1, \quad \ldots, \quad \bar{x}_p \pm \sqrt{\chi^2_p(\alpha)}\sqrt{\frac{s_{pp}}{n}} \text{ contains } \mu_p$$

hold jointly with probability approximately $1-\alpha$.

Proof. [Supplement] As in the proof of Theorem 5.12, $\mathbf{a}^\top\boldsymbol{\mu}$ lies in the interval for *every* $\mathbf{a}$ if and only if $\max_{\mathbf{a}} t(\mathbf{a})^2 \le \chi^2_p(\alpha)$, where $t(\mathbf{a})^2 = n(\mathbf{a}^\top\bar{\mathbf{X}} - \mathbf{a}^\top\boldsymbol{\mu})^2/\mathbf{a}^\top\mathbf{S}\mathbf{a}$. By the Cauchy–Schwarz argument there, $\max_{\mathbf{a}} t(\mathbf{a})^2 = n(\bar{\mathbf{X}}-\boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{X}}-\boldsymbol{\mu})$. Hence the simultaneous coverage probability equals $P(n(\bar{\mathbf{X}}-\boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{X}}-\boldsymbol{\mu}) \le \chi^2_p(\alpha)) \to 1-\alpha$ by Theorem 4.12. ∎

However, the question of what is a large sample size is not easy to answer.

Example 5.18: Golub gene expression data

From Golub's gene expression data (dimension $d = 7129$), take the first and the 1674th variables (genes), so that $p = 2$. There are two populations: 11 observations from AML and 27 from ALL. For each group, the 95% and 99% elliptical confidence regions for the mean are drawn (Figure 1 of the slides), and the elliptical region is compared with the simultaneous intervals for $\mathbf{a}_1 = (1,0)^\top$ and $\mathbf{a}_2 = (0,1)^\top$, i.e. the rectangle they form (Figure 2). The regions of the two groups are well separated, so the two groups clearly have different mean expression.

Chapter 6. Comparing Mean Vectors and MANOVA

Paired comparisons, contrasts, two populations, one- and two-way MANOVA, simultaneous intervals

1 Paired Comparisons and Repeated Measures

1.1 Paired comparisons

[Supplement] When two treatments are applied to the same unit and p responses are measured, analyze the differences $\mathbf{D}_j = \mathbf{X}_{1j} - \mathbf{X}_{2j}$, $j = 1, \ldots, n$; unit effects cancel, increasing precision.

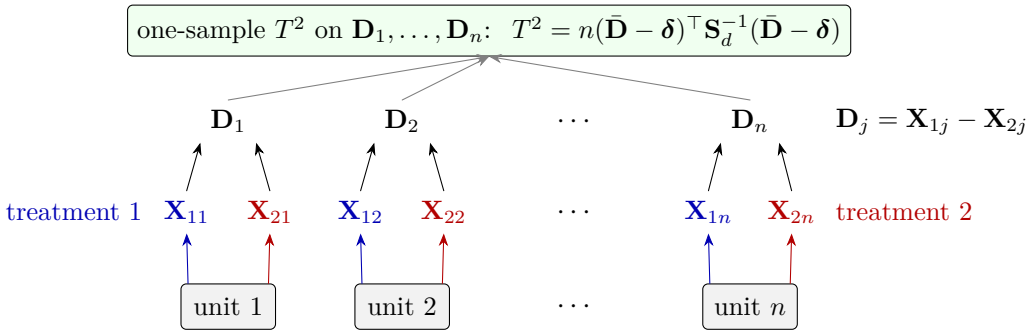


Figure. *[Supplement] Paired design: both treatments are applied to each of n units and p responses are recorded under each. Taking differences within a unit removes the unit effect, and the comparison of the two treatments becomes a one-sample problem for the mean difference $\boldsymbol{\delta} = \mathrm{E}\mathbf{D}_j$.*

Theorem 6.1 (Result 6.1): paired T^2

Let the differences $\mathbf{D}_1, \ldots, \mathbf{D}_n$ be a random sample from an $N_p(\boldsymbol{\delta}, \boldsymbol{\Sigma}_d)$ population. Then

$$T^2 = n(\bar{\mathbf{D}} - \boldsymbol{\delta})^\top \mathbf{S}_d^{-1}(\bar{\mathbf{D}} - \boldsymbol{\delta})$$

is distributed as an $\frac{(n-1)p}{n-p} F_{p,n-p}$ random variable, whatever the true $\boldsymbol{\delta}$ and $\boldsymbol{\Sigma}_d$. If n and $n-p$ are both large, T^2 is approximately distributed as a χ^2_p random variable, regardless of the form of the underlying population of differences.

Note that $(n-1)\mathbf{S}_d$ follows a Wishart distribution [Homework]. (The slide prints $(n-p)\mathbf{S}_d$; the degrees of freedom are $n-1$, as shown below.)

Homework: $(n-1)\mathbf{S}_d$ is Wishart, and the proof of Result 6.1 [Supplement]. **Step 1.** $\mathbf{D}_1, \ldots, \mathbf{D}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\delta}, \boldsymbol{\Sigma}_d)$, so by Theorem 4.4, $\sqrt{n}(\bar{\mathbf{D}} - \boldsymbol{\delta}) \sim N_p(\mathbf{0}, \boldsymbol{\Sigma}_d)$, $(n-1)\mathbf{S}_d = \sum_j (\mathbf{D}_j - \bar{\mathbf{D}})(\mathbf{D}_j - \bar{\mathbf{D}})^\top \sim W_p(n-1, \boldsymbol{\Sigma}_d)$, and $\bar{\mathbf{D}} \perp\!\!\!\perp \mathbf{S}_d$.

Step 2. Hence $T^2 = \{\sqrt{n}(\bar{\mathbf{D}} - \boldsymbol{\delta})\}^\top \mathbf{S}_d^{-1} \{\sqrt{n}(\bar{\mathbf{D}} - \boldsymbol{\delta})\} \sim T^2_p(n-1)$, and Theorem 5.2 with $m = n-1$ gives $\frac{n-p}{(n-1)p} T^2 \sim F_{p,n-p}$; nothing depends on $\boldsymbol{\delta}$ or $\boldsymbol{\Sigma}_d$.

Step 3. Without normality, Theorem 4.12 applied to the i.i.d. vectors $\mathbf{D}_j$ gives $T^2 \Rightarrow \chi^2_p$. ■

[Supplement] The test of $H_0 : \boldsymbol{\delta} = \mathbf{0}$, the confidence region, simultaneous intervals $\bar{d}_i \pm \sqrt{\frac{(n-1)p}{n-p} F_{p,n-p}(\alpha)} \sqrt{s^2_{d_i}/n}$ and Bonferroni intervals $\bar{d}_i \pm t_{n-1}(\alpha/2p) \sqrt{s^2_{d_i}/n}$ are as in Ch. 5.

1.2 Contrasts and repeated measures designs

[Supplement] With q repeated measurements $\mathbf{X}_j \in \mathbb{R}^q$ on each unit, "all treatment means are equal" is a hypothesis $\mathbf{C}\boldsymbol{\mu} = \mathbf{0}$, e.g. $\mathbf{C} = \begin{pmatrix} -1 & 1 & 0 & \cdots \\ -1 & 0 & 1 & \cdots \\ & & \ddots & \end{pmatrix}$ ($(q-1) \times q$, rows are contrasts).

Theorem 6.2: test for contrasts

If $\mathbf{X}_j \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{C} \in \mathbb{R}^{q \times p}$ has rank q, then under $H_0 : \mathbf{C}\boldsymbol{\mu} = \mathbf{b}$

$$T_c^2 = n(\mathbf{C}\bar{\mathbf{x}} - \mathbf{b})^\top (\mathbf{C}\mathbf{S}\mathbf{C}^\top)^{-1} (\mathbf{C}\bar{\mathbf{x}} - \mathbf{b}) \sim \frac{(n-1)q}{n-q} F_{q,n-q} = \frac{q(n-1)}{(n-1)-(q-1)} F_{q,n-q}.$$

Proof. (i) $\sqrt{n}(\mathbf{C}\bar{\mathbf{X}} - \mathbf{C}\boldsymbol{\mu}) \sim N_q(\mathbf{0}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\top)$;

(ii) $(n-1)\mathbf{C}\mathbf{S}\mathbf{C}^\top \sim W_q(n-1, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\top)$ (Theorem 4.2(i)), independent of $\mathbf{C}\bar{\mathbf{X}}$ since $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$. Apply Theorem 5.2 in dimension q with $m = n-1$. ■

[Supplement] By invariance (Theorem 5.4), T_c^2 does not depend on which contrast matrix with the same row space is used.

2 Comparing Mean Vectors from Two Populations

Motivation. Two-group comparisons generalize the two-sample t-test. With a common covariance matrix, the pooled $\mathbf{S}$ gives an exact T^2 test; with unequal covariances, large-sample approximations are used.

Remark 6.3: comparing two distributions versus two means

The general question is $H_0 : F_1 = F_2$ versus $H_1 : F_1 \neq F_2$ for the two population distributions. If both populations are normal with a common covariance, $F_\ell = N_p(\boldsymbol{\mu}_\ell, \boldsymbol{\Sigma})$, then $F_1 = F_2 \iff \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$, so the problem reduces to $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ versus $H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$. Without these assumptions, equal means does not imply equal distributions.

Remark 6.4: the univariate case $p = 1$

With samples $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_n$, the pooled two-sample t statistic is

$$T = \left(\frac{1}{m} + \frac{1}{n}\right)^{-1/2} \frac{\bar{X}_m - \bar{Y}_n}{S_p}, \qquad S_p^2 = \frac{\sum_{i=1}^m (X_i - \bar{X}_m)^2 + \sum_{i=1}^n (Y_i - \bar{Y}_n)^2}{m+n-2},$$

and $T \sim t_{m+n-2}$ under H_0. For $p \geq 1$ the pooled covariance $\mathbf{S}_p = (s_{ij})$ has entries

$$s_{ij} = \frac{\sum_{k=1}^m (X_{ki} - \bar{X}_{\cdot i})(X_{kj} - \bar{X}_{\cdot j}) + \sum_{k=1}^n (Y_{ki} - \bar{Y}_{\cdot i})(Y_{kj} - \bar{Y}_{\cdot j})}{m+n-2},$$

and the multivariate statistic is the squared norm generalizing T^2:

$$T^2 = \left(\frac{1}{m} + \frac{1}{n}\right)^{-1} \left\| \mathbf{S}_p^{-1/2} (\bar{\mathbf{X}}_m - \bar{\mathbf{Y}}_n) \right\|_2^2 \sim \frac{(m+n-2)p}{m+n-p-1} F_{p,m+n-p-1}.$$

For $p = 1$ this is exactly the square of the t statistic, since $t_\nu^2 = F_{1,\nu}$. The p-value is $P(T^2 > T^2_{\text{obs}} \mid H_0) = P(F_{p,m+n-p-1} \geq F_{\text{obs}})$ with $F_{\text{obs}} = \frac{m+n-p-1}{(m+n-2)p} T^2_{\text{obs}}$. This follows from

the general form $T^2 = \mathbf{z}^\top (\mathbf{W}/\nu)^{-1}\mathbf{z} \sim \frac{p\nu}{\nu-(p-1)} F_{p,\nu-(p-1)}$ for $\mathbf{z} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ independent of $\mathbf{W} \sim W_p(\nu, \boldsymbol{\Sigma})$ (Theorem 5.2 with $m = \nu$), here with $\nu = m + n - 2$.

2.1 Common covariance: the pooled T^2

$\mathbf{X}_{11}, \ldots, \mathbf{X}_{1n_1} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and an independent sample $\mathbf{X}_{21}, \ldots, \mathbf{X}_{2n_2} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$. Pooled covariance:

$$\mathbf{S}_{\text{pooled}} = \frac{(n_1 - 1)\mathbf{S}_1 + (n_2 - 1)\mathbf{S}_2}{n_1 + n_2 - 2}.$$

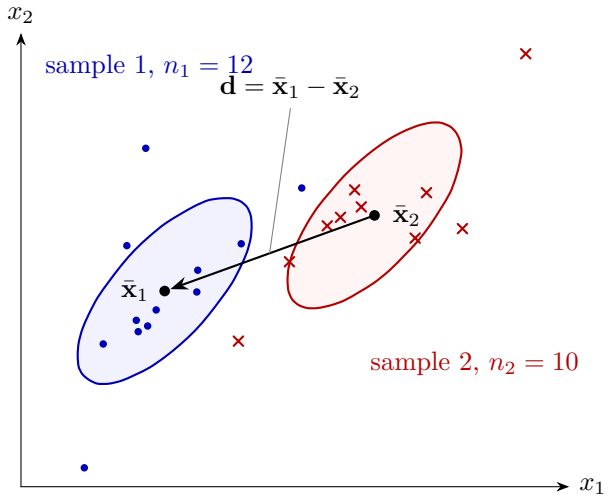


Figure. *Two samples drawn from normal populations with a* common *covariance* $\boldsymbol{\Sigma}$. *The two ellipses are the same contour of* $\mathbf{S}_{pooled}$ *centred at* $\bar{\mathbf{x}}_1$ *and at* $\bar{\mathbf{x}}_2$*: because the shapes are assumed equal, both samples are pooled to estimate one* $\boldsymbol{\Sigma}$*, and* T^2 *measures the length of* $\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$ *in the metric of* $(\frac{1}{n_1} + \frac{1}{n_2})\mathbf{S}_{pooled}$ *(here* $T^2 = 57.1$*).*

Theorem 6.5 (Result 6.2): two-sample T^2

If $\mathbf{X}_{11}, \ldots, \mathbf{X}_{1n_1}$ is a random sample of size n_1 from $N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and $\mathbf{X}_{21}, \ldots, \mathbf{X}_{2n_2}$ is an independent random sample of size n_2 from $N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$, then

$$T^2 = (\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2))^\top \left[\left(\tfrac{1}{n_1} + \tfrac{1}{n_2}\right)\mathbf{S}_{\text{pooled}}\right]^{-1} (\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2))$$
$$\sim \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p,n_1+n_2-p-1}.$$

Consequently $P(T^2 \le c_\alpha^2) = 1 - \alpha$ with $c_\alpha^2 = \frac{(n_1+n_2-2)p}{n_1+n_2-p-1} F_{p,n_1+n_2-p-1}(\alpha)$.

Note that $(n_1 + n_2 - 2)\mathbf{S}_{\text{pooled}}$ follows a Wishart distribution [Homework]; this is step (ii) below.

Proof. (i) $\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 \sim N_p(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2, (\frac{1}{n_1} + \frac{1}{n_2})\boldsymbol{\Sigma})$, so $\mathbf{z} = (\frac{1}{n_1} + \frac{1}{n_2})^{-1/2}\{\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)\} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$.
(ii) $(n_i - 1)\mathbf{S}_i \sim W_p(n_i - 1, \boldsymbol{\Sigma})$ independently, so $(n_1 + n_2 - 2)\mathbf{S}_{\text{pooled}} \sim W_p(n_1 + n_2 - 2, \boldsymbol{\Sigma})$ by Theorem 4.2(iii).

(iii) $\bar{\mathbf{X}}_i \perp\!\!\!\perp \mathbf{S}_i$ within each sample and the samples are independent, so $\mathbf{z} \perp\!\!\!\perp \mathbf{S}_{\text{pooled}}$. Hence $T^2 = \mathbf{z}^\top \mathbf{S}_{\text{pooled}}^{-1}\mathbf{z} = (\frac{1}{n_1} + \frac{1}{n_2})^{-1} \|\mathbf{S}_p^{-1/2}(\bar{\mathbf{x}} - \bar{\mathbf{y}} - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2))\|_2^2 \sim T_p^2(n_1 + n_2 - 2)$, and Theorem 5.2 with $m = n_1 + n_2 - 2$ gives the F form. ∎

The p-value is $P(F_{p,n_1+n_2-p-1} \geq F_{\text{obs}})$ with $F_{\text{obs}} = \frac{n_1+n_2-p-1}{(n_1+n_2-2)p} T^2_{\text{obs}}$. For contrasts, $H_0 : \mathbf{C}\boldsymbol{\mu}_1 = \mathbf{C}\boldsymbol{\mu}_2$ ($\mathbf{C}$: $q \times p$):

$$\left(\tfrac{1}{n_1} + \tfrac{1}{n_2}\right)^{-1} \|(\mathbf{C}\mathbf{S}_p\mathbf{C}^\top)^{-1/2}\mathbf{C}(\bar{\mathbf{x}} - \bar{\mathbf{y}})\|_2^2 \sim \frac{q(n_1+n_2-2)}{n_1+n_2-q-1} F_{q,n_1+n_2-q-1}.$$

Theorem 6.6 (Result 6.3): simultaneous intervals with a pooled covariance

Let c_α^2 be as above. With probability $1-\alpha$,

$$\mathbf{a}^\top(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2) \pm c_\alpha \sqrt{\mathbf{a}^\top\left(\tfrac{1}{n_1} + \tfrac{1}{n_2}\right)\mathbf{S}_{\text{pooled}}\mathbf{a}}$$

covers $\mathbf{a}^\top(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ for all $\mathbf{a}$. In particular $\mu_{1j} - \mu_{2j}$ is covered by $(\bar{X}_{1j} - \bar{X}_{2j}) \pm c_\alpha\sqrt{(\frac{1}{n_1}+\frac{1}{n_2})[\mathbf{S}_{\text{pooled}}]_{jj}}$, $j = 1, \ldots, p$.

Proof. [Supplement] Write $\boldsymbol{\delta} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$, $\mathbf{d} = \bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2$ and $\mathbf{A} = (\frac{1}{n_1}+\frac{1}{n_2})\mathbf{S}_{\text{pooled}} \succ 0$.

Step 1. For a fixed $\mathbf{a} \neq \mathbf{0}$, $\mathbf{a}^\top\boldsymbol{\delta}$ lies in the interval if and only if $t(\mathbf{a})^2 := \dfrac{\{\mathbf{a}^\top(\mathbf{d}-\boldsymbol{\delta})\}^2}{\mathbf{a}^\top\mathbf{A}\mathbf{a}} \leq c_\alpha^2$.

Step 2. With $\mathbf{b} = \mathbf{A}^{1/2}\mathbf{a}$ and $\mathbf{v} = \mathbf{A}^{-1/2}(\mathbf{d}-\boldsymbol{\delta})$, Cauchy–Schwarz gives $t(\mathbf{a})^2 = (\mathbf{b}^\top\mathbf{v})^2/\mathbf{b}^\top\mathbf{b} \leq \mathbf{v}^\top\mathbf{v}$, with equality for $\mathbf{a} \propto \mathbf{A}^{-1}(\mathbf{d}-\boldsymbol{\delta})$, so

$$\max_{\mathbf{a}\neq\mathbf{0}} t(\mathbf{a})^2 = (\mathbf{d}-\boldsymbol{\delta})^\top\mathbf{A}^{-1}(\mathbf{d}-\boldsymbol{\delta}) = T^2.$$

Step 3. Therefore $P(\text{all intervals cover}) = P(\max_{\mathbf{a}} t(\mathbf{a})^2 \leq c_\alpha^2) = P(T^2 \leq c_\alpha^2) = 1-\alpha$ by Result 6.2. The componentwise statement is the choice $\mathbf{a} = \mathbf{e}_j$. ∎

2.2 Unequal covariances: large samples

Theorem 6.7 (Result 6.4): simultaneous intervals without a pooled covariance

If $n_1 - p$ and $n_2 - p$ are large, an approximate $100(1-\alpha)\%$ confidence ellipsoid for $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ is

$$[\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top\left(\tfrac{1}{n_1}\mathbf{S}_1 + \tfrac{1}{n_2}\mathbf{S}_2\right)^{-1}[\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)] \leq \chi^2_p(\alpha),$$

and $\mathbf{a}^\top(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ belongs to $\mathbf{a}^\top(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2) \pm \sqrt{\chi^2_p(\alpha)}\sqrt{\mathbf{a}^\top(\frac{1}{n_1}\mathbf{S}_1 + \frac{1}{n_2}\mathbf{S}_2)\mathbf{a}}$ simultaneously for all $\mathbf{a}$.

Sketch. (i) $\boldsymbol{\delta}_n := (\bar{\mathbf{x}}-\bar{\mathbf{y}}) - (\boldsymbol{\mu}_1-\boldsymbol{\mu}_2) \sim N_p(\mathbf{0}, \frac{1}{n_1}\boldsymbol{\Sigma}_1 + \frac{1}{n_2}\boldsymbol{\Sigma}_2)$ (exactly under normality, approximately by the CLT), so $\boldsymbol{\delta}_n^\top(\frac{1}{n_1}\boldsymbol{\Sigma}_1 + \frac{1}{n_2}\boldsymbol{\Sigma}_2)^{-1}\boldsymbol{\delta}_n \sim \chi^2_p$.

(ii) $(n_1-1)\mathbf{S}_1 \sim W_p(n_1-1, \boldsymbol{\Sigma}_1)$, $(n_2-1)\mathbf{S}_2 \sim W_p(n_2-1, \boldsymbol{\Sigma}_2)$, and $\mathbf{S}_i \xrightarrow{p} \boldsymbol{\Sigma}_i$.

(iii) For $n_1 = n_2 = n$: $\mathbf{S}_1 + \mathbf{S}_2 \to \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2$, so $(\mathbf{S}_1+\mathbf{S}_2)^{-1} \to (\boldsymbol{\Sigma}_1+\boldsymbol{\Sigma}_2)^{-1}$ and

$$T^2 = n\boldsymbol{\delta}_n^\top(\mathbf{S}_1+\mathbf{S}_2)^{-1}\boldsymbol{\delta}_n = n\boldsymbol{\delta}_n^\top(\boldsymbol{\Sigma}_1+\boldsymbol{\Sigma}_2)^{-1}\boldsymbol{\delta}_n + n\boldsymbol{\delta}_n^\top[(\mathbf{S}_1+\mathbf{S}_2)^{-1} - (\boldsymbol{\Sigma}_1+\boldsymbol{\Sigma}_2)^{-1}]\boldsymbol{\delta}_n \Rightarrow \chi^2_p,$$

the second term being $O_p(1)\cdot o_p(1)$. ∎

3 One-Way MANOVA

3.1 Data and assumptions

Random samples from g populations: population ℓ: $\mathbf{X}_{\ell 1}, \mathbf{X}_{\ell 2}, \ldots, \mathbf{X}_{\ell n_\ell}$, $\ell = 1, \ldots, g$. MANOVA (multivariate analysis of variance) asks whether the mean vectors are equal and, if not, which components differ.

1. $\mathbf{X}_{\ell 1}, \ldots, \mathbf{X}_{\ell n_\ell}$ is a random sample of size n_ℓ from a population with mean $\boldsymbol{\mu}_\ell$; samples from different populations are **independent**.
2. All populations have a **common** covariance matrix $\boldsymbol{\Sigma}$.
3. Each population is **multivariate normal**.

Condition 3 can be relaxed by appealing to the central limit theorem (Result 4.13; Theorem 4.11).

3.2 Review: univariate ANOVA

In the univariate case $X_{\ell 1}, \ldots, X_{\ell n_\ell}$ is a random sample from $N(\mu_\ell, \sigma^2)$, $\ell = 1, \ldots, g$, the samples being independent. Populations usually correspond to different experimental conditions (treatments), so it is convenient to study the deviations $\tau_\ell = \mu_\ell - \mu$ associated with the ℓth population; the reparameterization $\mu_\ell = \mu + \tau_\ell$ restates equality of means as $H_0 : \tau_1 = \cdots = \tau_g = 0$. The model is $X_{\ell j} = \mu + \tau_\ell + e_{\ell j}$ (μ: overall mean, τ_ℓ: treatment effect, $e_{\ell j}$ independent $N(0, \sigma^2)$ errors), with the constraint $\sum_\ell n_\ell \tau_\ell = 0$ and $H_0 : \tau_1 = \cdots = \tau_g = 0$ ($\mu_\ell = \mu + \tau_\ell$, $\tau_\ell = \mu_\ell - \mu$). The decomposition $x_{\ell j} = \bar{x} + (\bar{x}_\ell - \bar{x}) + (x_{\ell j} - \bar{x}_\ell)$ ($\bar{x}$ estimates μ, $\hat{\tau}_\ell = \bar{x}_\ell - \bar{x}$ estimates τ_ℓ, $x_{\ell j} - \bar{x}_\ell$ estimates $e_{\ell j}$) gives

$$\underbrace{\sum_\ell \sum_j (x_{\ell j} - \bar{x})^2}_{SS_{\text{cor}}\ \text{(total)}} = \underbrace{\sum_\ell n_\ell (\bar{x}_\ell - \bar{x})^2}_{SS_{\text{tr}}\ \text{(between)}} + \underbrace{\sum_\ell \sum_j (x_{\ell j} - \bar{x}_\ell)^2}_{SS_{\text{res}}\ \text{(within)}},$$

obtained by squaring and summing both sides over j and ℓ ([Supplement] the cross term is $2\sum_\ell (\bar{x}_\ell - \bar{x}) \sum_j (x_{\ell j} - \bar{x}_\ell) = 0$). Alternatively, $\sum_\ell \sum_j x_{\ell j}^2 = (n_1 + \cdots + n_g)\bar{x}^2 + \sum_\ell n_\ell (\bar{x}_\ell - \bar{x})^2 + \sum_\ell \sum_j (x_{\ell j} - \bar{x}_\ell)^2$, i.e. $SS_{\text{obs}} = SS_{\text{mean}} + SS_{\text{tr}} + SS_{\text{res}}$. The usual F-test rejects $H_0 : \tau_1 = \cdots = \tau_g = 0$ at level α if $F = \dfrac{SS_{\text{tr}}/(g-1)}{SS_{\text{res}}/(\sum n_\ell - g)} > F_{g-1, \sum n_\ell - g}(\alpha)$, the upper (100α)th percentile of the F-distribution with $g - 1$ and $\sum n_\ell - g$ degrees of freedom; this is equivalent to rejecting H_0 for large values of the F-statistic.

Source of variation	Sum of squares (SS)	d.f.
Treatments (SS_{tr})	$\sum_\ell n_\ell (\bar{x}_\ell - \bar{x})^2$	$g - 1$
Residual (error) (SS_{res})	$\sum_\ell \sum_j (x_{\ell j} - \bar{x}_\ell)^2$	$\sum n_\ell - g$
Total (corrected for the mean) (SS_{cor})	$\sum_\ell \sum_j (x_{\ell j} - \bar{x})^2$	$\sum n_\ell - 1$

3.3 The MANOVA model and the SSP decomposition

Definition 6.8: one-way MANOVA model

$\mathbf{X}_{\ell j} = \boldsymbol{\mu} + \boldsymbol{\tau}_\ell + \mathbf{e}_{\ell j}$, $j = 1, \ldots, n_\ell$, $\ell = 1, \ldots, g$, with $\mathbf{e}_{\ell j}$ independent $N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\sum_\ell n_\ell \boldsymbol{\tau}_\ell = \mathbf{0}$. (Indeed, $\boldsymbol{\mu} = \mathrm{E}[\bar{\mathbf{X}}] \iff \sum_\ell \sum_j \mathrm{E}(\mathbf{X}_{\ell j}) = n\boldsymbol{\mu} \iff n\boldsymbol{\mu} + \sum n_\ell \boldsymbol{\tau}_\ell = n\boldsymbol{\mu} \iff \sum n_\ell \boldsymbol{\tau}_\ell = \mathbf{0}$.)

Here $\boldsymbol{\mu}$ is an overall mean (level) and $\boldsymbol{\tau}_\ell$ the ℓth treatment effect; $H_0 : \boldsymbol{\tau}_1 = \cdots = \boldsymbol{\tau}_g = \mathbf{0}$ is tested by comparing the relative sizes of the treatment and residual sums of squares and cross products.

Each component of the observation vector $\mathbf{X}_{\ell j}$ satisfies the univariate model; the errors for the components of $\mathbf{X}_{\ell j}$ are correlated, but the covariance matrix $\boldsymbol{\Sigma}$ is the same for all populations.

Theorem 6.9: SSP decomposition

From $\mathbf{x}_{\ell j} = \bar{\mathbf{x}} + (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) + (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)$ (overall mean $\hat{\boldsymbol{\mu}}$ + estimated treatment effect $\hat{\boldsymbol{\tau}}_\ell$ + residual $\hat{\mathbf{e}}_{\ell j}$),

$$\underbrace{\sum_{\ell=1}^{g}\sum_{j=1}^{n_\ell}(\mathbf{x}_{\ell j}-\bar{\mathbf{x}})(\mathbf{x}_{\ell j}-\bar{\mathbf{x}})^\top}_{\mathbf{T}=\mathbf{B}+\mathbf{W}} = \underbrace{\sum_{\ell=1}^{g} n_\ell(\bar{\mathbf{x}}_\ell-\bar{\mathbf{x}})(\bar{\mathbf{x}}_\ell-\bar{\mathbf{x}})^\top}_{\mathbf{B}\text{: between}} + \underbrace{\sum_{\ell=1}^{g}\sum_{j=1}^{n_\ell}(\mathbf{x}_{\ell j}-\bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j}-\bar{\mathbf{x}}_\ell)^\top}_{\mathbf{W}\text{: within}}.$$

$\mathbf{W} = (n_1-1)\mathbf{S}_1 + \cdots + (n_g-1)\mathbf{S}_g$ generalizes $(n_1+n_2-2)\mathbf{S}_{\text{pooled}}$.

Proof. [Supplement] Write $\mathbf{x}_{\ell j} - \bar{\mathbf{x}} = (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) + (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)$ and expand the outer product. The cross terms are $\sum_\ell (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top = \mathbf{0}$ and its transpose, because $\sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell) = \mathbf{0}$ for every ℓ. ■

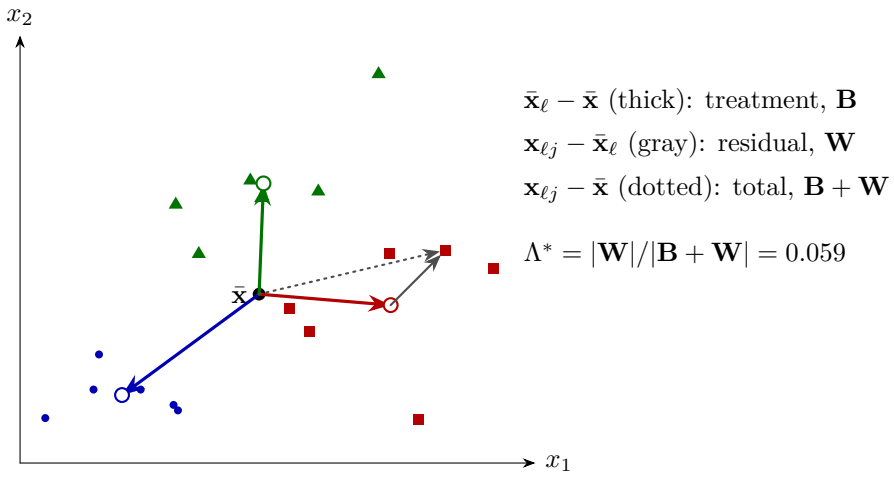


Figure. *One-way MANOVA with $g = 3$ groups of $n_\ell = 6$ observations ($p = 2$). Each deviation from the grand mean splits as $\mathbf{x}_{\ell j} - \bar{\mathbf{x}} = (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) + (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)$, shown for one point of group 2. Summing outer products of the thick arrows (weighted by n_ℓ) gives $\mathbf{B}$; those of the within-group deviations give $\mathbf{W}$. Here the groups are far apart relative to their spread, so $\mathbf{B}$ dominates and Wilks' Λ^* is small.*

Source	Matrix of SSP	d.f.
Treatment	$\mathbf{B} = \sum_\ell n_\ell(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})^\top$	$g-1$
Residual (error)	$\mathbf{W} = \sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top$	$\sum n_\ell - g$
Total	$\mathbf{B}+\mathbf{W} = \sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}})(\mathbf{x}_{\ell j} - \bar{\mathbf{x}})^\top$	$\sum n_\ell - 1$

Remark 6.10: T = B + W as a mixture

Observe $(\mathbf{X}, G)$ with $G \in \{1,\dots,g\}$, $P(G=i)=\pi_i$, $\mathbf{X} \mid G=i \sim N_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$, so $f(\mathbf{x}) = \sum_i f(\mathbf{x} \mid G=i)P(G=i) = \sum_i \phi(\mathbf{x};\boldsymbol{\mu}_i,\boldsymbol{\Sigma})\pi_i$. Then $\mathrm{E}(\mathbf{X}) = \int \mathbf{x} \sum_i \phi(\mathbf{x};\boldsymbol{\mu}_i,\boldsymbol{\Sigma})\pi_i \, d\mathbf{x} = \sum_i \pi_i \boldsymbol{\mu}_i = \boldsymbol{\mu}$ and

$$\mathrm{Var}(\mathbf{X}) = \sum_i \pi_i \mathrm{E}\big[(\mathbf{x}-\boldsymbol{\mu}_i)(\mathbf{x}-\boldsymbol{\mu}_i)^\top \mid G=i\big] + \sum_i \pi_i(\boldsymbol{\mu}_i-\boldsymbol{\mu})(\boldsymbol{\mu}_i-\boldsymbol{\mu})^\top$$

$$= \boldsymbol{\Sigma} + \sum_i \pi_i(\boldsymbol{\mu}_i-\boldsymbol{\mu})(\boldsymbol{\mu}_i-\boldsymbol{\mu})^\top.$$

The first term is the population version of $\mathbf{W}/n$, the second of $\mathbf{B}/n$.

3.4 Wilks' lambda

Definition 6.11: Wilks' test

Reject $H_0 : \boldsymbol{\tau}_1 = \cdots = \boldsymbol{\tau}_g = \mathbf{0}$ if the ratio of generalized variances

$$\Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{B}+\mathbf{W}|} = \frac{\left|\sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top\right|}{\left|\sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}})(\mathbf{x}_{\ell j} - \bar{\mathbf{x}})^\top\right|}$$

is **too small**. Proposed by Wilks, it corresponds to the equivalent form of the F test in the univariate case, is convenient, and is related to the likelihood ratio criterion. The exact distribution of Λ^* can be derived only for special cases.

Theorem 6.12: Wilks' lambda is the likelihood ratio statistic

With $n = \sum n_\ell$, $\Lambda^{2/n} = \Lambda^*$ and $-2\log\Lambda = n\log\dfrac{|\mathbf{B}+\mathbf{W}|}{|\mathbf{W}|} = n\log\dfrac{|\mathbf{T}|}{|\mathbf{W}|} \overset{\cdot}{\sim} \chi^2_{p(g-1)}$.

Proof. Under H_1: $\hat{\boldsymbol{\mu}}_\ell^{\text{MLE}} = \bar{\mathbf{x}}_\ell$, $\hat{\boldsymbol{\Sigma}}^{\text{MLE}} = \mathbf{S}_p = \dfrac{\sum_\ell (n_\ell - 1)\mathbf{S}_\ell}{n} = \dfrac{\mathbf{W}}{n}$. Under H_0: $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, $\hat{\boldsymbol{\Sigma}} = \dfrac{\mathbf{T}}{n}$. The maximized likelihoods are $(2\pi)^{-np/2}|\hat{\boldsymbol{\Sigma}}|^{-n/2}e^{-np/2}$, so $\Lambda = (|\mathbf{W}|/|\mathbf{T}|)^{n/2}$ and $-2\log(L_0/L_1) = n\log\frac{|\mathbf{T}|}{|\mathbf{W}|}$. Reject H_0 if $n\log\frac{|\mathbf{T}|}{|\mathbf{W}|} > \chi^2_{p(g-1),\alpha}$. ■

Theorem 6.13: distribution of Wilks' lambda for multivariate normal data

No. of variables	No. of groups	Sampling distribution
$p = 1$	$g \geq 2$	$\left(\dfrac{\sum n_\ell - g}{g-1}\right)\left(\dfrac{1-\Lambda^*}{\Lambda^*}\right) \sim F_{g-1,\sum n_\ell - g}$
$p = 2$	$g \geq 2$	$\left(\dfrac{\sum n_\ell - g - 1}{g-1}\right)\left(\dfrac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim F_{2(g-1),2(\sum n_\ell - g - 1)}$
$p \geq 1$	$g = 2$	$\left(\dfrac{\sum n_\ell - p - 1}{p}\right)\left(\dfrac{1-\Lambda^*}{\Lambda^*}\right) \sim F_{p,\sum n_\ell - p - 1}$
$p \geq 1$	$g = 3$	$\left(\dfrac{\sum n_\ell - p - 2}{p}\right)\left(\dfrac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim F_{2p,2(\sum n_\ell - p - 2)}$

The proof is omitted.

Derivation: the $g = 2$ row — relation to the two-sample T^2

[Supplement] For $g = 2$, $\bar{\mathbf{x}} = (n_1\bar{\mathbf{x}}_1 + n_2\bar{\mathbf{x}}_2)/n$, so $\bar{\mathbf{x}}_1 - \bar{\mathbf{x}} = \frac{n_2}{n}\mathbf{d}$ and $\bar{\mathbf{x}}_2 - \bar{\mathbf{x}} = -\frac{n_1}{n}\mathbf{d}$ with $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$, and

$$\mathbf{B} = n_1\tfrac{n_2^2}{n^2}\mathbf{d}\mathbf{d}^\top + n_2\tfrac{n_1^2}{n^2}\mathbf{d}\mathbf{d}^\top = \tfrac{n_1 n_2}{n}\mathbf{d}\mathbf{d}^\top.$$

By the determinant lemma $|\mathbf{W}+\mathbf{B}| = |\mathbf{W}|(1 + \frac{n_1 n_2}{n}\mathbf{d}^\top\mathbf{W}^{-1}\mathbf{d})$, and $\mathbf{W} = (n-2)\mathbf{S}_{\text{pooled}}$ gives $\frac{n_1 n_2}{n}\mathbf{d}^\top\mathbf{W}^{-1}\mathbf{d} = \frac{T^2}{n-2}$. Hence

$$\Lambda^* = \left(1 + \frac{T^2}{n-2}\right)^{-1}, \qquad \frac{1-\Lambda^*}{\Lambda^*} = \frac{T^2}{n-2},$$

and Theorem 6.5 ($\frac{n-p-1}{(n-2)p}T^2 \sim F_{p,n-p-1}$) gives the third row. Wilks' test with $g = 2$ is the two-sample T^2 test.

Theorem 6.14: Bartlett's test for Wilks' lambda

If H_0 is true and $\sum n_\ell = n$ is large,

$$-\Big(n-1-\frac{p+g}{2}\Big)\ln\Lambda^* = -\Big(n-1-\frac{p+g}{2}\Big)\ln\Big(\frac{|\mathbf{W}|}{|\mathbf{B}+\mathbf{W}|}\Big) \;\dot\sim\; \chi^2_{p(g-1)},$$

so reject H_0 at level α if the left side exceeds $\chi^2_{p(g-1)}(\alpha)$.

[Supplement] The degrees of freedom $p(g-1)$ are the difference in the number of mean parameters (g mean vectors versus one); the factor $n-1-(p+g)/2$ improves the chi-square approximation.

Remark 6.15: other test statistics

With the eigenvalues $\eta_1 \ge \cdots \ge \eta_s$ of $\mathbf{W}^{-1}\mathbf{B}$ ($s=\min(p,g-1)$),

$$\frac{|\mathbf{T}|}{|\mathbf{W}|} = |\mathbf{I}_p + \mathbf{W}^{-1/2}\mathbf{B}\mathbf{W}^{-1/2}| = \prod_i (1+\eta_i) = \frac{1}{\Lambda^*}, \qquad \text{Lawley–Hotelling } \operatorname{tr}(\mathbf{W}^{-1}\mathbf{B}) = \sum \eta_i,$$

$$\text{Pillai } \operatorname{tr}\{\mathbf{B}(\mathbf{B}+\mathbf{W})^{-1}\} = \sum \frac{\eta_i}{1+\eta_i}, \qquad \text{Roy } \eta_1 = \|\mathbf{W}^{-1}\mathbf{B}\|_2.$$

[Supplement] All summarize "$\mathbf{B}$ is large relative to $\mathbf{W}$". The η_i are also the eigenvalues of Fisher's discriminant analysis (Ch. 10).

Example 6.16 (Example 6.10): Wisconsin nursing home data

The Wisconsin Department of Health and Social Services reimburses nursing homes for the services provided, using a cost formula based on factors such as the level of care, the mean wage rate and the average wage in the state. Nursing homes are classified by ownership (private, non-profit, government) and by certification (skilled nursing facility, intermediate care facility, or a combination). The study investigates the effect of ownership (and certification) on costs. Four costs per patient-day were measured: X_1 nursing labour, X_2 dietary labour, X_3 plant operation and maintenance labour, X_4 housekeeping and laundry labour.

Group	n_ℓ	$\bar{\mathbf{x}}_\ell^\top$
$\ell=1$ (private)	271	(2.066, .480, .082, .360)
$\ell=2$ (non-profit)	138	(2.167, .596, .124, .418)
$\ell=3$ (government)	107	(2.273, .521, .125, .383)
	$\sum n_\ell = 516$	

Sample covariance matrices (symmetric; lower triangles shown):

$$\mathbf{S}_1 = \begin{pmatrix} .291 & & & \\ -.001 & .011 & & \\ .002 & .000 & .001 & \\ .010 & .003 & .000 & .010 \end{pmatrix}, \quad \mathbf{S}_2 = \begin{pmatrix} .561 & & & \\ .011 & .025 & & \\ .001 & .004 & .005 & \\ .037 & .007 & .002 & .019 \end{pmatrix},$$

$$\mathbf{S}_3 = \begin{pmatrix} .261 & & & \\ .030 & .017 & & \\ .003 & -.000 & .004 & \\ .018 & .006 & .001 & .013 \end{pmatrix}.$$

Since the $\mathbf{S}_\ell$ seem to be reasonably compatible, they are pooled, $\mathbf{W} = (n_1-1)\mathbf{S}_1 + (n_2 -$

$1)\mathbf{S}_2+(n_3-1)\mathbf{S}_3$, the pooled mean is $\bar{\mathbf{x}}=\frac{n_1\bar{\mathbf{x}}_1+n_2\bar{\mathbf{x}}_2+n_3\bar{\mathbf{x}}_3}{n_1+n_2+n_3}$, and $\mathbf{B}=\sum_{\ell=1}^{3} n_\ell(\bar{\mathbf{x}}_\ell-\bar{\mathbf{x}})(\bar{\mathbf{x}}_\ell-\bar{\mathbf{x}})^\top$:

$$\mathbf{W}=\begin{pmatrix}182.962 & & & \\ 4.408 & 8.200 & & \\ 1.695 & 0.633 & 1.484 & \\ 9.581 & 2.428 & 0.934 & 6.538\end{pmatrix},\qquad \mathbf{B}=\begin{pmatrix}3.475 & & & \\ 1.111 & 1.225 & & \\ 0.821 & 0.453 & 0.235 & \\ 0.584 & 0.610 & 0.230 & 0.304\end{pmatrix}$$

(symmetric; lower triangles shown), $\bar{\mathbf{x}}=(2.136,0.519,0.102,0.380)^\top$. With $\Lambda^*=0.7714$, $p=4$, $g=3$:

$$\Big(\frac{\sum n_\ell-p-2}{p}\Big)\Big(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\Big)=\Big(\frac{516-4-2}{4}\Big)\Big(\frac{1-\sqrt{.7714}}{\sqrt{.7714}}\Big)=17.67>F_{8,1020}(.01)\approx 2.51.$$

Reject H_0 at the 1% level: average costs differ depending on type of ownership. ([Supplement] Bartlett's test agrees: $-(516-1-3.5)\ln 0.7714=132.8>\chi^2_8(.01)=20.09$.)

4 Simultaneous Intervals for Treatment Effects

Let τ_{kj} be the jth component of $\boldsymbol{\tau}_k$. Since $\hat{\boldsymbol{\tau}}_k=\bar{\mathbf{X}}_k-\bar{\mathbf{X}}$, $\hat{\tau}_{kj}-\hat{\tau}_{\ell j}=\bar{X}_{kj}-\bar{X}_{\ell j}$ and a two-sample t-based interval is valid with an appropriately corrected α:

$$\mathrm{Var}(\hat{\tau}_{kj}-\hat{\tau}_{\ell j})=\Big(\frac{1}{n_k}+\frac{1}{n_\ell}\Big)\sigma_{jj},\qquad \widehat{\mathrm{Var}}(\bar{X}_{kj}-\bar{X}_{\ell j})=\Big(\frac{1}{n_k}+\frac{1}{n_\ell}\Big)\frac{\mathbf{W}_{jj}}{n-g}.$$

Apportioning the error rate: for p variables and $g(g-1)/2$ pairwise differences there are $m=p\,g(g-1)/2$ statements, so each interval uses $t_{n-g}(\alpha/2m)$.

Theorem 6.17 (Result 6.5): simultaneous intervals for differences in treatment effects

Let $n=\sum_k n_k$. With confidence at least $1-\alpha$,

$$\tau_{kj}-\tau_{\ell j}\ \text{ belongs to }\ \bar{x}_{kj}-\bar{x}_{\ell j}\pm t_{n-g}\Big(\frac{\alpha}{pg(g-1)}\Big)\sqrt{\frac{\mathbf{W}_{jj}}{n-g}\Big(\frac{1}{n_k}+\frac{1}{n_\ell}\Big)}$$

for all components $j=1,\dots,p$ and all differences $\ell<k=1,\dots,g$.

Proof. [Supplement] **Step 1 (one interval).** Under the MANOVA model, $\bar{X}_{kj}-\bar{X}_{\ell j}\sim N(\tau_{kj}-\tau_{\ell j},(\frac{1}{n_k}+\frac{1}{n_\ell})\sigma_{jj})$. The (j,j) entry of $\mathbf{W}=\sum_\ell(n_\ell-1)\mathbf{S}_\ell$ satisfies $W_{jj}/\sigma_{jj}\sim\chi^2_{n-g}$ (a sum of independent $\chi^2_{n_\ell-1}$), independent of the group means because $\bar{\mathbf{X}}_\ell \perp\!\!\!\perp \mathbf{S}_\ell$ within each sample. Hence

$$\frac{(\bar{X}_{kj}-\bar{X}_{\ell j})-(\tau_{kj}-\tau_{\ell j})}{\sqrt{\frac{W_{jj}}{n-g}\big(\frac{1}{n_k}+\frac{1}{n_\ell}\big)}}\sim t_{n-g},$$

and each interval, built with $t_{n-g}(\alpha/2m)$, has coverage exactly $1-\alpha/m$.

Step 2 (Bonferroni). There are $m=p\,g(g-1)/2$ statements, so by the Bonferroni inequality (Theorem 5.14) all hold with probability at least $1-m\cdot\alpha/m=1-\alpha$. Finally $\alpha/2m=\alpha/\{pg(g-1)\}$. ■

Example 6.18 (Example 6.11): nursing home cost comparison

Compare plant operation and maintenance (X_3) between private and government homes, i.e. estimate $\tau_{13} - \tau_{33}$: $\hat{\boldsymbol{\tau}}_1 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}) = (-.070, -.039, -.020, -.020)^\top$, $\hat{\boldsymbol{\tau}}_3 = (\bar{\mathbf{x}}_3 - \bar{\mathbf{x}}) = (.137, .002, .023, .003)^\top$, so $\hat{\tau}_{13} - \hat{\tau}_{33} = -.020 - .023 = -.043$. With $n = 271 + 138 + 107 = 516$: $\sqrt{(\frac{1}{271} + \frac{1}{107})\frac{1.484}{516-3}} = 0.00614$; for $p = 4$, $g = 3$ the 95% interval uses $t_{513}(.05/4(3)2) = t_{513}(.00208) \approx 2.87$. Hence $\tau_{13} - \tau_{33} \in -0.043 \pm 2.87 \times 0.00614 = (-0.061, -0.025)$: government homes have higher plant operation costs.

5 Two-Way MANOVA

5.1 Review: univariate two-way ANOVA

We extend the one-way approach to a two-way analysis. Two sets of experimental conditions are the levels of factor 1 and factor 2: factor 1 with g levels, factor 2 with b levels, n independent observations at each of the gb combinations; $X_{\ell kr}$ is the rth observation at level ℓ of factor 1 and level k of factor 2.

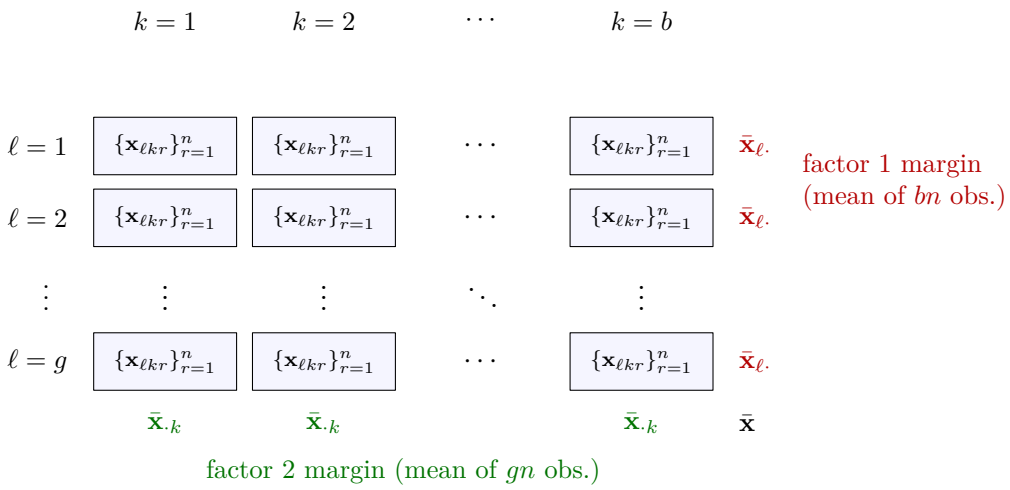


Figure. *Two-way layout: g levels of factor 1 (rows) by b levels of factor 2 (columns), with n independent replicates in each cell. The cell means $\bar{\mathbf{x}}_{\ell k}$, row means $\bar{\mathbf{x}}_{\ell\cdot}$, column means $\bar{\mathbf{x}}_{\cdot k}$ and grand mean $\bar{\mathbf{x}}$ give the decomposition $\mathbf{x}_{\ell kr} - \bar{\mathbf{x}} = (\bar{\mathbf{x}}_{\ell\cdot} - \bar{\mathbf{x}}) + (\bar{\mathbf{x}}_{\cdot k} - \bar{\mathbf{x}}) + (\bar{\mathbf{x}}_{\ell k} - \bar{\mathbf{x}}_{\ell\cdot} - \bar{\mathbf{x}}_{\cdot k} + \bar{\mathbf{x}}) + (\mathbf{x}_{\ell kr} - \bar{\mathbf{x}}_{\ell k})$ behind the SSP table.*

The univariate two-way fixed-effects model with interaction is

$$X_{\ell kr} = \underbrace{\mu}_{\text{overall mean}} + \underbrace{\tau_\ell}_{\text{factor 1 effect}} + \underbrace{\beta_k}_{\text{factor 2 effect}} + \underbrace{\gamma_{\ell k}}_{\text{interaction}} + \underbrace{e_{\ell kr}}_{\text{normal error}},$$

$\sum_\ell \tau_\ell = \sum_k \beta_k = \sum_\ell \gamma_{\ell k} = \sum_k \gamma_{\ell k} = 0$, and $SS_{\text{cor}} = SS_{\text{fac1}} + SS_{\text{fac2}} + SS_{\text{int}} + SS_{\text{res}}$:

Source	Sum of squares	d.f.
Factor 1	$SS_{\text{fac1}} = \sum_\ell bn(\bar{x}_{\ell\cdot} - \bar{x})^2$	$g - 1$
Factor 2	$SS_{\text{fac2}} = \sum_k gn(\bar{x}_{\cdot k} - \bar{x})^2$	$b - 1$
Interaction	$SS_{\text{int}} = \sum_\ell \sum_k n(\bar{x}_{\ell k} - \bar{x}_{\ell\cdot} - \bar{x}_{\cdot k} + \bar{x})^2$	$(g-1)(b-1)$
Residual	$SS_{\text{res}} = \sum_\ell \sum_k \sum_r (x_{\ell kr} - \bar{x}_{\ell k})^2$	$gb(n-1)$
Total	$SS_{\text{cor}} = \sum\sum\sum (x_{\ell kr} - \bar{x})^2$	$gbn - 1$

F-ratios: $F_{\text{fac1}} = \frac{SS_{\text{fac1}}/(g-1)}{SS_{\text{res}}/(gb(n-1))}$, $F_{\text{fac2}} = \frac{SS_{\text{fac2}}/(b-1)}{SS_{\text{res}}/(gb(n-1))}$, $F_{\text{int}} = \frac{SS_{\text{int}}/((g-1)(b-1))}{SS_{\text{res}}/(gb(n-1))}$.

5.2 Multivariate two-way fixed-effects model with interaction

Definition 6.19: two-way MANOVA

$\mathbf{X}_{\ell kr} = \boldsymbol{\mu}+\boldsymbol{\tau}_\ell+\boldsymbol{\beta}_k+\boldsymbol{\gamma}_{\ell k}+\mathbf{e}_{\ell kr}$ with $\mathbf{e}_{\ell kr}$ independent $N_p(\mathbf{0},\boldsymbol{\Sigma})$ and $\sum_\ell \boldsymbol{\tau}_\ell = \sum_k \boldsymbol{\beta}_k = \sum_\ell \boldsymbol{\gamma}_{\ell k} = \sum_k \boldsymbol{\gamma}_{\ell k} = \mathbf{0}$; all vectors are $p \times 1$.

The SSP decomposition $\mathrm{SSP}_{\mathrm{cor}} = \mathrm{SSP}_{\mathrm{fac1}} + \mathrm{SSP}_{\mathrm{fac2}} + \mathrm{SSP}_{\mathrm{int}} + \mathrm{SSP}_{\mathrm{res}}$ replaces each $(\cdot)^2$ by $(\cdot)(\cdot)^\top$, with the same degrees of freedom:

Source	Matrix of SSP	d.f.
Factor 1	$\mathrm{SSP}_{\mathrm{fac1}} = \sum_\ell bn(\bar{\mathbf{x}}_{\ell\cdot} - \bar{\mathbf{x}})(\bar{\mathbf{x}}_{\ell\cdot} - \bar{\mathbf{x}})^\top$	$g-1$
Factor 2	$\mathrm{SSP}_{\mathrm{fac2}} = \sum_k gn(\bar{\mathbf{x}}_{\cdot k} - \bar{\mathbf{x}})(\bar{\mathbf{x}}_{\cdot k} - \bar{\mathbf{x}})^\top$	$b-1$
Interaction	$\mathrm{SSP}_{\mathrm{int}} = \sum_\ell \sum_k n(\bar{\mathbf{x}}_{\ell k} - \bar{\mathbf{x}}_{\ell\cdot} - \bar{\mathbf{x}}_{\cdot k} + \bar{\mathbf{x}})(\cdot)^\top$	$(g-1)(b-1)$
Residual (error)	$\mathrm{SSP}_{\mathrm{res}} = \sum_\ell \sum_k \sum_r (\mathbf{x}_{\ell kr} - \bar{\mathbf{x}}_{\ell k})(\mathbf{x}_{\ell kr} - \bar{\mathbf{x}}_{\ell k})^\top$	$gb(n-1)$
Total (corrected)	$\mathrm{SSP}_{\mathrm{cor}} = \sum_\ell \sum_k \sum_r (\mathbf{x}_{\ell kr} - \bar{\mathbf{x}})(\mathbf{x}_{\ell kr} - \bar{\mathbf{x}})^\top$	$gbn-1$

Here $(\cdot)^\top$ repeats the preceding factor.

Theorem 6.20: two-way MANOVA tests (Bartlett-corrected)

Interaction, $H_0 : \boldsymbol{\gamma}_{11} = \boldsymbol{\gamma}_{12} = \cdots = \boldsymbol{\gamma}_{gb} = \mathbf{0}$: with $\Lambda^* = \dfrac{|\mathrm{SSP}_{\mathrm{res}}|}{|\mathrm{SSP}_{\mathrm{int}} + \mathrm{SSP}_{\mathrm{res}}|}$, reject if

$$-\Big[gb(n-1) - \frac{p+1-(g-1)(b-1)}{2}\Big] \ln \Lambda^* > \chi^2_{(g-1)(b-1)p}(\alpha).$$

Main effects of factor 1, $H_0 : \boldsymbol{\tau}_1 = \cdots = \boldsymbol{\tau}_g = \mathbf{0}$: with $\Lambda^* = \dfrac{|\mathrm{SSP}_{\mathrm{res}}|}{|\mathrm{SSP}_{\mathrm{fac1}} + \mathrm{SSP}_{\mathrm{res}}|}$, reject if

$$-\Big[gb(n-1) - \frac{p+1-(g-1)}{2}\Big] \ln \Lambda^* > \chi^2_{(g-1)p}(\alpha).$$

[Supplement] Factor 2 is tested in the same way with $\mathrm{SSP}_{\mathrm{fac2}}$ and $g-1$ replaced by $b-1$. Test the interaction first; if it is present, interpret the main effects with care.

Theorem 6.21: simultaneous intervals in two-way MANOVA

Simultaneous confidence intervals provide insight into the nature of the factor effects. The $100(1-\alpha)\%$ simultaneous intervals for $\tau_{\ell j} - \tau_{mj}$ are: with $\nu = gb(n-1)$ and E_{jj} the jth diagonal element of $\mathbf{E} = \mathrm{SSP}_{\mathrm{res}}$,

$$\tau_{\ell j} - \tau_{mj} \quad \text{belongs to} \quad (\bar{x}_{\ell\cdot j} - \bar{x}_{m\cdot j}) \pm t_\nu\Big(\frac{\alpha}{pg(g-1)}\Big)\sqrt{\frac{E_{jj}}{\nu}\cdot\frac{2}{bn}}.$$

Proof. [Supplement] **Step 1.** Averaging the model over k and r, using $\sum_k \boldsymbol{\beta}_k = \mathbf{0}$ and $\sum_k \boldsymbol{\gamma}_{\ell k} = \mathbf{0}$, gives $\bar{X}_{\ell\cdot j} = \mu_j + \tau_{\ell j} + \bar{e}_{\ell\cdot j}$, an average of bn independent errors. Hence $\bar{X}_{\ell\cdot j} - \bar{X}_{m\cdot j} \sim N(\tau_{\ell j} - \tau_{mj}, \frac{2}{bn}\sigma_{jj})$ for $\ell \neq m$.

Step 2. $E_{jj} = \sum_{\ell,k}\sum_r (x_{\ell krj} - \bar{x}_{\ell kj})^2$ is a sum of gb independent within-cell sums of squares, so $E_{jj}/\sigma_{jj} \sim \chi^2_\nu$, independent of the cell means and hence of $\bar{X}_{\ell\cdot j} - \bar{X}_{m\cdot j}$. Thus the studentized difference with $\sqrt{(E_{jj}/\nu)(2/bn)}$ is t_ν.

Step 3. There are $m = p\,g(g-1)/2$ statements (all components, all pairs of factor-1 levels); each

uses $t_\nu(\alpha/2m) = t_\nu(\alpha/\{pg(g-1)\})$, and the Bonferroni inequality gives joint coverage at least $1-\alpha$. Intervals for $\beta_{kj} - \beta_{qj}$ are analogous with $\alpha/\{pb(b-1)\}$ and $2/(gn)$. ■

Chapter 7. Principal Component Analysis

Population and sample principal components, SVD, choosing the number of components

Ch. 1, §6, introduced PCA through sample geometry. Here it is formulated at the population level, with the Lagrange-multiplier derivation, the SVD and the choice of the number of components.

1 Population Principal Components

Motivation. With many correlated variables, most of the variation often lies in a few directions. Principal components are the orthogonal directions of maximal variance, the eigenvectors of $\boldsymbol{\Sigma}$, and they allow us to summarize the data in fewer dimensions.

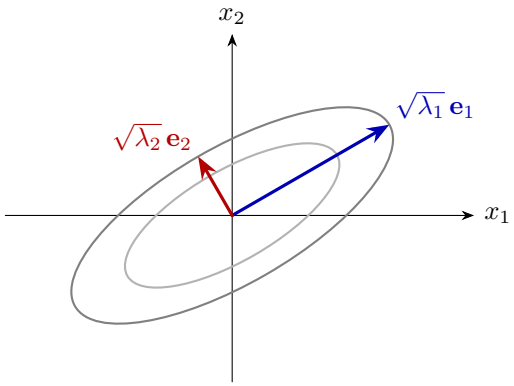


Figure. *Contours of a bivariate normal density are ellipses whose axes are the eigenvectors $\mathbf{e}_1, \mathbf{e}_2$ of $\boldsymbol{\Sigma}$, with half-lengths proportional to $\sqrt{\lambda_1}, \sqrt{\lambda_2}$. The first principal component is the projection on the long axis, the direction of maximal variance.*

Let $\mathbf{X} \in \mathbb{R}^p$ with $\mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$, $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$. A linear combination $Z(\mathbf{a}) = \mathbf{a}^\top \mathbf{X}$ has variance $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$. (For $d = 1$: $\mathbf{a}_1 = \arg\max_{\|\mathbf{a}\|=1} \mathrm{Var}(Z(\mathbf{a}))$.)

Definition 7.1: principal components

First PC direction vector: $\mathbf{u}_1 = \arg\max_{\|\mathbf{u}\|=1} \mathrm{Var}(Z(\mathbf{u}))$, with $\lambda_1 = \mathrm{Var}(Z_1)$. Second: $\mathbf{u}_2 = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_1 = 0} \mathrm{Var}(\mathbf{u}^\top \mathbf{X})$, with $\lambda_2 = \mathrm{Var}(Z_2)$; and so on:

$$\mathbf{u}_k = \arg \max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_j = 0 \ (j<k)} \mathrm{Var}(\mathbf{u}^\top \mathbf{X}), \qquad Z_k = \mathbf{u}_k^\top \mathbf{X}.$$

Then $\lambda_1 \geq \lambda_2 \geq \cdots$ and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Theorem 7.2: solution

$\mathbf{u}_k$ is the kth eigenvector of $\boldsymbol{\Sigma}$, $\mathrm{Var}(Z_k) = \lambda_k$, and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Lagrange multipliers. **1.** $\Phi(\mathbf{u}, \lambda) = \mathrm{Var}(\mathbf{u}^\top \mathbf{X}) - \lambda(\mathbf{u}^\top \mathbf{u} - 1) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1)$. $\frac{1}{2} \frac{\partial}{\partial \mathbf{u}} \Phi(\mathbf{u}, \lambda) = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} = \mathbf{0}$, so $\boldsymbol{\Sigma}\mathbf{u} = \lambda \mathbf{u}$ and $\lambda = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}$: the objective equals the eigenvalue, so take the largest, λ_1.

2. With $\boldsymbol{\gamma}^k = (\gamma_1, \dots, \gamma_{k-1})$,

$$\Phi(\mathbf{u}, \lambda, \boldsymbol{\gamma}^k) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1) - \sum_{j=1}^{k-1} 2\gamma_j \mathbf{u}_j^\top \mathbf{u},$$

$$\frac{1}{2}\frac{\partial \Phi}{\partial \mathbf{u}} = \boldsymbol{\Sigma}\mathbf{u} - \lambda\mathbf{u} - \sum_{j=1}^{k-1}\gamma_j\mathbf{u}_j = \mathbf{0}, \quad \frac{\partial \Phi}{\partial \gamma_j} = \mathbf{u}_j^\top \mathbf{u} = 0.$$

Premultiplying by $\mathbf{u}_i^\top$ $(i<k)$: $\mathbf{u}_i^\top\boldsymbol{\Sigma}\mathbf{u} = \gamma_i$, and $\mathbf{u}_i^\top\boldsymbol{\Sigma}\mathbf{u} = \lambda_i\mathbf{u}_i^\top\mathbf{u} = 0$, so $\gamma_i = 0$. Hence $\boldsymbol{\Sigma}\mathbf{u} = \lambda\mathbf{u}$ again, and among eigenvectors orthogonal to $\mathbf{u}_1, \ldots, \mathbf{u}_{k-1}$ the largest eigenvalue λ_k is chosen: $\mathbf{u}_k, \lambda_k$. Finally $\mathrm{Cov}(Z_j, Z_k) = \mathbf{u}_j^\top\boldsymbol{\Sigma}\mathbf{u}_k = \lambda_k\mathbf{u}_j^\top\mathbf{u}_k = 0$. ■

Theorem 7.3: total variance and correlations

1. $\sum_{j=1}^p \mathrm{Var}(X_j) = \mathrm{tr}\boldsymbol{\Sigma} = \sum_{k=1}^p \lambda_k = \sum_k \mathrm{Var}(Z_k)$; the proportion of variance explained by PC k is $\lambda_k / \sum_j \lambda_j$.
2. $\mathrm{Corr}(Z_k, X_j) = \dfrac{u_{jk}\sqrt{\lambda_k}}{\sqrt{\sigma_{jj}}}$, where u_{jk} is the jth entry of $\mathbf{u}_k$.

Proof. (1) $\mathrm{tr}(\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top) = \mathrm{tr}(\boldsymbol{\Lambda}\mathbf{U}^\top\mathbf{U}) = \mathrm{tr}\boldsymbol{\Lambda}$, and $\mathrm{Var}(Z_k) = \lambda_k$ by Theorem 7.2.
(2) $\mathrm{Cov}(Z_k, X_j) = \mathbf{u}_k^\top\boldsymbol{\Sigma}\mathbf{e}_j = \lambda_k\mathbf{u}_k^\top\mathbf{e}_j = \lambda_k u_{jk}$, so $\mathrm{Corr} = \lambda_k u_{jk}/(\sqrt{\lambda_k}\sqrt{\sigma_{jj}})$. ■

Theorem 7.4: equicorrelation structure

If $\boldsymbol{\Sigma} = \sigma^2\{(1-\rho)\mathbf{I}_p + \rho\mathbf{1}_p\mathbf{1}_p^\top\}$ with $0 < \rho < 1$, then $\lambda_1 = \sigma^2\{1+(p-1)\rho\}$ with $\mathbf{u}_1 = \mathbf{1}_p/\sqrt{p}$, and $\lambda_2 = \cdots = \lambda_p = \sigma^2(1-\rho)$ with any orthonormal basis of $\mathbf{1}_p^\perp$ as $\mathbf{u}_2, \ldots, \mathbf{u}_p$. The first PC $Z_1 = \frac{1}{\sqrt{p}}\sum_j X_j$ is a scaled total and explains the proportion $\rho + (1-\rho)/p$ of the total variance.

Proof. $\boldsymbol{\Sigma}\mathbf{1}_p = \sigma^2\{(1-\rho)+p\rho\}\mathbf{1}_p$, and for $\mathbf{v} \perp \mathbf{1}_p$, $\boldsymbol{\Sigma}\mathbf{v} = \sigma^2(1-\rho)\mathbf{v}$ because $\mathbf{1}_p^\top\mathbf{v} = 0$. Since $\mathbb{R}^p = \mathrm{span}(\mathbf{1}_p) \oplus \mathbf{1}_p^\perp$, these are all the eigenpairs, and $1+(p-1)\rho > 1-\rho$ for $\rho > 0$. Finally $\lambda_1/\mathrm{tr}\boldsymbol{\Sigma} = \{1+(p-1)\rho\}/p = \rho + (1-\rho)/p$. ■

With ρ close to 1, a single component summarizes all p variables; with $\rho = 0$ every direction is a principal component and PCA reduces nothing.

Theorem 7.5: Ky Fan maximum principle — q-dimensional subspaces

[Supplement]

$$\max_{\mathbf{V}\in\mathbb{R}^{p\times q},\ \mathbf{V}^\top\mathbf{V}=\mathbf{I}_q} \mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V}) = \lambda_1 + \cdots + \lambda_q,$$

attained at $\mathbf{V} = [\mathbf{u}_1, \ldots, \mathbf{u}_q]$ (and its rotations).

Proof [Supplement]. $\mathbf{P} = \mathbf{V}\mathbf{V}^\top$ is a rank-q projection with eigenvalues 1 (q times) and 0. By von Neumann's upper bound, $\mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V}) = \mathrm{tr}(\boldsymbol{\Sigma}\mathbf{P}) \le \sum_i \lambda_i(\boldsymbol{\Sigma})\lambda_i(\mathbf{P}) = \lambda_1 + \cdots + \lambda_q$. ■

Remark 7.6: PCA as optimal approximation

[Supplement] Since $\mathbf{I} - \mathbf{V}\mathbf{V}^\top$ is symmetric idempotent, $\mathrm{E}\|(\mathbf{X}-\boldsymbol{\mu}) - \mathbf{V}\mathbf{V}^\top(\mathbf{X}-\boldsymbol{\mu})\|^2 = \mathrm{tr}\{(\mathbf{I}-\mathbf{V}\mathbf{V}^\top)\boldsymbol{\Sigma}\} = \mathrm{tr}\boldsymbol{\Sigma} - \mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V})$, so among q-dimensional subspaces through the mean, the first q PCs minimize the expected squared reconstruction error, the minimum being $\lambda_{q+1} + \cdots + \lambda_p$.

2 Sample Principal Components and the SVD

2.1 Sample PCs

Let $\tilde{\mathbf{X}} = \mathbf{H}_n\mathbf{X} = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top$ be the centered data matrix, $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top$, and

$$\hat{\mathbf{u}}_k = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top\hat{\mathbf{u}}_j=0,\ j=1,\dots,k-1} \mathbf{u}^\top\mathbf{S}\mathbf{u}, \qquad \text{scores } t_{ik} = \hat{\mathbf{u}}_k^\top(\mathbf{x}_i - \bar{\mathbf{x}}).$$

The sample PC scores have sample variance $\hat{\lambda}_k$ and are mutually uncorrelated.

Proof [Supplement]. The score vectors are the columns of $\tilde{\mathbf{X}}\hat{\mathbf{U}}$, which are centered because $\mathbf{1}_n^\top\tilde{\mathbf{X}} = \mathbf{0}^\top$. Their sample covariance matrix is $\frac{1}{n-1}\hat{\mathbf{U}}^\top\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}\hat{\mathbf{U}} = \hat{\mathbf{U}}^\top\mathbf{S}\hat{\mathbf{U}} = \hat{\mathbf{\Lambda}}$, which is diagonal with entries $\hat{\lambda}_k$. ■

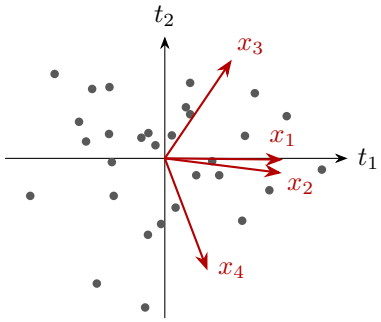


Figure. *Biplot of synthetic standardized data ($n = 30$, $p = 4$; $\hat{\lambda}_1 = 2.29$, $\hat{\lambda}_2 = 1.52$, together 95% of $\mathrm{tr}\mathbf{R} = 4$). Dots are the scores (t_{i1}, t_{i2}); arrow j has coordinates $(\mathrm{Corr}(Z_1, X_j), \mathrm{Corr}(Z_2, X_j))$, scaled by 2.6. x_1, x_2 load only on the first PC; x_3 and x_4 point to opposite sides of the second, which is a contrast between them.*

2.2 Relation to the singular value decomposition

Write the SVD of the $p \times n$ matrix $\tilde{\mathbf{X}}^\top$ $(n > p)$ as

$$\tilde{\mathbf{X}}^\top = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \underbrace{[\mathbf{u}_1 \cdots \mathbf{u}_p]}_{p\times p} \underbrace{\begin{pmatrix} d_1 & & \\ & \ddots & \\ & & d_p \end{pmatrix}}_{p\times p} \underbrace{\begin{pmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_p^\top \end{pmatrix}}_{p\times n}, \qquad \mathbf{U}^\top\mathbf{U} = \mathbf{I}_p,\ \mathbf{V}^\top\mathbf{V} = \mathbf{I}_p,\ d_1 \geq \cdots \geq d_p \geq 0.$$

Then

$$\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \mathbf{U}\mathrm{diag}\Big(\frac{d_1^2}{n-1}, \dots, \frac{d_p^2}{n-1}\Big)\mathbf{U}^\top, \qquad \begin{pmatrix} \mathbf{z}_{(1)}^\top \\ \vdots \\ \mathbf{z}_{(p)}^\top \end{pmatrix} = \mathbf{Z} = \mathbf{U}^\top\tilde{\mathbf{X}}^\top = \mathbf{D}\mathbf{V}^\top = \begin{pmatrix} d_1\mathbf{v}_1^\top \\ \vdots \\ d_p\mathbf{v}_p^\top \end{pmatrix}.$$

The PC directions are the left singular vectors, the eigenvalues are $d_k^2/(n-1)$, and the kth PC score vector is $\mathbf{z}_{(k)} = \hat{\mathbf{u}}_k^\top\tilde{\mathbf{X}}^\top = d_k\mathbf{v}_k^\top$. Numerically the SVD is preferred to forming $\mathbf{S}$ (computation of PCA).

3 PCA and the Best Approximating Affine Subspace

Motivation. A second view of PCA: the first r components span the r-dimensional affine subspace closest to the data in squared distance. Maximizing variance and minimizing reconstruction error are the same problem.

Theorem 7.7: claim

PCA with r components is equivalent to finding the best r-dimensional affine subspace (hyperplane) approximating the data.

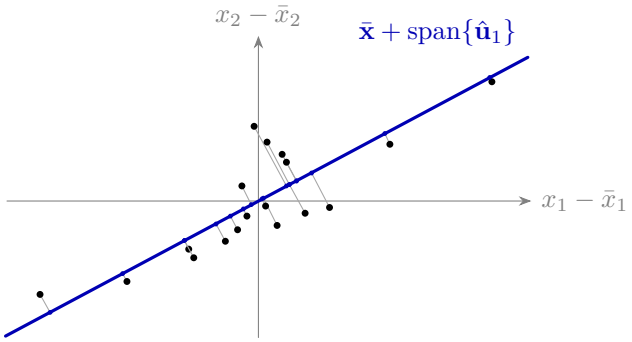


Figure. *Synthetic centered data ($n = 18$, $p = 2$, $r = 1$). Each point $\mathbf{x}_i$ (black) is replaced by its orthogonal projection $\bar{\mathbf{x}} + (\hat{\mathbf{u}}_1^\top \tilde{\mathbf{x}}_i)\hat{\mathbf{u}}_1$ (blue) on the first-PC line, $\hat{\mathbf{u}}_1 = (0.88, 0.47)^\top$. The sum of the squared gray segments is $\text{SSE} = (n-1)\hat{\lambda}_2$, the smallest over all lines; the spread of the blue points along the line is the retained variance $\hat{\lambda}_1$.*

3.1 The problem

Let the data be $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^p$ and $r \in \{1, \ldots, p\}$. We wish to approximate $\mathbf{x}_1, \ldots, \mathbf{x}_n$ by $\bar{\mathbf{x}} + \mathbf{a}_1, \ldots, \bar{\mathbf{x}} + \mathbf{a}_n$ where $\text{rank}([\mathbf{a}_1, \ldots, \mathbf{a}_n]) = r$. The approximation of $\mathbf{x}_i$ by $\bar{\mathbf{x}} + \mathbf{a}_i$ is better when the error $\|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller, and the approximation of $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ by $\{\bar{\mathbf{x}} + \mathbf{a}_i\}$ is better when $\text{SSE} = \sum_{i=1}^n \|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller. That is, we wish to find $\mathbf{A}_r = [\mathbf{a}_1, \ldots, \mathbf{a}_n]$ $(p \times n)$ solving

$$(\star) \qquad \min_{\mathbf{A}_r} \sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 \quad \text{subject to} \quad \text{rank}(\mathbf{A}_r) = r.$$

Since every column of a rank-r matrix lies in an r-dimensional subspace, the points $\bar{\mathbf{x}} + \mathbf{a}_i$ lie on an r-dimensional affine subspace through $\bar{\mathbf{x}}$; conversely any such subspace gives feasible $\mathbf{A}_r$.

3.2 Observations

1. With $\tilde{\mathbf{X}}^\top = [\mathbf{x}_1 - \bar{\mathbf{x}}, \ldots, \mathbf{x}_n - \bar{\mathbf{x}}]$ $(p \times n)$,
$$\sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 = \|\tilde{\mathbf{X}}^\top - \mathbf{A}_r\|_F^2,$$
because for a $p \times n$ matrix $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_n]$ the Frobenius norm is $\|\mathbf{B}\|_F^2 = \sum_{i=1}^p \sum_{j=1}^n b_{ij}^2 = \sum_{j=1}^n \|\mathbf{b}_j\|_2^2 = \text{tr}(\mathbf{B}\mathbf{B}^\top) = \text{tr}(\mathbf{B}^\top\mathbf{B})$.
2. The sample PCA of $(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is given by
   (1) the eigendecomposition $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top = \sum_{i=1}^p \hat{\lambda}_i \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{\lambda}_i$ the ith PC variance, $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_r > \hat{\lambda}_{r+1} \geq \cdots \geq \hat{\lambda}_p$;
   (2) the SVD $\tilde{\mathbf{X}}^\top_{(p\times n)} = \hat{\mathbf{U}}\hat{\mathbf{D}}\hat{\mathbf{V}}^\top = \sum_{i=1}^p \hat{d}_i \hat{\mathbf{u}}_i \hat{\mathbf{v}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{d}_i\hat{\mathbf{v}}_i$ the ith PC score vector;
   (3) $\hat{\mathbf{u}}_i^\top \tilde{\mathbf{X}}^\top = \hat{d}_i \hat{\mathbf{v}}_i^\top$, i.e.
$$\begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix}.$$

Theorem 7.8: solution (Eckart–Young–Mirsky)

The $\mathbf{A}_r$ that minimizes $(\star)$ is

$$\mathbf{A}_r = \sum_{i=1}^{r} \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r][\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r]^\top \tilde{\mathbf{X}}^\top,$$

the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace, and the minimum is $\sum_{i>r} \hat{d}_i^2 = (n-1)\sum_{i=r+1}^{p} \hat{\lambda}_i$.

Proof. Step 1 (parametrize rank-r matrices). Since $\operatorname{rank}(\mathbf{A}_r) = r$, we can write $\mathbf{A}_r = \mathbf{E}\mathbf{C}^\top$ with $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_r]$ $(p \times r)$ satisfying $\mathbf{E}^\top\mathbf{E} = \mathbf{I}_r$ (an orthonormal basis of the column space of $\mathbf{A}_r$) and $\mathbf{C} = \begin{pmatrix} \mathbf{c}_1^\top \\ \vdots \\ \mathbf{c}_n^\top \end{pmatrix}$ an $n \times r$ matrix. Writing $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}$, $(\star)$ is equivalent to $\min_{\mathbf{E},\mathbf{C}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2$, where

$$(\star\star) \qquad \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2 = \big\|(\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n) - \mathbf{E}[\mathbf{c}_1, \dots, \mathbf{c}_n]\big\|_F^2 = \sum_{i=1}^{n} \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2.$$

Step 2 (optimal $\mathbf{C}$ for fixed $\mathbf{E}$). For any given $\mathbf{E}$, $(\star\star)$ separates over i and its minimizer is

$$\hat{\mathbf{c}}_i = \arg\min_{\mathbf{c}_i} \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2 = (\mathbf{E}^\top\mathbf{E})^{-1}\mathbf{E}^\top\tilde{\mathbf{x}}_i = \mathbf{E}^\top\tilde{\mathbf{x}}_i,$$

the projection scores of $\tilde{\mathbf{x}}_i$ onto $\operatorname{span}(\mathbf{E})$ (least squares). Stacking, $\hat{\mathbf{C}} = \begin{pmatrix} \tilde{\mathbf{x}}_1^\top\mathbf{E} \\ \vdots \\ \tilde{\mathbf{x}}_n^\top\mathbf{E} \end{pmatrix} = \tilde{\mathbf{X}}\mathbf{E}$, i.e. $\hat{\mathbf{C}}^\top = \mathbf{E}^\top\tilde{\mathbf{X}}^\top$.

Step 3 (profile out $\mathbf{C}$). Plugging $\hat{\mathbf{C}}$ into $(\star\star)$ (profiling, Theorem 2.26), and using that $\mathbf{I} - \mathbf{E}\mathbf{E}^\top$ is symmetric idempotent, $(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)^2 = \mathbf{I} - \mathbf{E}\mathbf{E}^\top$,

$$\begin{aligned}
\min_{\mathbf{E}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{E}^\top\tilde{\mathbf{X}}^\top\|_F^2 &= \min_{\mathbf{E}} \|(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top\|_F^2 = \min_{\mathbf{E}} \operatorname{tr}(\tilde{\mathbf{X}}(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top) \\
&= \min_{\mathbf{E}} \Big\{ \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1)\operatorname{tr}\Big(\mathbf{E}^\top \frac{\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}}{n-1}\mathbf{E}\Big) \Big\} \\
&= \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1)\max_{\mathbf{E}} \operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top).
\end{aligned}$$

Step 4 (von Neumann). Recall $\mathbf{S} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top$. $\mathbf{E}\mathbf{E}^\top$ is symmetric with eigenvalues 1 (r times) and 0 ($p - r$ times), so von Neumann's trace inequality (Theorem 2.14) gives

$$\operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top) \le \hat{\lambda}_1 + \cdots + \hat{\lambda}_r,$$

with equality when $\mathbf{E} = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r]$ (or any $\mathbf{E}$ spanning the same space).

Step 5 (the minimizer). Therefore

$$\mathbf{A}_r = \mathbf{E}\hat{\mathbf{C}}^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix} = \sum_{i=1}^{r} \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top$$

by Observation 2(3): the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace. Since $\operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) = (n-1)\operatorname{tr}\mathbf{S} = (n-1)\sum_i \hat{\lambda}_i$, the minimum is $(n-1)\sum_{i>r} \hat{\lambda}_i = \sum_{i>r} \hat{d}_i^2$. ∎

Remark 7.9: interpretation

[Supplement]

- The best approximating hyperplane is $\bar{\mathbf{x}}+\text{span}\{\hat{\mathbf{u}}_1,\dots,\hat{\mathbf{u}}_r\}$ and each $\mathbf{x}_i$ is approximated by its orthogonal projection $\bar{\mathbf{x}}+\sum_{k\le r}(\hat{\mathbf{u}}_k^\top\tilde{\mathbf{x}}_i)\hat{\mathbf{u}}_k$: the PC scores are the coordinates on the hyperplane.
- The fitted values satisfy $\hat{\mathbf{c}}_i=\mathbf{E}^\top\tilde{\mathbf{x}}_i$, i.e. reconstruction uses projection scores, exactly the decomposition $\mathbf{x}=\bar{\mathbf{x}}+\sum s_{\mathbf{v}_j}(\mathbf{x}-\bar{\mathbf{x}})\mathbf{v}_j+r_{\mathcal{L}_q}(\mathbf{x})$ of Ch. 1.
- Minimizing squared error and maximizing retained variance (the $\arg\max\mathbf{u}^\top\mathbf{S}\mathbf{u}$ definition) are the same problem, since $\text{SSE}=(n-1)\{\text{tr}\mathbf{S}-\text{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top)\}$.
- If $\hat{\lambda}_r=\hat{\lambda}_{r+1}$ the optimal subspace is not unique; the strict gap $\hat{\lambda}_r>\hat{\lambda}_{r+1}$ makes it unique.

3.3 Correlation PCA

If the units differ, use PCA of standardized data: $\mathbf{R}=\mathbf{D}_{\boldsymbol{\Sigma}}^{-1/2}\boldsymbol{\Sigma}\mathbf{D}_{\boldsymbol{\Sigma}}^{-1/2}=\mathbf{U}_R\boldsymbol{\Lambda}_R\mathbf{U}_R^\top$ (sample: $\mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2}$). Since $\text{tr}\mathbf{R}=p$, the proportion explained is λ_k^R/p. Covariance and correlation PCs are in general not simple transformations of each other (scale dependence).

4 How Many Components to Keep?

- **Scree plot / cumulative scree plot**: plot $\hat{\lambda}_k$ against k and look for the elbow, or take the q at which the cumulative proportion reaches a target (e.g. 80–90%).
- **Kaiser's rule**: keep components with $\hat{\lambda}_k>\bar{\lambda}=\frac{1}{p}\sum_j\hat{\lambda}_j$; for correlation PCA, $\hat{\lambda}_k>1$.
- **Likelihood ratio testing**: under normality test $H_0(k):\lambda_{k+1}=\cdots=\lambda_p$ (the remaining space is spherical). With arithmetic mean $\bar{a}$ and geometric mean $\bar{g}$ of the last $p-k$ sample eigenvalues,

$$-2\log\Lambda=n(p-k)\log\frac{\bar{a}}{\bar{g}}\ \dot{\sim}\ \chi^2_{(p-k+2)(p-k-1)/2};$$

since $\bar{a}\ge\bar{g}$ (AM–GM), the statistic is small when the eigenvalues are nearly equal.

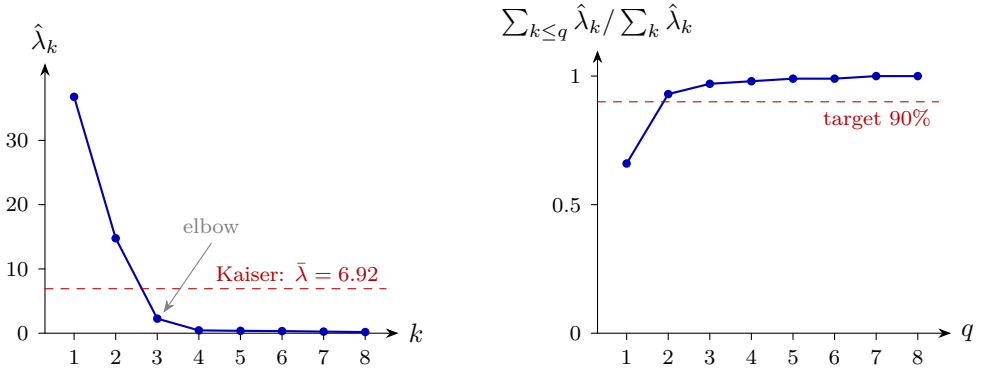


Figure. *Scree plot (left) and cumulative scree plot (right) for synthetic data ($n=200$, $p=8$, generated with three strong directions). The elbow is at $k=3$, Kaiser's rule keeps $q=2$ ($\hat{\lambda}_1,\hat{\lambda}_2>\bar{\lambda}$), and the 90% target is reached at $q=2$ (93%).*

Remark 7.10: cautions in interpretation

- The sign of an eigenvector is arbitrary; what matters is the sign pattern (same signs: "size"; mixed signs: "contrast"). In the scores example $\mathbf{u}_1 \propto (1,1)$ was overall level and $\mathbf{u}_2 \propto (1,-1)$ the contrast between subjects.
- With tied eigenvalues the directions within the eigenspace are not unique.
- Sample eigenvalues are biased: large ones are overestimated and small ones underestimated. Under normality, for distinct λ_k, $\sqrt{n}(\hat{\lambda}_k - \lambda_k) \Rightarrow N(0, 2\lambda_k^2)$.
- High explained variance does not mean usefulness for classification or prediction (the $X_1 = 10Z$, $X_2 = Y$ example of Ch. 1).

5 Example: Scores in Two Subjects Revisited

For $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$ the characteristic equation $(125-\lambda)^2 - 75^2 = 0$ gives $\lambda = 200, 50$; the null space of $\mathbf{S} - 200\mathbf{I} = \begin{pmatrix} -75 & 75 \\ 75 & -75 \end{pmatrix}$ is spanned by $(1,1)^\top$. $\mathrm{Corr}(Z_1, X_1) = u_{11}\sqrt{\lambda_1}/\sqrt{s_{11}} = \frac{1}{\sqrt{2}}\sqrt{200}/\sqrt{125} = 0.894$, $\mathrm{Corr}(Z_2, X_1) = \frac{1}{\sqrt{2}}\sqrt{50}/\sqrt{125} = 0.447$, and $0.894^2 + 0.447^2 = 1$: each variable's variance is fully decomposed over the PCs.

6 Selected exercises

Motivation. PCA is worthwhile only if the variables are correlated, and the eigenvalue tests above compare a covariance matrix with a spherical one. The first two exercises derive the corresponding likelihood ratio tests; the third is a covariance whose components can be found by hand.

Exercise 7.1 (Testing that all variables are independent; J&W 8.9(a))

Let $\mathbf{x}_1, \ldots, \mathbf{x}_n$ be a sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $n > p$. Show that the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma} = \mathrm{diag}(\sigma_{11}, \ldots, \sigma_{pp})$ is

$$\Lambda = \frac{|\mathbf{S}|^{n/2}}{\prod_{j=1}^{p} s_{jj}^{n/2}} = |\mathbf{R}|^{n/2},$$

that $0 < \Lambda \le 1$, and that $-2\log\Lambda$ is approximately $\chi^2_{p(p-1)/2}$ for large n.

Solution.

Step 1 (unrestricted maximum). By Theorem 4.7 and the remark after it, $\max_{\boldsymbol{\mu},\boldsymbol{\Sigma}} L = (2\pi)^{-np/2}|\mathbf{S}_0|^{-n/2}e^{-np/2}$ with $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$.

Step 2 (maximum under H_0). If $\boldsymbol{\Sigma}$ is diagonal, the density of $\mathbf{x}_i$ is the product of univariate $N(\mu_j, \sigma_{jj})$ densities, so $L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \prod_{j=1}^{p} L_j(\mu_j, \sigma_{jj})$ with L_j the likelihood of the jth coordinate. The factors have separate parameters, so each is maximized separately; Theorem 4.7 with $p = 1$ gives $\hat{\mu}_j = \bar{x}_j$, $\hat{\sigma}_{jj} = (\mathbf{S}_0)_{jj}$ and

$$\max_{H_0} L = \prod_{j=1}^{p} (2\pi)^{-n/2}(\mathbf{S}_0)_{jj}^{-n/2}e^{-n/2} = (2\pi)^{-np/2}\Big\{\prod_{j}(\mathbf{S}_0)_{jj}\Big\}^{-n/2}e^{-np/2}.$$

Step 3 (the ratio). Dividing,

$$\Lambda = \Big(\frac{|\mathbf{S}_0|}{\prod_j (\mathbf{S}_0)_{jj}}\Big)^{n/2} = \Big(\frac{|\mathbf{S}|}{\prod_j s_{jj}}\Big)^{n/2},$$

since $|\mathbf{S}_0| = (\frac{n-1}{n})^p |\mathbf{S}|$ and $\prod_j (\mathbf{S}_0)_{jj} = (\frac{n-1}{n})^p \prod_j s_{jj}$. With $\mathbf{D} = \operatorname{diag}(s_{11}, \dots, s_{pp})$, $\mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2}$ and $|\mathbf{R}| = |\mathbf{S}| / \prod_j s_{jj}$.

Step 4. $\mathbf{S} \succ 0$ a.s. (as $n > p$), so $\mathbf{R} \succ 0$ and Exercise 2.2 gives $0 < |\mathbf{R}| \le 1$, with $\Lambda = 1$ iff $\mathbf{R} = \mathbf{I}_p$.

Step 5 (degrees of freedom). The unrestricted model has $p + p(p+1)/2$ free parameters and H_0 has $2p$; the large-sample likelihood ratio theory used in Ch. 5 gives $-2\log\Lambda \overset{\cdot}{\sim} \chi^2_\nu$ with $\nu = p(p+1)/2 - p = p(p-1)/2$, the number of off-diagonal correlations. ■

Exercise 7.2 (Sphericity test; J&W 8.9(b))

In the setting of Exercise 7.1, show that the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_p$ ($\sigma^2 > 0$ unknown) is

$$\Lambda = \Big(\frac{|\mathbf{S}|}{(\operatorname{tr}\mathbf{S}/p)^p}\Big)^{n/2} = \Big(\frac{\bar g}{\bar a}\Big)^{np/2},$$

where $\bar a$ and $\bar g$ are the arithmetic and geometric means of the eigenvalues $\hat\lambda_1, \dots, \hat\lambda_p$ of $\mathbf{S}$. Show $-2\log\Lambda \ge 0$ and find its degrees of freedom; compare with the test of $H_0(k)$ in the last section.

Solution.

Step 1 (log-likelihood under H_0). With $Q(\boldsymbol{\mu}) = \sum_i \|\mathbf{x}_i - \boldsymbol{\mu}\|^2$,

$$\log L(\boldsymbol{\mu}, \sigma^2 \mathbf{I}) = -\frac{np}{2}\log(2\pi\sigma^2) - \frac{Q(\boldsymbol{\mu})}{2\sigma^2}.$$

$Q(\boldsymbol{\mu}) = Q(\bar{\mathbf{x}}) + n\|\bar{\mathbf{x}} - \boldsymbol{\mu}\|^2$ is minimized at $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ for every σ^2, and $Q(\bar{\mathbf{x}}) = n\operatorname{tr}\mathbf{S}_0$.

Step 2 ($\hat\sigma^2$). Let $c = Q(\bar{\mathbf{x}})/(np) = \operatorname{tr}\mathbf{S}_0/p$ and $x = c/\sigma^2$. Then

$$\log\sigma^2 + \frac{Q(\bar{\mathbf{x}})}{np\,\sigma^2} = \log c - \log x + x \;\ge\; \log c + 1$$

by Theorem 2.15, with equality iff $x = 1$. Hence $\hat\sigma^2 = \operatorname{tr}\mathbf{S}_0/p$ and $\max_{H_0} L = (2\pi\hat\sigma^2)^{-np/2} e^{-np/2}$.

Step 3 (the ratio). With Step 1 of Exercise 7.1,

$$\Lambda = \frac{|\mathbf{S}_0|^{n/2}}{(\operatorname{tr}\mathbf{S}_0/p)^{np/2}} = \Big(\frac{|\mathbf{S}|}{(\operatorname{tr}\mathbf{S}/p)^p}\Big)^{n/2},$$

because numerator and denominator are both homogeneous of degree $np/2$ in $\mathbf{S}_0$. Since $|\mathbf{S}| = \prod_i \hat\lambda_i = \bar g^p$ and $\operatorname{tr}\mathbf{S}/p = \bar a$, $\Lambda = (\bar g/\bar a)^{np/2}$.

Step 4. $-2\log\Lambda = np\log(\bar a/\bar g) \ge 0$ by the AM–GM inequality, with equality iff all $\hat\lambda_i$ are equal.

Step 5 (degrees of freedom). The free parameters are $p + p(p+1)/2$ versus $p+1$ under H_0, so $\nu = p(p+1)/2 - 1 = (p+2)(p-1)/2$. This is the statistic $n(p-k)\log(\bar a/\bar g)$ with $\chi^2_{(p-k+2)(p-k-1)/2}$ of the last section for $k = 0$; for $k \ge 1$ the same comparison is applied to the last $p-k$ eigenvalues. ■

Exercise 7.3 (A tridiagonal covariance; J&W 8.4)

Find the principal components and the proportion of total variance explained by each for

$$\boldsymbol{\Sigma} = \sigma^2 \begin{pmatrix} 1 & \rho & 0 \\ \rho & 1 & \rho \\ 0 & \rho & 1 \end{pmatrix}, \qquad 0 < \rho < \frac{1}{\sqrt{2}}.$$

Solution.

Step 1 (eigenvalues). Let $\mathbf{C} = \boldsymbol{\Sigma}/\sigma^2$. Expanding along the first row,

$$|\mathbf{C} - \lambda \mathbf{I}| = (1-\lambda)\{(1-\lambda)^2 - \rho^2\} - \rho \cdot \rho(1-\lambda) = (1-\lambda)\{(1-\lambda)^2 - 2\rho^2\},$$

so the eigenvalues of $\boldsymbol{\Sigma}$ are $\sigma^2(1+\sqrt{2}\rho) > \sigma^2 > \sigma^2(1-\sqrt{2}\rho)$, all positive because $\rho < 1/\sqrt{2}$.

Step 2 (eigenvectors). For $\lambda = 1 \pm \sqrt{2}\rho$ the first and third rows of $(\mathbf{C} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ read $\mp\sqrt{2}\rho x_1 + \rho x_2 = 0$ and $\rho x_2 \mp \sqrt{2}\rho x_3 = 0$, so $x_2 = \pm\sqrt{2}x_1$ and $x_3 = x_1$. For $\lambda = 1$ they give $x_2 = 0$ and the middle row gives $x_1 = -x_3$. Normalizing,

$$\mathbf{u}_1 = \tfrac{1}{2}(1, \sqrt{2}, 1)^\top, \qquad \mathbf{u}_2 = \tfrac{1}{\sqrt{2}}(1, 0, -1)^\top, \qquad \mathbf{u}_3 = \tfrac{1}{2}(1, -\sqrt{2}, 1)^\top.$$

Step 3 (components). $Z_1 = \tfrac{1}{2}(X_1 + \sqrt{2}X_2 + X_3)$, $Z_2 = \tfrac{1}{\sqrt{2}}(X_1 - X_3)$, $Z_3 = \tfrac{1}{2}(X_1 - \sqrt{2}X_2 + X_3)$. Since $\mathrm{tr}\boldsymbol{\Sigma} = 3\sigma^2$, the proportions are $(1+\sqrt{2}\rho)/3$, $1/3$ and $(1-\sqrt{2}\rho)/3$.

Z_1 is a weighted level with the middle variable counted most; Z_2 contrasts the two ends and does not depend on ρ. As $\rho \uparrow 1/\sqrt{2}$, $\mathrm{Var}(Z_3) \to 0$ and $\boldsymbol{\Sigma}$ becomes singular. ■

Chapter 8. Canonical Correlation Analysis

Maximal correlation between two sets of variables

1 Setting

We want to summarize the association between two sets of variables $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ with $\operatorname{Cov}\begin{pmatrix}\mathbf{X}\\ \mathbf{Y}\end{pmatrix} = \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix}$, $\boldsymbol{\Sigma}_{11}, \boldsymbol{\Sigma}_{22} \succ 0$. Instead of inspecting pq correlations, maximize the correlation of linear combinations

$$\xi = \mathbf{g}^\top \mathbf{X}, \qquad \omega = \mathbf{h}^\top \mathbf{Y}.$$

For $q = 1$ this is the multiple correlation (Ch. 3). PCA maximizes variance *within* one set; CCA maximizes correlation *between* two sets.

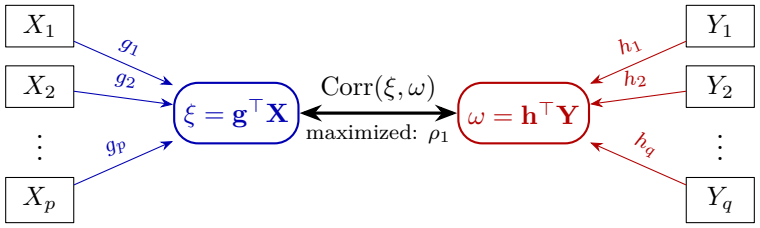


Figure. *CCA projects each set onto one direction, $\mathbf{X} \mapsto \xi = \mathbf{g}^\top \mathbf{X}$ and $\mathbf{Y} \mapsto \omega = \mathbf{h}^\top \mathbf{Y}$, and chooses $(\mathbf{g}, \mathbf{h})$ so that the two projections are as correlated as possible. Later pairs repeat this subject to being uncorrelated with the earlier variates of their own set.*

Definition 8.1: canonical variates and canonical correlations

$$(\mathbf{g}_1, \mathbf{h}_1) = \arg \max_{\mathbf{g} \in \mathbb{R}^p,\, \mathbf{h} \in \mathbb{R}^q} \operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}),$$

$$(\mathbf{g}_k, \mathbf{h}_k) = \arg \max_{\substack{\mathbf{g} \in \mathbb{R}^p,\, \mathbf{h} \in \mathbb{R}^q \\ \mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}_j = 0,\ \mathbf{h}^\top \boldsymbol{\Sigma}_{22} \mathbf{h}_j = 0,\ j = 1, \dots, k-1}} \operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}).$$

$\xi_k = \mathbf{g}_k^\top \mathbf{X}$, $\omega_k = \mathbf{h}_k^\top \mathbf{Y}$ form the kth pair of canonical variates and $\rho_k = \operatorname{Corr}(\xi_k, \omega_k)$ is the kth canonical correlation. Correlation is scale-free, so normalize $\operatorname{Var}(\xi_k) = \operatorname{Var}(\omega_k) = 1$. The constraints mean $\operatorname{Cov}(\xi_k, \xi_j) = \operatorname{Cov}(\omega_k, \omega_j) = 0$, and $\rho_k \le \rho_j$ for $j < k$.

$\rho(\mathbf{X}, \mathbf{Y}) := \max \operatorname{Corr}(\mathbf{a}^\top \mathbf{X}, \mathbf{b}^\top \mathbf{Y}) = \rho_1$ is the **largest canonical coefficient**.

2 Solution via the Singular Value Decomposition

Motivation. After each set is whitened to identity covariance, correlation becomes a bilinear form $\mathbf{a}^\top \mathbf{M} \mathbf{b}$ in two unit vectors, and maximizing a bilinear form over unit vectors is exactly what the singular value decomposition does.

Theorem 8.2: solution of CCA

Let $\mathbf{M} = \boldsymbol{\Sigma}_{11}^{-1/2} \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1/2}$ $(p \times q)$ have SVD $\mathbf{M} = \sum_{k=1}^{r} d_k \mathbf{a}_k \mathbf{b}_k^\top$, $d_1 \ge \cdots \ge d_r > 0$, $r = \operatorname{rank} \boldsymbol{\Sigma}_{12}$. Then

$$\rho_k = d_k, \qquad \mathbf{g}_k = \boldsymbol{\Sigma}_{11}^{-1/2} \mathbf{a}_k, \qquad \mathbf{h}_k = \boldsymbol{\Sigma}_{22}^{-1/2} \mathbf{b}_k.$$

Equivalently $\mathbf{a}_k$ is the kth eigenvector of $\mathbf{M}\mathbf{M}^\top = \boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}$, $\mathbf{b}_k$ that of $\mathbf{M}^\top\mathbf{M} = \boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2}$, and ρ_k^2 is the eigenvalue.

Proof. Substituting $\mathbf{a} = \boldsymbol{\Sigma}_{11}^{1/2}\mathbf{g}$, $\mathbf{b} = \boldsymbol{\Sigma}_{22}^{1/2}\mathbf{h}$,

$$\operatorname{Corr}(\mathbf{g}^\top\mathbf{X}, \mathbf{h}^\top\mathbf{Y}) = \frac{\mathbf{g}^\top\boldsymbol{\Sigma}_{12}\mathbf{h}}{\sqrt{\mathbf{g}^\top\boldsymbol{\Sigma}_{11}\mathbf{g}}\sqrt{\mathbf{h}^\top\boldsymbol{\Sigma}_{22}\mathbf{h}}} = \frac{\mathbf{a}^\top\mathbf{M}\mathbf{b}}{\sqrt{\mathbf{a}^\top\mathbf{a}}\sqrt{\mathbf{b}^\top\mathbf{b}}},$$

so we maximize $\mathbf{a}^\top\mathbf{M}\mathbf{b}$ over $\|\mathbf{a}\| = \|\mathbf{b}\| = 1$. With $\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, $\boldsymbol{\alpha} = \mathbf{U}^\top\mathbf{a}$, $\boldsymbol{\beta} = \mathbf{V}^\top\mathbf{b}$, Cauchy–Schwarz gives

$$\mathbf{a}^\top\mathbf{M}\mathbf{b} = \mathbf{a}^\top\mathbf{U}\mathbf{D}\mathbf{V}^\top\mathbf{b} = \sum_k d_k\alpha_k\beta_k \le d_1\sum_k|\alpha_k\beta_k| \le d_1\|\boldsymbol{\alpha}\|\|\boldsymbol{\beta}\| \le d_1,$$

with equality at $\mathbf{a} = \mathbf{a}_1$, $\mathbf{b} = \mathbf{b}_1$. Under $\mathbf{g}^\top\boldsymbol{\Sigma}_{11}\mathbf{g}_j = \mathbf{a}^\top\mathbf{a}_j = 0$ and $\mathbf{b}^\top\mathbf{b}_j = 0$ $(j < k)$ we have $\alpha_j = \beta_j = 0$ for $j < k$, and the same argument gives maximum d_k at $(\mathbf{a}_k, \mathbf{b}_k)$. ■

Theorem 8.3: properties of canonical variates

1. $\operatorname{Var}(\xi_k) = \operatorname{Var}(\omega_k) = 1$ and $\operatorname{Corr}(\xi_k, \xi_l) = \operatorname{Corr}(\omega_k, \omega_l) = 0$ for $k \ne l$.
2. $\operatorname{Corr}(\xi_k, \omega_l) = 0$ for $k \ne l$, $\operatorname{Corr}(\xi_k, \omega_k) = \rho_k$.
3. Invariance: canonical correlations are unchanged by $\mathbf{X} \mapsto \mathbf{D}_1\mathbf{X} + \mathbf{c}_1$, $\mathbf{Y} \mapsto \mathbf{D}_2\mathbf{Y} + \mathbf{c}_2$ with $\mathbf{D}_i$ non-singular; covariance and correlation matrices give the same ρ_k.
4. For $q = 1$, $\rho_1^2 = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}/\sigma_{22}$ is the squared multiple correlation.

Proof. (1) $\operatorname{Cov}(\xi_k, \xi_l) = \mathbf{g}_k^\top\boldsymbol{\Sigma}_{11}\mathbf{g}_l = \mathbf{a}_k^\top\mathbf{a}_l = \delta_{kl}$.

(2) $\operatorname{Cov}(\xi_k, \omega_l) = \mathbf{g}_k^\top\boldsymbol{\Sigma}_{12}\mathbf{h}_l = \mathbf{a}_k^\top\mathbf{M}\mathbf{b}_l = d_l\mathbf{a}_k^\top\mathbf{a}_l = d_l\delta_{kl}$.

(3) $\operatorname{Corr}(\mathbf{g}^\top\mathbf{D}_1\mathbf{X}, \mathbf{h}^\top\mathbf{D}_2\mathbf{Y}) = \operatorname{Corr}((\mathbf{D}_1^\top\mathbf{g})^\top\mathbf{X}, (\mathbf{D}_2^\top\mathbf{h})^\top\mathbf{Y})$ and $\mathbf{g} \mapsto \mathbf{D}_1^\top\mathbf{g}$ is a bijection, so the maxima agree.

(4) For $q = 1$ the only nonzero eigenvalue of $\mathbf{M}\mathbf{M}^\top$ is $\mathbf{M}^\top\mathbf{M} = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}/\sigma_{22}$. ■

Theorem 8.4: interpreting the canonical variates

1. With $\mathbf{G} = [\mathbf{g}_1, \dots, \mathbf{g}_r]$, $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_r]$, $\boldsymbol{\xi} = \mathbf{G}^\top\mathbf{X}$ and $\boldsymbol{\omega} = \mathbf{H}^\top\mathbf{Y}$,

$$\begin{aligned} \operatorname{Cov}(\mathbf{X}, \boldsymbol{\xi}) &= \boldsymbol{\Sigma}_{11}\mathbf{G}, & \operatorname{Cov}(\mathbf{Y}, \boldsymbol{\omega}) &= \boldsymbol{\Sigma}_{22}\mathbf{H}, \\ \operatorname{Cov}(\mathbf{X}, \boldsymbol{\omega}) &= \boldsymbol{\Sigma}_{12}\mathbf{H}, & \operatorname{Cov}(\mathbf{Y}, \boldsymbol{\xi}) &= \boldsymbol{\Sigma}_{21}\mathbf{G}, \end{aligned}$$

   and the correlations follow by dividing row i by $\sqrt{\sigma_{ii}}$ (the variates have unit variance). For standardized variables these are the **canonical loadings** used to name the variates.
2. $\rho_1 \ge \max_{i,j}|\operatorname{Corr}(X_i, Y_j)|$, and more generally $\rho_1 \ge$ the multiple correlation of any single X_i with $\mathbf{Y}$ (and of any Y_j with $\mathbf{X}$).

Proof. (1) $\operatorname{Cov}(\mathbf{X}, \mathbf{G}^\top\mathbf{X}) = \operatorname{Cov}(\mathbf{X})\mathbf{G} = \boldsymbol{\Sigma}_{11}\mathbf{G}$ and $\operatorname{Cov}(\mathbf{X}, \mathbf{H}^\top\mathbf{Y}) = \operatorname{Cov}(\mathbf{X}, \mathbf{Y})\mathbf{H} = \boldsymbol{\Sigma}_{12}\mathbf{H}$; the other two are the same computation.

(2) ρ_1 is the maximum of $\operatorname{Corr}(\mathbf{g}^\top\mathbf{X}, \mathbf{h}^\top\mathbf{Y})$ over all $(\mathbf{g}, \mathbf{h})$. Taking $\mathbf{g} = \mathbf{e}_i$ and maximizing over $\mathbf{h}$ only gives the multiple correlation of X_i with $\mathbf{Y}$, a value attained in a smaller set; taking also $\mathbf{h} = \pm\mathbf{e}_j$ gives $|\operatorname{Corr}(X_i, Y_j)|$. ■

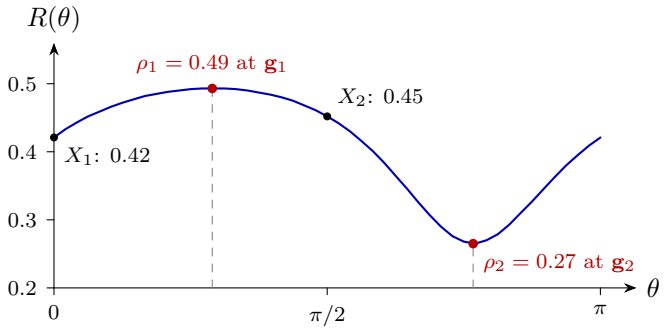


Figure. *A synthetic population with $p = q = 2$ (unit variances, within-set correlations 0.4 in $\mathbf{\Sigma}_{11}$ and 0.2 in $\mathbf{\Sigma}_{22}$, $\mathbf{\Sigma}_{12} = \left(\begin{smallmatrix} 0.30 & 0.35 \\ 0.05 & 0.45 \end{smallmatrix}\right)$). For $\mathbf{g}(\theta) = (\cos\theta, \sin\theta)^\top$, $R(\theta) = \max_{\mathbf{h}} \operatorname{Corr}(\mathbf{g}(\theta)^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y})$ is the multiple correlation of $\mathbf{g}(\theta)^\top \mathbf{X}$ with $\mathbf{Y}$. Its maximum is ρ_1, attained in the direction of $\mathbf{g}_1$, and for $p = 2$ its minimum is ρ_2, attained at $\mathbf{g}_2$. The single variables X_1 ($\theta = 0$) and X_2 ($\theta = \pi/2$) reach only 0.42 and 0.45, as Theorem 8.4(2) requires.*

3 Sample Canonical Correlations

Replace the covariances by $\mathbf{S}_{11} = \widehat{\operatorname{Cov}}(\mathbf{X})$, $\mathbf{S}_{22} = \widehat{\operatorname{Cov}}(\mathbf{Y})$, $\mathbf{S}_{12} = \widehat{\operatorname{Cov}}(\mathbf{X}, \mathbf{Y})$:

$$(\hat{\mathbf{g}}_k, \hat{\mathbf{h}}_k) = \arg \max_{\substack{\mathbf{g} \in \mathbb{R}^p,\, \mathbf{h} \in \mathbb{R}^q \\ \mathbf{g}^\top \mathbf{S}_{11} \hat{\mathbf{g}}_j = 0,\ \mathbf{h}^\top \mathbf{S}_{22} \hat{\mathbf{h}}_j = 0,\ j = 1, \dots, k-1}} \frac{\mathbf{g}^\top \mathbf{S}_{12} \mathbf{h}}{\sqrt{\mathbf{g}^\top \mathbf{S}_{11} \mathbf{g}\, \mathbf{h}^\top \mathbf{S}_{22} \mathbf{h}}}.$$

With $\mathbf{a} = \mathbf{S}_{11}^{1/2} \mathbf{g}$, $\mathbf{b} = \mathbf{S}_{22}^{1/2} \mathbf{h}$ this is maximizing $\dfrac{\mathbf{a}^\top \mathbf{S}_{11}^{-1/2} \mathbf{S}_{12} \mathbf{S}_{22}^{-1/2} \mathbf{b}}{\sqrt{\mathbf{a}^\top \mathbf{a}\, \mathbf{b}^\top \mathbf{b}}} = \max_{\|\mathbf{a}\| = \|\mathbf{b}\| = 1} \mathbf{a}^\top \mathbf{M} \mathbf{b} = \mathbf{a}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top \mathbf{b}$, so

$$\mathbf{a}_i = \mathbf{v}_i(\mathbf{S}_{11}^{-1/2} \mathbf{S}_{12} \mathbf{S}_{22}^{-1} \mathbf{S}_{21} \mathbf{S}_{11}^{-1/2}), \qquad \mathbf{b}_i = \mathbf{v}_i(\mathbf{S}_{22}^{-1/2} \mathbf{S}_{21} \mathbf{S}_{11}^{-1} \mathbf{S}_{12} \mathbf{S}_{22}^{-1/2}),$$

where $\mathbf{v}_i(\mathbf{M})$ is the ith eigenvector of $\mathbf{M}$.

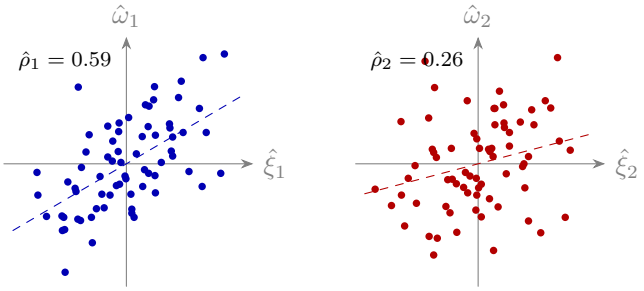


Figure. *Sample canonical variates for $n = 70$ draws from the population of the previous figure ($p = q = 2$). Each variate has sample variance 1, so the dashed least-squares line has slope $\hat{\rho}_k$. The first pair is clearly related; the second, uncorrelated with the first pair, carries the remaining weaker association.*

3.1 Regularized CCA

If p or q is large relative to n, $\mathbf{S}_{11}, \mathbf{S}_{22}$ are singular or unstable and sample canonical correlations are biased towards 1. **Regularized CCA** adds a ridge term,

$$\mathbf{S}_{11} \mapsto \mathbf{S}_{11} + \lambda_1 \mathbf{I}, \qquad \mathbf{S}_{22} \mapsto \mathbf{S}_{22} + \lambda_2 \mathbf{I},$$

with λ_1, λ_2 chosen by cross-validation.

4 Testing Canonical Correlations

Under multivariate normality the likelihood ratio statistic for $H_0 : \mathbf{\Sigma}_{12} = \mathbf{0}$ (all $\rho_k = 0$) is

$$\Lambda^{2/n} = \frac{|\mathbf{S}|}{|\mathbf{S}_{11}||\mathbf{S}_{22}|} = \prod_{k=1}^{\min(p,q)} (1 - \hat{\rho}_k^2),$$

and Bartlett's approximation gives $-(n-1-\frac{p+q+1}{2}) \sum_k \ln(1-\hat{\rho}_k^2) \dot{\sim} \chi^2_{pq}$. The hypothesis that only the first k are nonzero, $H_0^{(k)} : \rho_{k+1} = \cdots = 0$, is tested sequentially with

$$-\Big(n - 1 - \frac{p+q+1}{2}\Big) \sum_{i=k+1}^{\min(p,q)} \ln(1-\hat{\rho}_i^2) \;\dot{\sim}\; \chi^2_{(p-k)(q-k)}.$$

Derivation: the determinant identity

[Supplement] By the block determinant formula, $|\mathbf{S}| = |\mathbf{S}_{22}||\mathbf{S}_{11} - \mathbf{S}_{12}\mathbf{S}_{22}^{-1}\mathbf{S}_{21}| = |\mathbf{S}_{22}||\mathbf{S}_{11}|\,|\mathbf{I} - \hat{\mathbf{M}}\hat{\mathbf{M}}^\top|$, and the eigenvalues of $\hat{\mathbf{M}}\hat{\mathbf{M}}^\top$ are $\hat{\rho}_k^2$, so $|\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|) = \prod(1-\hat{\rho}_k^2)$.

5 Relations to PCA, Multiple Correlation and Discriminant Analysis

- **Multiple correlation**: CCA with $q = 1$.
- [Supplement] **Discriminant analysis**: with $\mathbf{Y}$ a set of group indicator (dummy) variables, the canonical variates of $\mathbf{X}$ give the same directions as Fisher's discriminants (Ch. 10).
- **PCA**: variance maximization within one set (eigendecomposition of $\mathbf{\Sigma}$) versus correlation maximization between two sets (SVD of $\mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1/2}$).

6 Selected exercises

Motivation. The exercises give a square-root-free way to compute canonical variates, two structures where the canonical correlation is found in closed form, and the likelihood ratio derivation of the test of $\mathbf{\Sigma}_{12} = \mathbf{0}$ quoted above.

Exercise 8.1 (Computing without square roots; J&W 10.4)

Let $\mathbf{M} = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1/2}$ as in Theorem 8.2 and $\mathbf{K} = \mathbf{\Sigma}_{11}^{-1}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}$.
(a) Show that if $\mathbf{M}\mathbf{M}^\top\mathbf{e} = \lambda\mathbf{e}$ with $\mathbf{e} \neq \mathbf{0}$, then $\mathbf{K}(\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e}) = \lambda\,\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e}$, and that $\mathbf{K}$ and $\mathbf{M}\mathbf{M}^\top$ have the same characteristic polynomial.
(b) Deduce that $\mathbf{K}\mathbf{g}_k = \rho_k^2\mathbf{g}_k$ and that $\mathbf{h}_k = \rho_k^{-1}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}\mathbf{g}_k$ for $k \le r$.

Solution.

(a) $\mathbf{K} = \mathbf{\Sigma}_{11}^{-1/2}(\mathbf{M}\mathbf{M}^\top)\mathbf{\Sigma}_{11}^{1/2}$, because $\mathbf{M}\mathbf{M}^\top = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}\mathbf{\Sigma}_{11}^{-1/2}$. Hence $\mathbf{K}\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e} = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{M}\mathbf{M}^\top\mathbf{e} = \lambda\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e}$, and $\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e} \neq \mathbf{0}$ since $\mathbf{\Sigma}_{11}^{-1/2}$ is invertible. Moreover

$$|\mathbf{K} - \lambda\mathbf{I}| = |\mathbf{\Sigma}_{11}^{-1/2}(\mathbf{M}\mathbf{M}^\top - \lambda\mathbf{I})\mathbf{\Sigma}_{11}^{1/2}| = |\mathbf{M}\mathbf{M}^\top - \lambda\mathbf{I}|,$$

so the eigenvalues of the non-symmetric $\mathbf{K}$ are real and equal to $\rho_1^2, \ldots, \rho_r^2$ and zeros.

(b)

Step 1. With $\mathbf{e} = \mathbf{a}_k$ and $\lambda = d_k^2 = \rho_k^2$ in (a), $\mathbf{K}\mathbf{g}_k = \rho_k^2 \mathbf{g}_k$ for $\mathbf{g}_k = \boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{a}_k$.

Step 2. From $\mathbf{M} = \sum_j d_j \mathbf{a}_j \mathbf{b}_j^\top$ and $\mathbf{a}_j^\top \mathbf{a}_k = \delta_{jk}$, $\mathbf{M}^\top \mathbf{a}_k = d_k \mathbf{b}_k$. Hence for $k \le r$

$$\mathbf{h}_k = \boldsymbol{\Sigma}_{22}^{-1/2}\mathbf{b}_k = \frac{1}{\rho_k}\boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{a}_k = \frac{1}{\rho_k}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\mathbf{g}_k.$$

So one solves the eigenproblem of $\mathbf{K}$, normalizes $\mathbf{g}_k^\top \boldsymbol{\Sigma}_{11}\mathbf{g}_k = 1$, and obtains $\mathbf{h}_k$ by regression of the $\mathbf{Y}$-set on ξ_k. ■

Exercise 8.2 (Equicorrelation between and within sets; J&W 10.7)

Let $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ be standardized with every correlation (within and between the sets) equal to $\rho \neq 0$:

$$\mathbf{R}_{11} = (1-\rho)\mathbf{I}_p + \rho \mathbf{1}_p \mathbf{1}_p^\top, \quad \mathbf{R}_{22} = (1-\rho)\mathbf{I}_q + \rho \mathbf{1}_q \mathbf{1}_q^\top, \quad \mathbf{R}_{12} = \rho \mathbf{1}_p \mathbf{1}_q^\top,$$

with the joint matrix positive definite. Find the canonical correlations and the first pair of canonical variates. Specialize to $p = q = 2$.

Solution.

Step 1 (action on 1). Let $\alpha_p = 1 + (p-1)\rho$. Then $\mathbf{R}_{11}\mathbf{1}_p = \alpha_p \mathbf{1}_p$ with $\alpha_p > 0$ (positive definiteness), so $\mathbf{1}_p$ is an eigenvector of $\mathbf{R}_{11}$ and, by Theorem 2.12, $\mathbf{R}_{11}^{-1/2}\mathbf{1}_p = \alpha_p^{-1/2}\mathbf{1}_p$. Likewise $\mathbf{R}_{22}^{-1/2}\mathbf{1}_q = \alpha_q^{-1/2}\mathbf{1}_q$.

Step 2 (the matrix M). Using symmetry of $\mathbf{R}_{22}^{-1/2}$,

$$\mathbf{M} = \mathbf{R}_{11}^{-1/2}\rho \mathbf{1}_p \mathbf{1}_q^\top \mathbf{R}_{22}^{-1/2} = \frac{\rho}{\sqrt{\alpha_p \alpha_q}} \mathbf{1}_p \mathbf{1}_q^\top = \frac{\rho\sqrt{pq}}{\sqrt{\alpha_p \alpha_q}}\, \mathbf{a}\mathbf{b}^\top, \qquad \mathbf{a} = \frac{\mathbf{1}_p}{\sqrt{p}},\ \mathbf{b} = \frac{\mathbf{1}_q}{\sqrt{q}}.$$

This is a rank-one SVD (replace $\mathbf{b}$ by $-\mathbf{b}$ if $\rho < 0$), so by Theorem 8.2

$$\rho_1 = |\rho| \sqrt{\frac{pq}{\{1+(p-1)\rho\}\{1+(q-1)\rho\}}}, \qquad \rho_2 = \cdots = 0.$$

Step 3 (variates). $\mathbf{g}_1 = \mathbf{R}_{11}^{-1/2}\mathbf{a} = \mathbf{1}_p/\sqrt{p\alpha_p}$, so $\xi_1 = \sum_i X_i/\sqrt{p\alpha_p}$, the standardized sum (indeed $\operatorname{Var}(\mathbf{1}_p^\top \mathbf{X}) = \mathbf{1}_p^\top \mathbf{R}_{11}\mathbf{1}_p = p\alpha_p$); similarly $\omega_1 = \operatorname{sign}(\rho)\sum_j Y_j/\sqrt{q\alpha_q}$. Check: $\operatorname{Corr}(\mathbf{1}^\top\mathbf{X}, \mathbf{1}^\top\mathbf{Y}) = \rho pq/\sqrt{p\alpha_p\, q\alpha_q}$, which is ρ_1 up to sign.

For $p = q = 2$: $\rho_1 = 2|\rho|/(1+\rho)$, $\xi_1 = (X_1 + X_2)/\sqrt{2(1+\rho)}$ and $\omega_1 = \operatorname{sign}(\rho)(Y_1 + Y_2)/\sqrt{2(1+\rho)}$. ■

Exercise 8.3 (Correlation for angular data; J&W 10.8)

An angle θ_2 is recorded as $\mathbf{Y} = (\cos\theta_2, \sin\theta_2)^\top$; assume $\operatorname{Cov}(\mathbf{Y}) \succ 0$.
(a) Show $\mathbf{b}^\top\mathbf{Y} = \sqrt{b_1^2 + b_2^2}\,\cos(\theta_2 - \beta)$, where $\cos\beta = b_1/\sqrt{b_1^2+b_2^2}$ and $\sin\beta = b_2/\sqrt{b_1^2+b_2^2}$.
(b) For a scalar X, show that the canonical correlation is $\rho_1 = \max_\beta \operatorname{Corr}\{X, \cos(\theta_2 - \beta)\}$.
(c) If also $\mathbf{X} = (\cos\theta_1, \sin\theta_1)^\top$, show $\rho_1 = \max_{\alpha,\beta} \operatorname{Corr}\{\cos(\theta_1 - \alpha), \cos(\theta_2 - \beta)\}$.

Solution.

(a) With $c = \sqrt{b_1^2 + b_2^2} > 0$, $\mathbf{b}^\top\mathbf{Y} = c(\cos\beta\cos\theta_2 + \sin\beta\sin\theta_2) = c\cos(\theta_2 - \beta)$.

(b)

Step 1. For scalar X and $a \neq 0$, $\mathrm{Corr}(aX, \mathbf{b}^\top \mathbf{Y}) = \mathrm{sign}(a)\mathrm{Corr}(X, \mathbf{b}^\top \mathbf{Y}) = \mathrm{Corr}(X, \mathrm{sign}(a)\mathbf{b}^\top \mathbf{Y})$, so $\rho_1 = \max_{\mathbf{b}\neq\mathbf{0}} \mathrm{Corr}(X, \mathbf{b}^\top \mathbf{Y})$.

Step 2. By (a) and scale invariance of correlation, $\mathrm{Corr}(X, \mathbf{b}^\top \mathbf{Y}) = \mathrm{Corr}\{X, \cos(\theta_2 - \beta)\}$; as $\mathbf{b}$ ranges over $\mathbb{R}^2 \setminus \{\mathbf{0}\}$, β ranges over $[0, 2\pi)$. Both variances are positive because $\mathrm{Cov}(\mathbf{Y}) \succ 0$.

Choosing the canonical variate therefore means choosing a new origin β for the angle.

(c) Apply (a) to both $\mathbf{a}^\top \mathbf{X}$ and $\mathbf{b}^\top \mathbf{Y}$ and use scale invariance; the maximum over $(\mathbf{a}, \mathbf{b})$ becomes a maximum over the two origins (α, β). ■

Exercise 8.4 (The likelihood ratio test of $\boldsymbol{\Sigma}_{12} = \mathbf{0}$)

For a sample of size $n > p + q$ from $N_{p+q}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, derive the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma}_{12} = \mathbf{0}$ and show $\Lambda^{2/n} = |\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|) = \prod_k (1 - \hat{\rho}_k^2)$, with pq degrees of freedom.

Solution [Supplement].

Step 1 (unrestricted). By Theorem 4.7, $\max L = (2\pi)^{-n(p+q)/2} |\mathbf{S}_0|^{-n/2} e^{-n(p+q)/2}$.

Step 2 (under H_0). If $\boldsymbol{\Sigma}_{12} = \mathbf{0}$, then $|\boldsymbol{\Sigma}| = |\boldsymbol{\Sigma}_{11}||\boldsymbol{\Sigma}_{22}|$ and $\boldsymbol{\Sigma}^{-1} = \mathrm{diag}(\boldsymbol{\Sigma}_{11}^{-1}, \boldsymbol{\Sigma}_{22}^{-1})$, so the density of each observation factorizes and $L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = L_1(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11}) L_2(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$, the likelihoods of the $\mathbf{X}$- and $\mathbf{Y}$-samples. The two factors have separate parameters; Theorem 4.7 applied to each gives

$$\max_{H_0} L = (2\pi)^{-n(p+q)/2} |\mathbf{S}_{0,11}|^{-n/2} |\mathbf{S}_{0,22}|^{-n/2} e^{-n(p+q)/2}.$$

Step 3 (the ratio). $\Lambda^{2/n} = |\mathbf{S}_0|/(|\mathbf{S}_{0,11}||\mathbf{S}_{0,22}|)$, and the factors $(\frac{n-1}{n})^{p+q}$ cancel, so $\Lambda^{2/n} = |\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|)$. The determinant identity above turns this into $\prod_k (1 - \hat{\rho}_k^2)$.

Step 4 (degrees of freedom). H_0 removes exactly the pq free entries of $\boldsymbol{\Sigma}_{12}$, so $-2 \log \Lambda \dot{\sim} \chi^2_{pq}$. For $q = 1$ the test is the test of zero multiple correlation; with blocks of size one, Exercise 7.1 is the analogous test for p blocks. ■

Chapter 9. Factor Analysis

Model, estimation (principal components, principal factors, MLE), rotation, factor scores

1 The Orthogonal Factor Model

Motivation. Factor analysis explains correlations by a few unobserved common factors: $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}$. Unlike PCA, it is a model with specific error variances, and the loadings are determined only up to rotation.

The covariance of p observed variables is explained by a few, k, unobserved common factors.

Definition 9.1: factor model

$$X_i = \mu_i + \sum_{\ell=1}^{k} \lambda_{i\ell} F_\ell + U_i \quad (i = 1, \dots, p), \qquad \mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F} + \mathbf{U},$$

where $\boldsymbol{\Lambda} = (\lambda_{i\ell})$ is the $p \times k$ **loading matrix**, $\mathbf{F} \in \mathbb{R}^k$ the common factors and $\mathbf{U} \in \mathbb{R}^p$ the specific factors, with

$$\mathrm{E}(\mathbf{F}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{F}) = \mathbf{I}_k, \quad \mathrm{E}(\mathbf{U}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{U}) = \boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \dots, \psi_p), \quad \mathrm{Cov}(\mathbf{F}, \mathbf{U}) = \mathbf{0}.$$

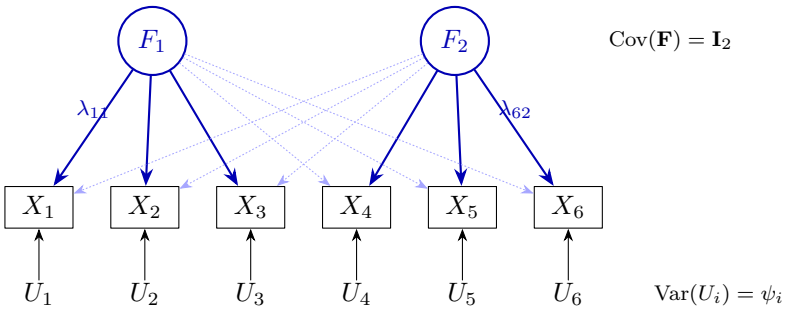


Figure. *Path diagram of a two-factor model for six variables. Every factor may load on every variable (dotted arrows carry small loadings in a simple structure), each variable has its own specific factor U_i, and the U_i are uncorrelated with each other and with $\mathbf{F}$. So any correlation between two X's must pass through the common factors:* $\mathrm{Cov}(X_i, X_j) = \sum_\ell \lambda_{i\ell}\lambda_{j\ell}$.

Theorem 9.2: covariance structure

$$\mathrm{E}(\mathbf{X}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}, \qquad \mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \boldsymbol{\Lambda},$$

$$\mathrm{Var}(X_i) = \underbrace{\sum_{\ell=1}^{k} \lambda_{i\ell}^2}_{h_i^2:\ \text{communality}} + \underbrace{\psi_i}_{\text{specific variance}}, \qquad \mathrm{Cov}(X_i, X_j) = \sum_{\ell} \lambda_{i\ell}\lambda_{j\ell} \ (i \neq j).$$

Proof. $\mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}) = \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F})\boldsymbol{\Lambda}^\top + \mathrm{Cov}(\mathbf{U}) + \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F}, \mathbf{U}) + \mathrm{Cov}(\mathbf{U}, \mathbf{F})\boldsymbol{\Lambda}^\top = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$, and $\mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}, \mathbf{F}) = \boldsymbol{\Lambda}$. ■

Because $\boldsymbol{\Psi}$ is diagonal, covariances between variables come *entirely* from the common factors. The model explains the $p(p+1)/2$ entries of $\boldsymbol{\Sigma}$ with $pk + p$ parameters.

1.1 Non-identifiability and rotation

For orthogonal $\mathbf{T}$ $(k \times k)$, $\mathbf{\Lambda}^* = \mathbf{\Lambda}\mathbf{T}$ and $\mathbf{F}^* = \mathbf{T}^\top \mathbf{F}$ give $\mathbf{\Lambda}^*\mathbf{F}^* = \mathbf{\Lambda}\mathbf{F}$, $\mathrm{Cov}\mathbf{F}^* = \mathbf{I}_k$ and $\mathbf{\Lambda}^*\mathbf{\Lambda}^{*\top} = \mathbf{\Lambda}\mathbf{\Lambda}^\top$: loadings are determined only up to rotation. If $\hat{\mathbf{\Lambda}}$ is an MLE, so is $\hat{\mathbf{\Lambda}}\mathbf{R}$ for any orthogonal $\mathbf{R}$. Removing the $k(k-1)/2$ rotational degrees of freedom leaves $pk + p - k(k-1)/2$ parameters and

$$\nu = \frac{p(p+1)}{2} - \left\{pk + p - \frac{k(k-1)}{2}\right\} = \frac{(p-k)^2 - p - k}{2},$$

which must be positive for the model to impose a real restriction.

1.2 Invariance to scale changes

If $Y_i = c_i X_i$, i.e. $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\mathbf{X}$, then $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\boldsymbol{\mu} + \mathrm{diag}(\mathbf{c})\mathbf{\Lambda}\mathbf{F} + \mathrm{diag}(\mathbf{c})\mathbf{U}$ is again a factor model with $\mathbf{\Lambda}_Y = \mathrm{diag}(\mathbf{c})\mathbf{\Lambda}$ and $\mathbf{\Psi}_Y = \mathrm{diag}(\mathbf{c})^2\mathbf{\Psi}$. For standardized variables ($\mathrm{Var}X_i = 1$),

$$\lambda_{i\ell} = \mathrm{Cov}(X_i, F_\ell) = \mathrm{Corr}(X_i, F_\ell), \qquad \mathrm{Corr}(Y_i, F_\ell) = \lambda_{i\ell},$$

so loadings are read as variable–factor correlations. In practice factor analysis is often run on the correlation matrix $\mathbf{R}$.

2 Estimation

2.1 Principal component method

From $\widehat{\mathrm{Corr}}(\mathbf{X}) = \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} = \sum_{j=1}^{p} l_j \mathbf{u}_j \mathbf{u}_j^\top \approx \sum_{j=1}^{k} l_j \mathbf{u}_j \mathbf{u}_j^\top$ (largest k eigenvalues),

$$\hat{\mathbf{\Lambda}} = [\sqrt{l_1}\mathbf{u}_1, \ldots, \sqrt{l_k}\mathbf{u}_k], \qquad \hat{\mathbf{\Psi}} = \mathrm{diag}(\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top).$$

The diagonal of $\mathbf{R}$ is reproduced exactly, and the sum of squared off-diagonal residuals of $\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top - \hat{\mathbf{\Psi}}$ is at most $l_{k+1}^2 + \cdots + l_p^2$. Factor ℓ explains a proportion l_ℓ/p of the total variance. Since eigenvalues do not change, increasing k leaves earlier loadings unchanged.

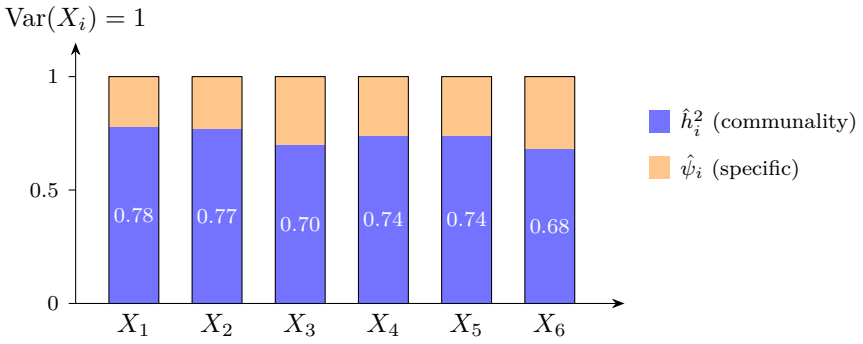


Figure. *Principal component solution with $k = 2$ for a synthetic correlation matrix ($n = 300$, $p = 6$; $l_1 = 2.43$, $l_2 = 1.96$, $l_3 = 0.48$). Each unit bar is $1 = \hat{h}_i^2 + \hat{\psi}_i$: the diagonal of $\mathbf{R}$ is reproduced exactly, and the two factors explain $(l_1 + l_2)/p = 73\%$ of the total standardized variance.*

Remark 9.3: number of factors with the principal component method

Heuristics are used when PCA is used to estimate the factor loadings:

1. use a **scree plot** or a **cumulative scree plot** of $l_1 \geq l_2 \geq \cdots \geq l_p$;

2. or use **Kaiser's rule**: retain factors $1, \dots, k$ satisfying

$$l_k > \bar{l} = \frac{1}{p}\sum_{i=1}^{p} l_i.$$

When correlation PCA is used, $\sum_i l_i = \mathrm{tr}\mathbf{R} = p$, so $\bar{l} = 1$ and the rule keeps factors with $l_k > 1$ (a factor must explain more than one standardized variable's worth of variance).

These are the same rules as for choosing the number of principal components (Ch. 7); the likelihood ratio test of §2.4 is available only under the normal MLE approach.

2.2 Principal factor method

Starting from initial communality estimates $\hat{h}_i^2$ (e.g. $1 - 1/r^{ii}$, r^{ii} the diagonal of $\mathbf{R}^{-1}$), form the reduced correlation matrix $\mathbf{R} - \hat{\boldsymbol{\Psi}}$ and iterate

1. $\mathbf{R} - \hat{\boldsymbol{\Psi}} \approx \hat{\boldsymbol{\Lambda}}_{(k)}\hat{\boldsymbol{\Lambda}}_{(k)}^{\top}$ (top-k eigendecomposition of the reduced matrix);
2. $\hat{\psi}_{(k)i} = 1 - \sum_{\ell=1}^{k} \hat{\lambda}_{(k)i\ell}^2$

until convergence.

2.3 Maximum likelihood

Assume $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Lambda}\boldsymbol{\Lambda}^{\top} + \boldsymbol{\Psi})$. Profiling $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, twice the negative log-likelihood is, up to constants and a factor n,

$$\ell(\hat{\boldsymbol{\mu}}, \boldsymbol{\Lambda}, \boldsymbol{\Psi}) = \log|\boldsymbol{\Lambda}\boldsymbol{\Lambda}^{\top} + \boldsymbol{\Psi}| + \mathrm{tr}\{\mathbf{S}_0(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^{\top} + \boldsymbol{\Psi})^{-1}\} + c.$$

For identifiability impose that $\boldsymbol{\Delta} = \boldsymbol{\Lambda}^{\top}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$ is diagonal (if $\hat{\boldsymbol{\Lambda}}$ is an MLE, so is $\hat{\boldsymbol{\Lambda}}\mathbf{R}$).

Theorem 9.4: (*) $\boldsymbol{\Lambda}$ given $\boldsymbol{\Psi}$

Given $\boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \dots, \psi_p)$, let $\mathbf{A} = \boldsymbol{\Psi}^{-1/2}\mathbf{S}_0\boldsymbol{\Psi}^{-1/2} = \sum_{i=1}^{p} d_i \mathbf{e}_i\mathbf{e}_i^{\top}$ ($d_1 \geq \cdots \geq d_p$) and $\delta_i = d_i - 1 > 0$ ($i \leq k$). Then

$$\hat{\boldsymbol{\Lambda}} = \boldsymbol{\Psi}^{1/2}[\mathbf{e}_1, \dots, \mathbf{e}_k]\mathrm{diag}(\delta_1^{1/2}, \dots, \delta_k^{1/2}).$$

Proof [Supplement]. Let $\boldsymbol{\Lambda}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}$ and $\boldsymbol{\Sigma}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Sigma}\boldsymbol{\Psi}^{-1/2} = \boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} + \mathbf{I}_p$. Then $\ell = \log|\boldsymbol{\Psi}| + \log|\boldsymbol{\Sigma}^*| + \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1})$. If $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top}$ has eigenvalues $\gamma_1 \geq \cdots \geq \gamma_k \geq 0 = \cdots = 0$, $\boldsymbol{\Sigma}^*$ has eigenvalues $1 + \gamma_j$ and 1. Von Neumann's lower bound (pair the small eigenvalues of $\boldsymbol{\Sigma}^{*-1}$ with the large ones of $\mathbf{A}$) gives

$$\log|\boldsymbol{\Sigma}^*| + \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1}) \geq \sum_{j=1}^{k}\left\{\log(1+\gamma_j) + \frac{d_j}{1+\gamma_j}\right\} + \sum_{j>k} d_j,$$

with equality when the eigenvectors of $\boldsymbol{\Sigma}^*$ coincide with the $\mathbf{e}_j$ in the same order. Each term $\log x + d/x = \log d + \{d/x - \log(d/x)\} \geq \log d + 1$ (Theorem 2.15) is minimized at $x = d$, so $1 + \gamma_j = d_j$, i.e. $\gamma_j = \delta_j$. Hence $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} = \sum_{j \leq k} \delta_j \mathbf{e}_j\mathbf{e}_j^{\top}$ and $\boldsymbol{\Lambda}^* = [\mathbf{e}_1, \dots, \mathbf{e}_k]\mathrm{diag}(\delta_j^{1/2})$ up to rotation; this choice makes $\boldsymbol{\Lambda}^{*\top}\boldsymbol{\Lambda}^* = \boldsymbol{\Delta}$ diagonal. ■

Theorem 9.5: (**) $\boldsymbol{\Psi}$ given $\boldsymbol{\Lambda}$

The MLE satisfies $\operatorname{diag}(\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}) = \operatorname{diag}(\mathbf{S}_0)$ (the model reproduces the sample variances). Accordingly, given $\boldsymbol{\Lambda}$, update

$$\hat{\boldsymbol{\Psi}} = \operatorname{diag}(\hat{\psi}_1, \dots, \hat{\psi}_p), \qquad \hat{\psi}_i = (\mathbf{S}_0 - \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)_{ii}.$$

Alternate the two steps. If some $\hat{\psi}_i \le 0$ (a Heywood case), bound it below by a small positive number.

Logic for (*) and (**): the likelihood equations

We evaluate $\dfrac{\partial \ell(\boldsymbol{\Lambda}, \boldsymbol{\Psi})}{\partial \boldsymbol{\Psi}} = 0$ and $\dfrac{\partial \ell(\boldsymbol{\Lambda}, \boldsymbol{\Psi})}{\partial \boldsymbol{\Lambda}} = 0$ for

$$\ell(\boldsymbol{\Lambda}, \boldsymbol{\Psi}) = \log|\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}| + \operatorname{tr}\{\mathbf{S}_0(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1}\}, \qquad \boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi},$$

using the facts that for a matrix $\mathbf{A} = \mathbf{A}(x)$,

$$\frac{\partial \mathbf{A}^{-1}}{\partial x} = -\mathbf{A}^{-1}\frac{\partial \mathbf{A}}{\partial x}\mathbf{A}^{-1}, \qquad \frac{\partial \log|\mathbf{A}|}{\partial x} = \operatorname{tr}\Big(\mathbf{A}^{-1}\frac{\partial \mathbf{A}}{\partial x}\Big)$$

(the second is the scalar form of Theorem 2.23; the first follows from differentiating $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$).

Derivative in ψ_i. With $\mathbf{e}_i$ the ith unit vector, $\frac{\partial}{\partial \psi_i}\operatorname{diag}(\psi_1, \dots, \psi_p) = \mathbf{e}_i\mathbf{e}_i^\top$, so $\frac{\partial \boldsymbol{\Sigma}}{\partial \psi_i} = \mathbf{e}_i\mathbf{e}_i^\top$ and

$$\begin{aligned}\frac{\partial \ell}{\partial \psi_i} &= \underbrace{\operatorname{tr}\Big(\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \psi_i}\Big)}_{\partial \log|\boldsymbol{\Sigma}|/\partial\psi_i} + \underbrace{\operatorname{tr}\Big(\mathbf{S}_0\frac{\partial \boldsymbol{\Sigma}^{-1}}{\partial \psi_i}\Big)}_{\partial \operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1})/\partial\psi_i} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{e}_i^\top) - \operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}) \\ &= \mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{e}_i - \mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{e}_i = \mathbf{e}_i^\top(\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1})\mathbf{e}_i = 0, \qquad i = 1, \dots, p.\end{aligned}$$

$$\textcircled{1} \qquad \operatorname{diag}(\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}) = \mathbf{0}.$$

Derivative in λ_{ij}. With $\boldsymbol{\Lambda} = [\mathbf{l}_1, \dots, \mathbf{l}_k]$ (columns), $\dfrac{\partial \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top}{\partial \lambda_{ij}} = \mathbf{l}_j\mathbf{e}_i^\top + \mathbf{e}_i\mathbf{l}_j^\top$. Then

$$\begin{aligned}\frac{\partial \ell}{\partial \lambda_{ij}} &= \operatorname{tr}\Big(\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top}{\partial \lambda_{ij}}\Big) - \operatorname{tr}\Big(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \lambda_{ij}}\boldsymbol{\Sigma}^{-1}\Big) \\ &= \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{l}_j\mathbf{e}_i^\top) + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{l}_j^\top) - \operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{l}_j\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}) - \operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{l}_j^\top\boldsymbol{\Sigma}^{-1}) \\ &= 2\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{l}_j - 2\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{l}_j = 0 \qquad (i = 1, \dots, p,\ j = 1, \dots, k),\end{aligned}$$

using symmetry of $\boldsymbol{\Sigma}^{-1}$ and $\boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}$ (each pair of traces is equal). Collecting over i, j:

$$\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} \overset{\text{premultiply } \boldsymbol{\Sigma}}{\Longrightarrow} \boldsymbol{\Lambda} = \mathbf{S}_0\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} = \mathbf{S}_0(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1}\boldsymbol{\Lambda} = \mathbf{S}_0\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda})^{-1},$$

where the last step uses $\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}) = (\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$, i.e. $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1}\boldsymbol{\Lambda} = \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda})^{-1}$.

Write $\boldsymbol{\Delta} \equiv \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$ (and require $\boldsymbol{\Delta}_{k\times k}$ to be diagonal). Then

$$\textcircled{2}' \qquad \boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Delta}) = \mathbf{S}_0\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}.$$

Premultiplying by $\boldsymbol{\Psi}^{-1/2}$:

$$\textcircled{2} \qquad \underbrace{\boldsymbol{\Psi}^{-1/2}\mathbf{S}_0\boldsymbol{\Psi}^{-1/2}}_{=:\mathbf{A}}\ \underbrace{\boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}}_{=:\tilde{\mathbf{E}}} = \underbrace{\boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}}_{\tilde{\mathbf{E}}}\ \underbrace{(\mathbf{I} + \boldsymbol{\Delta})}_{=:\mathbf{D}=\operatorname{diag}(d_1,\dots,d_k)}, \qquad \tilde{\mathbf{E}} = [\tilde{\mathbf{e}}_1, \dots, \tilde{\mathbf{e}}_k].$$

Hence $\mathbf{A}\tilde{\mathbf{e}}_i = \tilde{\mathbf{e}}_i d_i$, $i = 1, \dots, k$: $(\tilde{\mathbf{e}}_i, d_i)$ is an eigenpair of $\mathbf{A}$, although $\tilde{\mathbf{e}}_i^\top \tilde{\mathbf{e}}_i \neq 1$ in general. Since $\tilde{\mathbf{e}}_i^\top \tilde{\mathbf{e}}_i = (\tilde{\mathbf{E}}^\top \tilde{\mathbf{E}})_{ii} = (\mathbf{\Lambda}^\top \mathbf{\Psi}^{-1} \mathbf{\Lambda})_{ii} = \Delta_{ii}$, let $\mathbf{e}_i \equiv \tilde{\mathbf{e}}_i / \Delta_{ii}^{1/2}$ so that $\mathbf{e}_i^\top \mathbf{e}_i = 1$ and, for $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_k]$, $\mathbf{E}^\top \mathbf{E} = \mathbf{I}_k$ ($\mathbf{\Delta}$ diagonal). Then $(\mathbf{e}_i, d_i)$ is a unit eigenvector–eigenvalue pair of $\mathbf{A}$, and

$$\mathbf{D} = \mathbf{I} + \mathbf{\Delta}, \qquad \tilde{\mathbf{E}} = \mathbf{\Psi}^{-1/2} \mathbf{\Lambda} = \mathbf{E} \mathbf{\Delta}^{1/2}.$$

$(*)$ **$\mathbf{\Psi}$ is fixed.** From ②, $\mathbf{A} = \mathbf{\Psi}^{-1/2} \mathbf{S}_0 \mathbf{\Psi}^{-1/2}$ is computable, with eigenpairs $(\mathbf{e}_i(\mathbf{A}), d_i(\mathbf{A}))$. Use $\mathbf{D} = \mathbf{I} + \mathbf{\Delta}$ to set $\hat{\mathbf{\Delta}} = \operatorname{diag}(d_i(\mathbf{A})) - \mathbf{I}$, and $\mathbf{E}\mathbf{\Delta}^{1/2} = \mathbf{\Psi}^{-1/2}\mathbf{\Lambda}$ to set

$$\hat{\mathbf{\Lambda}} = \mathbf{\Psi}^{1/2} \mathbf{E}(\mathbf{A}) \hat{\mathbf{\Delta}}^{1/2}, \qquad \mathbf{E}(\mathbf{A}) = [\mathbf{e}_1(\mathbf{A}), \dots, \mathbf{e}_k(\mathbf{A})],$$

taking the k largest eigenvalues (the von Neumann argument above shows these give the minimum).

$(**)$ **$\mathbf{\Lambda}$ is fixed.** From ①, $\operatorname{diag}(\mathbf{\Sigma}^{-1} - \mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1}) = \mathbf{0}$.

Claim: if ②′ holds, then

$$\mathbf{\Sigma}^{-1} - \mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1} = \mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Sigma}^{-1} = \mathbf{\Psi}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1} = \mathbf{\Psi}^{-1}(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi})\mathbf{\Psi}^{-1}.$$

Proof of claim [Supplement]. $\mathbf{\Sigma}\mathbf{\Psi}^{-1}\mathbf{\Lambda} = (\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})\mathbf{\Psi}^{-1}\mathbf{\Lambda} = \mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta})$, and by ②′ also $\mathbf{S}_0\mathbf{\Psi}^{-1}\mathbf{\Lambda} = \mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta})$, so $(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1}\mathbf{\Lambda} = \mathbf{0}$. By Woodbury, $\mathbf{\Sigma}^{-1} = \mathbf{\Psi}^{-1} - \mathbf{\Psi}^{-1}\mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta})^{-1}\mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}$, hence

$$(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Sigma}^{-1} = (\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1} - \underbrace{(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1}\mathbf{\Lambda}}_{=\mathbf{0}}(\mathbf{I} + \mathbf{\Delta})^{-1}\mathbf{\Lambda}^\top\mathbf{\Psi}^{-1} = (\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1},$$

and transposing, $\mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0) = \mathbf{\Psi}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)$. Therefore $\mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Sigma}^{-1} = \mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1} = \mathbf{\Psi}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1}$.

Since $\mathbf{\Psi}^{-1}$ is diagonal, the ith diagonal entry of $\mathbf{\Psi}^{-1}(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi})\mathbf{\Psi}^{-1}$ is $(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi})_{ii}/\psi_i^2$, so ① becomes

$$\operatorname{diag}(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi}) = \mathbf{0} \quad \Longrightarrow \quad \psi_i = (\mathbf{S}_0 - \mathbf{\Lambda}\mathbf{\Lambda}^\top)_{ii},$$

which is the update $(**)$ and also shows $\operatorname{diag}(\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}}) = \operatorname{diag}(\mathbf{S}_0)$ at the MLE.

2.4 Likelihood ratio test for the number of factors

$H_0 : \mathbf{\Sigma} = \mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi}$ (k factors) versus H_1: unrestricted $\mathbf{\Sigma}$. With $L_1 = L(\bar{\mathbf{x}}, \mathbf{S}_0)$ and $L_0 = L(\bar{\mathbf{x}}, \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}})$,

$$W = -2 \log \frac{L_0}{L_1} = n \log\Big(\frac{|\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}}|}{|\mathbf{S}_0|}\Big) \;\dot{\sim}\; \chi^2_\nu,$$

$$\nu = \frac{p(p+1)}{2} - \Big\{pk + p - \frac{k(k-1)}{2}\Big\} = \frac{(p-k)^2 - p - k}{2}.$$

(At the MLE $\operatorname{tr}\{\mathbf{S}_0\hat{\mathbf{\Sigma}}^{-1}\} = p$, so the trace terms cancel.) With Bartlett's correction, reject the k-factor model if $(n - 1 - \frac{2p+4k+5}{6}) \ln \frac{|\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}}|}{|\mathbf{S}_0|} > \chi^2_\nu(\alpha)$.

3 Factor Rotation

Non-identifiability is used to find interpretable loadings, ideally a simple structure in which each variable loads heavily on only one factor. Rotate $\mathbf{\Lambda}^* = \mathbf{\Lambda}\mathbf{J}$ with $\mathbf{J}$ orthogonal.

Definition 9.6: varimax and quartimax

Let $\tilde{\lambda}^*_{i\ell} = \lambda^*_{i\ell}/h_i$ (normalized by the communality).

$$\text{Varimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{\ell=1}^{k} \widehat{\text{Var}}\big((\tilde{\lambda}^*_{1\ell})^2, \dots, (\tilde{\lambda}^*_{p\ell})^2\big)$$

— spread the squared loadings within each *column* (factor), so each factor has some large and some near-zero loadings.

$$\text{Quartimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{i=1}^{p} \widehat{\text{Var}}\big((\lambda^*_{i1})^2, \dots, (\lambda^*_{ik})^2\big)$$

— spread the squared loadings within each *row* (variable), so each variable concentrates on one factor.

Rotation leaves the communalities h_i^2 and $\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top$ unchanged.

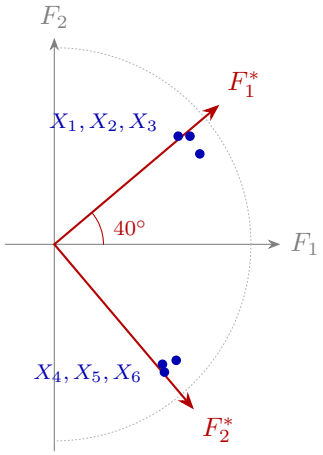


Figure. *Loadings of the six variables of the previous figure on the unrotated principal component factors (blue). Every variable loads about* 0.6 *on* F_1*, so* F_1 *looks like a general factor. Varimax rotates the axes by* 40° *(red); in the new coordinates the loadings are about* $(0.86, 0)$ *for variables 1–3 and* $(0, 0.85)$ *for variables 4–6, a simple structure. The dotted unit circle bounds the communalities, which rotation does not change.*

4 Factor Score Prediction

4.1 Weighted least squares (Bartlett) scores

(1) Choose $\hat{\mathbf{F}}$ to minimize the distance between $\mathbf{X}$ and $\boldsymbol{\mu} + \mathbf{\Lambda F}$: with the estimates fixed, regard $\mathbf{x}_i - \bar{\mathbf{x}} = \hat{\mathbf{\Lambda}}\mathbf{F}_i + \mathbf{U}_i$ as a heteroscedastic regression,

$$\hat{\mathbf{F}}_i^{\text{wls}} = \arg\min_{\mathbf{F}_i} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\mathbf{\Lambda}}\mathbf{F}_i)^\top \hat{\mathbf{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\mathbf{\Lambda}}\mathbf{F}_i) = (\hat{\mathbf{\Lambda}}^\top \hat{\mathbf{\Psi}}^{-1} \hat{\mathbf{\Lambda}})^{-1} \hat{\mathbf{\Lambda}}^\top \hat{\mathbf{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$$

(set the gradient of the quadratic form to zero).

4.2 Regression scores

(2) Since $\begin{pmatrix} \mathbf{X} \\ \mathbf{F} \end{pmatrix} \sim N_{p+k}\Big(\begin{pmatrix} \boldsymbol{\mu} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{\Lambda\Lambda}^\top + \mathbf{\Psi} & \mathbf{\Lambda} \\ \mathbf{\Lambda}^\top & \mathbf{I}_k \end{pmatrix} \Big)$, the conditional expectation (Theorem 3.12) gives

$$\hat{\mathbf{F}}_i^{\text{reg}} = \text{E}(\mathbf{F} \mid \mathbf{X} = \mathbf{x}_i) = \hat{\mathbf{\Lambda}}^\top (\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}})^{-1} (\mathbf{x}_i - \bar{\mathbf{x}}).$$

Theorem 9.7: relation between the two scores

With $\boldsymbol{\Delta} = \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$, $\hat{\mathbf{F}}^{\text{reg}} = (\mathbf{I}+\boldsymbol{\Delta})^{-1}\boldsymbol{\Delta}\,\hat{\mathbf{F}}^{\text{wls}} = (\mathbf{I}+\boldsymbol{\Delta}^{-1})^{-1}\hat{\mathbf{F}}^{\text{wls}}$. If $\boldsymbol{\Delta}$ is diagonal each component is shrunk by $\delta_\ell/(1+\delta_\ell) < 1$, so $|\hat{F}_\ell^{\text{wls}}| \geq |\hat{F}_\ell^{\text{reg}}|$.

Proof [Supplement]. By Woodbury, $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1} = \boldsymbol{\Psi}^{-1} - \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I}+\boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}$, so

$$\boldsymbol{\Lambda}^\top(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1} = \{\mathbf{I} - \boldsymbol{\Delta}(\mathbf{I}+\boldsymbol{\Delta})^{-1}\}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1} = (\mathbf{I}+\boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}$$

(as $\mathbf{I} - \boldsymbol{\Delta}(\mathbf{I}+\boldsymbol{\Delta})^{-1} = (\mathbf{I}+\boldsymbol{\Delta}-\boldsymbol{\Delta})(\mathbf{I}+\boldsymbol{\Delta})^{-1}$), while $\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}(\mathbf{x}-\bar{\mathbf{x}}) = \boldsymbol{\Delta}\hat{\mathbf{F}}^{\text{wls}}$. ■

Regression scores shrink towards 0, reflecting the prior $\mathbf{F} \sim N(\mathbf{0}, \mathbf{I})$ (a Bayes estimate); WLS scores treat $\mathbf{F}$ as a fixed parameter and are conditionally unbiased.

5 PCA versus Factor Analysis

[Supplement] The table collects the differences met in Ch. 7 and in this chapter.

	PCA	Factor analysis
Goal	linear combinations explaining total variance $\text{tr}\boldsymbol{\Sigma}$	latent factors explaining covariances (off-diagonal)
Model	none (descriptive transformation)	$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$
Direction	variables $\to$ components ($\mathbf{Z} = \mathbf{U}^\top\mathbf{X}$)	factors $\to$ variables ($\mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}$)
Uniqueness	unique if eigenvalues distinct	only up to rotation
Increasing k	earlier components unchanged	(MLE) earlier loadings change

6 Selected exercises

Motivation. The exercises prove two claims made in passing above, the residual bound of the principal component method and the simplification of the likelihood ratio statistic, and show by the smallest example why a factor model need not be identified.

Exercise 9.1 (Residuals of the principal component method; J&W 9.5)

Let $\mathbf{R} = \sum_{j=1}^p l_j \mathbf{u}_j\mathbf{u}_j^\top$ ($l_1 \geq \cdots \geq l_p$), $\hat{\boldsymbol{\Lambda}} = [\sqrt{l_1}\mathbf{u}_1, \ldots, \sqrt{l_k}\mathbf{u}_k]$, $\hat{\boldsymbol{\Psi}} = \text{diag}(\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top)$ and $\mathbf{E} = \mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top - \hat{\boldsymbol{\Psi}}$. Show

$$\sum_{i,j} E_{ij}^2 \leq l_{k+1}^2 + \cdots + l_p^2.$$

Solution.

Step 1 (zero diagonal). By the definition of $\hat{\boldsymbol{\Psi}}$, $E_{ii} = 0$, and $E_{ij} = (\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top)_{ij}$ for $i \neq j$. Hence

$$\|\mathbf{E}\|_F^2 = \sum_{i\neq j}(\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top)_{ij}^2 \leq \|\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top\|_F^2.$$

Step 2 (the discarded part). $\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top = \sum_{j\leq k} l_j\mathbf{u}_j\mathbf{u}_j^\top$, so $\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top = \mathbf{P}_2\mathbf{L}_2\mathbf{P}_2^\top$ with $\mathbf{P}_2 = [\mathbf{u}_{k+1}, \ldots, \mathbf{u}_p]$, $\mathbf{P}_2^\top\mathbf{P}_2 = \mathbf{I}$ and $\mathbf{L}_2 = \text{diag}(l_{k+1}, \ldots, l_p)$.

Step 3. By Theorem 2.3(ii)–(iii), $\|\mathbf{P}_2\mathbf{L}_2\mathbf{P}_2^\top\|_F^2 = \text{tr}(\mathbf{P}_2\mathbf{L}_2\mathbf{P}_2^\top\mathbf{P}_2\mathbf{L}_2\mathbf{P}_2^\top) = \text{tr}(\mathbf{L}_2^2\mathbf{P}_2^\top\mathbf{P}_2) = \sum_{j>k} l_j^2$.

The same argument applies with $\mathbf{S}$ in place of $\mathbf{R}$. ■

Exercise 9.2 (Non-uniqueness for $p = 2$, $k = 1$; J&W 9.7)

For $p = 2$ and $k = 1$ the model reads $\sigma_{11} = \lambda_1^2 + \psi_1$, $\sigma_{12} = \lambda_1\lambda_2$, $\sigma_{22} = \lambda_2^2 + \psi_2$. Given $\mathbf{\Sigma} \succ 0$ with $\sigma_{12} \neq 0$, find all solutions with $\psi_1, \psi_2 \geq 0$ and show that there are infinitely many. Relate this to the sign of ν.

Solution.

Step 1. $\sigma_{12} \neq 0$ forces $\lambda_1 = t \neq 0$ and $\lambda_2 = \sigma_{12}/t$; then $\psi_1 = \sigma_{11} - t^2$ and $\psi_2 = \sigma_{22} - \sigma_{12}^2/t^2$ are determined.

Step 2. $\psi_1 \geq 0$ and $\psi_2 \geq 0$ iff $\sigma_{12}^2/\sigma_{22} \leq t^2 \leq \sigma_{11}$. Since $\mathbf{\Sigma} \succ 0$, $\sigma_{12}^2 < \sigma_{11}\sigma_{22}$, so this is a non-degenerate interval for t^2, and every t in it (with either sign) gives a valid $(\mathbf{\Lambda}, \mathbf{\Psi})$.

Step 3. For $k = 1$ the only orthogonal $\mathbf{T}$ is ± 1, so rotation explains just the sign of t; the continuum of solutions is a genuine lack of identification. This matches $\nu = \{(p-k)^2 - p - k\}/2 = (1-3)/2 = -1 < 0$: three equations for four parameters. ■

Exercise 9.3 (Simplifying the likelihood ratio statistic)

Let $(\hat{\mathbf{\Lambda}}, \hat{\mathbf{\Psi}})$, $\hat{\mathbf{\Psi}} \succ 0$, solve the likelihood equations ① and ②′, and $\hat{\mathbf{\Sigma}} = \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}}$. Show $\operatorname{tr}(\mathbf{S}_0\hat{\mathbf{\Sigma}}^{-1}) = p$, and deduce $W = -2\log(L_0/L_1) = n\log(|\hat{\mathbf{\Sigma}}|/|\mathbf{S}_0|)$.

Solution [Supplement].

Step 1. The proof of the Claim above shows that ②′ implies $\hat{\mathbf{\Sigma}}^{-1}(\hat{\mathbf{\Sigma}} - \mathbf{S}_0) = \hat{\mathbf{\Psi}}^{-1}(\hat{\mathbf{\Sigma}} - \mathbf{S}_0)$, and ① then gives $\operatorname{diag}(\hat{\mathbf{\Sigma}}) = \operatorname{diag}(\mathbf{S}_0)$.

Step 2. Since $\hat{\mathbf{\Psi}}^{-1}$ is diagonal,

$$p - \operatorname{tr}(\hat{\mathbf{\Sigma}}^{-1}\mathbf{S}_0) = \operatorname{tr}\{\hat{\mathbf{\Sigma}}^{-1}(\hat{\mathbf{\Sigma}} - \mathbf{S}_0)\} = \operatorname{tr}\{\hat{\mathbf{\Psi}}^{-1}(\hat{\mathbf{\Sigma}} - \mathbf{S}_0)\} = \sum_{i=1}^{p} \frac{(\hat{\mathbf{\Sigma}} - \mathbf{S}_0)_{ii}}{\hat{\psi}_i} = 0.$$

Step 3. With $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, $L(\bar{\mathbf{x}}, \mathbf{\Sigma}) = (2\pi)^{-np/2}|\mathbf{\Sigma}|^{-n/2}\exp\{-\tfrac{n}{2}\operatorname{tr}(\mathbf{\Sigma}^{-1}\mathbf{S}_0)\}$, so

$$-2\log\frac{L_0}{L_1} = n\log\frac{|\hat{\mathbf{\Sigma}}|}{|\mathbf{S}_0|} + n\{\operatorname{tr}(\hat{\mathbf{\Sigma}}^{-1}\mathbf{S}_0) - \operatorname{tr}(\mathbf{S}_0^{-1}\mathbf{S}_0)\} = n\log\frac{|\hat{\mathbf{\Sigma}}|}{|\mathbf{S}_0|},$$

because both traces equal p. As $L_1 \geq L_0$, also $|\hat{\mathbf{\Sigma}}| \geq |\mathbf{S}_0|$. ■

Chapter 10. Discrimination and Classification

Bayes rule, LDA/QDA, Fisher's discriminant, logistic regression, assessing classifiers

1 The Bayes Classification Rule

Motivation. Classification is a decision problem: assign a new observation to a group so as to minimize expected misclassification cost. The Bayes rule compares prior-weighted densities and is optimal; every practical classifier estimates it.

1.1 Setting

Let $Y \in \{1, \dots, K\}$ be the group and $\mathbf{X} \in \mathbb{R}^p$ the features, with priors $P(Y = k) = \pi_k$ and $\mathbf{X} \mid Y = k \sim f_k$. For $K = 2$: $\mathbf{X} \mid Y = 1 \sim f_1(\mathbf{x})$, $\mathbf{X} \mid Y = 2 \sim f_2(\mathbf{x})$, $P(Y = 1) = \pi_1$, and

$$\eta(\mathbf{x}) := P(Y = 1 \mid \mathbf{X} = \mathbf{x}) = \frac{f_1(\mathbf{x})\pi_1}{f_1(\mathbf{x})\pi_1 + f_2(\mathbf{x})\pi_2}.$$

The misclassification probability of a classifier δ (or ϕ) : $\mathbb{R}^p \to \{1, \dots, K\}$ is $P(\phi(\mathbf{X}) \neq Y) = \sum_k P(\phi(\mathbf{X}) \neq k \mid Y = k) P(Y = k)$.

Theorem 10.1: optimality of the Bayes rule

$\phi_{\text{Bayes}}(\mathbf{x}) = \arg\max_k P(Y = k \mid \mathbf{X} = \mathbf{x}) = \arg\max_k \pi_k f_k(\mathbf{x})$ minimizes the misclassification probability: $P(\phi_{\text{Bayes}}(\mathbf{X}) \neq Y) \leq P(\phi(\mathbf{X}) \neq Y)$ for every ϕ.

$K = 2$.

$$\begin{aligned}
\text{Risk}(\delta) &= \mathrm{E}\big[\mathbf{1}_{\{Y \neq \delta(\mathbf{X})\}}\big] = \mathrm{E}_Y\Big[\mathrm{E}_{\mathbf{X}|Y}\big[\mathbf{1}_{\{Y \neq \delta(\mathbf{X})\}} \mid Y\big]\Big] \\
&= \pi_1 \mathrm{E}_{\mathbf{X}|Y=1}\big[\mathbf{1}_{\{\delta(\mathbf{X}) \neq 1\}} \mid Y = 1\big] + \pi_2 \mathrm{E}_{\mathbf{X}|Y=2}\big[\mathbf{1}_{\{\delta(\mathbf{X}) \neq 2\}} \mid Y = 2\big] \\
&= \int \underbrace{\Big[\pi_1 \mathbf{1}_{\{\delta(\mathbf{x}) \neq 1\}} f_1(\mathbf{x}) + \pi_2 \mathbf{1}_{\{\delta(\mathbf{x}) \neq 2\}} f_2(\mathbf{x})\Big]}_{=: h(\mathbf{x})} d\mathbf{x}.
\end{aligned}$$

At each $\mathbf{x}$, $\delta(\mathbf{x}) = 1$ gives $h(\mathbf{x}) = \pi_2 f_2(\mathbf{x})$ and $\delta(\mathbf{x}) = 2$ gives $h(\mathbf{x}) = \pi_1 f_1(\mathbf{x})$. To minimize the integral choose the smaller pointwise: if $\pi_2 f_2(\mathbf{x}) < \pi_1 f_1(\mathbf{x})$ choose $\delta(\mathbf{x}) = 1$; if $>$, choose $\delta(\mathbf{x}) = 2$. Thus $\phi_{\text{Bayes}}(\mathbf{x}) = \arg\max_k P(Y = k \mid \mathbf{X} = \mathbf{x})$. For general K the same argument minimizes $1 - P(Y = \phi(\mathbf{x}) \mid \mathbf{x})$ pointwise. ∎

Remark 10.2: misclassification costs

If $c(2 \mid 1)$ is the cost of classifying a member of group 1 into group 2, the rule minimizing expected cost classifies into group 1 when $\dfrac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \dfrac{c(1 \mid 2)}{c(2 \mid 1)} \dfrac{\pi_2}{\pi_1}$.

2 Normal Populations: LDA and QDA

2.1 Common covariance: linear discriminant analysis

If $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$ ($\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma}$), then $\log\{\pi_k f_k(\mathbf{x})\} = -\frac{1}{2}\mathbf{x}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \log \pi_k +$ const, and the first term does not depend on k, so

$$\phi(\mathbf{x}) = \arg\min_k \delta_k(\mathbf{x}), \qquad \delta_k(\mathbf{x}) = b_{0k} - \mathbf{b}_k^\top \mathbf{x}, \quad b_{0k} = \tfrac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k - \log(\pi_k), \quad \mathbf{b}_k = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k.$$

The discriminant function is linear in $\mathbf{x}$: **linear discriminant analysis (LDA)**. For $K = 2$, $\delta_1 < \delta_2$ rearranges to

$$\phi(\mathbf{x}) = 1 \;\text{ if }\; \mathbf{b}^\top\Big(\mathbf{x} - \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2}{2}\Big) < \log\Big(\frac{\pi_1}{\pi_2}\Big), \qquad \mathbf{b} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1).$$

$K = 2$. $\delta_1 < \delta_2 \iff (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_1^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2) > \log(\pi_2/\pi_1)$. Since $\boldsymbol{\mu}_1^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)$, this is $-\mathbf{b}^\top(\mathbf{x} - \frac{\boldsymbol{\mu}_1+\boldsymbol{\mu}_2}{2}) > \log(\pi_2/\pi_1)$. ■

The sample version plugs in $\boldsymbol{\mu}_k \to \bar{\mathbf{x}}_k$, $\boldsymbol{\Sigma} \to \mathbf{S}_p$ (pooled covariance), $\pi_k \to n_k/n$: $\hat{\phi}(\mathbf{x}) = \arg\min_k(\hat{b}_{0k} - \hat{\mathbf{b}}_k^\top\mathbf{x})$ with $\hat{\mathbf{b}}_k = \mathbf{S}_p^{-1}\bar{\mathbf{x}}_k$, $\hat{b}_{0k} = \frac{1}{2}\bar{\mathbf{x}}_k^\top\mathbf{S}_p^{-1}\bar{\mathbf{x}}_k - \log(n_k/n)$, and for $K = 2$, $\hat{\mathbf{b}} = \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$ with the rule "$\hat{\mathbf{b}}^\top(\mathbf{x} - \frac{\bar{\mathbf{x}}_1+\bar{\mathbf{x}}_2}{2}) < \log(n_1/n_2) \Rightarrow 1$".

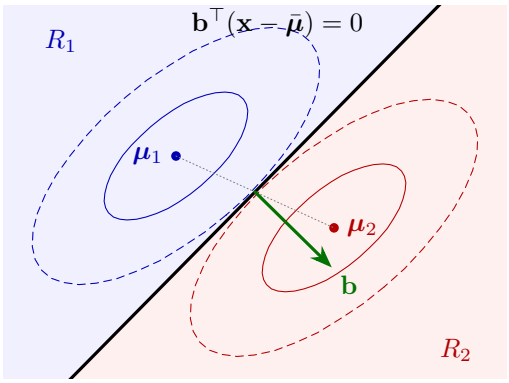


Figure. *LDA for two synthetic normal populations with common* $\boldsymbol{\Sigma} = \left(\begin{smallmatrix} 1 & 0.6 \\ 0.6 & 0.8 \end{smallmatrix}\right)$ *and equal priors; solid and dashed ellipses contain* 39% *and* 86% *of each population. The boundary is the line through* $\bar{\boldsymbol{\mu}} = (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)/2$ *orthogonal to Fisher's direction* $\mathbf{b} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)$ *(green), which is* not *the direction of* $\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1$ *(dotted) because* $\boldsymbol{\Sigma}$ *tilts the populations.*

Theorem 10.3: error rates of LDA

Let $K = 2$, $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$, $\Delta^2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ and $c = \log(\pi_2/\pi_1)$. The Bayes rule has

$$P(2 \mid 1) = \Phi\Big(\frac{c - \Delta^2/2}{\Delta}\Big), \qquad P(1 \mid 2) = \Phi\Big(\frac{-c - \Delta^2/2}{\Delta}\Big),$$

so with equal priors the optimal error rate is $\mathrm{OER} = \Phi(-\Delta/2)$, a decreasing function of the Mahalanobis distance between the populations.

Proof. By the proof above the rule assigns $\mathbf{x}$ to population 1 iff $W(\mathbf{x}) := (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \frac{\boldsymbol{\mu}_1+\boldsymbol{\mu}_2}{2}) > c$. W is linear in $\mathbf{X}$, so under population k it is normal with variance $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) = \Delta^2$ and mean

$$\mathrm{E}_1 W = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}\tfrac{\boldsymbol{\mu}_1-\boldsymbol{\mu}_2}{2} = \tfrac{\Delta^2}{2}, \qquad \mathrm{E}_2 W = -\tfrac{\Delta^2}{2}.$$

Hence $P(2 \mid 1) = P_1(W \le c) = \Phi\{(c - \Delta^2/2)/\Delta\}$ and $P(1 \mid 2) = P_2(W > c) = 1 - \Phi\{(c + \Delta^2/2)/\Delta\} = \Phi\{(-c - \Delta^2/2)/\Delta\}$. For $c = 0$ both equal $\Phi(-\Delta/2)$, and so does $\frac{1}{2}P(2 \mid 1) + \frac{1}{2}P(1 \mid 2)$. ■

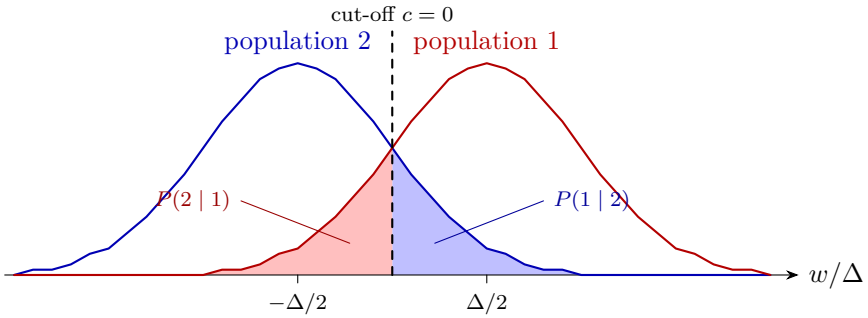


Figure. *Densities of the standardized score W/Δ under the two populations, $N(\pm\Delta/2, 1)$, drawn for $\Delta = 2$ and equal priors. The shaded tails are the two misclassification probabilities, each $\Phi(-\Delta/2) = 0.159$. A prior or cost ratio moves the cut-off c and trades one tail against the other; larger Δ pulls the curves apart and shrinks both.*

2.2 Different covariances: quadratic discriminant analysis

If $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$,

$$\phi(\mathbf{x}) = \arg\max_k \Big\{ -\tfrac{1}{2}\log|\boldsymbol{\Sigma}_k| - \tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\mathbf{x}-\boldsymbol{\mu}_k) + \log\pi_k \Big\} = \arg\max_k \left(\mathbf{x}^\top \mathbf{C}_k \mathbf{x} + \mathbf{b}_k^\top \mathbf{x} - b_{0k}\right)$$

with $\mathbf{C}_k = -\frac{1}{2}\boldsymbol{\Sigma}_k^{-1}$, $\mathbf{b}_k = \boldsymbol{\Sigma}_k^{-1}\boldsymbol{\mu}_k$ and $b_{0k} = \frac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}_k^{-1}\boldsymbol{\mu}_k + \frac{1}{2}\log|\boldsymbol{\Sigma}_k| - \log\pi_k$ (expand the quadratic form). The quadratic terms no longer cancel, so the boundaries are quadratic surfaces: **QDA**. It needs $(K-1)\,p(p+1)/2$ more covariance parameters than LDA, so with small samples LDA is often more stable (bias–variance trade-off).

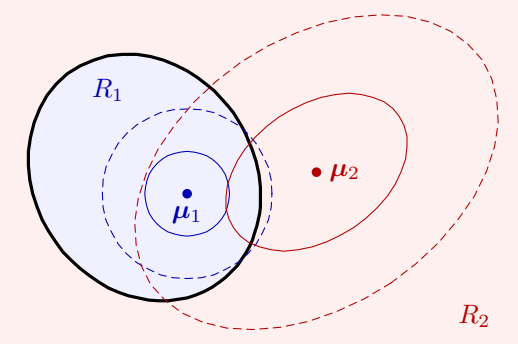


Figure. *QDA for two synthetic normal populations with equal priors, a tight spherical $\boldsymbol{\Sigma}_1 = 0.35\,\mathbf{I}$ (blue) and a wide $\boldsymbol{\Sigma}_2 = \left(\begin{smallmatrix} 1.6 & 0.5 \\ 0.5 & 1.2 \end{smallmatrix}\right)$ (red); ellipses as in the previous figure. The boundary $\{\mathbf{x} : \pi_1 f_1(\mathbf{x}) = \pi_2 f_2(\mathbf{x})\}$ is a closed conic: R_1 is a bounded region around $\boldsymbol{\mu}_1$, and points far away in* every *direction, even far to the left of $\boldsymbol{\mu}_1$, go to the more spread-out population 2.*

3 Fisher's Linear Discriminant

Motivation. Fisher's approach needs no distributional assumption: find the projection that best separates the groups relative to the within-group spread. Under normality with equal covariances it coincides with the Bayes rule.

Without distributional assumptions, find the projection that best separates the groups.

Definition 10.4: Fisher's LDA direction ($K = 2$)

$$\mathbf{u}_0 = \arg\max_{\mathbf{u}\in\mathbb{R}^p} \frac{(\mathbf{u}^\top \bar{\mathbf{x}}_1 - \mathbf{u}^\top \bar{\mathbf{x}}_2)^2}{\mathbf{u}^\top \mathbf{S}_p \mathbf{u}} :$$

squared difference of projected means over projected within-group variance.

Theorem 10.5: solution

$\mathbf{u}_0 \propto \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$ (equivalently $\mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$), and the maximum is $\mathbf{d}^\top \mathbf{S}_p^{-1}\mathbf{d}$ with $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$, the squared sample Mahalanobis distance.

Proof. In general, to maximize $\frac{\mathbf{u}^\top \mathbf{A}\mathbf{u}}{\mathbf{u}^\top \mathbf{B}\mathbf{u}}$ ($\mathbf{B} \succ 0$) use $L(\mathbf{u}, \lambda) = \mathbf{u}^\top \mathbf{A}\mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{B}\mathbf{u} - 1)$: $\partial L/\partial \mathbf{u} = \mathbf{0} \Rightarrow \mathbf{A}\mathbf{u} = \lambda \mathbf{B}\mathbf{u} \Rightarrow \mathbf{B}^{-1}\mathbf{A}\mathbf{u} = \lambda \mathbf{u}$, so the top eigenvector of $\mathbf{B}^{-1}\mathbf{A}$ is optimal. With $\mathbf{A} = \mathbf{d}\mathbf{d}^\top$ (rank 1) and $\mathbf{B} = \mathbf{S}_p$, $\mathbf{S}_p^{-1}\mathbf{d}\mathbf{d}^\top$ has a single nonzero eigenvalue $\mathbf{d}^\top \mathbf{S}_p^{-1}\mathbf{d}$ with unique eigenvector $\mathbf{S}_p^{-1}\mathbf{d}$ (rank 1). Directly: with $\mathbf{v} = \mathbf{S}_p^{1/2}\mathbf{u}$, Cauchy–Schwarz gives $(\mathbf{u}^\top \mathbf{d})^2/\mathbf{u}^\top \mathbf{S}_p \mathbf{u} = (\mathbf{v}^\top \mathbf{S}_p^{-1/2}\mathbf{d})^2/\mathbf{v}^\top \mathbf{v} \le \mathbf{d}^\top \mathbf{S}_p^{-1}\mathbf{d}$. ∎

Remark 10.6: Fisher's rule = LDA with equal priors

With the discriminant score $y_i = \mathbf{b}_Y^\top \mathbf{x}_i$, $\mathbf{b}_Y = \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$, and the cut-off $b_{0Y} = \mathbf{b}_Y^\top(\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2)/2$,

$$\text{“}\mathbf{b}_Y^\top \mathbf{y} - b_{0Y} < 0 \Rightarrow 1\text{”} \iff \text{“}[\mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)]^\top \Big(\mathbf{x} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}\Big) < 0 \Rightarrow 1\text{”},$$

which is sample LDA with $\pi_1 = \pi_2$. A rule derived without normality coincides with the Bayes rule under normality.

Several groups ($K \ge 3$): $\max_{\mathbf{u}} \frac{\mathbf{u}^\top \mathbf{B}\mathbf{u}}{\mathbf{u}^\top \mathbf{W}\mathbf{u}}$ ($\mathbf{B}, \mathbf{W}$ the MANOVA between/within SSP matrices) is solved by the eigenvectors of $\mathbf{W}^{-1}\mathbf{B}$; there are at most $\min(p, K-1)$ nonzero eigenvalues, the same η_i that appear in the Wilks/Pillai statistics of Ch. 6.

4 Logistic Regression

For $K = 2$, $Y \in \{0, 1\}$, model the posterior directly:

$$\operatorname{logit}(\eta(\mathbf{x})) := \log\Big(\frac{\eta(\mathbf{x})}{1 - \eta(\mathbf{x})}\Big) = b + \boldsymbol{\beta}^\top \mathbf{x}, \qquad \eta(\mathbf{x}) = P(Y = 1 \mid \mathbf{X} = \mathbf{x}).$$

Classification: “$\eta(\mathbf{x}) > \frac{1}{2} \Rightarrow 1$” $\iff$ “$f(\mathbf{x}) = b + \boldsymbol{\beta}^\top \mathbf{x} > 0 \Rightarrow 1$” (a linear boundary). The log-likelihood is

$$\begin{aligned}\ell(b, \boldsymbol{\beta}) &= \sum_{i=1}^{n} \big[y_i \log(\eta(\mathbf{x}_i; b, \boldsymbol{\beta})) + (1 - y_i)\log\{1 - \eta(\mathbf{x}_i; b, \boldsymbol{\beta})\}\big] \\ &= \sum_{i=1}^{n} \big[y_i(b + \boldsymbol{\beta}^\top \mathbf{x}_i) - \log\{1 + \exp(b + \boldsymbol{\beta}^\top \mathbf{x}_i)\}\big],\end{aligned}$$

which is concave and maximized by Newton–Raphson (IRLS). In the equal-covariance normal model $\log \frac{\pi_1 f_1}{\pi_0 f_0}$ is exactly linear in $\mathbf{x}$, so the logistic model holds; LDA models the joint distribution, logistic regression only the conditional one.

Derivation: the second equality and the derivatives

[Supplement] $\log\eta = (b+\boldsymbol{\beta}^\top\mathbf{x}) - \log(1+e^{b+\boldsymbol{\beta}^\top\mathbf{x}})$ and $\log(1-\eta) = -\log(1+e^{b+\boldsymbol{\beta}^\top\mathbf{x}})$, so $y\log\eta+(1-y)\log(1-\eta) = y(b+\boldsymbol{\beta}^\top\mathbf{x})-\log(1+e^{b+\boldsymbol{\beta}^\top\mathbf{x}})$. The gradient is $\sum_i(y_i-\eta_i)(1,\mathbf{x}_i^\top)^\top$ and the Hessian $-\sum_i\eta_i(1-\eta_i)(1,\mathbf{x}_i^\top)^\top(1,\mathbf{x}_i^\top)\preceq 0$.

5 Simple Classifiers in High Dimensions

If $p > n$, $\mathbf{S}_p$ is singular and LDA cannot be used (e.g. the Golub data).

- **Nearest centroid**: LDA with $\boldsymbol{\Sigma}=\mathbf{I}$. $\phi(\mathbf{x})=1$ if $\mathbf{b}^\top(\mathbf{x}-\frac{\bar{\mathbf{x}}_1+\bar{\mathbf{x}}_2}{2})<0$, $\mathbf{b}=\bar{\mathbf{x}}_2-\bar{\mathbf{x}}_1$ (assign to the closer mean).
- **Naive Bayes classifier (diagonal LDA)**: assume independent variables and replace $\mathbf{S}_p$ by $\mathbf{D}_p=\operatorname{diag}(\mathbf{S}_p)$: $\phi(\mathbf{x})=1$ if $\mathbf{b}^\top(\mathbf{x}-\frac{\bar{\mathbf{x}}_1+\bar{\mathbf{x}}_2}{2})<0$, $\mathbf{b}=\mathbf{D}_p^{-1}(\bar{\mathbf{x}}_2-\bar{\mathbf{x}}_1)$.
- **k-nearest neighbors**: majority vote among the k training points closest to $\mathbf{x}$; non-parametric, high variance for small k and high bias for large k.

6 Assessment of Classifiers

- True error rate $P(\phi(\mathbf{X})\neq Y)=\sum_{k=1}^K P(\phi(\mathbf{X})\neq k\mid Y=k)P(Y=k)$; ϕ is optimal if $P(\phi(\mathbf{X})\neq Y)\le P(\varphi(\mathbf{X})\neq Y)$ for all φ.
- Empirical misclassification rate $\hat{P}(\hat{\phi}_D(\mathbf{X})\neq Y\mid D)=\frac{1}{M}\sum_{i=1}^M \mathbf{1}_{\{\hat{\phi}(\mathbf{x}_i)\neq y_i\}}$.
- **Training and test sets**: build $\hat{\phi}$ from $\mathbf{X}_{\text{tr}}$ and estimate the misclassification rate on $\mathbf{X}_{\text{test}}$. The training error (APER) is optimistically biased.
- **Confusion matrix**: actual group by predicted group (training and testing confusion matrices).
- **K-fold cross-validation**: split into K parts, use each in turn as the test set and average. $K\uparrow\Rightarrow$ bias $\downarrow$, variance $\uparrow$; $K\downarrow\Rightarrow$ the reverse. $K=n$ is LOOCV (Lachenbruch's holdout).

7 Selected exercises

Motivation. The exercises prove the cost-weighted Bayes rule stated above, work out classification regions for non-normal densities, show how the total separation of several groups splits over Fisher's discriminants, and explain when logistic regression has no maximum likelihood estimate.

Exercise 10.1 (The minimum expected cost rule; J&W 11.3, 11.4)

For two populations with densities f_1, f_2, priors π_1,π_2 and costs $c(2\mid 1), c(1\mid 2)\ge 0$, a rule is a partition $\mathbb{R}^p = R_1\cup R_2$ and

$$\mathrm{ECM}(R_1) = c(2\mid 1)\pi_1\int_{R_2} f_1(\mathbf{x})\,d\mathbf{x} + c(1\mid 2)\pi_2\int_{R_1} f_2(\mathbf{x})\,d\mathbf{x}.$$

(a) Show that ECM is minimized by $R_1^* = \{\mathbf{x} : c(2\mid 1)\pi_1 f_1(\mathbf{x}) \ge c(1\mid 2)\pi_2 f_2(\mathbf{x})\}$.
(b) With $c(2\mid 1)=50$, $c(1\mid 2)=100$, $\pi_2=0.2$, classify an item with $f_1(\mathbf{x})=0.3$ and $f_2(\mathbf{x})=0.5$.

Solution.

(a)

Step 1. Since $\int_{R_2} f_1 = 1 - \int_{R_1} f_1$,

$$\mathrm{ECM}(R_1) = c(2 \mid 1)\pi_1 + \int_{R_1} h(\mathbf{x})\, d\mathbf{x}, \qquad h(\mathbf{x}) = c(1 \mid 2)\pi_2 f_2(\mathbf{x}) - c(2 \mid 1)\pi_1 f_1(\mathbf{x}).$$

Step 2. For any measurable R_1, splitting R_1 according to the sign of h,

$$\int_{R_1} h = \int_{R_1 \cap \{h \le 0\}} h + \int_{R_1 \cap \{h > 0\}} h \;\ge\; \int_{R_1 \cap \{h \le 0\}} h \;\ge\; \int_{\{h \le 0\}} h = \int_{R_1^*} h,$$

because adding the points of $\{h \le 0\}$ outside R_1 can only decrease the integral. Hence $\mathrm{ECM}(R_1) \ge \mathrm{ECM}(R_1^*)$. Where $f_2 > 0$ this is the ratio form $f_1/f_2 \ge \{c(1 \mid 2)/c(2 \mid 1)\}(\pi_2/\pi_1)$; points with $h = 0$ may be assigned either way.

(b) The threshold is $(100/50)(0.2/0.8) = 0.5$ and $f_1/f_2 = 0.6 \ge 0.5$, so the item is assigned to population 1, although $f_2 > f_1$: the larger prior of population 1 outweighs the doubled cost. ■

Exercise 10.2 (Triangular densities; J&W 11.7)

Let $f_1(x) = 1 - |x|$ for $|x| \le 1$ and $f_2(x) = 1 - |x - 0.5|$ for $-0.5 \le x \le 1.5$ (both 0 elsewhere), with equal costs.
(a) Find the classification regions for $\pi_1 = \pi_2$ and the total probability of misclassification.
(b) Find the regions for $\pi_1 = 0.2$.

Solution. By Exercise 10.1 with equal costs, assign x to population 1 iff $\pi_1 f_1(x) \ge \pi_2 f_2(x)$. On $[-1, -0.5)$ only f_1 is positive (population 1) and on $(1, 1.5]$ only f_2 (population 2); it remains to treat $[-0.5, 1]$, where $f_2(x) = 0.5 + x$ for $x \le 0.5$ and $f_2(x) = 1.5 - x$ for $x \ge 0.5$.

(a)

Step 1. On $[-0.5, 1]$, $f_1 \ge f_2 \iff |x| \le |x - 0.5| \iff x \le 0.25$. So $R_1 = [-1, 0.25]$ and $R_2 = (0.25, 1.5]$.

Step 2. $P(2 \mid 1) = \int_{0.25}^{1} (1 - x)\, dx = 0.75^2/2 = 0.28125$ and $P(1 \mid 2) = \int_{-0.5}^{0.25} (0.5 + x)\, dx = 0.75^2/2 = 0.28125$, so the total probability of misclassification is $\frac{1}{2}(0.28125 + 0.28125) = 0.28125$.

(b) Now population 1 needs $0.2 f_1 \ge 0.8 f_2$, i.e. $f_1 \ge 4 f_2$.

Step 1. On $[-0.5, 0]$: $1 + x \ge 4(0.5 + x) \iff x \le -1/3$.

Step 2. On $[0, 0.5]$: $1 - x \ge 2 + 4x \iff x \le -0.2$, impossible. On $[0.5, 1]$: $1 - x \ge 6 - 4x \iff x \ge 5/3$, impossible.

Step 3. Hence $R_1 = [-1, -1/3]$ and $R_2 = (-1/3, 1.5]$: the smaller prior shifts the boundary from 0.25 to $-1/3$. ■

Exercise 10.3 (Total separation and Fisher's discriminants; J&W 11.22)

For g groups with means $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_g$, common covariance $\boldsymbol{\Sigma} \succ 0$ and $\bar{\boldsymbol{\mu}} = \frac{1}{g}\sum_i \boldsymbol{\mu}_i$, let $\mathbf{B}_\mu = \sum_i (\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})(\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})^\top$ and define the total separation

$$\Delta_S^2 = \sum_{i=1}^{g} (\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}}).$$

Let $\lambda_1 \ge \cdots \ge \lambda_s > 0$ be the nonzero eigenvalues of $\boldsymbol{\Sigma}^{-1}\mathbf{B}_\mu$ and $Y_k = \mathbf{e}_k^\top \mathbf{X}$ the discriminants, with $\boldsymbol{\Sigma}^{-1}\mathbf{B}_\mu \mathbf{e}_k = \lambda_k \mathbf{e}_k$ and $\mathbf{e}_k^\top \boldsymbol{\Sigma} \mathbf{e}_l = \delta_{kl}$. Show $\Delta_S^2 = \lambda_1 + \cdots + \lambda_s$, and that the first r

discriminants carry separation $\sum_{k\le r}\sum_i(\mathrm{E}_i Y_k-\bar\mu_{Y_k})^2=\lambda_1+\cdots+\lambda_r$.

Solution.

Step 1. $\Delta_S^2=\sum_i \mathrm{tr}\{\boldsymbol\Sigma^{-1}(\boldsymbol\mu_i-\bar{\boldsymbol\mu})(\boldsymbol\mu_i-\bar{\boldsymbol\mu})^\top\}=\mathrm{tr}(\boldsymbol\Sigma^{-1}\mathbf B_\mu)=\mathrm{tr}(\mathbf A)$, where $\mathbf A=\boldsymbol\Sigma^{-1/2}\mathbf B_\mu\boldsymbol\Sigma^{-1/2}$ is symmetric and positive semidefinite.

Step 2. $\boldsymbol\Sigma^{-1}\mathbf B_\mu=\boldsymbol\Sigma^{-1/2}\mathbf A\boldsymbol\Sigma^{1/2}$ is similar to $\mathbf A$, so they have the same eigenvalues, and tr$\mathbf A$ is the sum of the eigenvalues; the zero ones contribute nothing, so $\Delta_S^2=\lambda_1+\cdots+\lambda_s$.

Step 3. Let $\mathbf A=\sum_k\lambda_k\mathbf f_k\mathbf f_k^\top$ with orthonormal $\mathbf f_k$ and $\mathbf e_k=\boldsymbol\Sigma^{-1/2}\mathbf f_k$. Then $\boldsymbol\Sigma^{-1}\mathbf B_\mu\mathbf e_k=\boldsymbol\Sigma^{-1/2}\mathbf A\mathbf f_k=\lambda_k\mathbf e_k$ and $\mathbf e_k^\top\boldsymbol\Sigma\mathbf e_l=\mathbf f_k^\top\mathbf f_l=\delta_{kl}$, so these are the discriminants: $\mathrm{Var}(Y_k)=1$ and $\mathrm{Cov}(Y_k,Y_l)=0$.

Step 4. The group means of Y_k are $\mathbf e_k^\top\boldsymbol\mu_i$ with average $\mathbf e_k^\top\bar{\boldsymbol\mu}$, so

$$\sum_{i=1}^{g}\{\mathbf e_k^\top(\boldsymbol\mu_i-\bar{\boldsymbol\mu})\}^2=\mathbf e_k^\top\mathbf B_\mu\mathbf e_k=\mathbf f_k^\top\mathbf A\mathbf f_k=\lambda_k.$$

Summing over $k\le r$ gives $\lambda_1+\cdots+\lambda_r$; for $r=s$ this is all of Δ_S^2. Thus $(\lambda_1+\cdots+\lambda_r)/\Delta_S^2$ is the share of separation kept by the first r discriminants. ■

Exercise 10.4 (Logistic regression: linearity and separation)

(a) If $\mathbf X\mid Y=k\sim N_p(\boldsymbol\mu_k,\boldsymbol\Sigma)$, $k=0,1$, show $\mathrm{logit}\,\eta(\mathbf x)=b+\boldsymbol\beta^\top\mathbf x$ with $\boldsymbol\beta=\boldsymbol\Sigma^{-1}(\boldsymbol\mu_1-\boldsymbol\mu_0)$ and $b=\log(\pi_1/\pi_0)-\frac12(\boldsymbol\mu_1+\boldsymbol\mu_0)^\top\boldsymbol\Sigma^{-1}(\boldsymbol\mu_1-\boldsymbol\mu_0)$.
(b) Suppose the data are completely separated: there is $(b_0,\boldsymbol\beta_0)$ with $z_i=b_0+\boldsymbol\beta_0^\top\mathbf x_i>0$ when $y_i=1$ and $z_i<0$ when $y_i=0$. Show that the log-likelihood $\ell(b,\boldsymbol\beta)$ has no maximizer.

Solution [Supplement].

(a) $\mathrm{logit}\,\eta(\mathbf x)=\log\dfrac{\pi_1 f_1(\mathbf x)}{\pi_0 f_0(\mathbf x)}=\log\dfrac{\pi_1}{\pi_0}-\dfrac12\{(\mathbf x-\boldsymbol\mu_1)^\top\boldsymbol\Sigma^{-1}(\mathbf x-\boldsymbol\mu_1)-(\mathbf x-\boldsymbol\mu_0)^\top\boldsymbol\Sigma^{-1}(\mathbf x-\boldsymbol\mu_0)\}$. The terms $\mathbf x^\top\boldsymbol\Sigma^{-1}\mathbf x$ cancel, leaving $(\boldsymbol\mu_1-\boldsymbol\mu_0)^\top\boldsymbol\Sigma^{-1}\mathbf x-\frac12(\boldsymbol\mu_1^\top\boldsymbol\Sigma^{-1}\boldsymbol\mu_1-\boldsymbol\mu_0^\top\boldsymbol\Sigma^{-1}\boldsymbol\mu_0)$, and $\boldsymbol\mu_1^\top\boldsymbol\Sigma^{-1}\boldsymbol\mu_1-\boldsymbol\mu_0^\top\boldsymbol\Sigma^{-1}\boldsymbol\mu_0=(\boldsymbol\mu_1+\boldsymbol\mu_0)^\top\boldsymbol\Sigma^{-1}(\boldsymbol\mu_1-\boldsymbol\mu_0)$ by symmetry of $\boldsymbol\Sigma^{-1}$.

(b)

Step 1 ($\ell<0$). Every term $y_i\log\eta_i+(1-y_i)\log(1-\eta_i)$ is negative because $0<\eta_i<1$, so $\ell(b,\boldsymbol\beta)<0$ for every parameter.

Step 2 ($\sup\ell=0$). Along $(tb_0,t\boldsymbol\beta_0)$, $t\to\infty$: if $y_i=1$, $\eta_i=1/(1+e^{-tz_i})\to1$ since $z_i>0$; if $y_i=0$, $1-\eta_i=1/(1+e^{tz_i})\to1$ since $z_i<0$. Every term tends to 0, so $\ell(tb_0,t\boldsymbol\beta_0)\to0$.

Step 3. The supremum 0 is approached but, by Step 1, never attained, so no MLE exists. Newton–Raphson then drives $\|\boldsymbol\beta\|$ to infinity; in practice one uses a penalty (e.g. ridge) or reports the separation. LDA, which models $\mathbf X$ given Y, is unaffected. ■

Chapter 11. Cluster Analysis

K-means, K-medoids, hierarchical clustering, Gaussian mixtures and the EM algorithm, choosing K

Unlike classification (Ch. 10), clustering groups similar observations without labels (unsupervised learning). A clustering is an encoder $C : \{1, \dots, n\} \to \{1, \dots, K\}$.

1 Dissimilarity and Within-Cluster Scatter

Motivation. Clustering looks for groups without labels, so we must decide what makes a grouping good. Small within-cluster scatter is the usual criterion; because total scatter is fixed, this is the same as large between-cluster scatter.

Definition 11.1: within-cluster scatter

For a dissimilarity $d(i, j) \geq 0$,

$$W(C) = \frac{1}{2} \sum_{k=1}^{K} \frac{1}{n_k} \sum_{i:C(i)=k} \sum_{j:C(j)=k} d(i, j), \qquad n_k = \#\{i : C(i) = k\}.$$

Smaller W is better.

Theorem 11.2: with squared Euclidean distance $W(C)$ is the within-cluster sum of squares

If $d(i, j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, then

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2, \qquad \bar{\mathbf{x}}_{(k)} = \frac{1}{n_k} \sum_{C(i)=k} \mathbf{x}_i,$$

and $T = \sum_i \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2 = W(C) + B(C)$ with $B(C) = \sum_k n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$.

Proof [Supplement]. Within cluster k, $\mathbf{x}_i - \mathbf{x}_j = (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) - (\mathbf{x}_j - \bar{\mathbf{x}}_{(k)})$ and $\sum_{C(i)=k} (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) = \mathbf{0}$, so

$$\sum_{C(i)=k} \sum_{C(j)=k} \|\mathbf{x}_i - \mathbf{x}_j\|^2 = 2n_k \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2;$$

multiply by $\frac{1}{2} \cdot \frac{1}{n_k}$. $T = W + B$ is the trace of the MANOVA SSP decomposition (Ch. 6). ∎

Since T is fixed, minimizing W is maximizing B.

1.1 Limits of combinatorial algorithms

The number of ways to partition n items into K non-empty clusters (Stirling number of the second kind) is

$$A(n, K) = \frac{1}{K!} \sum_{k=1}^{K} (-1)^{K-k} \binom{K}{k} k^n \sim \frac{K^n}{K!}.$$

E.g. $A(19, 4) \approx 10^{10}$: exhaustive search is impossible, so greedy iterative algorithms are used.

2 K-means and K-medoids

Motivation. Minimizing within-cluster scatter exactly is combinatorially hard. K-means alternates two easy steps, assigning points and updating centres, each of which decreases the criterion, so it converges to a local optimum.

2.1 K-means

With $d(i,j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, any point $\boldsymbol{m}$ satisfies $\sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}\|^2 \geq \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, so

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2 = \min_{\boldsymbol{m}_1,\ldots,\boldsymbol{m}_K} L(C \mid \boldsymbol{m}), \qquad L(C \mid \boldsymbol{m}) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}_k\|_2^2.$$

Definition 11.3: K-means algorithm

(i) Minimize over C with $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ fixed: update $C(i) = k$ for the closest center.

(ii) Minimize over $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ with C fixed: update $\boldsymbol{m}_k$ by the new average of cluster k.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Theorem 11.4: convergence

K-means stops after finitely many iterations at a local minimum.

Proof [Supplement]. Each step minimizes L in one block of variables with the other fixed, so L is non-increasing. There are finitely many encoders C and step (ii) is unique for a given C, so while L strictly decreases no C can recur; the algorithm stops in finitely many steps. ■

The result depends on the starting values, so run several starts and keep the smallest W. K-means is scale-dependent; consider standardizing.

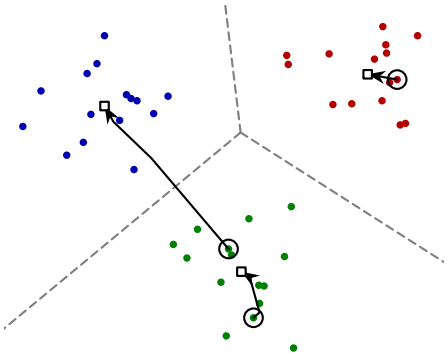


Figure. *K-means ($K = 3$) on synthetic data, 45 points from three groups. The starting centres (circles) are three observations, two of them in the bottom group. Black arrows trace the centres over the iterations until C stops changing (squares, after three updates). The dashed lines are the perpendicular bisectors of the final centres: step (i) assigns each point to the Voronoi cell of its nearest centre, so the cells are convex.*

2.2 K-medoids

Use an arbitrary dissimilarity and restrict centers to observations.

(i) Minimize over C: update $C(i) = k$ for the closest medoid $\boldsymbol{m}_k$.

(ii) Minimize over $\boldsymbol{m}_1, \dots, \boldsymbol{m}_K$: update $\boldsymbol{m}_k$ by the item in the kth cluster that minimizes the total dissimilarity to the cluster.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Using actual observations as representatives makes it robust to outliers and usable for categorical data.

3 Hierarchical Clustering

3.1 Agglomerative

Start with singletons and repeatedly merge the two closest clusters. Between-cluster distances (linkage):

Single linkage	$d(G_1, G_2) = \min_{i \in G_1, j \in G_2} d_{ij}$
Complete linkage	$d(G_1, G_2) = \max_{i \in G_1, j \in G_2} d_{ij}$
Average linkage	$d(G_1, G_2) = \text{average}_{i \in G_1, j \in G_2} d_{ij}$

Single linkage tends to produce long chained clusters, complete linkage compact round ones. The result is drawn as a dendrogram and cut at a chosen height.

Ward's method merges, at each step, the two clusters whose union increases the within-cluster sum of squares $W(C)$ the least.

Theorem 11.5: Ward's merging cost

Let $\mathrm{ESS}(G) = \sum_{i \in G} \|\mathbf{x}_i - \bar{\mathbf{x}}_G\|^2$. For disjoint clusters G, H of sizes n_G, n_H,

$$\mathrm{ESS}(G \cup H) - \mathrm{ESS}(G) - \mathrm{ESS}(H) = \frac{n_G n_H}{n_G + n_H} \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|^2.$$

Proof [Supplement].

Step 1. Applying $T = W + B$ (Theorem 11.2) to the set $G \cup H$ split into the two clusters G, H, with $N = n_G + n_H$ and $\bar{\mathbf{x}} = (n_G \bar{\mathbf{x}}_G + n_H \bar{\mathbf{x}}_H)/N$ the mean of $G \cup H$,

$$\mathrm{ESS}(G \cup H) = \mathrm{ESS}(G) + \mathrm{ESS}(H) + n_G \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}\|^2 + n_H \|\bar{\mathbf{x}}_H - \bar{\mathbf{x}}\|^2.$$

Step 2. $\bar{\mathbf{x}}_G - \bar{\mathbf{x}} = \frac{n_H}{N}(\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H)$ and $\bar{\mathbf{x}}_H - \bar{\mathbf{x}} = -\frac{n_G}{N}(\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H)$, so the last two terms equal $\frac{n_G n_H^2 + n_H n_G^2}{N^2} \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|^2 = \frac{n_G n_H}{N} \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|^2$. ■

So Ward's method is the greedy version of K-means' criterion: it is centroid distance weighted by cluster sizes, which penalizes merging two large clusters and favours compact clusters of similar size.

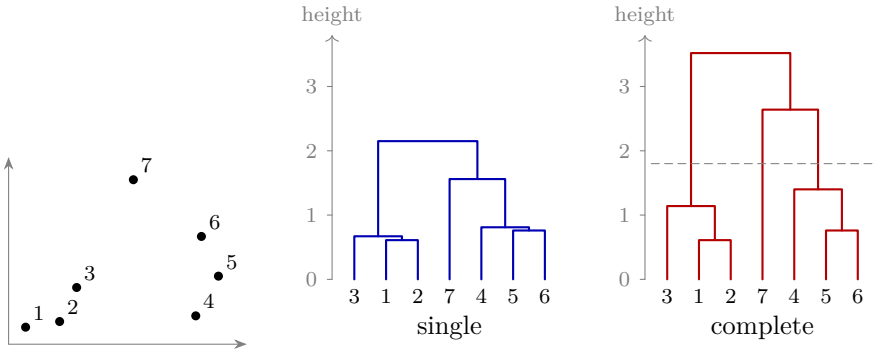


Figure. *Seven synthetic points (left) and their single- and complete-linkage dendrograms (Euclidean distance). Both merge* $\{1,2\}$ *first at height* 0.61*, but single linkage then chains the isolated point* 7 *onto* $\{4,5,6\}$ *at the smallest gap* 1.56*, while complete linkage merges* 7 *only at* 2.64*. Cutting the complete-linkage tree at the dashed height gives the three clusters* $\{1,2,3\}$*,* $\{4,5,6\}$*,* $\{7\}$*. In both trees the heights increase upwards (Exercise 11.4).*

Distances between observations:

$$p\text{-norm: } d(\mathbf{x},\mathbf{y}) = \|\mathbf{x}-\mathbf{y}\|_p = \Big(\sum_{l=1}^{d} |x_l - y_l|^p\Big)^{1/p},$$

$$\text{standardized distance: } d^2(\mathbf{x},\mathbf{y}) = \sum_{l=1}^{d} \frac{(x_l - y_l)^2}{s_l^2} = (\mathbf{x}-\mathbf{y})^\top \mathbf{D}^{-1}(\mathbf{x}-\mathbf{y}).$$

3.2 Divisive

Start from a single cluster and split repeatedly (e.g. $K = 2$ means at each step).

4 Gaussian Mixtures and the EM Algorithm

4.1 Model

$\mathbf{X}_1,\dots,\mathbf{X}_n \overset{\text{iid}}{\sim} g(\mathbf{x})$ with latent labels $\delta_1,\dots,\delta_n \overset{\text{iid}}{\sim}$, $P(\delta_i = k) = \pi_k$, and $\mathbf{X} \mid \delta = k \sim N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$:

$$\mathbf{X} \sim g(\mathbf{x}) = \sum_{k=1}^{K} \text{pdf}(\mathbf{x} \mid \delta = k) P(\delta = k) = \sum_{k=1}^{K} \pi_k g_k(\mathbf{x}),$$

$$L(\boldsymbol{\theta} \mid \mathbf{X}) = \prod_{i=1}^{n} g(\mathbf{x}_i) = \prod_{i=1}^{n} \Big\{ \sum_{k=1}^{K} g_k(\mathbf{x}_i)\pi_k \Big\},$$

giving $\hat{\pi}_k, \hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k$. Then $\hat{P}(\delta = k \mid \mathbf{x}) = \hat{g}_k(\mathbf{x})\hat{P}(\delta = k)/\hat{g}(\mathbf{x})$ and $C : \mathcal{X} \to \{1,\dots,K\}$, $C(\mathbf{x}) = k$ iff $\hat{P}(\delta = k \mid \mathbf{x}) \geq \hat{P}(\delta = j \mid \mathbf{x})$ for all j (soft clustering). With all $\boldsymbol{\Sigma}_k = \sigma^2\mathbf{I}$ and $\sigma \to 0$ this becomes K-means.

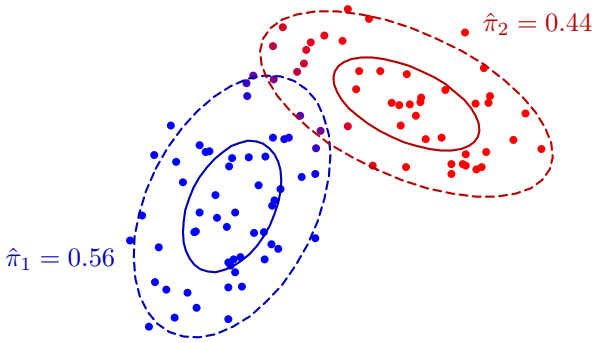


Figure. *A two-component Gaussian mixture fitted by EM to* 100 *synthetic points (*60 *and* 40 *drawn from two bivariate normals). Ellipses are the* 1- *and* 2*-Mahalanobis-distance contours of the fitted components. Each point is coloured by its responsibility* $\hat P(\delta = 1 \mid \mathbf{x})$*, from blue (*1*) to red (*0*); points in the overlap get intermediate colours, the soft assignment that* K*-means lacks.*

4.2 The EM algorithm ($K = 2$)

$\mathbf{X}_1, \dots, \mathbf{X}_n \sim \pi_1 \phi(\mathbf{x}; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \pi_0 \phi(\mathbf{x}; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, $\pi_0 = 1 - \pi_1$. The observed-data log-likelihood

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log\{\pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) + \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)\}$$

has no closed-form maximizer (log of a sum). If $\delta_1, \dots, \delta_n \in \{0, 1\}$ were observed, $L(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \prod_i \mathrm{pdf}(\mathbf{x}_i, \delta_i \mid \boldsymbol{\theta}) = \prod_i \mathrm{pdf}(\mathbf{x}_i \mid \delta_i, \boldsymbol{\theta}) \mathrm{pdf}(\delta_i)$ and

$$\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \sum_{i:\delta_i = 1} \log \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \sum_{i:\delta_i = 0} \log \pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$$

$$= \sum_{i=1}^{n} \left[\delta_i \log \pi_1 \phi_1(\mathbf{x}_i) + (1 - \delta_i) \log \pi_0 \phi_0(\mathbf{x}_i)\right],$$

which separates and gives closed forms such as $\hat\pi_1^{\mathrm{MLE}} = n_1/n$.

Definition 11.6: EM algorithm

(i) $\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n)$ has an easy MLE, but $\delta_1, \dots, \delta_n$ are unobserved;
(ii) iteratively approximate the maximum. At $\boldsymbol{\theta}^{(m)}$:

- **E-step**: predict δ_i by $\hat\delta_i = \mathrm{E}(\delta_i \mid \mathbf{x}_i, \boldsymbol{\theta}^{(m)})$, i.e. compute $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) := \mathrm{E}[\ell(\boldsymbol{\theta} \mid D) \mid D_{\mathrm{obs}}, \boldsymbol{\theta}^{(m)}]$. Since the complete-data log-likelihood is linear in δ_i,

$$\hat\delta_i = \frac{\pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})}{\pi_0^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_0^{(m)}, \boldsymbol{\Sigma}_0^{(m)}) + \pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})} \quad \text{(responsibility)}.$$

- **M-step**: $\boldsymbol{\theta}^{(m+1)} = \arg\max_{\boldsymbol{\theta}} L(\boldsymbol{\theta} \mid \mathbf{X}^n, \hat{\boldsymbol{\delta}}^n)$, i.e. $\arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$:

$$\pi_1 = \frac{\sum \hat\delta_i}{n}, \quad \boldsymbol{\mu}_1 = \frac{\sum \hat\delta_i \mathbf{x}_i}{\sum \hat\delta_i}, \quad \boldsymbol{\Sigma}_1 = \frac{\sum \hat\delta_i (\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^\top}{\sum \hat\delta_i},$$

  and similarly for component 0 with weights $1 - \hat\delta_i$.

Repeat until $d(\boldsymbol{\theta}^{(m)}, \boldsymbol{\theta}^{(m+1)})$ is small.

Theorem 11.7: monotonicity of EM

$\ell(\boldsymbol{\theta}^{(m+1)}) \geq \ell(\boldsymbol{\theta}^{(m)})$.

Proof [Supplement]. $\ell(\boldsymbol{\theta}) = Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$ with $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \mathrm{E}[\log p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta}) \mid \mathbf{X}, \boldsymbol{\theta}^{(m)}]$. By Gibbs' (Jensen's) inequality $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) \leq H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})$ (KL divergence ≥ 0). The M-step does not decrease Q, so $\ell(\boldsymbol{\theta}^{(m+1)}) - \ell(\boldsymbol{\theta}^{(m)}) = \{Q(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - Q(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} - \{H(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} \geq 0$. ■

EM may converge to a local maximum (or saddle point), and the likelihood is unbounded as some $\boldsymbol{\Sigma}_k$ becomes singular, so use several starts and regularization.

5 Choosing K and Other Topics

- **Scatter decomposition**: $W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, $B(C) = \sum_{k=1}^{K} n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$. W always decreases as K grows, so it cannot choose K by itself (elbow method: where the decrease flattens).
- Indices: $\mathrm{CI}(K, c) = W(K, c)/T$ (smaller is better) and Calinski–Harabasz $\mathrm{CH}(K, c) = \dfrac{B(K, c)/(K-1)}{W(K, c)/(n-K)}$ (larger is better; the analogue of the one-way MANOVA F).
- **Gap statistic**: compare with the expected $\log W_K^*$ under a reference distribution (uniform over the data range),

$$\mathrm{Gap}(K) = \mathrm{E}^*[\log W_K^*] - \log W_K, \qquad s(K) = \text{(adjusted) s.d. of } \log W_K^*,$$

  and choose the smallest K with $\mathrm{Gap}(K) \geq \mathrm{Gap}(K+1) - s(K+1)$.
- **Clustering variables**: define $d(v_i, v_j) = 1 - r_{ij}$ or $1 - r_{ij}^2$ and apply hierarchical clustering.

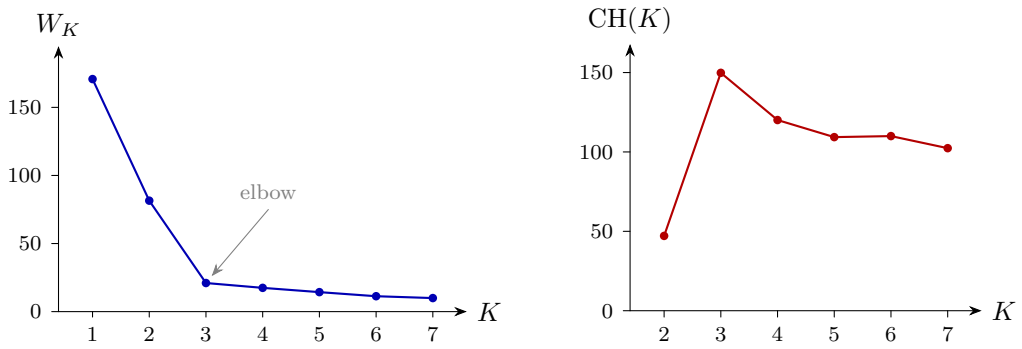


Figure. *Choosing K for the K-means data of the earlier figure (best of 20 random starts for each K). Left: W_K keeps decreasing, but the decrease flattens after $K = 3$ (elbow). Right: the Calinski–Harabasz index, which divides the between and within scatter by their degrees of freedom, has its maximum at $K = 3$.*

6 Selected exercises

Motivation. The exercises prove facts used above without proof: the count of partitions, the closed-form M-step, the unbounded mixture likelihood, and the monotone merge heights that make a dendrogram well defined.

Exercise 11.1 (Counting partitions)

Show that the number of partitions of $n \geq 1$ items into K non-empty clusters is

$$A(n,K) = \frac{1}{K!}\sum_{k=0}^{K}(-1)^{K-k}\binom{K}{k}k^n,$$

and that $A(n,K)\,K!/K^n \to 1$ as $n \to \infty$ for fixed K.

Solution [Supplement].

Step 1 (encoders). A partition into K non-empty blocks together with a labelling of the blocks by $1,\dots,K$ is the same as a surjective encoder $C : \{1,\dots,n\} \to \{1,\dots,K\}$. Each partition has $K!$ labellings, so $A(n,K) = N/K!$ with N the number of surjections.

Step 2 (inclusion–exclusion). Let E_j be the set of encoders that never use label j. For $J \subseteq \{1,\dots,K\}$ with $|J| = m$, $|\bigcap_{j\in J} E_j| = (K-m)^n$. By inclusion–exclusion,

$$N = \sum_{J}(-1)^{|J|}\Big|\bigcap_{j\in J}E_j\Big| = \sum_{m=0}^{K}(-1)^m\binom{K}{m}(K-m)^n,$$

and the substitution $k = K - m$ gives the formula. The $k = 0$ term vanishes for $n \geq 1$, which gives the form used above.

Step 3 (asymptotics). $K!\,A(n,K)/K^n = \sum_{k=0}^{K}(-1)^{K-k}\binom{K}{k}(k/K)^n \to 1$, since $(k/K)^n \to 0$ for $k < K$. For example $A(19,4) \approx 4^{19}/24 \approx 1.1 \times 10^{10}$. ■

Exercise 11.2 (Deriving the M-step)

Given responsibilities $\hat\delta_i \in (0,1)$, let

$$Q(\boldsymbol\theta) = \sum_{i=1}^{n}\big[\hat\delta_i \log\{\pi_1\phi(\mathbf{x}_i;\boldsymbol\mu_1,\boldsymbol\Sigma_1)\} + (1-\hat\delta_i)\log\{\pi_0\phi(\mathbf{x}_i;\boldsymbol\mu_0,\boldsymbol\Sigma_0)\}\big], \qquad \pi_0 = 1-\pi_1.$$

Assuming the weighted scatter matrices below are positive definite, show that Q is maximized by the M-step formulas of the EM algorithm.

Solution [Supplement].

Step 1 (separation). With $N_1 = \sum_i \hat\delta_i \in (0,n)$ and $N_0 = n - N_1$,

$$Q = \{N_1\log\pi_1 + N_0\log(1-\pi_1)\} + Q_1(\boldsymbol\mu_1,\boldsymbol\Sigma_1) + Q_0(\boldsymbol\mu_0,\boldsymbol\Sigma_0), \qquad Q_1 = \sum_i \hat\delta_i \log\phi(\mathbf{x}_i;\boldsymbol\mu_1,\boldsymbol\Sigma_1),$$

and Q_0 likewise with weights $1-\hat\delta_i$; the three parts have separate parameters.

Step 2 (π_1). $N_1\log\pi + N_0\log(1-\pi)$ is strictly concave on $(0,1)$ with derivative $N_1/\pi - N_0/(1-\pi)$, which vanishes only at $\pi = N_1/n$.

Step 3 ($\boldsymbol\mu_1$). $-2Q_1 = N_1 p\log 2\pi + N_1\log|\boldsymbol\Sigma_1| + \sum_i \hat\delta_i(\mathbf{x}_i-\boldsymbol\mu_1)^\top\boldsymbol\Sigma_1^{-1}(\mathbf{x}_i-\boldsymbol\mu_1)$. With $\boldsymbol m_1 = \sum_i \hat\delta_i\mathbf{x}_i/N_1$ we have $\sum_i\hat\delta_i(\mathbf{x}_i-\boldsymbol m_1) = \mathbf{0}$, so

$$\sum_i \hat\delta_i(\mathbf{x}_i-\boldsymbol\mu_1)^\top\boldsymbol\Sigma_1^{-1}(\mathbf{x}_i-\boldsymbol\mu_1) = \sum_i \hat\delta_i(\mathbf{x}_i-\boldsymbol m_1)^\top\boldsymbol\Sigma_1^{-1}(\mathbf{x}_i-\boldsymbol m_1) + N_1(\boldsymbol m_1-\boldsymbol\mu_1)^\top\boldsymbol\Sigma_1^{-1}(\boldsymbol m_1-\boldsymbol\mu_1),$$

minimized by $\boldsymbol\mu_1 = \boldsymbol m_1$ whatever $\boldsymbol\Sigma_1$ is (profiling, Theorem 2.26).

Step 4 ($\boldsymbol\Sigma_1$). With $\mathbf{V}_1 = \sum_i \hat\delta_i(\mathbf{x}_i-\boldsymbol m_1)(\mathbf{x}_i-\boldsymbol m_1)^\top/N_1 \succ 0$, the profiled criterion is $N_1\{\log|\boldsymbol\Sigma_1| + \operatorname{tr}(\boldsymbol\Sigma_1^{-1}\mathbf{V}_1)\} + \text{const}$. Let $\mathbf{B} = \mathbf{V}_1^{1/2}\boldsymbol\Sigma_1^{-1}\mathbf{V}_1^{1/2} \succ 0$ with eigenvalues β_j. Then

$$\log|\boldsymbol\Sigma_1| + \operatorname{tr}(\boldsymbol\Sigma_1^{-1}\mathbf{V}_1) - \log|\mathbf{V}_1| - p = \operatorname{tr}\mathbf{B} - \log|\mathbf{B}| - p = \sum_{j=1}^{p}(\beta_j - \log\beta_j - 1) \geq 0$$

by Theorem 2.15, with equality iff all $\beta_j = 1$, i.e. $\mathbf{B} = \mathbf{I}$, i.e. $\boldsymbol{\Sigma}_1 = \mathbf{V}_1$.

Step 5. The same argument with weights $1 - \hat{\delta}_i$ gives $\boldsymbol{\mu}_0$ and $\boldsymbol{\Sigma}_0$. These are the formulas of the EM algorithm: weighted versions of the MLEs of Theorem 4.7. ■

Exercise 11.3 (The mixture likelihood is unbounded)

Let $x_1, \dots, x_n$ be distinct reals and $\ell(\boldsymbol{\theta}) = \sum_i \log\{\pi\phi(x_i; \mu_1, \sigma_1^2) + (1-\pi)\phi(x_i; \mu_0, \sigma_0^2)\}$.
(a) Show $\sup_{\boldsymbol{\theta}} \ell = +\infty$.
(b) Show that if the variances are constrained to be equal, $\sigma_0 = \sigma_1 = \sigma$, and $n \geq 3$, then ℓ is bounded above.

Solution [Supplement].

(a) Fix $\pi \in (0,1)$, μ_0 and σ_0, and set $\mu_1 = x_1$. The $i = 1$ term is at least $\log\{\pi/(\sqrt{2\pi}\sigma_1)\} \to \infty$ as $\sigma_1 \downarrow 0$, while each term $i \geq 2$ is at least $\log\{(1-\pi)\phi(x_i; \mu_0, \sigma_0^2)\}$, a finite constant. Hence $\ell \to \infty$: a component collapses onto one observation, and no global MLE exists.

(b)

Step 1. Let $\delta = \min_{i \neq j} |x_i - x_j| > 0$. A mean μ_k is within distance $< \delta/2$ of at most one x_i, so for any (μ_0, μ_1) at least $n - 2 \geq 1$ observations, say x_{i_0}, satisfy $|x_{i_0} - \mu_k| \geq \delta/2$ for $k = 0, 1$.

Step 2. Every term is at most $\log\{1/(\sqrt{2\pi}\sigma)\}$, and the i_0 term is at most $\log\{1/(\sqrt{2\pi}\sigma)\} - \delta^2/(8\sigma^2)$. So

$$\ell \leq u(\sigma) := -n \log(\sqrt{2\pi}\,\sigma) - \frac{\delta^2}{8\sigma^2}.$$

Step 3. u is continuous on $(0, \infty)$ and tends to $-\infty$ as $\sigma \downarrow 0$ and as $\sigma \to \infty$, so $\sup_\sigma u(\sigma) < \infty$, which bounds ℓ. This is why EM is run with constrained or regularized covariances. ■

Exercise 11.4 (Monotone merge heights)

In agglomerative clustering let h_t be the linkage distance of the two clusters merged at step t.
(a) Show $h_1 \leq h_2 \leq \cdots$ for single, complete and average linkage.
(b) Show by example that centroid linkage, $d(G, H) = \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|$, can produce $h_2 < h_1$.

Solution [Supplement].

(a)

Step 1 (update rule). If G_1, G_2 are merged into $G = G_1 \cup G_2$, then for every other cluster H

$$d(G, H) = \min\{d(G_1, H), d(G_2, H)\}, \quad \max\{d(G_1, H), d(G_2, H)\}, \quad \frac{|G_1| d(G_1, H) + |G_2| d(G_2, H)}{|G_1| + |G_2|}$$

for single, complete and average linkage respectively, since the pairs $(i, j) \in G \times H$ are the union of $G_1 \times H$ and $G_2 \times H$.

Step 2. At step t the merged pair has the smallest distance among current clusters, so $d(G_1, H), d(G_2, H) \geq h_t$ and all distances between untouched clusters are $\geq h_t$. A minimum, maximum or weighted average of numbers $\geq h_t$ is $\geq h_t$, so every distance after the merge is $\geq h_t$, and $h_{t+1} \geq h_t$.

(b) Take the vertices of an equilateral triangle with side 1. First two vertices merge at $h_1 = 1$; their centroid is the midpoint of a side, at distance $\sqrt{3}/2$ from the third vertex, so $h_2 = \sqrt{3}/2 < h_1$ and the dendrogram has an inversion. ■